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A1 The Effect of Phenylephrine on the Cerebral Oxygenation

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Introduction: Cerebral autoregulation is defined as a constant cerebral blood flow despite changes in arterial blood pressure. However, several studies found that a α_1 -mediated induced increase in blood pressure caused a small but significant decline in cerebral oxygenation (cO_2Hb). It is unclear whether this decline in cO_2Hb is induced by the concomitant decrease in cardiac output or by activation of α_1 receptors known to be present in cerebral vessels.

Material and Methods: In 7 patients undergoing a CABG, we monitored cO_2Hb with near infra-red spectroscopy (INVOS 5100, Somanetics Corp.) during an increase in blood pressure with 100 μg phenylephrine (an α_1 -receptor agonist) and 0.3 I.E. vasopressin (a V_1 -receptor agonist) while cardiopulmonary bypass (CPB) flow was clamped at $2.1 \text{ l} \cdot \text{min}^{-1} \cdot \text{m}^2$. To determine the influence of systemic blood flow on the cerebral oxygenation, we measured cO_2Hb during a 20% increase and a 20% decrease in CPB flow.

Results: Phenylephrine and vasopressin increased blood pressure with 17 ± 7 and 17 ± 4 mm Hg (mean $\pm$ SEM), respectively. The decrease in cO_2Hb of $2\% \pm 1\%$ and $2\% \pm 2\%$ was comparable. Elevation of CPB flow increased blood pressure with 12 ± 5 mm Hg and cO_2Hb with $3\% \pm 1\%$ while a decrease in CPB flow decreased blood pressure with 13 ± 6 mm Hg and cO_2Hb with $2\% \pm 1\%$.

Conclusions: In patients on CPB, cO_2Hb follows changes in flow. The observation that an α_1 -mediated blood pressure rise caused a decrease in cO_2Hb comparable to a nonmediated α_1 -receptor blood pressure elevation, suggests that the decline in cO_2Hb cannot be explained by α_1 -receptor mediated vasoconstriction of cerebral vessels.

A2 The Influence of Prone Position on Cerebral Oxygen Saturation

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Background: Regional cerebral oxygenation (rSO_2) may be assessed noninvasively using near infrared spectroscopy (NIRS). A reliable, noninvasive, and easy to use tool would be desirable to monitor patient's cerebral oxygenation intraoperatively. Various NIRS monitors are available with different technologies. INVOS uses 2 wavelengths (730 and 810 nm) and assesses a trend value whereas Fore Sight monitor uses 4 wavelength and provides an absolute value. Therefore, the interchangeability of the 2 monitors is unclear. In the present study, these 2 monitoring systems were compared in patients turned from supine into prone position, which may provoke hemodynamic changes and may influence global and regional cerebral perfusion and oxygenation. This

aspect of the study is of additional interest because only little is known about the effect of prone position on cerebral oxygenation.

Methods: After ethical care committee approval and informed consent, 15 patients undergoing elective discectomy in prone position were included. Anesthetic management was standardized using sufentanil 0.3 $\mu\text{g}/\text{kg}/\text{body weight}$, propofol 2 to 2.5 $\text{mg}/\text{kg}/\text{body weight}$ and atracurium 0.5 $\text{mg}/\text{kg}/\text{body weight}$ for induction and sevoflurane 0.7 to 1.0 MAC for maintenance. Regional cerebral oxygenation (rSO_2) was assessed using the INVOS and Fore-Sight monitor. Mean arterial pressure (MAP), heart rate, arterial oxygen saturation (SpO_2), etCO_2 , FiO_2 , end-tidal sevoflurane concentration, and rSO_2 were measured before patient positioning, during surgical procedure and in supine position after surgery. A decrease of rSO_2 for more than 20% from baseline and a MAP under 70 mm Hg were treated after a predetermined protocol using phenylephrine bolus of 0.1 mg or volume bolus (HES 0.4/130 500 mL) as clinically required. Data were analyzed using Pearson correlation (SPSS 15.0).

Results: Positioning in prone position showed a significant influence on cerebral oxygenation measured with NIRS ($77.3\% + 9.5\%/77.8\% + 5.6\%$ INVOS/Fore-Sight before and $70.2\% + 6.9\%/71.5\% + 4.7\%$ after positioning). MAP did not change with positioning (86.6 mm Hg $\pm$ 12.5 before and 85.3 mm Hg $\pm$ 13.3 after positioning). Changes detected with INVOS or Fore-Sight cerebral oximeter during preoxygenation and positioning were comparable.

Conclusions: Prone position in the anesthetized patient evokes only a slight decrease ($< 10\%$) of cerebral oxygenation and might, therefore, be safe even for long-term application of about 2.5 hours. INVOS and Fore-Sight do not differ in measurement of cerebral oxygenation in the present study and thus it seems to be justified to use both monitoring system equally.

A3 Cerebral Oxygenation Measured by NIRS Fore-Sight Technology Reveals Adequacy of Cerebral Perfusion During Cardiopulmonary Resuscitation

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Introduction: Neurological outcome in out-of-hospital cardiac arrest (OHCA) may be influenced by cerebral perfusion during cardiopulmonary resuscitation (CPR). Near-infrared spectroscopy (NIRS) is a noninvasive monitoring method to assess cerebral oxygenation. Recent developments included the use of 4 different wavelengths to improve differentiation between oxygenated and de-oxygenated hemoglobin (Fore-Sight technology), not relying on a pulse signal anymore. This technology provides absolute cerebral tissue oxygen saturation (SctO_2) and is not just a trend-only monitor anymore. In this study, we assessed the feasibility and the additive information of obtaining and recording field SctO_2 readings in patients with OHCA.

Patients and Methods: With IRB approval, we implemented a protocol in which for every emergency call for OHCA, a Fore-Sight monitor was transported along to the scene of the OHCA (by a third person not

taking part in the medical rescue team). SctO₂ data were automatically collected every 2 seconds from arrival at the scene until end of the rescue intervention (or until transfer to the hospital). All obtained information was blinded to the medical intervention team and could by no means interfere with the medical decisions.

Results: As far as today, data on 12 patients were collected. In all patients, CPR was already started (at least 5 min) before arrival of the medical rescue team. In all 12 patients, we were able to obtain immediate and reliable bilateral SctO₂ values, readily after arrival at the scene and after application of the forehead sensors [and before return of spontaneous circulation (ROSC)]. In all 12 patients, initial SctO₂ values before ROSC were between 35% and 50% (SctO₂ threshold for cerebral ischemia is set at 55%). SctO₂ values increased significantly at ROSC, reaching SctO₂ values between 60% and 70%. In case of no ROSC, SctO₂ values did not increase, remained between 30% and 50%, and finally decreased to the lowest observed value of 15%, when the patient was declared dead. In 2 patients, ventricular fibrillation re-occurred and SctO₂ values decreased immediately below 50%, while increasing again to values above 70% after defibrillation (and after ROSC).

Conclusions: These first observations revealed that the newest NIRS technology gives useful real-time information on cerebral oxygenation (and adequacy of cerebral perfusion) during cardiac arrest and CPR. Future availability of Fore-Sight transport monitors will facilitate further investigations on the adequacy of cerebral perfusion during CPR.

A4 Low Frequency Oscillations of NIRS in Brain Injury: Relation to Flow, Intracranial Pressure and Blood Pressure

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Introduction: Spontaneous oscillations (0.05 to 0.01 Hz) of cerebral vasculature might represent autoregulatory phenomena. Characterizing this is challenging because of the nonlinear, nonstationary relationships. Pressure passive oscillations may signify impaired vasoreactivity and indices of cerebrovascular reactivity, such as the pressure reactivity index (PRx), have been described using these relationships.¹ Such methods focus on a wide frequency range (0.05 Hz–0.003 Hz) and do not characterize specific processes, which might be of physiological or pathological significance.² Near-infrared spectroscopy (NIRS) is a noninvasive optical technique that measures regional cerebral hemodynamics and identifies cerebral slow wave activity. We explored the interrelation between slow waves (0.05 to 0.01 Hz) identified by NIRS, transcranial Doppler flow velocity (FV), intracranial pressure (ICP), and arterial pressure (ABP) to understand their physiological origin.

Methods: One hour recordings were analyzed from 20 ventilated, sedated brain injured patients. Monitoring included NIRS-derived total hemoglobin index (THI; NIRO 100, Hamamatsu Photonics), FV, ICP, and ABP. Signals were compared in the time and frequency domains using wavelet-based analysis techniques.³ These included measures of coherence and phase synchronization, allowing comparison of interrelationships between signals. Synchronization through the region of interest (0.05 Hz) was correlated with the PRx.

Results: Wavelet analysis identified strong bands of coherence (> 0.5) within the 0.05 Hz to 0.01 Hz frequency range between THI, ICP, FV, and ABP in 13 of 20 patients. Figure 1 is a typical example of the dynamic relationship of wavelet coherence between THI and ABP, indicated by changes in coherence and frequency distribution with time, organized within a specific frequency band. Similar dynamic changes were present in other patients, suggesting the possibility of a vasogenic autoregulatory origin for the oscillations. Phase synchronization between ABP/THI at 0.05 Hz correlated with PRx ($r = 0.48$ $P = 0.02$) and synchronization between ABP/ICP ($r = 0.72$ $P < 0.01$).

Discussion: We have identified a frequency band where a dynamic relationship exists between THI, ICP, FV, and ABP. The physiological significance of this is unclear, but it might represent a manifestation of impaired vascular reactivity. Individualized analysis of NIRS-derived variables might provide sensitive information regarding vascular

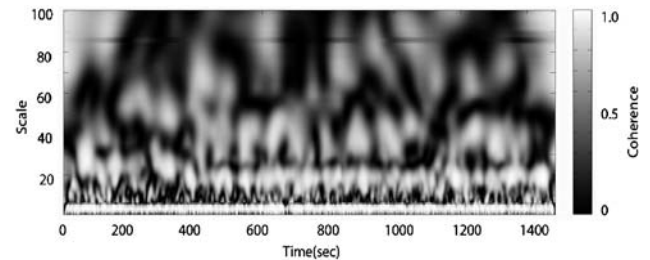


FIGURE 1. Demonstrates dynamic bands of coherence between ABP and THI using wavelet coherence. A 30 minute time period is represented on the x-axis with wavelet scale (representing increasing frequency from 1 Hz to 0.01 Hz) on the y-axis.

performance and could focus assessment to specific intracerebral processes, thereby reducing signal contamination. Synchronization between THI and ABP within the dominant activity in this frequency band may quantify the efficiency of pressure vasoreactivity, and therefore of autoregulatory processes, as suggested by its relationship to PRx. However, the complexity of these interrelationships precludes simple analysis techniques and might be best addressed by mathematical models of the cerebral circulation. Further work has the potential to deliver qualitative and quantitative information from NIRS, describing cerebrovascular performance that is of clinical relevance.

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A5 Shivering During Induced Hypothermia After Cardiac Arrest Results in Significant Decreases in Noninvasive Cerebral Oxygenation

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Introduction: Induced hypothermia after cardiac arrest improves both survival and neurological outcome. Most recently introduced cerebral oximetry techniques relying on near-infrared spectroscopy (NIRS) (Fore-Sight technology) using 4 wavelengths to allow a better differentiation between oxygenated and de-oxygenated hemoglobin. In this report, we wanted to monitor any changes in cerebral tissue oxygen saturation (SctO₂) during induced hypothermia after cardiac arrest.

Patients and Methods: With IRB approval, 15 patients were monitored during induction and maintenance (24 h) of therapeutic hypothermia (TH) (33 degrees) after successful resuscitation after cardiac arrest. Cold saline (30 mL/kg) was administered as soon as possible after hospital admission and further TH was induced by endovascular cooling (Coolgard system). All patients were sedated (propofol/remifentanyl for the duration of TH). SctO₂ monitoring was applied immediately on arrival to the hospital, before the start of TH.

Results: In 2 of 15 patients, SctO₂ values remained below 50% (40% to 50%). In both patients, SctO₂ did not change significantly during induction of TH and both patients died within first 24 hours of admission due to refractory hemodynamic shock. In the other 13 patients, starting SctO₂ values were between 60% and 70%. In 6 of 13 patients, SctO₂ values decreased below 50% at induction of TH, probably indicating an imbalance between oxygen supply and demand of the brain. This decrease in SctO₂ was not associated to a major change in any hemodynamic variable. SctO₂ values further normalized within 2 to 3 hours after induction of TH, without any specific intervention. In 8 of 13 patients, despite sedation, shivering was observed during the maintenance phase of TH and these shivering periods went along with significant decreases in SctO₂ (below 50%). Increased sedation and

administration of muscle relaxants immediately resulted in normalization of SctO₂ (between 60% and 70%). Rewarming at the end of TH (max 0.3 degrees/h) did not cause any significant change in SctO₂ values.

Conclusions: Noninvasive monitoring of cerebral oxygenation could indicate any imbalance between oxygen supply and oxygen demand of the brain during induced hypothermia. This might help to improve the neuroprotective effects of therapeutic hypothermia after cardiac arrest.

A6 Monitoring of Effect of Supplemental Oxygen Administration During Locoregional Anesthesia on Cerebral Oxygenation as Measured by NIRS (Fore-Sight Technology)

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Introduction: During surgical procedures under locoregional anesthesia, it is common practice to administer supplemental O₂ (by O₂ mask) to the patient. The Fore-Sight cerebral oximeter uses 4 wavelengths to determine absolute cerebral oxygen saturation (SctO₂) at the microvascular level. Using this new technology, few data are available on the influence of supplemental O₂ administration on cerebral oxygenation. Therefore, in this study we compared SctO₂ values with O₂ mask to SctO₂ values without O₂ mask.

Patients and Methods: With IRB approval, 25 (ASA I-II) patients scheduled for elective knee surgery under epidural anesthesia (with propofol sedation) were included. Bilateral SctO₂ monitoring was started after patient's installation (supine position) and under stable conditions (epidural analgesia and conscious sedation with propofol). For the first 15 minutes, O₂ mask (4L O₂/min) was applied. After 15 minutes, O₂ mask was then removed. However, if peripheral O₂ saturation (SpO₂) ever decreased below 90%, O₂ mask was immediately re-applied. Otherwise, after 15 minutes without O₂, O₂ mask was reinstalled. All these observations occurred under stable conditions, without any surgical intervention. In all patients, hemodynamic and respiratory parameters were monitored.

Results: Mean SpO₂ (with O₂) was 98.2% (96% to 100%) and decreased to m91.1% (80% to 96%) without O₂. Mean SctO₂ (with O₂) was 75.7% (66% to 89%) and decreased to m66.4% (56% to 81%) without O₂. Right to left difference in SctO₂ with O₂ was m3.8% (0% to 9%); without O₂ we observed a m4.3% difference (1% to 13%) between right and left side. In 4 patients, SpO₂ decreased below 90% and O₂ was immediately re-administered. In none of these 4 patients, SctO₂ decreased below 55% (= set as critical threshold for cerebral ischemia) [m61.2%, (57% to 69%)]. In the other 21 patients, SpO₂ remained above 90%, without supplemental O₂ administration, and O₂ mask was re-installed after 15-minute period, mSpO₂ increased to 98.4% (95% to 100%), whereas mSctO₂ increased to 76.1% (69% to 86%). We did not observe any significant change in hemodynamic parameters (heart rate, blood pressure) during the whole study period.

Conclusions: Supplemental O₂ administration during locoregional anesthesia (with sedation) is an effective intervention increasing peripheral as well as cerebral oxygen saturation. In normal individuals, cerebral oxygen saturation reciprocally follows changes in peripheral oxygen saturation. If oxygen is immediately re-administered whenever SpO₂ drops below 90%, no critical decreases in cerebral oxygenation (SctO₂ below 55%) could be observed.

A7 Noninvasive Monitoring of Absolute Cerebral Tissue Oxygen Saturation (Fore-Sight Technology) in Healthy Volunteers

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Introduction: Most recently introduced cerebral oximeter relying on near-infrared spectroscopy (NIRS) (Fore-Sight technology) uses 4 wavelengths to determine absolute cerebral oxygen saturation (SctO₂) at the microvascular level. Very few data are currently available on

normal SctO₂ values (using this technology) in healthy volunteers. In the present study, we wanted to assess normal SctO₂ values in healthy (awake) volunteers in supine positioning. The validation of this monitoring technology as to normal values in healthy (awake) volunteers seems extremely important as to any further clinical implication of this NIRS technology.

Material and Methods: With IRB approval and written informed consent, 102 adult (18 to 30y) ASA I volunteers were monitored by bilateral frontal SctO₂ monitoring (right and left SctO₂/2 s). Additional monitoring included pulse rate (/min), pulse oximetry (SpO₂/min), and noninvasive blood pressure monitoring (/3 min). In all volunteers, SctO₂ monitoring was started in supine position and monitored for a 15-minute period.

Results: In all 102 volunteers, we were able to obtain reliable signal monitoring (and bilateral SctO₂ values). In supine position, mean SctO₂ for all 102 volunteers (at start, mean of first minute of monitoring) was 74.25%, with SD 4.603% and range between 62% and 85%. In every individual volunteer, mean SctO₂ change over 15 minutes period was 3.08% (range between 1% and 8%). Mean right to left differences in SctO₂ for all 102 volunteers at the start (first minute) was 3.49%, with range between 0% and 9%. In every individual volunteer, mean right to left difference in SctO₂ over 15 minutes period was 2.87%, with range 2% to 10%.

Conclusions: Using the Fore-Sight technology, normal absolute cerebral oxygen saturation values in healthy volunteers are between 62% and 85%. Right to left differences are insignificant and changes in SctO₂ over a 15-minute period in every individual patient are insignificant. These validation data, obtained in healthy volunteers, could guide a better interpretation of significant changes in SctO₂ occurring in intraoperative (anesthesia) conditions.

A8 The Influence of Beach Chair Position on Cerebral Oxygen Saturation

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Background: Beach chair position (BCP) is frequently used for shoulder surgery. In anesthetized patients it may affect hemodynamic function and may thus influence systemic and regional perfusion and oxygenation. Its influence on cerebral oxygenation is discussed controversially. Adverse neurologic events have been observed in patients undergoing surgery in BCP. The etiology of these events is not yet definitely characterized, but cerebral hypoxemia can be hypothesized. Studies dealing with the effect of BCP on cerebral oxygenation show controversial results. Cerebral oxygenation can be monitored noninvasively using near-infrared spectroscopy (NIRS). The aim of the present study was to determine the influence of BCP on cerebral oxygenation. Second, 2 frequently used monitors of cerebral oxygenation, INVOS and FORE-SIGHT were compared.

Methods: After ethical care committee approval and informed consent, data were collected on 35 patients undergoing elective shoulder surgery in BCP. Anesthetic management was standardized using sufentanil 0.3 µg/kg/body weight, propofol 2 to 2.5 mg/kg/body weight and atracurium 0.5 mg/kg/body weight for induction and sevoflurane 0.7 to 1.0 MAC for maintenance in all patients. Regional cerebral oxygenation (rSO₂) was assessed using both INVOS and Fore-Sight monitor in each patient. Mean arterial pressure (MAP), heart rate, arterial oxygen saturation (SpO₂), petCO₂, FiO₂, concentration of end-tidal sevoflurane concentration, and rSO₂ were measured during induction of anesthesia, before patient positioning, during surgical procedure, and in supine position after surgery. An RSO₂ decrease of more than 20% from baseline and a MAP under 70 mm Hg were treated following a predetermined protocol using phenylephrine bolus of 0.1 mg or volume bolus (HES 0.4/130 500 mL) as clinically required. Data were analyzed using Pearson correlation (SPSS).

Results: A significant decrease in cerebral oxygenation could be observed following BCP (INVOS: from 78.5% ± 8.9% to 66.1% ± 12.7% and Fore-Sight: from 77.1% ± 4.9% to 68.7% ± 7.4%). This decrease did not correlate with changes in MAP (87.5 ± 18.2 mm Hg before and

78.3 ± 10.2 mm Hg after positioning). When changing to supine position after surgery, rSO₂ increased to baseline levels. All other measured variables were not influenced by beach chair position. Changes detected with INVOS or Fore-Sight cerebral oximeter did not differ.

Conclusions: In the present study, BCP is associated with decrease in cerebral oxygenation in anesthetized patients. However, this decrease did not reach the critical lower limit of cerebral oxygen saturation (<20% of baseline or absolute <50%). Furthermore, this decrease did not correlate with MAP changes as MAP was kept >70 mm Hg constantly to maintain adequate cerebral perfusion. It may be hypothesized that a decreasing cardiac output may be held responsible for cerebral saturation decrease in BCP. INVOS and Fore-Sight cerebral oximeter seem to give comparable result and, therefore, might be used equally.

A9 Novel Rat Model of Post-Stroke Depression Using Middle Cerebral Artery Occlusion

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Introduction: Poststroke depression (PSD) is the most frequent psychiatric complication after ischemic stroke. PSD affects up to 60% of all patients and is associated with an increase in morbidity and mortality after ischemic stroke. The pathophysiology of PSD remains elusive and appears to be multifactorial rather than purely biological or psychosocial. Thus, valid animal models of PSD would contribute to the study its etiology and treatment. This study describes a novel rat model for PSD using middle cerebral artery occlusion (MCAO).

Materials and Methods: Rats were randomly divided into 2 groups (25 animals in each group). One group was used as a sham control while the second group was subjected to MCAO using an uncoated 4-0 monofilament. The 2-way shuttle avoidance task, Porsolt forced swim test and sucrose preference test were employed to assess depression-like behavior 30 days after surgery. Localized brain expressions of BDNF protein levels were assessed to examine possible involvement of the brain neuronal plasticity in the observed behavioral syndrome. The raw data were subjected to unsupervised fuzzy clustering (UFC) algorithms to assess the sensitivity of bio-behavioral measures indicative of depressive symptoms post-MCAO operation.

Results: Fifty-six percent of the rats developed significant depressive-like behavioral disruptions as the result of MCAO, compared with 4% in the sham-operated control rats. A pattern of depressive-like behavioral response was common to all affected MCAO animals, characterized by significantly more escape failures and reduced number of total avoidance shuttles, a significant elevation in immobility duration, and reduced sucrose preference. A significant down-regulation in BDNF protein levels in the hippocampal subregions, frontal cortex and hypothalamus were observed in all affected MCAO animals. The UFC analysis supports the behavioral analysis and thus lends validity to our results.

Conclusions: The findings of this study provide evidence that long-term behavioral and molecular changes after cerebral ischemia in rats are demonstrable. Our model of depression in rats fully emulates the depression syndrome which can be observed in patients suffering from PDS. Therefore, this model of PDS contributes to both our understanding of the pathophysiology of stroke and the development of new therapies.

A10 Comparison of the Effect of Propofol-Induced EEG Silence With Hypothermia and Normothermic Controls With an Active EEG on Early Ischemic Markers During Transient Global Cerebral Hypoperfusion: A Microdialysis Study in the Rabbit Brain

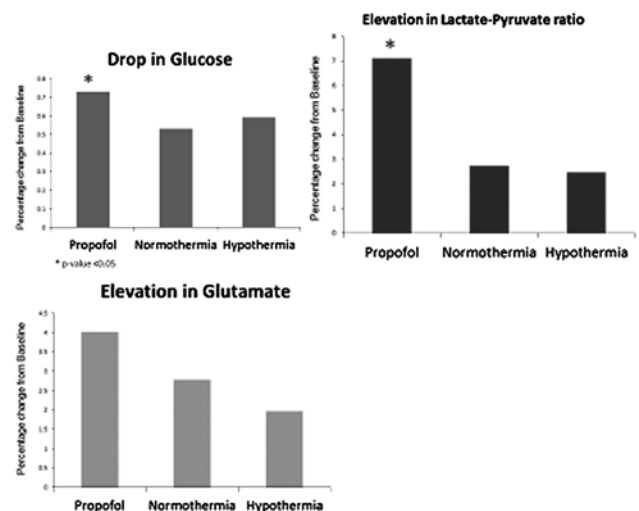
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Introduction: The neuroprotective benefit of intraoperative anesthetics is widely described and is aimed to invoke electroencephalographic (EEG) silence in anticipation of transient cerebral ischemia. The metabolic significance of achieving EEG silence during transient global cerebral hypoperfusion (TGCH) is still unclear.

Methods: The study involved New Zealand white rabbits who were anesthetized with ketamine and mechanically ventilated. The monitoring included bilateral EEG, laser Doppler probes to measure cerebral blood flow (CBF), and temperature probe in tympanic membrane. Microdialysis probes (CMA Microdialysis, Stockholm, Sweden), 4 mm membrane length, perfusion rate of 1 µL/min in a 2.5 mL syringe were placed through a hole drilled 2 mm anterior and 2 mm lateral to the bregma on either side. Thereafter, animals were randomized to normothermic controls under baseline propofol anesthesia with an active EEG, intravenous 1% propofol (20 mg) sufficient to result in EEG silence or hypothermic group (core temp 32 to 34°C). TGCH was achieved with bilateral internal carotid artery occlusion and pharmacologically induced systemic hypotension on each of the 3 randomized groups for 3 to 5 minutes. Samples collected every 10 minutes during the whole experiment were analyzed for Glucose, Lactate, Pyruvate, and Glutamate. All peak values are presented as a percentage change from the baseline values and analyzed using Student *t* test. A *P* value of ≤0.05 was considered to be significant.

Results: The 3 randomized groups completing the transient ischemia protocol included normothermia (N = 20), propofol (N = 20), and hypothermia (N = 18). Hemodynamics (heart rate, mean arterial pressure, and end-tidal CO₂) and CBF changes were comparable in all 3 groups. The earliest event was a significant drop in glucose followed by elevation in lactate-pyruvate ratio (LPR) and then a rise in glutamate in all 3 groups after TGCH that reached statistical significance. The propofol group had significantly higher drop in glucose (0.73 ± 0.06 vs. 0.53 ± 0.07 vs. 0.59 ± 0.08) and elevation in LPR (7.1 ± 2.08 vs. 2.72 ± 0.84 vs. 2.45 ± 0.67) compared with normothermic controls and hypothermia group respectively with a *P* value <0.05. Glutamate rise was found to be highest in propofol group, followed by normothermic controls and hypothermia (4.01 ± 1.5 vs. 2.78 ± 1.2 vs. 1.96 ± 0.55) but did not reach statistical significance due to high variability.

Conclusions: We observe a significant difference in the elevation of early ischemia markers (Glucose, LPR) during transient global cerebral hypoperfusion with propofol-induced EEG silence compared with hypothermia. This is consistent with previous rat survival studies showing that EEG silence was not required for full neuroprotective effects of Pentothal anesthesia.



A11 MicroRNA Influences Cell Injury in Cerebral Ischemia Through Regulating a Molecular Chaperone

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The discovery of microRNAs (miRNAs) has added a new level of posttranscriptional regulatory control to our understanding of the regulation of gene expression. These 21 to 23 nucleotide long, noncoding RNA molecules modulate protein expression by binding to complementary or partially complementary sequences commonly in the 3' untranslated region (3'UTR) of target messenger RNAs (mRNAs) causing either degradation or translational inhibition of the mRNA. Recent studies have profiled miRNAs after cerebral ischemia in animal models and humans and found extensive changes with ischemic brain injury. These findings suggest miRNAs are potential biomarkers and therapeutic targets in stroke. However, there is currently little known about individual miRNAs in this setting. We are only beginning to understand the functions of individual miRNAs. To date none have addressed the role of miRNA in the regulation of chaperones or the cellular stress response in stroke. Heat shock protein 70 (HSP70) family members are important neuroprotective proteins, so we used bioinformatics to identify miR-181 as a possible regulator of several family members. We have previously studied the protective roles of several HSP70 family members in ischemic brain injury both in vitro and in vivo. GRP78 is the HSP70 family member that is expressed in endoplasmic reticulum, and plays a critical role in the unfolded protein response. We validated GRP78/BIP as a target by dual luciferase assay. This places the 3'UTR of GRP78 after the luciferase gene so changes in expression are detected as changes in luciferase activity. As binding is via a seed sequence of 8 base pairs we used controls mutated in the seed sequence. We used a mouse focal ischemia model, transient middle cerebral artery occlusion, to assess changes in miR-181. In response to stroke we find that miR-181 increases in the core, where cells die, but decreases in the penumbra, where cells survive. We assessed protein levels by western and found that increased levels of miR-181a are associated with decreased GRP78 protein levels. As miRNA can either cause translational arrest or degradation of the mRNA we assessed levels of mRNA and found them increased. This suggests that translational arrest is the mechanism in this setting. We then manipulated levels of miR-181a using plasmid overexpression of pri-miR-181ab or mimic to increase, and antagomir or inhibitor to specifically reduce levels of miR-181. Increased miR-181 exacerbated injury both in vitro in primary astrocyte cultures subjected to glucose deprivation and increased infarct volume significantly in mouse stroke. Conversely, reduced levels were associated with reduced injury. As we previously observed that protection by overexpression of GRP78 was associated with reduced oxidative stress and preserved mitochondrial

function we assessed these parameters. We observed that protection due to reduction of miR-181 was accompanied by reduced oxidative stress and better preserved mitochondrial membrane potential. In conclusion, our data suggest that miR-181 contributes to cerebral ischemic injury. Knockdown of endogenous miR-181 provides protection against cerebral ischemia through up-regulating chaperone GRP78.

A12 Sex Differences in Neuronal Caspase-3 and Poly (ADP Ribose) Polymerase Expression After 45 Minutes of Deep Hypothermic Circulatory Arrest in Rats

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Background: Female sex is known as an independent risk factor especially within the setting of cardiopulmonary bypass (CPB).¹ Aim of the study was to investigate sex, hormonal status, and hippocampal expression of Caspase-3 and Poly (ADP ribose) polymerase after 45 minutes of CPB with deep hypothermic circulatory arrest (DHCA) in normal and neutered female and male rats.

Methods: With institutional review board approval, 40 rats were assigned to 4 groups (n = 10 each): female-normal, female-neutered, male-normal, male-neutered. At 12 weeks of age, animals were neutered or sham-operated according to group assignment. After 28 days to allow for elimination of sex hormones, rats were anesthetized, cannulated, connected to CPB, cooled down within 30 minutes (15 to 18°C) and exposed to 45 minutes of DHCA. After reinstitution of flow and rewarming (40 min), animals were weaned from CPB at 35.5°C rectal temperature. One hour after DHCA, rats were allowed to recover from anesthesia. Fourteen days after DHCA, rats were sacrificed, serum 17 β -estradiol, progesterone, and testosterone levels determined and hippocampal activated Caspase-3 and PARP assessed immunohistochemically. Data were analyzed using Kruskal-Wallis and post-hoc Mann Whitney U ($P < 0.05$).

Results: 17 β -estradiol, progesterone, and testosterone differed between groups as expected. 17 β -estradiol and progesterone levels were highest in the female-normal group. Testosterone was present in the male-normal group only. Neuronal expression of activated Caspase-3 was significantly higher in females compared with both male groups (number of total neurons: female-neutered 227.8 ± 26.3 , female-normal 235.6 ± 11.8 , male-neutered 183.8 ± 20.3 , male-normal 186.0 ± 21.3). Neuronal expression of PARP was significantly higher in males compared with both female groups (number of total neurons: female-neutered 92.9 ± 6.9 , female-normal 103.9 ± 11.2 , male-neutered 280.3 ± 45.1 , male-normal 233.7 ± 26.4).

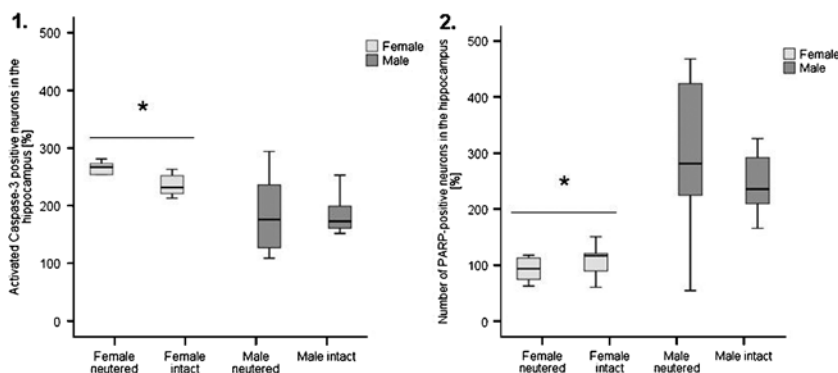


FIGURE 1. Shows the number of activated Caspase-3 positive neurons in the hippocampus at 14 days after 45 minutes of DHCA. Female groups show significantly more activated Caspase-3 positive neurons compared to male groups ($*P < 0.05$).

FIGURE 2. Shows the number of Poly (ADP ribose) polymerase (PARP) positive neurons in the hippocampus at 14 days after 45 minutes of DHCA. Female groups show significantly less PARP positive neurons compared to male groups ($*P < 0.05$).

Conclusions: Fourteen days after 45 min of DHCA, female groups equally show more activated Caspase-3 positive neurons in the hippocampus compared with male animals. Both male groups show equally elevated numbers of PARP-positive neurons in the hippocampus compared with the female groups. The expression of activated Caspase-3 and PARP seems up-regulated independent of sex hormones in this particular setting. Whether this observation is limited to the hippocampus or extends to other organs such as the heart needs to be determined by further studies.

Reference:

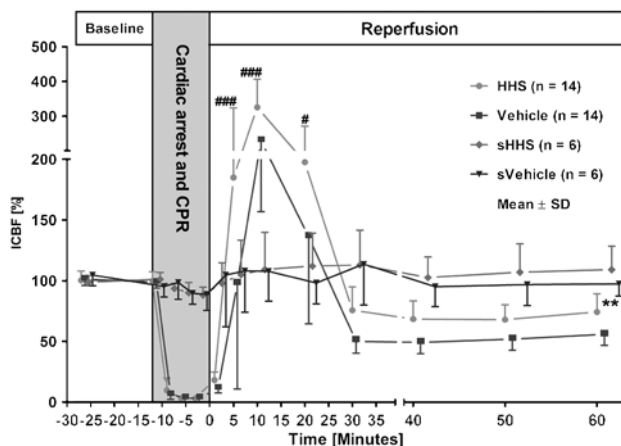
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A13 A Single Dose of Hypertonic Saline Hydroxyethyl Starch Has No Effects on Long-term Survival, Neurologic Outcome and Neurogenesis After Experimental Cardiac Arrest and CPR

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Introduction: Beneficial effects of hypertonic saline hydroxyethyl starch (HHS) on neuronal survival and on cerebral blood flow have been demonstrated in several animal models of global and focal brain ischemia.^{1,2} However, only very little is known about effects of HHS on long-term outcome and neuronal survival after CA and CPR. The present study investigated the effect of 7.5% saline plus 6% hydroxyethyl starch (HyperHAES, Fresenius, Germany) on local cerebral blood flow, long-term survival, neurologic outcome, and neurogenesis after experimental CA and CPR.

Methods: After governmental animal care committee approval (Koblenz, Germany), 40 male Sprague-Dawley rats (275 to 325 g) were included. Twenty-eight animals underwent asphyxial cardiac arrest and CPR, randomized to a group with HHS treatment [HHS, $n = 14$, bolus (4 mL/kg)] or isotonic saline treatment [Vehicle, $n = 14$ (equivalent fluid volume)]. Twelve animals received surgical intervention without cardiac arrest and CPR (sham) for technical control (sHHS, $n = 6$; sVehicle, $n = 6$). After 9 minutes of asphyxia and CA, CPR was initiated by ventilation with 100% oxygen, chest compressions, and intravenous administration of epinephrine (10 µg/kg). Experimental treatment was started 2 minutes after return of spontaneous circulation (ROSC). Local cerebral blood flow (ICBF, laser Doppler flowmetry), hemodynamic parameters, and tympanic temperature were recorded continuously during baseline, CA/CPR, and reperfusion. Survival, weight gain, and neurological assessment (neuro-deficit score, NDS) were evaluated over a 7-day observation period. Furthermore, 100 mg/kg BrdU (bromodeoxyuridine) was administered daily intraperitoneal during this period.



Histopathological neuronal injury was assessed using HE-staining (expressed as numbers of viable neurons in cortex and hippocampal formation) and neurogenesis was quantified in the dentate gyrus via BrdU/DCX (Doublecortin) immunofluorescence labeling.

Results: ICBF increased in animals treated with HHS after ROSC during the reperfusion period compared with the Vehicle group (HHS vs. Vehicle: $P = 0.0074$, Fig. 1). No differences were detected among the CA groups, also between the sham groups, concerning hemodynamic parameters and tympanic temperature. Seven animals did not survive the observation period in each of the CA groups. No animals died in the sham groups. NDS (HHS vs. Vehicle, $P = 0.2701$), neuronal injury [hippocampal formation CA1: HHS 48 (3, 165) vs. Vehicle 48 (6, 168) viable neurons, [median (min;max)]; cortex: HHS 382 (267, 720) vs. Vehicle 487 (273, 782) viable neurons] and neurogenesis (HHS 11977 (10177, 19647) vs. Vehicle 16991 (10900, 20500) BrdU/DCX positive cells) showed no differences among the CA groups.

Conclusions: Although the administration of HHS caused an improvement of ICBF after CA and CPR, no beneficial effects on long-term survival, neurological outcome and neurogenesis were detected. A pronounced neurological injury generated by this model of CA and CPR, based on clinical reality, may anticipate neuroprotective effects of HHS.

References:

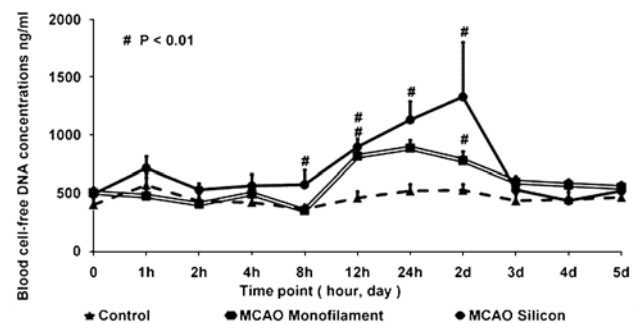
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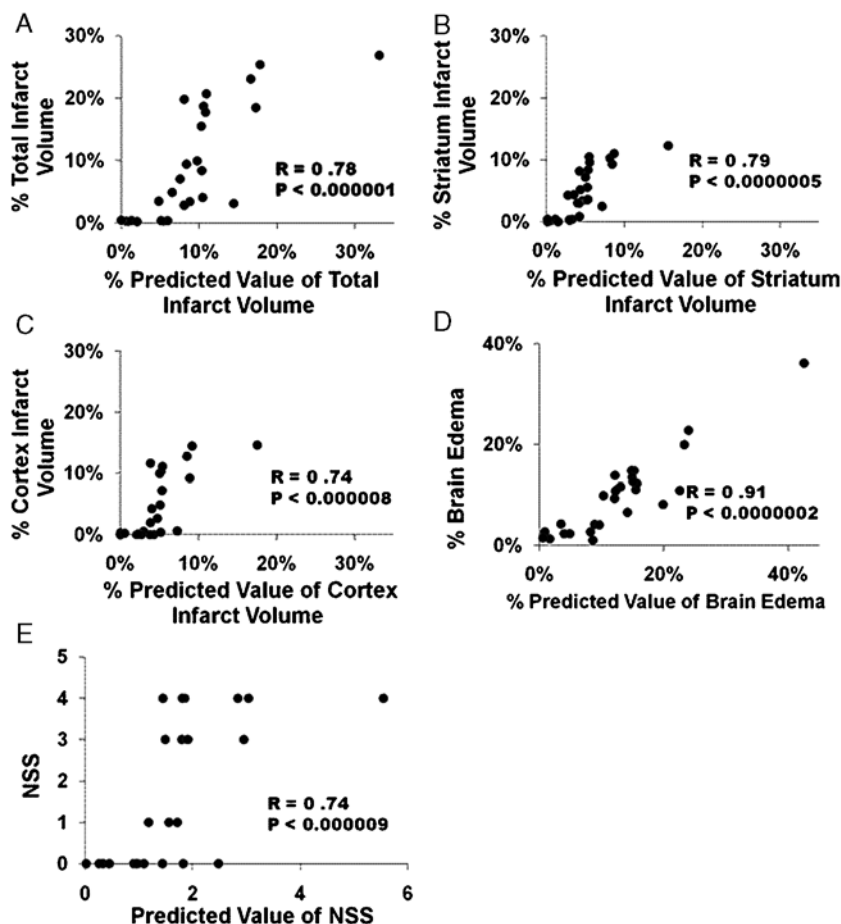
A14 Cell-free DNA: A Marker to Predict Ischemic Brain Damage in a Rat Stroke Experimental Model

Matthew Boyko, MSc, Yoram Shapira, MD, PhD, Sharon Ohayon, MD, Benjamin F. Gruenbaum, BA, Alexander Zlotnik, MD, PhD. *Soroka University Medical Center.*

Introduction: The animal model of stroke most frequently used is a rat model of focal brain ischemia caused by middle cerebral artery occlusion (MCAO). Several studies have demonstrated a link between levels of cell-free DNA (CFD) and neurological outcome in human stroke. The purpose of this study was to assess brain injury and measure CFD levels in 2 models of MCAO in rats and to determine whether brain injury correlates with CFD.

Materials and Methods: Twenty rats underwent a sham procedure, 20 rats had MCAO using a monofilament, and 20 rats had MCAO with a silicon-coated filament. Groups were further divided into 2 subgroups. In 1 subgroup of 10 rats, neurological performance (measured by a neurological severity score, NSS) was measured at 1 and 24 hours after the procedure, and brain edema and infarct volume were determined at 24 hours. To determine the correlation between histological findings and CFD levels, CFD was measured at 0 and 24 hours. In the second subgroup of 10 rats, CFD was measured at 0, 1, 2, 4, 8, 12, and 24 hours and at 2, 3, 4, and 5 days to describe the pattern of stroke-related changes in CFD levels. Neurological performance (measured by NSS) was measured at 1 and 24 hours after the procedure.





Results: CFD levels were increased at 24 and 48 hrs after the onset of MCAO and returned to baseline by 72 hours (Fig. 1). The correlation between the total infarct volume and CFD levels of the 3 groups was $R = 0.78$, $P < 0.0001$ (Fig. 2). Brain edema and NSS also were strongly correlated with CFD levels at 24 hours after MCAO ($R = 0.91$, $P < 0.0001$ and $R = 0.73$, $P < 0.0001$, respectively).

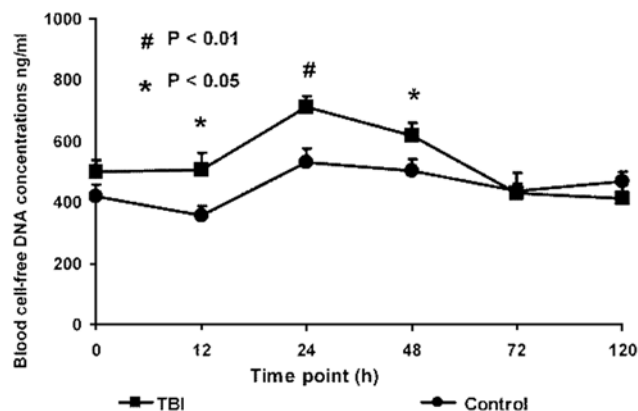
Conclusions: This study demonstrated a positive correlation between CFD levels and the extent of ischemic injury, brain edema, and neurological outcome in rats 24 hours after MCAO. Furthermore, there is a positive correlation between CFD levels and the expected temporal progression of ischemic injury. These findings suggest that CFD may be a useful biomarker for stroke in both experimental and clinical contexts.

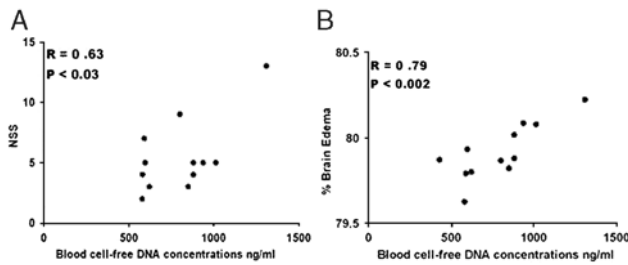
Methods: Thirty-nine Sprague-Dawley rats were randomly assigned to 2 groups: rats in group 1 (sham group) were anesthetized and had a scalp incision without TBI, and rats in group 2 were anesthetized and had a scalp incision with TBI. A neurological severity score (NSS) was assessed at 1, 24, and 48 hours after TBI. CFD was measured via blood samples drawn at $t = 0$ (baseline), 12, 24, 48, 72, and 120 hours after TBI. At 48 hours after TBI, brain edema was determined in a subgroup of 11 rats by calculating the difference between rats' wet and dry brain weight.

A15 The Role of Cell-free DNA as a Prognostic Indicator of Neurological Outcomes After Traumatic Brain Injury in Rats

Akiva Leibowitz, MD, Matthew Boyko, MSc., Sharon Ohayon, MD, Benjamin F. Gruenbaum, BA, Alexander Zlotnik, MD, PhD. Soroka University Medical Center.

Introduction: Traumatic brain injury (TBI) is a major cause of morbidity and mortality, and early predictors of neurological outcomes are of great clinical importance. Cell-free DNA (CFD), a biomarker used for the diagnosis and monitoring of several diseases, has been implicated as a possible prognostic indicator after TBI. The purpose of this study was to determine the pattern and timing of CFD levels after TBI, and whether a relationship exists between the level of CFD and brain edema and neurological outcomes.





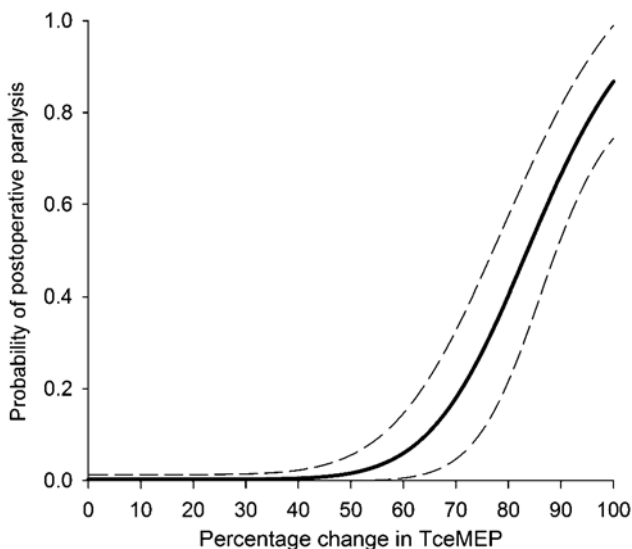
Results: Compared with baseline levels, the CFD levels in rats subjected to TBI were significantly increased at 24, and 48 hours after TBI, returning to baseline levels by 72 hrs after TBI ($P < 0.01$, $P < 0.05$, respectively; Fig. 1). A strong positive correlation was demonstrated between CFD levels and extent of brain edema ($r = 0.63$, $P < 0.05$; Fig. 2B), as well as between CFD levels and the NSS ($r = 0.79$, $P < 0.005$; Fig. 2A).

Discussion: In this study, we demonstrated an increase in CFD levels after TBI, as well as a correlation between CFD levels and brain edema and NSS. CFD levels may provide a quick, reliable, and simple prognostic indicator of neurological outcome after TBI. Its role in humans has not been clearly elucidated, but has potentially significant clinical implications.

A16 Redefining the Warning Criteria for Transcranial Electric Motor Evoked Potential

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Introduction: Changes of signal amplitude in transcranial electric motor evoked potential (TceMEP) monitoring are commonly used to assess the integrity of the spinal cord during surgery. However, a wide range of values (50% to 100% decrease) has been proposed to indicate significant neurologic injury. In a porcine model of spinal cord injury, we aimed to define the minimum change in TceMEP amplitude, above which irreversible postoperative weakness will occur.



Methods: The study was approved by the Animal Experimentation Ethics Committee. TceMEP monitoring was established in 31 pigs (25 to 30 kg) during propofol infusion using high voltage cortical stimulator. The motor cortex was stimulated by a train of 5 pulses of electrical current at 2 ms intervals through insulated screws placed transcranially. Compound muscle action potentials were recorded from quadriceps muscles. After baseline recordings, direct spinal cord injury was created using a purposely designed weight-loading compression device applied to the exposed spinal cord at T12 level for 30 minutes. After compression was released, another 30 minutes was allowed before the animals were recovered from anesthesia. TceMEP was monitored throughout surgery. Changes of TceMEP amplitude were expressed as percentage of baseline. This was correlated with postoperative motor function (according to Tarlov scale¹) using logistic regression.

Results: Reproducible TceMEPs were recorded in all animals. Nine (29%) animals had postoperative paralysis (Tarlov score ≤ 2). The maximum decrease in TceMEP amplitude was higher in paralyzed animals, median (range): 100% (91% to 100%), than those without, 38% (9% to 64%), $P < 0.001$ (Mann-Whitney test). Time to maximum decrease in TceMEP amplitude after injury was also faster in the paralyzed animals, 1 (1 to 11) minutes versus 13 (1 to 30) minutes, $P < 0.001$. The risk of postoperative paralysis increased when TceMEP amplitude was decreased by $\geq 60\%$ (Fig. 1).

Conclusions: There was substantial risk of postoperative paralysis when TceMEP amplitude was decreased by $\geq 60\%$ of baseline. A deterioration of TceMEP, reaching its minimum value < 5 minutes, and a lack of signal recovery > 30 minutes after injury implied poor prognosis with postoperative paralysis.

Reference:

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A17 Evaluation of Feasibility and Reliability of Visual Evoked Potentials Monitoring During Neurosurgical Procedures

Kyoko Hasuwa, MD, Masahiko Kawaguchi, MD, Hironobu Hayashi, MD, Ryuichi Abe, MD, Hitoshi Furuya, MD. *Nara Medical University.*

Introduction: The visual evoked potentials (VEPs) monitoring can be used to prevent the postoperative visual dysfunction during the neurosurgical procedures, in which the optic pathway is at a risk of injury. However, it has been difficult to obtain the stable VEP recording due to the influences of anesthetic agents and surgical procedures. Recently, to improve the feasibility and reliability of VEP monitoring, we have introduced a new light-stimulating device to improve the retinal light stimulation and electroretinography (ERG) to ascertain the arrival of the stimulus at the retina. The present study was conducted to investigate the feasibility and reliability of VEP monitoring, using a combination of a new light-stimulating device and ERG, during the neurosurgical procedures.

Methods: After the institutional approval and written informed consent were obtained, 25 patients between January 2010 and March 2011 were enrolled in this study. All patients were at a risk for postoperative visual impairment. Anesthesia was maintained with propofol and fentanyl/remifentanyl. For the VEP monitoring, the new light-stimulating device (Unique Medical, Tokyo, Japan), which consists of 16 red high-luminosity LEDs embedded in a soft silicone disc, was placed on both closed eyelids. The luminosity of the device is changeable from 500 to 20,000 Lx. The VEP electrodes were inserted at the Oz, Ro and Lo (international 10 to 20 method) and at the A1 and A2 (reference). For ERG monitoring, the needle electrodes were inserted subcutaneously at the lateral canthus and the reference electrodes were placed in the contralateral canthus. A signal processor (Neuropack, Nihonkoden, Tokyo, Japan) was used to record the VEP. For VEP assessment (amplitudes and latency), the largest positive peak that appeared around 120 ms after the stimulus onset was used. More than 10 ms prolongation in VEP latency or 50% decrease in VEP amplitude compared with the control level was defined as “VEP change.” VEPs were recorded at the beginning of operation (control), during the surgical procedures and at the end of the operation. Assessments of visual function were performed by ophthalmologists before and after the surgery.

Result: Stable ERG and VEPs data could be obtained in 24 of 25 patients (96%), but not in 1 patient with severe preoperative visual dysfunction. Of 24 patients with the monitoring of VEPs, VEPs were changed in 6 patients and remained unchanged in 18 patients. In 22 patients, postoperative visual function tests could be performed. Visual function was postoperatively deteriorated in 3 patients (quadrantanopsia, scotoma central or trivial bitemporal hemianopia). To predict the postoperative deterioration of visual function, the sensitivity and specificity of VEP monitoring were 67% and 79%, respectively. Positive and negative predictive values were 33% and 95%, respectively.

Conclusions: The results in the present study indicated that the intraoperative VEP monitoring using a combination of a new light-stimulating device and ERG was feasible except the patient with severe preoperative visual dysfunction. Further study with more patients would be required to conclude the reliability of VEP monitoring.

A18 Effect of Adenosine on Transcranial Motor Evoked Potentials During Intracranial Aneurysm Clipping

Laura B. Hemmer, MD, Vijay K. Ramaiah, MD, John F. Bebawy, MD, Antoun Koht, MD, Dhanesh K. Gupta, MD. *Northwestern University Feinberg School of Medicine.*

Introduction: Adenosine-induced flow arrest is an alternative technique to facilitate clip ligation of complex intracranial aneurysms and as a rescue maneuver during intraoperative rupture.¹ However, unlike temporary arterial occlusion that can produce focal cerebral ischemia, adenosine has the potential to produce transient global cerebral ischemia. Transcranial motor evoked potentials (MEPs) is a sensitive modality for cerebral ischemia monitoring. We reviewed peri-adenosine administration MEP amplitudes to evaluate if global hypoperfusion from adenosine elicited cerebral ischemia was detected by a change in MEPs.

Methods: An IRB approved intracranial aneurysm database was reviewed to identify patients who received adenosine for aneurysm clip ligation. Patients received an initial dose of adenosine (approximately 0.34 mg/kg ideal body weight) as previously described.¹ MEP tracings were generated by multipulse transcranial electrical stimulation of each hemisphere and recorded from the corresponding contralateral upper and lower extremities. From MEPs recorded from within 15 minutes prefinal and 15 minutes postfinal adenosine administration, percentage change in peak-to-valley amplitude (pre-post) was analyzed for each extremity within groups stratified by presence or absence of a new postoperative motor neurologic deficit. Non-normally distributed percentage change in MEP amplitude data was described by the median and 25% to 75% confidence intervals. Homogeneity of the percentage change for each limb in each deficit group was tested with the Wilcoxon signed rank test. Difference between deficit groups for each limb was determined with the Mann-Whitney *U* rank-sum test. $P < 0.05$ was considered statistically significant.

Results: Of our 35 patients who underwent adenosine-induced flow arrest, 5 had a motor neurological deficit on postoperative day 1.

Variation in percentage change in MEP amplitude of each limb for each deficit group was not statistically different (Table 1 *P* values in each individual cell). Percentage change in MEP amplitude for each limb pre-adenosine and post-adenosine administration was also not different between deficit groups (*P* values in Table column).

Discussion: The sensitivity of MEP monitoring for changes in cerebral blood flow to motor pathways allows MEP monitoring to be an early warning of potential ischemic injury compared with other neurophysiological monitoring modalities employed in aneurysm surgery.² Although global cerebral hypoperfusion from adenosine-induced flow arrest may produce cerebral ischemia, in this small observational study, we were unable to identify a relationship between percentage change in MEP amplitude (pre-post adenosine) and development of a new motor deficit. In addition, no individual extremity was more sensitive to adenosine-induced flow arrest cerebral hypoperfusion. Larger patient samples are required to confirm our findings of the relative neurological safety of adenosine-induced flow arrest.

Conclusions: Global cerebral hypoperfusion from adenosine-induced flow arrest used to facilitate intracranial aneurysm clipping does not appear to cause a significant change in transcranial motor evoked potential amplitudes.

References:

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A19 Effects of Tetanic Stimulation of Median Nerve Prior to Transcranial Stimulation on the Threshold of Myogenic Motor Evoked Potentials During Craniotomy

Ryuichi Abe, MD, Masahiko Kawaguchi, MD, Hironobu Hayashi, MD, Kyoko Hasuwa, MD, Satoki Inoue, MD, Hitoshi Huruya, MD. *Nara Medical University.*

Introduction: To elicit myogenic motor evoked potentials (MEPs) during craniotomy, transcranial or direct electrical stimulation can be applied. Both techniques have advantages and disadvantages. Especially, false negative result of MEPs elicited by transcranial stimulation is one of the major issues, which should be resolved. Transcranial stimulation at a high intensity can stimulate the brain tissues located at the deeper parts beyond the targets surgically manipulated. So an attempt to reduce stimulus intensity for MEP monitoring is an important clinical challenge. Recently, we developed a new MEP technique, called as “posttetanic MEP (p-MEP),” in which tetanic stimulation of peripheral nerve is applied before transcranial stimulation.¹ By using p-MEP, MEP amplitude is enlarged compared with that of conventional MEPs. We hypothesized that application of tetanic stimulation before transcranial stimulation may reduce the threshold to elicit myogenic motor evoked potentials. A reduction of threshold may contribute to a reduction in the false negative rates of MEP monitoring by reducing the stimulus intensity required for intraoperative monitoring. In this preliminary study, we investigated the effects of tetanic stimulation of median nerve before transcranial stimulation on the threshold of myogenic MEPs during craniotomy.

Methods: After IRB approval and informed consent, 21 patients undergoing elective craniotomy were enrolled in this study. In all patients, anesthesia was maintained with remifentanyl, fentanyl, and a target-controlled infusion of propofol. For conventional MEP (c-MEP), transcranial train-pulse stimulation was delivered to C3 (or C4; International 10 to 20) as an anodal stimulation and the compound muscle action potentials were bilaterally recorded from the abductor pollicis brevis (APB), abductor hallucis, and tibialis anterior muscles. For p-MEP, tetanic stimulation (50 Hz, 50 mA of stimulus intensity) for 5 seconds was applied to the median nerve at the contralateral side to craniotomy with an interstimulus interval of 1 second (between the end of tetanic stimulation and transcranial stimulation). Threshold (T; V) was defined as a stimulus intensity when MEP from the contralateral APB could be elicited with an amplitude of more than 30 mV. T values of c-MEP and p-MEP (Tc, Tp, respectively) were compared. MEP measurements were performed at an intensity of T + 20 V. Amplitude and latency of p-MEPs elicited at an intensity of Tp + 20 V or Tc + 20 V were compared with those of c-MEP at an intensity of Tc + 20 V.

Attachment:

MEP Monitored Location	Neuro Intact Group (N=30) MEP Amplitude % Change Pre - Post Adenosine Median (25%, 75% CI)	Neuro Motor Deficit Group (N=5) MEP Amplitude % Change Pre - Post Adenosine Median (25%, 75% CI)	P Value (Between Deficit Groups)
Right Upper Extremity	2.7% (-30.6%, 39.2%) p=0.681	1.1% (-29.7%, 27.8%) p=1.000	0.906
Right Lower Extremity	15.3% (-17.4%, 25.2%) p=0.244	16.4% (-84.9%, 38.9%) p=1.000	1.000
Left Upper Extremity	-0.9% (-48.3%, 51.9%) p=0.812	11.1% (-191.2%, 15.7%) p=0.625	0.906
Left Lower Extremity	3.1% (-29.6%, 50.5%) p=0.762	6.2% (-77.4%, 50.2%) p=1.00	0.795

MEP=motor evoked potential

Results: In all patients, c-MEPs and p-MEPs could be recorded without any complications. Tp values were significantly lower compared with those of Tc (117 ± 46 , vs. 125 ± 47 V, respectively; $P < 0.0001$). Amplitudes of p-MEP at Tp+20 V were similar to those of c-MEP at Tc+20 V, whereas amplitudes of p-MEP at Tc+20 V were significantly higher compared with those of c-MEP at Tc+20 V. Latencies of p-MEP at Tp+20 or Tc+20 V were similar to those of c-MEP at Tc+20 V.

Conclusions: Application of tetanic stimulation to median nerve before transcranial stimulation reduced the threshold to elicit myogenic MEPs in patients undergoing craniotomy. Validity of such reduction in threshold needs to be elucidated.

A20 Post-tetanic Motor Evoked Potentials can Detect Spinal Cord Ischemia and Have no Harmful Effects on Ischemic Spinal Cord in Rabbits

Masahiko Kawaguchi, MD, Yuri Yamamoto, MD, Hironobu Hayashi, MD, Ryuichi Abe, MD, Hitoshi Furuya, MD. *Nara Medical University.*

Introduction: Recently, posttetanic motor evoked potentials (p-MEP) have been developed to augment MEPs under general anesthesia. For p-MEP recording, tetanic stimulation of peripheral nerve is applied before transcranial stimulation. Application of tetanic stimulation induces the excitation of nervous systems and therefore may affect MEP accuracy to detect spinal cord injury and susceptibility to neuronal injury after spinal cord ischemia (SCI). However, there has been no data available regarding p-MEP monitoring in a model of SCI. The present study was therefore conducted to investigate whether p-MEP can detect SCI as well as conventional-MEP (c-MEP) and whether p-MEP monitoring has harmful effects on ischemic spinal cord injury in rabbits.

Methods: Thirty male rabbits were randomly assigned to 1 of 3 groups ($n = 10$ /each). In p-MEP group, p-MEPs were recorded from 30 minutes before an aortic occlusion to 1 hour after reperfusion. In c-MEP group, c-MEPs were recorded in the same way. In no MEP group, MEP recording was not performed. SCI was induced by an occlusion of infrarenal aorta for 13 minutes. MEP responses were recorded from muscles at proximal and distal sites to SCI. Hind limb motor function was assessed at 2 hours and 2, 4, and 7 days after reperfusion using Tarlov score. For histological assessments, gray and white matter injury was evaluated 7 days after reperfusion using the number of normal neurons and the extents of vacuolations in the white matter, respectively.

Results: After an aortic occlusion, the amplitudes of p-MEPs and c-MEPs from distal site were reduced similarly and latencies of p-MEPs and c-MEPs from distal site were prolonged, whereas the amplitudes and latencies of p-MEPs and c-MEPs from proximal site remained unchanged. There were no significant differences in Tarlov scores, the number of normal neurons, and the percentage areas of vacuolations among the 3 groups.

Conclusions: The results indicated that p-MEP could detect SCI as well as c-MEP, and both p-MEP and c-MEP monitoring had no harmful effects on ischemic spinal cord in rabbits.

Reference:

1. *Anesthesiology*. 2003;99:1112–1117.

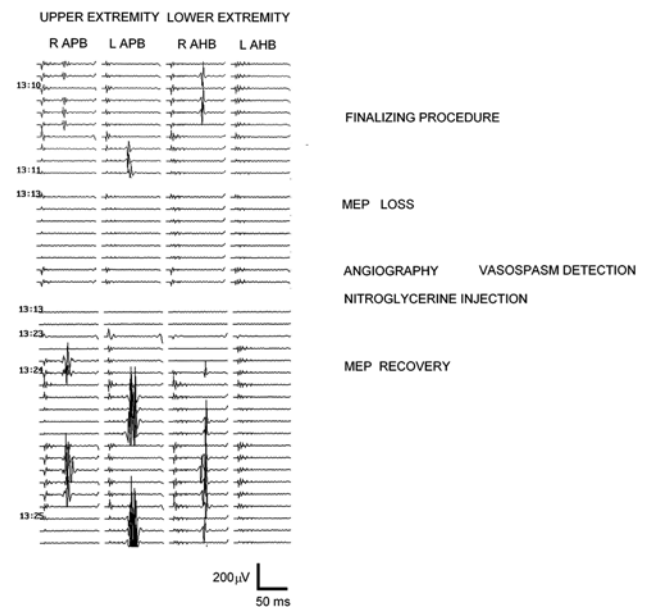
A21 Loss of Motor Evoked Potentials at the End of Embolization of Posterior Communicating Artery Aneurysm in a 4-Year-Old Girl

Amitabh Bhandari, MD, Jolie Narang, MD, Yasunari Niimi, MD, Svetlana Lenina, MD, Sedat Ulkatan, MD. *St. Lukes-Roosevelt Hospital Center.*

Case Report: A 4-year-old girl presented for cerebral angiogram and embolization of a right posterior cerebral artery (P1 segment) aneurysm. She had no past medical history, but developed an acute onset of right eye ptosis 4 days before admission. Diagnosis was acute otitis media with ocular spread. Amoxicillin yielded no improvement so an emergent computed tomography scan was ordered showing no intracranial pathology. Further imaging with magnetic resonance imaging (MRI) revealed a right posterior cerebral artery (P1 segment) aneurysm. The decision was made for a cerebral angiogram and embolization. The

patient had intravenous access before induction of anesthesia, which was placed for the diagnostic MRI. Induction was done with a combination of fentanyl, midazolam, and propofol. Endotracheal intubation was unremarkable, and arterial cannulation was done along with standard monitors placed. Maintenance of anesthesia was with TIVA (remifentanyl and propofol infusions). Neurophysiological monitoring (NPM) included cortical somatosensory evoked potentials (SEP) from the upper (median nerve) and lower (posterior tibial nerve) extremities and motor evoked potentials (MEP) elicited via transcranial and direct cortical stimulation recorded from abductor pollicis brevis (APB) bilaterally and extensor (RE) muscles from the right arm. Intraoperative monitoring demonstrated smaller MEP amplitudes on left upper and lower extremities, prolonged latency, and higher thresholds than on the right side. SEP of the left leg was significantly worse than the right side. The aneurysm was successfully embolized and a 3-dimensional angiogram confirmed complete occlusion of the aneurysm. At closing baselines, the left lower extremity MEPs decreased in amplitude with significant increment of thresholds. The same was followed by loss of MEPs in all extremities for approximately 10 minutes. At this time, the left tibial SEPs amplitude decreased significantly while upper extremities and right tibial SEPs remained stable. Repeat angiography was done to ascertain the cause of these changes. A decrease in caliber of the distal basilar artery was noted, possibly due to a vasospasm. Total, 25mcg of nitroglycerin was administered intra-arterially and systolic arterial blood pressure rose from 70 to 100 mm Hg. This resulted in recovery of MEPs in the upper extremities bilaterally, however lower extremity MEPs recovered asymmetrically with the left being worse. The patient woke up without any new neurological deficits. The patient remained stable throughout hospital stay and was discharged with improvement in right eye ptosis. In this case, unexpected changes in NPM at the end of the procedure urged prompt intervention. A change in MEP has to be investigated as it is very rare to get false negative results. Loss of MEP, could be explained by the mechanical presence of a catheter in a perforator distal to basilar artery, embolism, or vasospasm. On the basis of this criterion, the loss of MEP modalities prompted preventive measure. After reviewing the films, the decrease in caliber of basilar artery was 30%, which is less likely to explain vasospasm, but could also be drifting of the catheter in the perforator. The duration of changes of greater than 10 minutes could have been significant and could have caused a permanent disability.

MOTOR EVOKED POTENTIALS



A22 Intraoperative Neurophysiological Monitoring Poses Needlestick Injury Risk to Operating Room Personnel and Patient

A. Elisabeth Abramowicz, MD, Allison B. Spinelli, DO, Matthew Wecksell, MD, Singh Nair, PhD, Michela Catalano, MD. *Montefiore Medical Center.*

Introduction: Intraoperative multimodality neurophysiological monitoring (NPhm) may cause needlestick injury risk. For MEP monitoring, nonhollow disposable stainless steel subdermal needle electrodes, 27 G (0.4 mm in diameter) and 13 mm long (Axon Systems) are inserted subcutaneously over muscles with the anesthetized patient in the supine position. They are taped over with a 2-inch wide, 2-inch long silk tape layer; a stress loop is taped to the skin below the tape overlying the needle. OR personnel come in contact with an anesthetized patient's skin during prone positioning. Occupational exposure from needlesticks is frequently underreported, especially by physicians.^{1,2} To assess the prevalence of NPhm needlestick injuries and their importance as an occupational exposure risk factor, we performed the following study.

Methods: In February 2010 we conducted a survey of 25 participants: 6 members of the monitoring group (Neurology), 8 neuro and orthopedic anesthesiologists, 5 neurosurgery attendings and senior residents, and 7 neurosurgical OR nurses. Years in practice ranged from 3 to 25. In addition, in April 2011, we manually searched the Employee Health Services needlestick reports for NPhm-related events.

Results: The survey identified 9 NPhm contaminated needlesticks; 5 occurred during patient positioning, 2 during needle removal and 2 during placement. Seven out of 8 anesthesiologists surveyed recalled needlestick injuries; 1 had 3 NPhm needlesticks. Two out of 5 neurologists had needlesticks—all 5 injuries were related to NPhm needles. Three of 5 neurosurgeons and 2/7 nurses had needlesticks. One neurosurgeon had an NPhm needlestick. No nurses' exposure was due to NPhm needles. Between January 2010 until April 2011, there were 5 needlestick injuries recorded by Employee Health Services that were caused by contaminated (ie, previously implanted) neurophysiological monitoring needles.

Discussion: The NPhm needlestick injuries are due to either a loose needle (on insertion, positioning, during/after removal), or a protrusion of the tip of the implanted needle during skin stretch during the turn to prone position. It then retracts back under the patient's skin, exposing the patient to the provider blood. The majority of NPhm needlesticks in the survey (5/9) occurred during patient positioning. At least 3 resulted in cross-contamination. To our knowledge, this has not been previously reported. The NPhm needlestick protocol now addresses this aspect of the exposure and prospectively tracks it. The findings have several implications: neurophysiological monitoring needles cause a risk of occupational exposure and to patients. Postexposure prophylaxis for patients must be considered. Nontangential angle of insertion of NPhm needles may minimize the risk of skin puncture by the embedded needle. Covering the skin over the needle with puncture-resistant material might lower the risk of needle protrusion, provider exposure and retraction of the provider-contaminated tip under the patient's skin.

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A23 Comparison of Clinician Practices for Measuring Cerebral Perfusion Pressure: A Review of the Literature and Survey of Members of the Neurocritical Care Society

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Introduction: Cerebral perfusion pressure is an integral parameter to the neurointensivist. Despite its importance, we suspected variance in the clinical practices used to measure it. In particular, due to the effects of gravity, MAP measured relative to the level of the heart may indicate higher arterial blood pressure experienced at the level of the head by up to 14 mm Hg (1 mm Hg for each 1.25 cm vertical distance)¹ in a 6-foot tall person with the head of the bed elevated to 45 degrees. We therefore examined the practices of those authors whose papers are cited by the Brain Trauma Foundation (BTF) guidelines for CPP, and

surveyed members of the Neurocritical Care Society (NCS) about their practices.

Methods: The papers cited by the BTF for CPP guidelines were examined to determine whether the MAP was measured relative to the heart or tragus. If descriptions were missing, authors were contacted via email. A survey was also sent to the members of the NCS via email with regard to their practices.

Results: The BTF cites 11 papers in their guidelines for the management of CPP. In 3 papers, MAP was measured relative to the heart; in 3 papers (2 patient cohorts), the tragus was used; in 1 the patients were nursed flat; and in the remaining 4, the authors could not be reached. The 2 groups that reported worse outcomes with CPP < 60 mm Hg measured MAP relative to the right atrium. There were 235 respondents from 30 different countries to our email survey. This included 180 MDs, 4 DOs, 17 CRNPs, 4 ARNPs, 25 RNs, 2 PAs, 1 PharmD, and 2 others. Forty-one percent reported measuring MAP at the level of the tragus, and 59% measured at the level of the heart. No statistically significant differences were seen when the MD, CRNP, and RN subgroups were compared.

Discussion/Conclusion: This survey indicates that a foundational element of neurocritical care is not followed in a uniform manner in both the cited evidence and in the clinical practices of CPP management. The magnitude of difference in CPP that can arise can produce a CPP > 14 mm Hg lower than the actual brain level optimal or lower limit of autoregulation. With a heart-level sitting MAP, the actual CPP may be < 45 mm Hg. The importance of this is underscored by (a) Drummond² suggestion that the normal lower limit of autoregulation (LLA) may be > 60 mm Hg. (b) Observations of optimal CPP in humans with TBI to be 60 to 85 mm Hg.³ (c) Suggestions that the LLA may be elevated to > 60 mm Hg in TBI.^{3,4} (d) Reports of stroke in sitting shoulder surgery with (heart level) MAP in the 60 mm Hg range,¹ and (e) BTF-cited articles regarding use of CPP tend to have worse outcomes when lower CPP is based on heart level blood pressure measurement. Accordingly, we argue that the possibility exists of widespread induction of cerebral hypoperfusion based on imprecisely described and adopted practices regarding CPP management. These results highlight the importance of specifying the method for assessing CPP in future clinical trials, and argue for standardization of CPP assessment practices.

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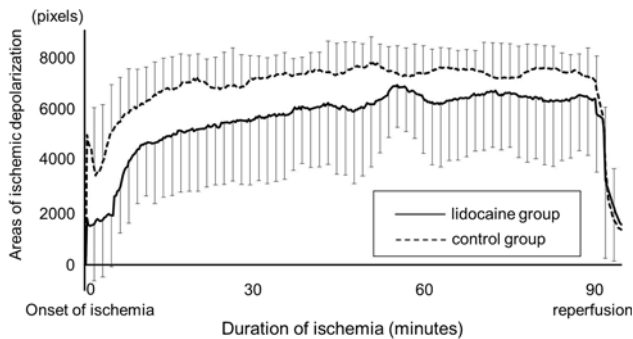
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A24 Effect of Lidocaine on the Area of Ischemic Depolarization During Focal Ischemia in Rats

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Background: The neuroprotective effects of lidocaine in animals have been shown in many previous studies.^{1,2} One of the mechanisms of its neuroprotective effect is the suppression of ischemic depolarization.² In our laboratory, reduced nicotinamide adenine dinucleotide (NADH) fluorescence imaging of the brain surface in vivo showed the area of depolarization of the neuronal cells because of the energy failure.³ In the mitochondria, NADH and oxygen are used in the respiratory chain to produce ATP. When ischemic depolarization is initiated under limited oxygen supply, NADH accumulates in the mitochondria because of the accentuation of anaerobic glycolysis. As NADH shows a blue fluorescence at 460 nm when excited with ultraviolet (UV) light of 360 nm, the images of NADH fluorescence can be used as indicators of energy failure. In this study, we used NADH fluorescence images to examine the effect of lidocaine on the area of ischemic depolarization during focal ischemia in rats.

Materials and Methods: Twelve male SD rats were anesthetized with 4% isoflurane in oxygen. After oral tracheal intubation, anesthesia was maintained by artificial ventilation with 2% isoflurane in 40% oxygen and 60% nitrogen. The animals were randomly assigned to either a



group receiving intravenous bolus (1.5 mg/kg) followed by an intravenous infusion (2 mg/kg/h) of lidocaine (lidocaine group, $n = 6$) or receiving the same volume of saline (control group, $n = 6$). Infusion was initiated 30 minutes before focal ischemia and continued until 30 minutes after ischemia. A large cranial window was made on the left parietal-temporal bone for measuring the area and the intensity of NADH fluorescence. To analyze the correlation between changes in NADH fluorescence and direct current (DC) potential, DC electrodes were placed in the cortex at a depth of 750 μm . Ischemia was initiated by occlusion of bilateral common carotid arteries and the left middle cerebral artery for 90 minutes. The cortical surface was illuminated with ultraviolet light (360-nm). The images of NADH fluorescence were obtained using a charge-coupled device camera with a 460 nm bandpass filter. The temperatures of the brain surface and rectum were maintained at $37.0 \pm 0.5^\circ\text{C}$ during the experiment. Statistical analysis was performed using the Student t test.

Results: A correlation between NADH fluorescence and DC potential were observed ($R^2 = 0.74$). Areas of ischemic depolarization in the lidocaine group and the control group are shown in Figure 1. Areas of ischemic depolarization were significantly smaller in the lidocaine group until 7 minutes ($P < 0.05$) and tended to be smaller after 7 minutes.

Conclusion: Intravenous infusion of lidocaine decreased the area of energy failure probably due to the suppression of ischemic depolarization. The therapeutic effect of lidocaine on energy failure was significant during 7 minutes after the onset of ischemia.

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A25 Isoflurane Postconditioning Induces Neuroprotection via Akt Activation and Reduced Mitochondrial Membrane Permeation

Zhiyi Zuo, MD, PhD, Liaoliao Li, PhD. *University of Virginia*.

We have shown that application of isoflurane at the onset of reperfusion (postconditioning) reduces brain ischemic injury in rats. This study was designed to determine whether this protection involved activation of survival protein kinases and maintenance of normal mitochondrial membrane permeation. Three-month-old male rats were subjected to a 90-minute middle cerebral arterial occlusion. They were then exposed or were not exposed to 2% isoflurane for 1 hour. Ischemic penumbral cerebral cortex was harvested immediately. Mitochondrial and cytosolic fractions were prepared. The results showed that mitochondrial nicotinamide adenine dinucleotide content in the penumbral cortex was significantly reduced, suggesting an increased mitochondrial membrane permeability. This increase was partly attenuated by isoflurane postconditioning. The content of the mitochondrial adenosine

diphosphate, but not adenosine triphosphate, in the penumbral cortex was also reduced no matter whether the animals were postconditioned with isoflurane. The phospho-Akt in the cytosolic and mitochondrial fractions of the ischemic penumbral cortex was higher than that in control cortex. This increase tended to be higher with isoflurane postconditioning. A similar change pattern was observed in the mitochondrial phospho-glycogen synthase kinase 3 β , an enzyme that is downstream of Akt and can regulate the mitochondrial membrane permeability. Although isoflurane postconditioning reduced oxygen-glucose deprivation-induced injury of rat cortical neuronal cultures, it failed to preserve the mitochondrial membrane potential. The isoflurane postconditioning-induced protection in neuronal cultures was attenuated by the Akt inhibitor LY294002. These results suggest that isoflurane postconditioning effects may be mediated by Akt and involve reduced mitochondrial membrane permeation.

A26 Continuous PKM ζ Activity is Required for Sevoflurane Induced Preconditioning Protection Against Hypoxic Damage in Rat Hippocampal Slices

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Introduction: Our previous study demonstrated that sevoflurane preconditioning improves recovery from hypoxia in CA1 pyramidal neurons of rat hippocampal slices. However, the protective mechanisms of sevoflurane preconditioning against hypoxic neuronal damage remains unclear. We have demonstrated that a protein kinase M zeta (PKM ζ) inhibitory peptide used before and during sevoflurane preconditioning reduced its protection in rat hippocampal slices. In this study, we examined the effects of Zeta inhibitory peptide (ZIP) applied after sevoflurane preconditioning but before and during hypoxia.

Methods: Intracellular recordings were made from CA1 pyramidal neurons in rat hippocampal slices. The slices were treated with sevoflurane 4% for 30 minutes followed by a 30-minute washout then 10 minutes of hypoxia. ZIP 5 μM was started 15 minutes after the end of the sevoflurane treatment and continued during 10 minutes of hypoxia. Resting and action potential were recorded before, during, and 60 minutes after hypoxia. An analysis of variance followed by the Newman-Keuls test was used.

Results: None of the neurons recovered their resting or action potentials after 10 minutes of hypoxia in the untreated group. Sevoflurane preconditioning significantly increased the amplitude of initial hyperpolarization, delayed the onset of the rapid hypoxic depolarization, attenuated the amplitude of the rapid depolarization and improved physiologic recovery after 10 minutes of hypoxia compared with the untreated group (75% vs. 0%, respectively, $P < 0.05$). The PKM ζ inhibitor, ZIP, started 15 minutes after sevoflurane treatment attenuated the protection of sevoflurane preconditioning; only 22% of the neurons recovered their resting and acting potentials (Table 1).

Conclusions: Sevoflurane preconditioning improves physiologic recovery after 10 minutes of hypoxia in CA1 pyramidal neurons. Continuous activity of PKM ζ is required for sevoflurane preconditioning protection. Blockade of its activity, even after the initiation of preconditioning reduces sevoflurane's protection.

TABLE 1. Effect of Zeta Inhibitory Peptide (ZIP) on Sevoflurane Induced Preconditioning

Groups	Amplitude of the initial hyperpolarization (mV)	Latency of rapid depolarization (sec)	The level of depolarization at 10 min hypoxia (mV)	Recovery (%)
Untreated	2.3 ± 0.2	249 ± 12	-0.43 ± 0.2	0
Sevoflurane 4%	$4.5 \pm 0.3^*$	$454 \pm 35^*$	$-15.6 \pm 2.8^*$	75*
Sevo + ZIP 5 μM	$1.1 \pm 0.2^{* \#}$	$326 \pm 26^{\#}$	$-4.7 \pm 2.1^{\#}$	22 $\#$

* $P < 0.05$ vs. untreated group; $\#P < 0.05$ vs. sevoflurane preconditioning group. (mean $\pm$ SE)

A27 pH Sensitive NMDA Inhibitors Improve Outcome in a Murine Model of SAH

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Introduction: Despite intensive research, neurological morbidity from delayed cerebral ischemia remains common after aneurismal subarachnoid hemorrhage (SAH). In the current study, we evaluate the neuroprotective effects of a pH-dependent NMDA receptor antagonist in a murine model of SAH.

Materials and Methods: After induction of SAH, C57-BL/6 mice received either NP 10075, a pH-dependent NMDA receptor antagonist, or vehicle. In a separate series of experiments, NP 10075, and a non-pH-dependent NMDA antagonist was administered to normoglycemic and hyperglycemic mice. Functional assessments were performed by serial assessment of Rotorod latencies. Vasospasm was assessed quantifying basilar and middle cerebral artery diameters.

Results: Administration of NP10075 was tolerated and had minimal hemodynamic effects after SAH. NP10075, but not the non-pH-dependent NMDA antagonist, was associated with a durable reduction in the functional deficits of both normoglycemic and hyperglycemic animals. NP 10075 was also associated with a reduction in vasospasm in the MCA associated with hemorrhage, but not the basilar artery, which was measured as a control.

Discussion: Although no neuroprotective agent has ever proven to be of benefit in the setting of SAH, the ability to initiate pretreatment before the development of delayed ischemic deficit would make this an attractive strategy. Moreover, a pH-sensitive NMDA receptor antagonist would be expected to be selectively activated at areas of local tissue acidosis and thus have less effect on level of consciousness and cognition, as compared with nonselective NMDA receptor antagonists.

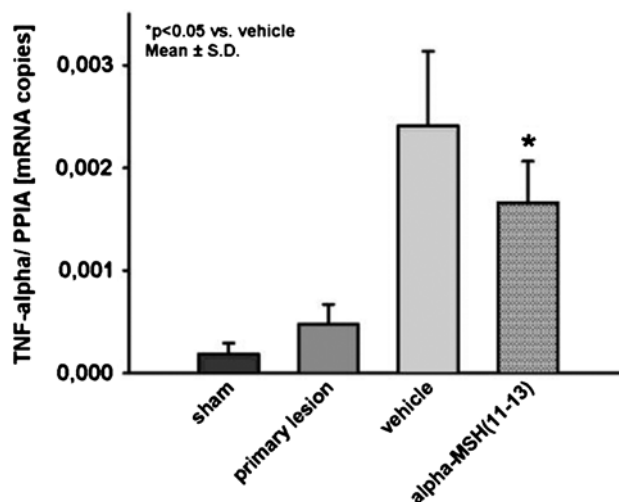
Summary: NP 10075, a pH-dependent NMDA receptor antagonist, reduces vasospasm and improves functional outcome in a murine model of subarachnoid hemorrhage and may be associated with fewer of the behavioral side effects associated with NMDA antagonism.

expansion of the secondary brain lesion volume. In a rat model of cerebral ischemia a reduction of α -MSH plasma concentration was observed after injury.³ Possibly, the substitution of α -MSH compensates this effect and maintains tissue stabilization. The data demonstrate a further starting point in the anti-inflammatory therapy after traumatic brain injury.

Summary: The tripeptide fragment (11-13) of the alpha-melanocyte-stimulating hormone (α -MSH) prevents secondary brain damage after experimental traumatic brain injury by reducing the posttraumatic inflammatory response.

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A28 Alpha-MSH(11-13) Improves Posttraumatic Inflammation and Brain Damage After Experimental TBI

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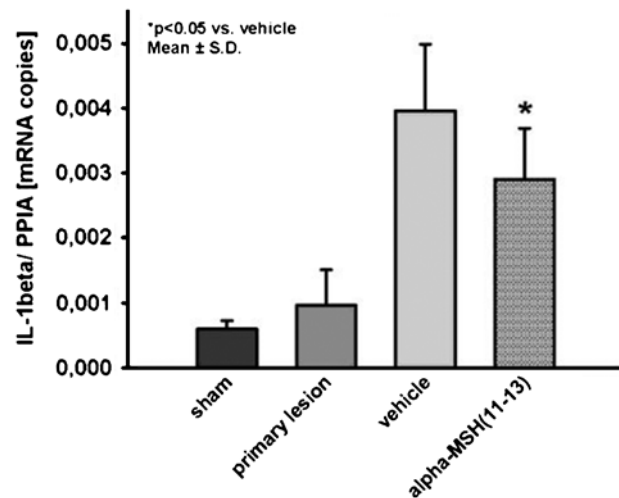
Introduction: Alpha-melanocyte-stimulating hormone (α -MSH) is a neuropeptide with immunomodulatory qualities. Its C-terminal tripeptide α -MSH(11-13) is also well known for anti-inflammatory effects¹ by reducing TNF- α , IL-6, and NO-induced proinflammatory processes.² The present study investigates the impact of the tripeptide α -MSH(11-13) on secondary brain damage and inflammatory gene regulation after experimental traumatic brain injury (TBI).

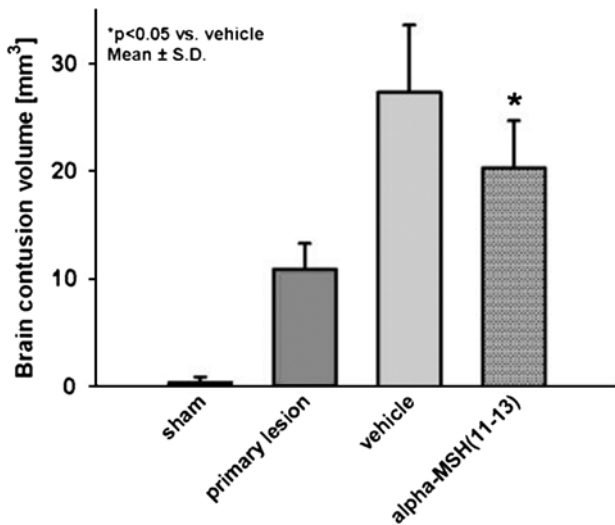
Methods: In male C57Bl/6N mice, anesthetized with isoflurane, a pneumatic brain trauma was induced on the right parietal cortex (controlled cortical impact, CCI). The animals were randomized to a sham operated group (n = 10) or a CCI group and were treated 30 min after insult with intraperitoneal injection of vehicle solution (0.9% NaCl, n = 10) or α -MSH(11-13) (1 mg/kg, n = 9). Contusion volume was determined in Nissl-stained sections after 24 hours. The primary lesion was quantified in a separate set of animals 15 minutes after CCI (n = 10). The mRNA expression of TNF- α , IL-1 β , and cyclophilin A (PPIA, housekeeping gene) were determined in contused brain tissue by real-time RT-PCR 24 hours post-CCI.

Statistics: ANOVA on ranks, Mann-Whitney rank-sum test, $P < 0.05$.

Results: See Figures.

Discussion: A single administration of the tripeptide acetate salt of α -MSH effectively reduced inflammation and secondary lesion after traumatic brain injury. In line with previous data, the application of α -MSH(11-13) after cerebral injury exerted an anti-inflammatory effect. The down-regulation of TNF- α and IL-1 β subsequently leads to limited





A29 Xenon Protects Against Brain Injury in Both Ischemic and Hemorrhagic Stroke

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Introduction: Low dose xenon (Xe, eg, 30% inhaled) offers rapid blood brain barrier (BBB) penetration, negligible effects on blood and intracranial pressure, and weak neuroprotective efficacy. Moderate therapeutic hypothermia (33 to 34°C) has shown benefit against a variety of brain insults, but requires mechanical ventilation. Mild hypothermia (eg, 36°C), while clinically feasible to achieve in awake patients, offers negligible neuroprotection.¹ In this study, we hypothesized that combined post-ischemic administration of Xe and mild hypothermia would offer neuroprotection superior to either treatment alone in a rat transient middle cerebral artery occlusion (MCAO) model. This would allow simple therapeutic intervention to be available to a larger fraction of patients. Because Xe could be used as an emergency stroke treatment, it is important to determine whether it could exacerbate intracerebral hemorrhage (ICH), which could initially present as occlusive vascular stroke. Xe has no known effect on coagulation. Therefore we hypothesized that xenon would not worsen outcome from experimental ICH.

Methods: Male Wistar rats (250 to 275 g) were subjected to 70 minutes MCAO with pericranial temperature held at 37.5°C. At 90 minutes post-MCAO, rats (n = 20 per group) were randomly assigned to experimental groups to breath 30% Xe in 30% O₂/40% N₂ or 30% O₂/70% N₂ for 20 hours while with pericranial temperature maintained at 37.5°C or 36°C. Neurological function and total cerebral infarct volumes were evaluated at 4 weeks post-ischemia. Male C57BL/6J mice (10 to 12 wk) were used to study ICH. Type IV-S Clostridial collagenase (0.075 U in 0.4 µL NS) was microinjected into the left striatum. At 2 hours after injection, mice were randomly treated with 30% Xe in 30% O₂/40% N₂ or 30% O₂/70% N₂ for 20 hours. Brain water content (n = 4 to 5 per group) was assessed 24 hours after injury. Rotorod latencies (n = 7 to 9 per group) were measured for 7 days after injury.

Results: There was a main effect for treatment group for MCAO infarct size [(37.5/No Xe = 224 ± 51 mm³, 36.0/No Xe = 177 ± 72 mm³, 37.5+Xe = 212 ± 76 mm³, 36.0+Xe = 157 ± 82 mm³, P = 0.015). Between group differences were present only for 37.5/No Xe vs. 36.0+Xe, P = 0.04). An identical pattern was present for neurologic score (37.5/No Xe vs. 36.0+Xe, P = 0.005). In the ICH study, compared with air, Xe significantly decreased whole brain water content from

80.19% ± 0.14% to 79.3% ± 0.29% (P = 0.038), and improved neurologic function (P < 0.01, Fig. 1).

Conclusion: Sub-anesthetic doses of post-ischemic Xe served as a therapeutic adjuvant to mild hypothermia in the context of focal cerebral ischemia with long-term efficacy. Xe did not worsen ICH. In fact, outcome was improved. This is consistent with prior demonstration of attenuated perivascular inflammation by Xe.² Neuroprotective effects of Xe in both ischemic and hemorrhagic stroke provide evidence that Xe has therapeutic potential and could be started immediately before a computed tomography scan.

A30 Statins Modulate Neuroinflammatory Responses and Improve Functional and Histological Outcomes in Murine Models of Traumatic Brain Injury and Intracerebral Hemorrhage

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Background: Traumatic brain injury (TBI) remains a leading cause of death and disability in the United States, with estimated annual costs of 60 billion dollars. Similarly, intracerebral hemorrhage (ICH) is associated with high mortality rate and persistent disability in survivors. Current management strategies following TBI and ICH remain largely supportive, as no pharmacological intervention has been demonstrated to improve long-term functional outcomes. HMG-CoA reductase inhibitors (statins) are a class of drugs with a favorable safety profile and long clinical track record, and have been demonstrated to improve outcomes in preclinical models of TBI, ICH, and SAH. To facilitate translation into clinical practice, we now characterize the optimal statin and dosing in murine models of ICH and TBI.

Methods: After approval by Duke Animal Care and Use Committee, closed head injury was induced on 8 to 12 week old C57BL/6J mice with a pneumatic impactor, as described by Lynch et al, 2005. Sixty mice were randomized into 5 treatment groups; rosvastatin 5 mg/kg and 1 mg/kg; simvastatin 5 mg/kg and 1 mg/kg, and vehicle. Treatment was administered immediately after surgery, and twice daily for 5 days. Vestibulo-motor function was assessed daily on rotarod (post-injury day 1 to 28) and spatial memory assessed with Morris water maze on post-injury day 30 to 35. Mice were then sacrificed for histology to assess short-term neuronal injury (FluoroJade) and neuronal density (NeuN antibody). RNA was extracted after sacrifice to assess differential expression of genes associated with inflammation. ICH was induced via injection of collagenase into the left striatum of C57BL/6J mice (8 to 12 wk old). Seventy-two ICH mice were randomized into 3 treatment groups; rosvastatin 1 mg/kg, simvastatin 1 mg/kg, vehicle. Rotarod testing was carried out and the result was followed-up by measurement of hemorrhage volume on days 1, 7, and 14.

Results: After TBI, administration of statins were associated with improved performance on Rotarod (Fig. 1A; P value < 0.05) and Morris water maze; 1 mg/kg rosvastatin was the optimal regimen. Treatment with statin was associated with a reduction in neuronal degeneration at 24 hours post-TBI, as demonstrated by a reduction of fluorojade B cells in the CA3 and polymorphic region (Fig. 2, n = 6/grp, R1:11.4 ± 3.4; V:52 ± 14 cells/section; P value: 0.0098). Statin treatment was also associated with increased neuronal density (NeuN immunopositive neurons) in the CA3 region in rosvastatin-treated mice at 35 days post-injury (Fig. 3, n = 6/grp, R1 = 10⁻⁴ ± 4 × 10⁻⁶; V = 8 × 10⁻⁵ ± 9 × 10⁻⁶; P value = 0.02). Differential gene expression by real time PCR showed significant down-regulation of pro-inflammatory markers in statin-treated mice, at day 1 and day 7 post-TBI. This result will be followed-up with F4/80 staining to look at microgliosis at the injured area. After ICH, the group treated with simvastatin 1 mg/kg had the lowest mortality and best rotarod performance, although there was no difference in hemorrhage volume.

A31 Optical Tracking of Blood Brain Barrier Disruption and Intra-arterial Drug Delivery

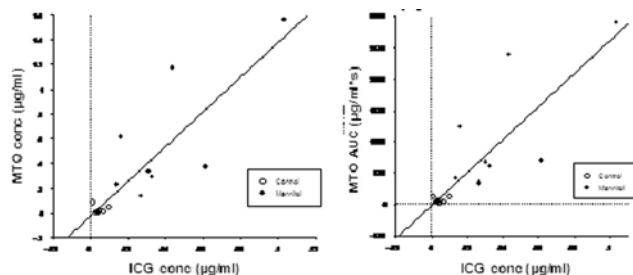
Shailendra Joshi, MD, Mei Wang, MPH, Aysegul Ergin, PhD, Irving J. Bigio, PhD. *Columbia University, Boston University.*

Background: By observing changes in spectrum of back-scattered visual-near infrared light, it is possible to measure in-vivo the tissue concentrations of certain drugs and tracers that have an absorption spectrum that is distinct from hemoglobin and deoxy-hemoglobin. Rapid drug/tracer concentration measurements are particularly useful in tracking intra-arterial drugs/tracers delivery that are beyond the temporal resolution of tissue biopsy or microdialysis. Osmotic disruption of the blood brain barrier (BBBD) by intra-arterial mannitol injection is often the key step in the delivery of chemotherapeutic drugs to brain tissue. BBBD with mannitol can be highly variable. Simultaneous quantification of BBBD and tissue drug concentrations using optical measurements could generate novel insights into pharmacokinetics of CNS drugs.

Methods: After IACUC approval the study was conducted on New-Zealand white rabbits. Surgical preparation included: tracheotomy for mechanical ventilation, femoral and selective internal carotid artery (ICA) catheterizations, skull screws for monitoring electrocortical activity, bilateral placement of laser Doppler probes and skull shaving to the inner table for the placement of fiber OP probe. For this experiment we measured brain tissue concentrations of indocyanine green (ICG, a marker of BBBD) and mitoxantrone (MTO, a chemotherapeutic drug effective in glioma treatment). Animals were randomized into 2 groups with (mannitol, intracarotid mannitol injection) or without (control, intravenous saline injection) BBBD. Immediately after mannitol injection 1 mg of ICG and mitoxantrone (3 mg) were injected via the ICA.

Results: The study involved 8 control and mannitol treated animals, each. Preliminary analysis of the data reveals that intracarotid ICG was rapidly cleared from the brain tissue, however, there was a small but significant retention of ICG concentrations after BBBD 0.005 ± 0.003 (control) versus $0.041 \pm 0.03 \mu\text{g/mL}$ (mannitol, $P < 0.005$) at the end of 5 minutes. The MTO concentrations were also significantly higher after BBBD [0.02 ± 0.03 (control) vs. 0.583 ± 0.51 ($\mu\text{g/mL}$, mannitol, $P < 0.008$]. There was a linear correlation between the tissue MTO concentrations and the area under the MTO concentration time curve and the end ICG concentrations, Figure 1.

Conclusion: This study shows the feasibility of optical monitoring of BBBD with intra-arterial ICG injections. However the study also revealed the problems of in-vivo measurements of ICG concentrations due to the shifts in the optical properties resulting from protein binding. Tissue MTO concentrations were linearly related to BBBD as assessed by the ICG. While most of the ICG dose was rapidly cleared from the brain tissue trace concentrations remained detectable in the brain tissue after BBBD. If concentrations of ICG could be detected in human brain tissue by NIRS spectroscopy then, optical monitoring of BBBD could help improve intra-arterial delivery of drugs. Inadequate disruption might require additional dose of mannitol while excessive disruption might result in reducing the dose of chemotherapeutic drugs.



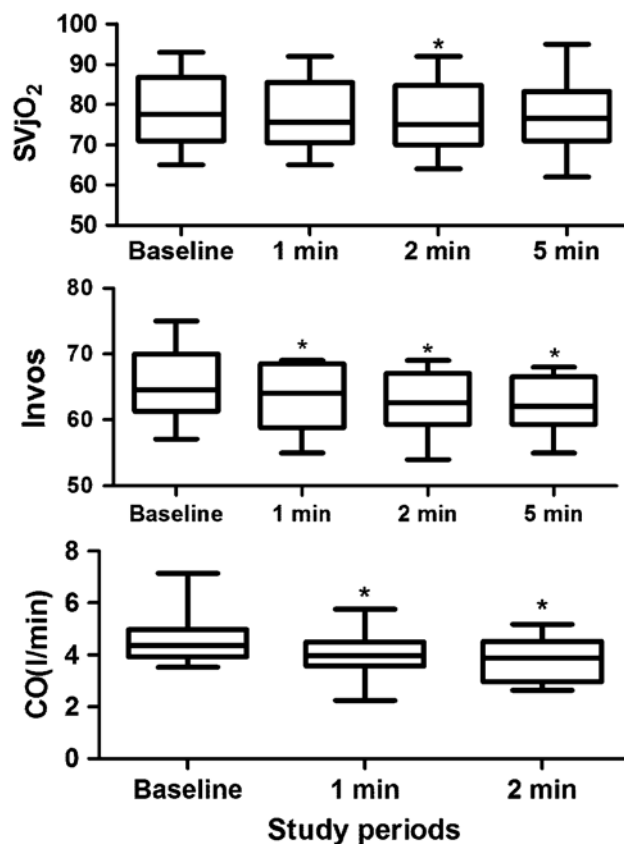
A32 INVOS, SVjO₂, and on EEG Derived Parameters in Pigs After a Remifentanyl Bolus

Aura Silva, DVM, MSc, Ana Liza Oliveira, DVM, Carlos Venâncio, MSc, Pedro Amorim, MD, David A. Ferreira, PhD. *UTAD, Escola Superior Agrária de Viseu.*

Background and Goal of the Study: Opioids may cause cerebral vasoconstriction¹ and suppression of the cerebral metabolism/cerebral blood flow.² This study addresses the effect of high remifentanyl doses on peripheral brain hemoglobin saturation, on brain venous blood oxygen saturation (SVjO₂), and on EEG derived parameters in pigs.

Materials and Methods: Ten healthy Large White male pigs, from a related investigation project, with $27.3 \pm 3.5 \text{ kg}$ were submitted to a propofol and remifentanyl constant infusion rates of 15 mg kg/L h/L , and $0.3 \mu\text{g kg/L h/L}$, respectively. Animals were under mechanical ventilation to keep EtCO₂ between 35 and 45 mm Hg. In a stimulus free period, a remifentanyl bolus (RemiBol) of $5 \mu\text{g kg/L}$ was administered at a rate of 600 mL h/L before surgical procedures. Invasive monitoring procedures included a Swan Ganz catheter in the sinus petrosus ventralis for measuring SVjO₂; a Swan Ganz catheter in the pulmonary artery for measuring cardiac output (CO) by the thermodilution method; invasive arterial blood pressure and heart rate (HR) in the femoral artery. An A2000XP BIS monitor was used to record bispectral index (BIS), spectral edge frequency 95% (SEF), suppression ratio (SR) and total power (TP). INVOS Cerebral Oximeter was used unilaterally (left side). BIS electrode and SVjO₂ were also placed in the left side. SVjO₂, Invos, BIS, SEF, SR, and TP were recorded at the moment right before the remifentanyl bolus, and 1, 2, and 5 minutes after the end of the bolus. CO was recorded 1 and 2 minutes after the end of the bolus. ANOVA was used to analyze data. Pearson r correlation analysis was performed between all parameters. Data are mean $\pm$ sd.

Results and Discussion: Before the RemiBol, MAP was $73 \pm 9.5 \text{ mm Hg}$, HR was $81 \pm 12.7 \text{ bpm}$, SVjO₂ was $78.7\% \pm 9\%$, CO was $4.6 \pm 1.1 \text{ l/min}$, Invos was 65 ± 5.6 and BIS was 63.9 ± 4.3 . Two minutes after the



RemiBol, MAP and HR decreased to 65 ± 7.2 mm Hg ($P < 0.01$) and to 72 ± 14 bpm ($P < 0.0001$), respectively; $SVjO_2$ decreased to 76.9 ± 9.3 ($P < 0.05$). Five minutes after the RemiBol, MAP and HR were 68 ± 7.4 mm Hg ($P = 0.01$) and 72.8 ± 12.5 bpm ($P < 0.001$), respectively. InvoS and CO decreased significantly throughout time ($P < 0.05$) (Fig. 1). A high correlation value was observed between InvoS and CO ($r = 0.74$; $P < 0.001$) and between HR and InvoS ($r = -0.57$; $P = 0.01$), but no correlation was found between InvoS and $SVjO_2$. BIS values along the study were 61.3 ± 5.9 . No significant effects were observed in any of the EEG derived parameters, and SR was always 0%.

Conclusions: The observed decrease in INVOS values supports the fact that high doses of remifentanyl may cause peripheral cerebral vasoconstriction, also followed by the decrease in $SVjO_2$. The high correlation observed between INVOS and CO indicates that the decrease in CO may have contributed to potentiate the effects of cerebral vasoconstriction on peripheral brain oxygen delivery.

Acknowledgments: Financial support by COMPETE: FCOMP-01-0124-FEDER-009525 and FCT (PTDC/CVT/101999/2008).

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A33 Influence of Propofol 2 μ g/mL on rCBF, GABA, and Glutamate Level—fMRI, MRS Study in Volunteers

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Introduction: Propofol as well sevoflurane 1 MAC cause an equivalent decrease in CMRO₂, but CBF effects are different—propofol decreases the CBF while sevoflurane causes no change in CBF (compared with awake state). Our fMRI group has demonstrated nonuniformity in rCBF with 0.25 MAC sevoflurane as well as 0.5 MAC sevoflurane. At the neurotransmitter level anesthetics act by either increasing the GABA level or decreasing the glutamate level. Our aim in this study was to measure the effect of propofol (2 μ g/mL—0.5 MAC equivalent) on rCBF, GABA, and glutamate level (in the thalamus) in healthy volunteers.

Methods: Yale University Human Investigation Committee approved the study protocol. In 19 healthy ASA I volunteers (19 to 35 y) propofol was administered using a STANPUMP (Marsch protocol) aiming for a target plasma propofol concentration of 2 μ g/mL (0.5 MAC equivalent). Standard ASA monitoring (EKG, NIBP, SpO₂, etCO₂) was carried out in all subjects. Regional CBF, GABA, and glutamate measurements were acquired on a 3 T Tim Trio scanner with a 12-channel phased-array head RF coil. Regional CBF was measured by pulsed arterial spin labeling (PASL) technique. GABA and glutamate/glutamine (glu/gln) and water level in the right thalamus (30 $\times$ 30 $\times$ 30 mm voxel) were measured using the MEGA-PRESS method. Two cycles of imaging was carried out—awake and anesthesia.

Data Analysis: From the magnetic resonance spectroscopy (MRS) trace in the thalamus generated in the awake state and anesthesia area under the curve (GABA and glu/gln) was measured and the difference between awake and propofol state was calculated. The GABA, glu/gln level in the thalamus was normalized by dividing the water level measured in the same voxel. The difference in rCBF in the various regions between awake and propofol state (δ CBF) was calculated. Anatomical regions were identified using the Talairache atlas.

Results: Vital signs of all the subjects were within normal limits during propofol infusion. With propofol 2 μ g/mL plasma level 18/19 subjects were asleep with Observer Assessment of Alertness and Sedation (OAA/S) score of ≤ 2 (no response to call) and no one had any memory of the event. With 2 μ g/mL plasma level propofol there was a significant drop in rCBF in most of the regions (frontal, parietal, occipital, temporal cortices, and thalamus) (Table 1). In the thalamus and occipital cortex decrease in CBF was close to 40% to 50%. In the anterior cingulate (ACC), insula and parahippocampal gyrus there was a modest rise in rCBF. Concentration of GABA (measured in ppm) decreased ($P < 0.05$) in the thalamus with propofol (0.25 ± 0.03 vs. 0.29 ± 0.04 a.u.). Glu/gln level increased in the thalamus with propofol but the rise was not significant.

Discussion: Propofol also causes a non uniformity in rCBF. Regions manifesting a rise in rCBF (ACC, insula, and parahippocampal gyrus) showed a similar response with 0.5 MAC sevoflurane also. The extent of decrease in rCBF was significantly more with propofol as compared with sevoflurane. ACC and insula are linked to pain processing, parahippocampal gyrus is related to memory while thalamus is linked to loss of consciousness. This study confirms that loss of consciousness with propofol correlates with deactivation of the thalamus ($\downarrow$ rCBF). Rise in GA.

Changes in rCBF with Propofol 2 mcg/ml

Region	Subregion	Talairach Coordinates	rCBF changes ml/100gm/min
Frontal Lobe	Precentral Gyrus R	(16 -31 61)	-17.46
	Middle Frontal G. R	(38 32 25)	-13.79
		(33 54 14)	-20.58
		(7 38 -13)	-12.65
	(BA 11)	(44 39 -8)	-21.17
	(BA 10)	(-18 47 16)	-18.36
	Middle Frontal G. L	(-37 32 28)	-15.48
Corpus Callosum		(0 16 15)	15.53
Insula		(35 5 -7)	5.66
Olfactory		(-6 15 -8)	9.22
Parietal Lobe	Inferior Parietal R (BA 7)	(35 -57 40)	-13.26
	Inferior Parietal L	(-35 -52 41)	-11.29
	Precuneus R (BA 7)	(9 -52 40)	-11.34
		(7 -63 40)	-16.55
	Supramarginal Gyrus R	(41 -49 29)	-11.23
	Sub-Gyral L	(-33 -61 29)	-10.84
Cingulate Gyrus		(6 -44 27)	-20.34
	Anterior	(3 7 27)	7.66
Occipital Lobe	Cuneus R (BA 7)	(21 -78 29)	-15.46
		(17 -89 20)	-19.14
	Cuneus L	(-26 -85 20)	-21.52
	Sub-Gyral L	(-30 -73 27)	-15.04
	Lingual G. L	(-1 -88 2)	-27.38
	(BA 17)	(-2 -90 2)	-30.85
	Middle Occip.G.R	(42 -76 3)	-18.55
	Middle Occip.G.L	(-39 -81 3)	-22.3
	Fusiform L	(-10 -88 -11)	-16.11
Temporal Lobe		(35 -36 1)	-6.6
		(39 -34 1)	-5.1
		(44 -48 -5)	-9.67
	Inf. Temp.G.L	(-55 -57 -10)	-21.4
		(-54 -25 -19)	-16.36
	Inf. Temp.G.R	(55 -55 11)	-16.88
		(56 -23 -18)	-14.73
	(BA 21)	(16 -35 -11)	-17.05
	(BA 20)	(-57 -54 -11)	-21.4
	ParaHippoCampa G R	(21 -26 -15)	27.31
	ParaHippoCampa G L	(-16 -26 -15)	29.26
Thalamus	Medial Dorsal Nucleus R	(9 -19 12)	-22.25
	Medial Dorsal Nucleus L	(-4 -21 10)	-25.11
Putamen R		(10 10 0)	-5.11
Putamen L		(-14 11 0)	-5.27
Cerebellum		(48 -46 32)	-16.97
		(-38 -41 -33)	-12.94
		(-13 -84 -18)	-11.22
		(2 -29 -22)	-11
	Pons	(30 -64 -22)	-11.03
	declive R	(-16 -65 -22)	-8.69
	declive L	(38 -70 -27)	-15.05
	Poster Lobe R	(-22 -73 -27)	-8.55
	Poster Lobe L	(-25 -71 -39)	-13.85
	Culmen R	(10 -46 0)	12.72
	Culmen L	(-10 -47 -4)	14.61

A34 The Effect of Furosemide on Fluid Balance and Electrolytes in Patients Receiving Mannitol

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Background: Many clinicians will add furosemide to mannitol to augment its forced diuresis. Over-aggressive forced diuresis may cause hypovolemia and hypoperfusion, hypokalemia, and hyponatremia. To assess the safety of furosemide combined with mannitol, we reviewed our initial patients that were enrolled in a randomized controlled trial to examine the effects of furosemide versus placebo when combined with mannitol on surgical brain relaxation and postoperative cerebral edema.

Methods: We unblinded the initial 24 patients that were enrolled in a double-blind, block randomized placebo-controlled study to examine the effects of furosemide (0.3 mg/kg) when combined with mannitol (1 g/kg) on surgical brain relaxation and postoperative cerebral edema. After induction of anesthesia, mannitol and the study drug were administered and arterial blood gases with electrolytes and urine output volume were recorded every 30 minutes for 3 hours, or until skin closure was completed. Systemic mean arterial pressure was maintained within 15% of baseline levels with a combination of hetastarch and phenylephrine per a treatment protocol. Plasma sodium, potassium, and lactic acid concentrations as well as interval urine output were compared across time and between furosemide-placebo assignment using a 2-factor repeated measures analysis of variance with the Holm-Sidak post-hoc test compared with control ($t = 0$) to isolate differences. A $P < 0.01$ was considered significant.

Results: Although mannitol produced a large volume diuresis (1533 $\pm$ 335 mL), the addition of a low dose of furosemide substantially increased the rate of production for the first 90 minutes after administration and the total volume of urine produced (2561 $\pm$ 611 mL, $P < 0.001$ compared with placebo group). There was neither a difference in the volume of intravenous crystalloid or hetastarch used between study drug groups, nor was there a difference in the cumulative dose of phenylephrine used. There were no differences in plasma sodium, potassium, or lactic acid concentrations between study drug assignments at any time point.

Discussion: Adding 0.3 mg/kg furosemide to 1 g/kg mannitol increased the rate of urine production over the initial 90 minutes and the total urine output by 67%, both of which caused many clinicians to be concerned about the safety of our study protocol. Therefore, we unblinded our initial patients in this double-blind, block randomized, placebo-controlled study to examine the effect of furosemide with mannitol on plasma sodium, potassium, and lactic acid concentrations. The large volume, forced diuresis did not result in any difference between plasma sodium or potassium between drug groups at any time point. There was no evidence of increased hypovolemia in the furosemide group, as measured by the volume of crystalloid or hetastarch used or the cumulative dose of phenylephrine used. In addition, there was no evidence of hypoperfusion from occult intravascular hypovolemia, as measured using lactic acid as a surrogate. In this small interim analysis to examine the safety of adding furosemide to mannitol, there appears to be no detrimental effect of the increased urine output on electrolytes or surrogate markers of inadequate intravascular volume.

A35 Complications of Stereoelectroencephalography (SEEG) Guided Adult Epilepsy Brain Surgery: Our Experience at the Cleveland Clinic

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Early epilepsy surgery is becoming a first line therapeutic intervention in patients with intractable epilepsy as studies have shown better outcome with early surgery.¹ Noninvasive techniques of epileptogenic zone identification often fail to localize the site of origin and spread of ictal discharge.^{2,3} Consequently, invasive intracranial EEG recording is necessary to prevent unwanted postoperative neurological deficits in these patients.³ Traditional invasive techniques included formal craniotomies or burr holes for placement of subdural grids or strips. More recently, stereoelectroencephalography (SEEG) is emerging as an increasingly popular technique of depth electrode placement that is being used to elucidate epileptogenic zones in patients with intractable epilepsy. Potential risks, in less than 1% of cases, include brain hemorrhage and infection, which can lead to permanent neurological impairment or death. However, there is paucity of data regarding the current applications and outcomes of SEEG electrode placement and SEEG guided epilepsy surgery along with anesthetic implications and perioperative morbidity.³ The aim of our study is to review the Cleveland Clinic patient profile, preoperative indications, perioperative complications, intraoperative anesthetic management and postoperative outcomes of SEEG guided adult epilepsy brain surgery. After obtaining consent from the Cleveland Clinic Institutional Review Board, the medical records of 50 patients undergoing SEEG guided adult epilepsy brain surgery from 2009 to 2010 were analyzed in this retrospective chart review. The variables studied included, type and duration of epilepsy, anti-epileptic drugs, complications from SEEG electrode placement, craniotomy and epilepsy site removed, presurgery and postsurgery hematocrit and coagulation parameters, blood loss, type of anesthesia (general vs. awake), use of intraoperative anesthetic and analgesic infusions (dexmedetomidine, remifentanyl, propofol), and vasopressor agents for maintenance of hemodynamics. Immediate and long-term postoperative complications including recurrence of seizures were also looked at. The preliminary data revealed no morbidity or mortality from the SEEG depth electrode implantation procedure. Subsequent surgery included 40% temporal lobectomies, followed by 32.5% frontal, 12.5% parietal, and 15% multilobar procedures. Total 85% patients underwent craniotomy under general anesthesia with either sevoflurane or isoflurane and 12.5% patients had awake craniotomies. During surgery estimated blood loss (EBL) was greater than 500 mL in 15% patients (5% frontal, 5% multilobar, 2.5% parietal, and 2.5% of temporal lobectomies). Intraoperative vasopressor infusion was needed in 27.5% of the patients (10% frontal, 7.5% temporal, 5% parietal, and 5% of multilobar lobectomies). Overall seizure recurrence during immediate or long-term follow-up was 27.5% and non-neurological complication rate was 2.5%. In patients with long standing epilepsy (preoperative diagnosis of epilepsy for greater than 20y), 36.36% had seizure recurrence, 16.66% had estimated blood loss (EBL) greater than 500 mL and 40% needed intraoperative vasopressor infusion.

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A36 Clinical Experience with Intraoperative Jugular Venous Oximetry During Pediatric Neurosurgery

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Background: Cerebral desaturation is associated with poor outcome in children with traumatic brain injury (TBI). Intraoperative utility of jugular venous oximetry in adult patients undergoing elective neurosurgical procedures is also well known. However, there is no information on intraoperative jugular venous oximetry in children undergoing elective neurosurgery.

Aim: To report our local experience with intraoperative jugular venous oximetry during neurosurgery and on the use of this technology to guide anesthetic care.

Methods: After IRB approval, we examined patient, equipment, placement and sampling characteristics, as well as interventions based on jugular saturation (SVjO₂) values in the pediatric neurosurgical patients aged <18 years who received intraoperative jugular venous oximetry. We determined the prevalence of intraoperative cerebral desaturation (defined as SVjO₂ <55%).

Results:

1. Data from 19 children (10 male and 9 female), aged 12 $\pm$ 1 years (range 7 to 17) were reviewed. These patients underwent craniotomy with arteriovenous malformation (AVM) resection (68%), tumor resection (21%), or aneurysm clipping (11%).
2. The side jugular bulb was cannulated on the right side in 17 children and left side in 2 children. Ultrasound guidance was used in all but 2 children.

TABLE 1. Representative Interventions and Responses to Jugular Venous Changes in 8 pediatric neurosurgical patients

Age (years)	Surgery	Initial SjVO ₂ (%) Initial PaCO ₂ (mmHg) Initial MAP (mmHg)	Observation	Most likely explanation of observation	Intervention	Resulting SjVO ₂ (%) PaCO ₂ (mmHg) MAP (mmHg)
8	Tumor	61 → 51 35 → 34 62 → 55	Cerebral Desaturation	Decrease in MAP	Simultaneous increase in PaCO ₂ and MAP	55 43 69
10	AVM	60 → 51 26 → 27 68 → 60	Cerebral Desaturation	Decrease in MAP	Increase in MAP	60 27 67
10	Tumor	54 → 44 40 → 34 65 → 65	Cerebral Desaturation	Decrease in PaCO ₂	Increase in PaCO ₂	57 44 60
16	AVM	57 → 35 40 → 28 57 → 60	Cerebral Desaturation	Decrease in PaCO ₂	Simultaneous increase in PaCO ₂ and MAP	62 38 70
16	Tumor	59 → 54 39 → 36 65 → 60	Cerebral Desaturation	Decrease in PaCO ₂ and MAP	Increase in PaCO ₂ and MAP	65 38 68
11	Aneurysm	50 35 60	Cerebral Desaturation	Low MAP	Increase in MAP	70 35 75
7	AVM	96/81/67/55 33/31/34/37 70/69/69/60	Progressive decrease in SjVO ₂	Progressive AVM resection	None	-
13	AVM	61 31 71	Brain swelling	Extreme lateral rotation of neck with increased JVP-CVP gradient	Reduced neck rotation	-

- The catheters most often used had calibers of 16 G, 20 G or 4Fr. The length of catheter inserted ranged from 5 to 13 cms.
- The median sampling frequency was 4 ± 2 (range 1 to 9) during neurosurgery. On average, patients underwent sampling once every anesthetic hour (range 1 to 3). Nineteen subjects contributed a total of corresponding 88 SvjO₂, PaCO₂, and mean arterial pressure (MAP) data points.
- Of the 88 samples, there were a total of 25 (28%) episodes of cerebral desaturation. Out of these 25 episodes, 6 were attributed to relative hypotension and 4 to hypocarbia.
- Eleven (58%) of 19 patients experienced at least 1 episode of cerebral desaturation (6 AVMs, 4 tumors and 1 aneurysm), 4 had expected decrease in SvjO₂ with AVM resection and no abnormal values were observed in the remaining 4 patients.
- There were no intraoperative or immediate postoperative (first 24 h) complications (carotid puncture, neck hematoma, pneumothorax) due to jugular venous oximetry.

Summary: Findings from this series indicate that (1) cerebral desaturation detect by jugular venous oximetry is common during pediatric neurosurgical procedures, (2) the use of SvjO₂ data can impact intraoperative interventions to optimize cerebral physiology and (3) intraoperative jugular venous oximetry is feasible and safe in the hands of experienced anesthesiologists.

A37 Anesthetic Management of Intraoperative Internal Carotid Artery Rupture and Massive Blood Loss During Clipping of a Distal Middle Cerebral Artery Aneurysm

Eugenia Ayrian, MD, Vladimir Zelman, MD, PhD. *University of Southern California, Keck School of Medicine.*

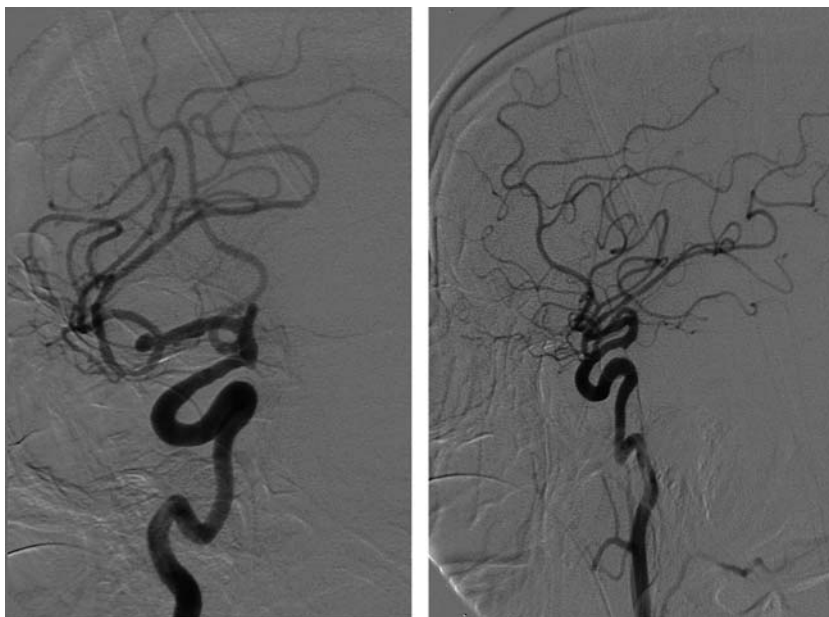
Intraoperative cerebral aneurysm rupture is one of the most disastrous complications. The majority of ruptures occur during its dissection or clipping.

Case Presentation: A 43 year old Male with a past medical history of ANCA associated vasculitis was referred to our center for intracranial hemorrhage. A cerebral angiogram confirmed right MCA aneurysm and mild vasospasm of ICA. Attempt to coil the aneurysm was unsuccessful, and the patient was brought to the OR for decompressive craniotomy and aneurysm clipping. The aneurysm was located. A temporary clip was then placed on the proximal MCA. Spontaneous rupture of the right ICA occurred during blunt dissection. Four liters of blood were lost within minutes. As a result of the patient's severe vasculitis, placement of a permanent clip was very challenging. With each attempt to place a clip the artery wall would rupture. A decision was made to place the clip on the right ICA permanently. Despite permanent clipping of the right ICA, the patient began to follow simple command on postoperative day 1, but had a significant left sided paralysis. He was extubated on postoperative day 2. Postoperative CT scan showed no change in the hemorrhage presentation. Therefore, it was determined that initial hemorrhage was not from MCA aneurysm rupture, but from rupture of right ICA proximally to the clip placement. Patient was discharged to a rehabilitation facility with minimal left-sided deficit.

Discussion: During cerebral aneurysm clipping the goal is to prevent aneurysm rupture and maintain CPP2. In the event of IOR clear communication of the blood pressure goals should be maintained between the surgeon and anesthesiologist. Surgical outcomes are made worse by hypotension. Nevertheless, if the aneurysm ruptures, mean arterial pressure can be decreased to 60 mm Hg or lower, to allow localization and placement of temporary clip. Then the blood pressure should be normalized to reperfuse the collateral vessels.

This patient displayed remarkable postoperative recovery despite severe intraoperative hemorrhage and permanent clipping of his right ICA. Several conditions led to his improved outcome.

- Ventilation was adjusted to maintain normocapnea.
- Diureses was induced to ease the tension on the brain parenchyma.
- As soon as dura was opened, the burst suppression with propofol infusion was introduced, and patient was placed on 100% O₂.



- (4) PRBC transfusion was initiated before the incision, during ICA rupture 9 U of PRBCs and 4 U of FFP were transfused immediately. The patient was kept normovolemic at all times.
- (5) The intraoperative neurophysiological monitoring indicated that the patient remained in burst suppression throughout the procedure. This provided additional cerebral protection and contributed to patient's better than expected neurological outcome.
- (6) Evoked potential monitoring provided an optimistic outlook for the patient. Even though the evoked potentials disappeared after placement of ICA clip, there was its surprising partial return towards the end of procedure.
- (7) Collateral circulation has been shown to develop with chronic vasculitis. These potential collateral vessels came out to be crucial in compensating for the obstructed right sided cerebral circulation.

A38 Roles of Testosterone in the Formation of Intracranial Aneurysms in Male Mice

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Introduction: Experimental and clinical studies suggest that sex steroids modulate cardiovascular diseases. We have recently developed a mouse model of intracranial aneurysms. In this model, a combination of hypertension and a single stereotaxic injection of elastase into the cerebrospinal fluid results in aneurysm formations. Using this model, we studied roles of testosterone and estrogen in the formation of intracranial aneurysm in male mice. **Methods and results:** We used 8-week-old C57BL/6J male mice. DOCA-salt model was used to induce hypertension. Mice received unilateral nephrectomy. Seven days later deoxycorticosterone acetate (DOCA) pellet (2.4 mg/d) was implanted subcutaneously, and drinking water was substituted with 1% sodium chloride solution. On the same day of implantation of DOCA, elastase (25 milli-units) was stereotaxically injected into the cerebrospinal fluid at the right basal cistern to disrupt elastic lamina.

(A) Roles of testosterone in the formation of intracranial aneurysms As a first step, we tested whether testosterone reduces the incidence of intracranial aneurysms. Three groups of mice were studied. Mice in the first group underwent orchiectomy on the same day of nephrectomy ($n = 19$). The second group received testosterone therapy via a testosterone pellet (1.5 mg, 60-day release), that was implanted immediately after the orchiectomy ($n = 13$). Mice in the third group received

sham orchiectomy alone ($n = 14$). Four weeks after elastase injection, all mice were sacrificed. There was a significant difference in the incidence of intracranial aneurysms between orchiectomized mice without testosterone treatment (89%) and orchiectomized mice with testosterone treatment (54%) ($P = 0.02$). There was a trend for the incidence of intracranial aneurysms to be higher in orchiectomized mice without testosterone treatment than sham orchiectomized mice (64%) ($P = 0.08$). **(B) Roles of estrogen in the formation of intracranial aneurysms** Testosterone can be converted to estrogen by aromatase in males.

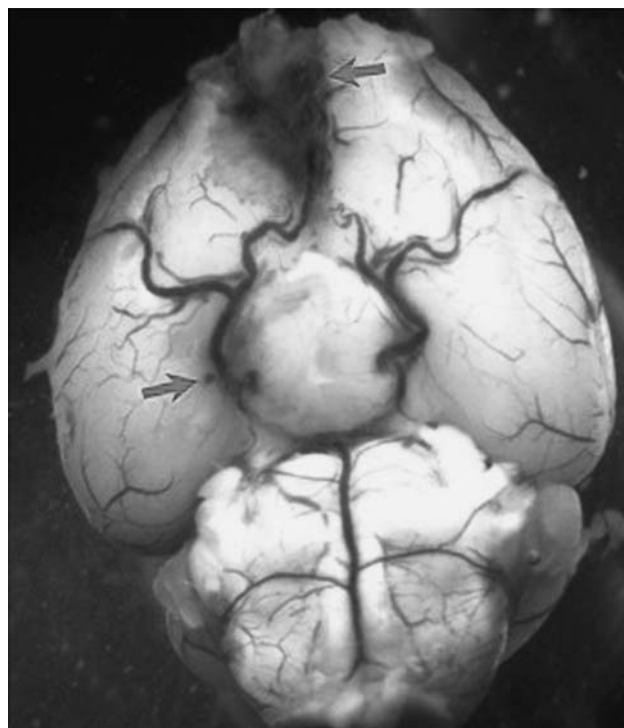


FIGURE 1. Representative intracranial aneurysm.

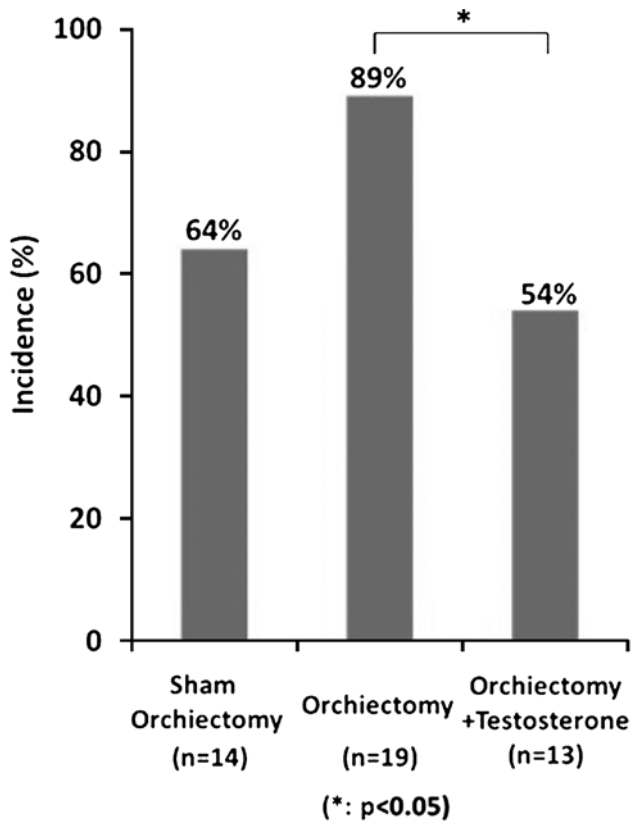


FIGURE 2. Effects of testosterone on the incidence of aneurysms.

Estrogen converted from testosterone may contribute to the protective effect of testosterone against the formation of intracranial aneurysms. Therefore, we tested whether endogenous estrogen that is converted from testosterone has protective effect against the formation of intracranial aneurysms in male mice. Two groups of mice were studied. Mice in the first group received anastrozole, an inhibitor of aromatase (10 mg/kg/d, gavage) ($n = 21$). Mice in the second group were given control vehicle instead of anastrozole ($n = 17$). We began the administration of anastrozole or vehicle 7 days before the elastase injection. Four weeks after elastase injection, all mice were sacrificed. There was no significant difference in the incidence of intracranial aneurysms between anastrozole group (81%) and control group (59%) ($P = 0.13$).

Conclusions: These findings suggest that testosterone has protective effects against the formation of intracranial aneurysms in male mice. The mechanism of these beneficial effects may go through androgen receptor, not via estrogen derived from testosterone.

A39 The Effects of Repeated Dose of 6% Hydroxyethyl Starch or 7.5% Sodium Chloride on Edema After Experimental Intracerebral Hemorrhage in Rats

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Objective: To observe the effects of repeated dose of 6% hydroxyethyl starch (130/0.4) or 7.5% sodium chloride on brain edema after experimental intracerebral hemorrhage (ICH) in rats.

Methods: Total 167 male Sprague-Dawley rats were divided into 4 groups randomly: Sham operation group (S, $n = 20$), ICH control group (M, $n = 38$), 7.5% sodium chloride group (N, $n = 55$) and 6% hydroxyethyl starch group (H, $n = 54$). The model of the ICH was established by stereotactically infusing 50 μ L of the autologous femoral artery blood into the right caudate nucleus in rats. The rats in N group

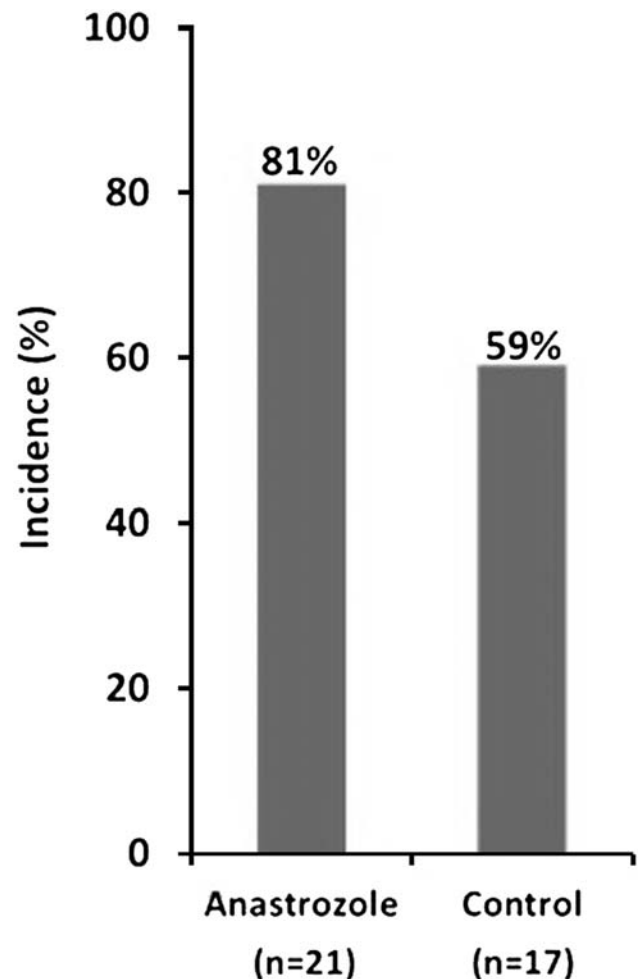


FIGURE 3. Effects of estrogen converted from testosterone on the incidence of aneurysms.

received 7.5% sodium chloride 5 mL/kg, while in H group received 6% hydroxyethyl starch 30 mL/kg at 2, 24, 48 and 72 hours after ICH respectively. The rats in S and M group didn't receive any infusion. The brain water content (BWC) was examined by dry-wet weight measurements before and after infusion at different time point; Rosenberger behavior scores and survive rate were also observed. The brain tissue content of tumor necrosis factor- α (TNF- α), interleukin-6 (IL-6), superoxide dismutase (SOD) and malondialdehyde (MDA) were measured at each infusion time point.

Results:

- (1) The peak BWC of the ipsilateral basal ganglia and cortex in M group was observed at 48 hours and lasting to 72 hours after ICH. After infusion, BWC of the ipsilateral cortex or basal ganglia at 2, 24, 48 and 72 hours in N group or in H group were less than that in M group ($P < 0.05$); Rosenberger behavior scores were less in N group and H group than that in M group at 24 and 48 hours after ICH ($P < 0.05$). Survive rate was lower in M group than that in H group at 72 hours ($P < 0.05$).
- (2) The brain content of IL-6 in N group was significantly more than that in M group at 24 and 72 hours after infusion ($P < 0.05$), and the content of TNF- α was less than that in M group at 24 and 48 hours after infusion ($P < 0.05$). The brain content of SOD in M group decreased to the lowest at 48 and 72 hours after ICH. SOD in N group and H group at 24, 48 and 72 hours after infusion was both significantly more than that in M group ($P < 0.05$). MDA in H group at 72 hours after infusion was less than that in M group ($P < 0.05$).

Conclusions: Repeated infusion of 6% hydroxyethyl starch (130/0.4) or 7.5% sodium chloride both can alleviate brain edema, which are closely related with their effects on inflammatory response and oxidative damage after experimental intracerebral hemorrhage in rats.

A40 Increase in NADH During Intra-arterial Infusion of Mannitol for the Disruption of the Blood Brain Barrier

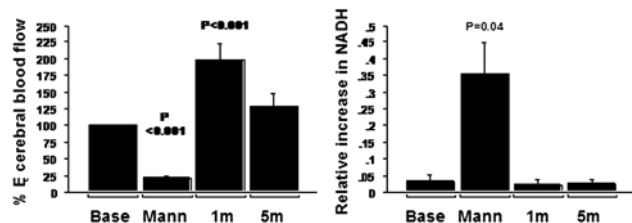
Mei Wang, MPH, Sachin Agarwal, MD, Rahul Rai, Shailendra Joshi, MD. *Columbia University.*

Introduction: Intracarotid infusion of mannitol is a key step for the disruption of the blood brain barrier (BBB) to augment delivery of chemotherapeutic drugs. Injection volumes of mannitol required for the procedure are sufficient to displace cerebral blood flow and expose the endothelial cells to the hyperosmotic agent for 30 seconds. Such concentrations of mannitol are likely to have significant cerebrovascular and metabolic effects and could contribute to the significant complications occasionally reported after treatment.

Methods: The study was conducted to assess acute hemodynamic and cerebrovascular effects of mannitol in New Zealand white rabbit that has a primate like separation of internal (ICA) and external carotid arteries under propofol/ketamine anesthesia. Systemic hemodynamic parameters, electrocerebral activity, cerebral blood flow (laser Doppler), and brain tissue redox state was continuously monitored. Relative changes in brain tissue redox state were assessed by ultraviolet spectroscopy by determine the tissue NADH/NAD ratio. Mannitol (25%, 8 mL) was injected over 30 seconds using a pneumatic syringe pump through a carefully isolated ICA. Data were analyzed at baseline, peak of mannitol injection and 1 and 5 minutes after injection and were analyzed by repeated measures ANOV with post-hoc Bonferroni-Dunn test.

Results: Injection of mannitol resulted in a significant decrease in cerebral blood flow (CBF) and rebound hyperemia in all 6 animals, figure, which lasted for at least 5 minutes. Consistent with the decrease in CBF was a significant increase in brain tissue NADH that promptly returned to baseline.

Discussion and Conclusions: The decrease in CBF observed in this study reflects the intentional displacement of blood from the cerebral circulation needed to disrupt the BBB by dehydrating the endothelial cells. However, this decrease in blood flow results in significant hyperemia (100% increase) that could affect subsequent drug delivery. It may assist the delivery of intravenous drugs by redistributing flow to target site. However, hyperemia could potentially decrease the delivery of intra-arterial drugs by decreasing the transit time and increasing washout. The study also shows that significant though transient cerebral ischemia also occurs during mannitol injection. Role of transient ischemia in rare by significant complications after mannitol injection needs to be addressed during further investigations.



A41 Carbonic Anhydrase 8 (Car8) is Polymorphic in Mice and Variable Dorsal Root Ganglia (DRG) Expression Alters Nociception

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Introduction: The identification of expression quantitative trait loci (eQTL) entails high-throughput expression profiling of mRNA transcripts from target tissue(s) using multiple inbred strains of mice. We hypothesized that differential gene expression may occur in the dorsal root ganglion (DRG) or cell bodies of nociceptors and alter neurobehavioral responses. We now present findings that carbonic anhydrase 8 (Car8) is a cis-regulated eQTL in the DRG of inbred mice and associated with altered mechanical thresholds (von Frey) after nerve injury. Car8 is a nonenzymatic member of the Car family with restricted expression (largely in certain neuronal cells) and an allosteric inhibitor of the IP3R1 that regulates neuronal calcium signaling. The role of Car8 in the nociceptor has not been characterized.

Methods: Total RNA was pooled from DRG of 25 healthy inbred strains of mice, amplified, and labeled using the Affymetrix (Santa Clara, CA) whole transcript sense target labeling assays for a gene array, providing whole-transcript coverage for 28,853 genes. Neurobehavior assessments included mechanical and thermal hypersensitivity after chronic constriction injury (CCI) of the sciatic nerve in Car8^{-/-} knockout (Car8 KO) mice (C57BLKS background) or wild type (WT) C57BLKS/J mice. Mechanical hypersensitivity was measured using withdrawal thresholds with von Frey filaments. Thermal hypersensitivity was measured using withdrawal latency with an infrared heat lamp device. Intergroup comparisons of the mechanical/thermal hypersensitivity response were facilitated by calculating the area under the curve (AUC), where the x-axis represents time (Pre-CCI, Days 1, 7, 14 and 21), and the y-axis represents the hypersensitivity response (threshold or latency) for each animal from each strain.

Results: Figure 1 shows a plot of the probability on the Y-axis ($-\log_{10}$ P value) that differential DRG expression of Car8 (probe 1424958_at) is regulated by a polymorphic element in the vicinity of the Car8 locus on chromosome 4, as indicated by the circle (cumulative position with the alternating colors indicating the chromosomes). Car8 KO mice had a 46% reduction in mechanical hypersensitivity compared with the WT mice (6.89 ± 2.22 vs. 12.64 ± 0.92 , $P < 0.05$), indicating decreased neuropathic pain in Car8 KO mice. No difference in thermal hypersensitivity was observed between these strains.

Discussion: These data indicate the Car8 locus is polymorphic and associated with biologic variability in Car8 DRG expression. Moreover, decreased Car8 expression is associated with lower mechanical hypersensitivity after nerve injury. Further investigations on the potential analgesic role of Car8 in the DRG are warranted.

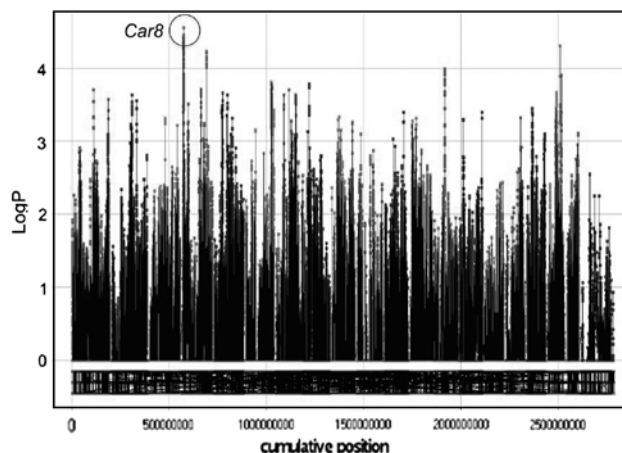


FIGURE 1. eQTL Results. The Y-axis is the $-\log_{10}$ of the P value. The Car8 locus is denoted by its location on the physical map of the X-axis.

A42 Effect of Esmolol on Ischemic Depolarization in Gerbils

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Introduction: In the current study, esmolol, a short-acting, highly selective beta1 adrenoreceptor antagonist, showed neuroprotective effects. We have reported that esmolol shortened the duration of ischemic depolarization in the 5-min forebrain ischemia model in gerbils. In this study, we studied the effect of esmolol on the duration of ischemic depolarization in the 3-min forebrain ischemia model.

Materials and Methods: Fourteen male Mongolian gerbils were used. The animals were anesthetized with 1% halothane in 30% oxygen and 70% nitrogen. They were randomly assigned to either the control group that was administered saline at 0.4 mL/h ($n = 7$) or the esmolol group that was administered esmolol at 200 mg/kg/min IV ($n = 7$). Infusion was initiated 30 minutes before forebrain ischemia by bilateral occlusion of common carotid arteries for 3 minutes and continued for 1 hour after forebrain ischemia. Direct-current potential was recorded from the hippocampal CA1 regions on both sides. Histological outcome was evaluated 5 d after the ischemia. Brain surface and rectal temperatures were maintained at $37.0 \pm 0.5^\circ\text{C}$ from 30 minutes before the ischemia to 60 minutes after the reperfusion. Statistical analysis was performed with the use of Student *t* test. $P < 0.05$ was considered significant.

Results: Onset times of ischemic depolarization were 1.60 ± 0.42 minutes in the esmolol group and 1.57 ± 0.37 minutes in the control group ($P = 0.80$). Recovery time (from the initiation of recirculation to 80% recovery from maximal DC deflection) was 2.05 ± 0.60 minutes in the esmolol group and 1.86 ± 0.46 minutes in the control group ($P = 0.35$). Duration of ischemic depolarization was 3.44 ± 0.88 minutes in the esmolol group and 3.29 ± 0.69 minutes in the control group ($P = 0.61$).

Conclusions: Continuous intravenous infusion of esmolol had no effect on the onset time, recovery time, and duration of ischemic depolarization in the 3-minutes forebrain ischemia model in gerbils.

A43 2011 Michenfelder Award Winner: Neuroligin Overexpression Drives Synapse Formation During Adult Hippocampal Neurogenesis

Eric Schnell, MD, PhD AeSoon L. Bensen, BA, Eric K. Washburn, BA, Gary L. Westbrook, MD *Portland VA Medical Center, Vollum Institute, OHSU.*

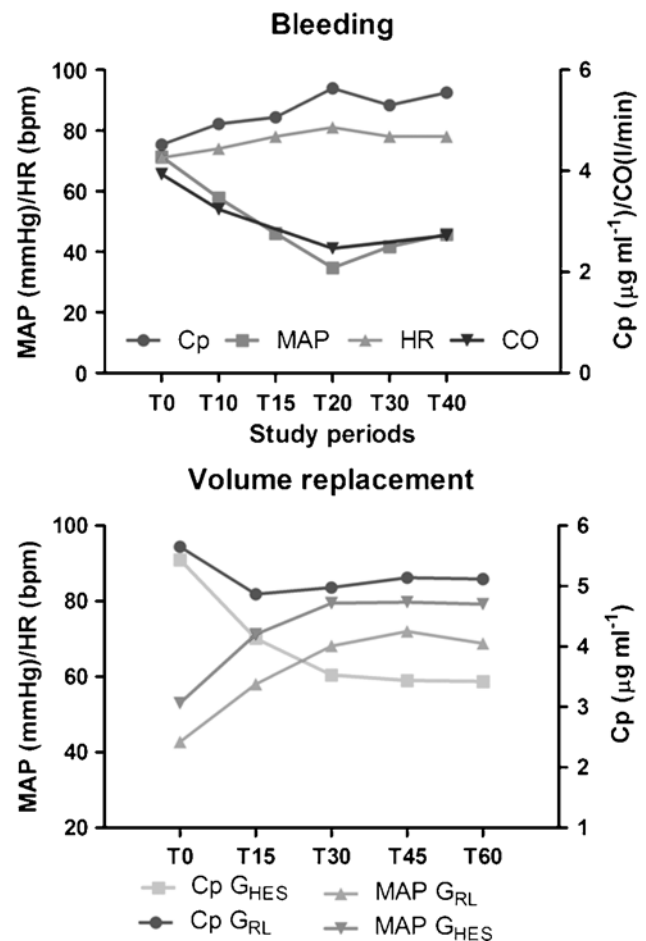
Constitutive neurogenesis continues throughout the life of adult mammals in at least 2 brain regions: the hippocampal dentate gyrus and the subventricular zone. Unlike neurons generated during embryonic development, newly born neurons in adults must navigate through the relatively hostile environment of mature tissue to extend neurites. As neuroligins are known to drive synapse formation during embryonic development, we examined whether neuroligins have a specific role in the formation of synapses by new adult-born neurons in the dentate gyrus. We used replication-deficient retroviruses to specifically infect dividing neuroblasts and manipulate neuroligin levels as these newborn granule cells differentiated in adult mice. Co-expression of GFP via an IRES sequence labeled infected neurons. Retrovirally expressed HA-tagged neuroligin proteins trafficked to punctate structures on granule cell dendrites in the dentate molecular layer. Over-expression of neuroligin-1 accelerated the development of dendritic spines in newborn granule cells, increasing the number of spines in neurons as they matured. These results are consistent with a role for neuroligin in the formation of excitatory synapses onto adult-generated granule cells. Whole cell recordings from infected cells demonstrated an increase in excitatory synaptic function, as well as a change in the ratio of synaptic AMPA-type to NMDA-type glutamate receptors. These experiments suggest that manipulations of neuroligin levels in other populations of newly differentiating adult neurons, such as those formed after brain injury or after stem cell transplantation, could modulate functional integration of these cells.

A44 Propofol Plasma Concentrations After Volume Replacement With HES 130/0.4 and Ringer Solution Following Severe Hemorrhage in Pigs

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Background and Goal of the Study: Profuse bleeding alters the volume capacity of the central compartment, and may change the intercompartmental constants. Additionally, volume replacement with crystalloids or colloids may yield different changes in the pharmacokinetics of the drugs. The objective of this study was to analyze the changes in propofol plasma concentrations (PropCp) during acute hemorrhage followed by volume replacement with Ringer lactate (RL) vs hydroxyethyl starch (HES) 130/0.4 in pigs.

Materials and Methods: Ten healthy Large White male pigs with 27.3 ± 3.5 kg were submitted to a propofol and remifentanyl constant infusion rates of 15 mg/kg/h/L, and $0.3 \mu\text{g/kg/h/L}$, respectively. One hour after the beginning of the propofol infusion, 25 mL/kg/L of blood were passively removed from the femoral artery during 20 minutes. After an extra 20 minutes waiting period, the volume was replaced by RL 25 mL/kg/L (GRL; $n = 5$) or HES 130/0.4 20 mL/kg/L (GHES; $n = 5$). Blood samples were taken at 0, 10, 15, 20, 30, 40 minutes after the beginning of bleeding, and 15, 30, 45 and 60 minutes after the beginning of volume replacement. During the bleeding and waiting periods, the 10 pigs were analyzed as a single group. Heart rate (HR), mean arterial pressure (MAP) and cardiac output (CO) were also analyzed. ANOVA was used to compare data. Data are mean $\pm$ SD. PropCp was determined by gas chromatography-mass spectrometry.



Results and Discussion: During the bleeding, MAP decreased from 71 ± 7.9 mm Hg to 45.8 ± 5.7 mm Hg $P < 0.001$; CO decreased from 3.9 ± 0.9 L min/L to 2.4 ± 1.1 L min/L; HR showed no significant changes (from 71 ± 14 bpm to 81 ± 29 bpm); PropCp was 4.5 ± 0.8 μ g mL/L at the beginning of the bleeding, and 5.6 ± 1.2 μ g mL/L at the end of the bleeding ($P < 0.05$). At the end of the waiting period, PropCp was 5.6 ± 1.3 μ g mL/L. During the volume replacement in GRL, MAP and CO increased 61% and 91%, respectively; in GHES, MAP and CO increased 53% and 101%, respectively; HR did not change consistently in any of the groups. There were no significant differences in MAP and CO between groups along the replacement period. At the beginning of volume replacement, PropCp was 5.6 ± 1.2 μ g/mL in GRL, and 5.4 ± 1.6 μ g/mL in GHES. Sixty minutes after the beginning of the volume infusion, PropCp was 5.1 ± 1.1 μ g mL/L in GRL and 3.4 ± 0.3 μ g mL/L in GHES ($P = 0.02$).

Conclusions: The increase in propofol concentrations during the bleeding may have been due to the reduced volume of the central compartment by bleeding, while propofol was still being administered. During the volume replacement, PropCp was lower in pigs that received HES 130/0.4, even after having received a lower amount of volume. This may have been due to the movement of interstitial fluids towards the central compartment, consequence of the increased oncotic pressure. These results indicate that extra attention is required when administering drugs by continuous infusions to patients receiving colloids.

Acknowledgments: Financial support by COMPETE: FCOMP-01-0124-FEDER-009525 and FCT (PTDC/CVT/101999/2008).

A45 Effect of Passive Lower Limb Elevation on Internal Jugular Vein Cross-sectional Area in Awake Patients

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Introduction: Cannulation of the Internal jugular vein (IJV) is associated with significant morbidity. Several maneuvers have been described to improve the success of IJV cannulation. This is the first large prospective study of the effect of passive lower limb elevation (PLLE) on IJV cross-sectional area (CSA) in patients scheduled to undergo surgery.

Method: After ethics approval and with informed consent, 45 adult ASA I-III patients scheduled to undergo elective or emergency surgery were selected. Before induction of anesthesia, the right IJV cross-section view was visualized (at the level of the cricoid with the head rotated 15 degrees to the left) using an ultrasound by a single experienced operator, with the patient in the following positions: body supine, supine with passive lower limb elevation (PLLE) at 20 degrees and 15 degree Trendelenburg position (TP). A 60 seconds time interval was allowed to elapse after change in each position before an image was stored. The stored images were arranged in a random order and analyzed by a separate blinded observer. The patient was then asked to choose between PLLE and TP based on comfort.

Results: A total of 45 ASA I-III patients were recruited (25 men, 20 women). Mean supine IJV CSA (1.14 cm² SD ± 0.71), increased in both 15 degree TP and 20 degree PLLE (1.39 cm² SD ± 0.7 and 1.35 cm² SD ± 0.75 respectively). There was no significant difference in IJV CSA between TP and PLLE ($P > 0.05$). Total 93% of patients preferred the PLLE position to TP based on comfort.

Conclusion: PLLE is a possible alternative to TP as a maneuver to aid IJV cannulation in patients where TP is not possible or hazardous. Moreover, PLLE is preferred over TP by most patients.

A46 Factors Affecting Response of Dynamic Maneuvers to Aid Internal Jugular Vein Cannulation in Awake Patients

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Introduction: Cannulation of the internal jugular vein (IJV) continues to be associated with significant morbidity and mortality. The routine use of an ultrasound (USG) is recommended to aid cannulation of the IJV,

but in real world conditions it is not always possible to have an USG available during all IJV cannulations. Maneuvers like the Trendelenburg position, are widely used to increase the IJV cross-sectional area (CSA) and thereby aid cannulation. In a recent study we have found that not all patients have a significant increase in IJV CSA when subjected to maneuvers like the Trendelenburg position. There is currently no scientific data about factors which may affect the response of IJV CSA to different maneuvers. The following is a secondary analysis of data from one of our previous study which attempts to identify factors which may affect the response of commonly used maneuvers to increase IJV CSA.

Method: After ethics approval and with informed consent, 45 adult ASA I-IV patients scheduled to undergo surgery were selected. Exclusion criteria included, inability to rotate or to access the neck, suspected/confirmed raised intracranial tension, skin hypersensitivity to ultrasound jelly, retinal detachment, pregnancy, age less than 21 years, patients unable to give consent, or the presence of a hip/lower limb/spine tenderness or fracture preventing patients from lying supine, 15 degree Trendelenburg or supine with legs raised at 20 degrees. Before induction of anesthesia, the right IJV cross-section was visualized (at the level of the cricoid with the head rotated to 15 degrees left) using an ultrasound by a single experienced operator, with the patient in the following positions: body supine, supine with passive lower limb elevation (PLLE) at 20 degrees and 15 degrees Trendelenburg position (TP). A 60-second time interval was allowed to elapse after change in each position before an image was stored. The stored images were arranged in a random order and analyzed by a separate blinded observer (unaware of the maneuver related to the image). The IJV CSA was determined by the ultrasound's planimetry software. Patients with an increase in IJV-CSA of more than 20% after a dynamic maneuver were designated as "responders." Discrete variables were analyzed using the χ^2 test while a t-test was used to analyze continuous variables. A P value of < 0.05 was taken to be significant.

Results: Total 25 patients had an increase in IJV CSA of $> 20\%$ after dynamic maneuvers (= Responders) while in 20 patients there was a $\leq 20\%$ change in IJV CSA (= nonresponders). There was significant difference in weight, sex and presence of maintenance fluids between the responders and nonresponders. While no difference was found in the comorbidities, fasting status, age or baseline IJV CSA between the responders and nonresponders. Patients with maintenance fluids running preprocedure had an adjusted Odds Ratio of 4.63 (95% CI: 1.0, 20.3) of being responders.

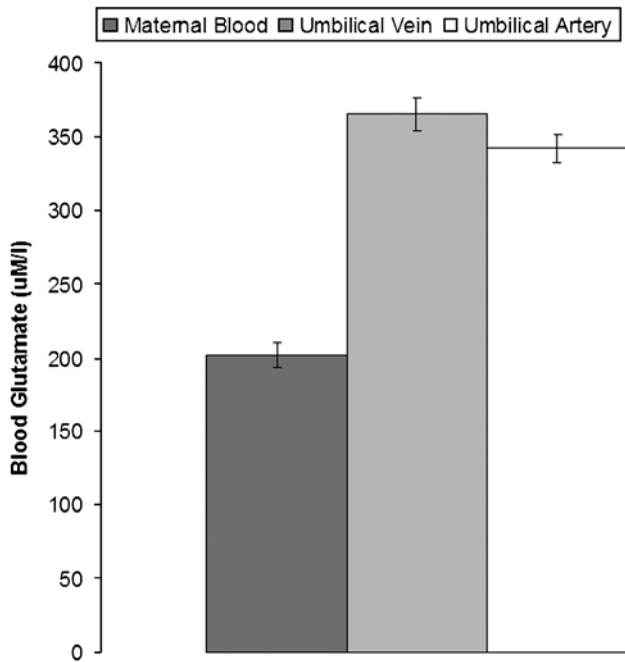
Conclusions: Our study suggests that presence of maintenance fluids preprocedure is associated with the success of maneuvers like the Trendelenburg position and leg elevation in significantly increasing the IJV CSA. But a larger study is needed to confirm these results.

A47 Relationship Between Maternal and Fetal Blood Glutamate Levels

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Introduction: It is well known that abnormally high levels of glutamate (G) in the brain's fluids after an acute brain injury is neurotoxic, and is associated with a poor neurological outcome. Excess glutamate in the brain is thought to be implicated in the pathophysiology of fetal anoxic brain injury, yet little is known about the mechanisms by which glutamate is regulated in the fetal brain. The purpose of this study was to examine whether there are differences between maternal and fetal glutamate concentrations, and whether a correlation between them exists.

Methods: Sixty-seven full-term pregnant women participated in the study. Immediately after delivery of the neonate (vaginal or cesarean section), venous blood was extracted simultaneously from the maternal peripheral vein, and the umbilical artery and vein. Samples were analyzed for levels of glutamate, glutamate oxaloacetate transaminase (GOT), and glutamate pyruvate transaminase (GPT).



Results: Maternal blood glutamate levels were found to be significantly lower than fetal blood glutamate levels in the umbilical artery ($P < 0.001$) and vein ($P < 0.001$) (Fig. 1). Similarly, GOT levels were found to be significantly lower in the maternal blood (19 ± 5 U/L) than in the umbilical artery (39 ± 19 U/L) ($P < 0.001$) and vein (38 ± 20 U/L), ($P < 0.001$). The difference in GPT levels between maternal and fetal blood were not statistically significant. There was no difference in fetal glutamate, GOT or GPT between the umbilical artery and vein. There was a significant positive correlation observed between glutamate levels in maternal blood and glutamate levels in both venous ($P < 0.01$) and arterial fetal blood ($P < 0.05$).

Discussion: This study demonstrated that higher concentrations of blood glutamate are present in fetal blood compared with maternal blood, and this was associated with a (likely compensatory) increase in GOT, but not GPT. A strong correlation was observed between maternal and fetal blood glutamate levels. This may offer new therapeutic insights into fetal blood glutamate reduction in utero by way of reducing maternal glutamate levels.

A48 Influence of PEEP on Cerebrovascular Autoregulation in Patients With Respiratory Insufficiency

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Background: To maintain adequate brain perfusion cerebral blood flow (CBF) is autoregulated within a wide range of arterial blood pressure changes to avoid the adverse effects of hypo-/hypertension on the brain. However, cerebrovascular autoregulation (AR) might be impaired by high levels of positive end expiratory pressure (PEEP) which is often needed in critically ill patients with acute respiratory insufficiency to prevent alveolar collapse and to maintain adequate oxygenation. Therefore, the aim of the present study was to characterize the influence of an elevated PEEP on AR.

Methods: After ethical care committee approval and informed consent, 8 adult patients with respiratory insufficiency and the need for invasive

ventilation were included. Cerebral blood flow velocity (CBFV) was measured using transcranial Doppler sonography and was correlated with the invasive arterial blood pressure curve to calculate the index of autoregulation Mx (Mx > 0.3 indicates impaired AR).^{1,2} Mx was measured during initially low PEEP levels (target PEEP: < 10 mbar) and after lung recruitment with higher level of PEEP (target PEEP: > 10 mbar). Physiological variables were also measured. The data were analyzed using a mixed model (SPSS 18.0) and presented as mean $\pm$ standard deviation.

Results: Eight patients (5 male, 3 female) with mean age of 68 ± 11 (range 49 to 77) years were included. With baseline PEEP (9 ± 1 mbar) the index of AR Mx was 0.32 ± 0.46 and with higher PEEP levels (14 ± 1 mbar) the Mx was 0.45 ± 0.41 (no statistical significance, $P = 0.55$). Five patients had an index of AR Mx > 0.3 at both PEEP levels (= impaired AR). The CBFV (53 ± 24 vs. 51 ± 25 cm/s, $P = 0.9$) and the paCO_2 (49.3 ± 9 vs. 50.1 ± 9 mm Hg, $P = 0.9$) have been comparable at both PEEP levels. The blood pressure, the respiratory frequency, and the sedatives did not change during the investigation within the individual patient.

Conclusions: The present data suggest that a PEEP of about 15 mbar does not further influence the AR. Surprisingly, AR was impaired at both PEEP levels in 5 patients, possibly due to the severity of the underlying illness, comparable to patients with severe sepsis.³ It might also be possible that in these 5 patients, a PEEP of 9 mm Hg already impaired the AR, however, in the other 3 patients AR was intact with a PEEP of 9. Furthermore, CBFV was also not influenced by higher PEEP levels. In conclusion an increase of PEEP from 9 to 15 mbar does not further influence AR.

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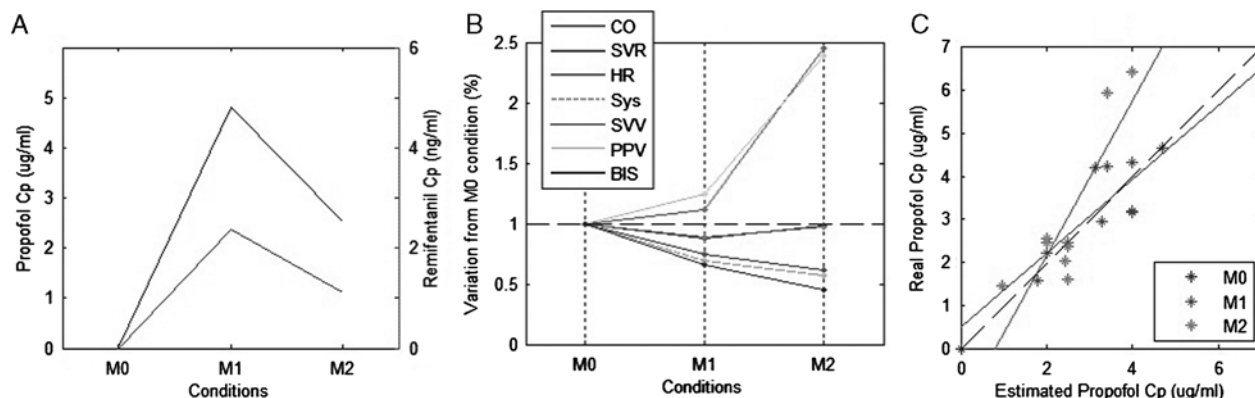
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A49 Hemodynamic Changes and Propofol Concentrations During Induction and Knee-Chest Positioning in Propofol and Remifentanyl TCI Guided Anesthesia

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Introduction: Cardiac output (CO) reduction caused by an induction agent¹ may change the pharmacokinetics of drugs, namely of the induction agent that caused the initial CO reduction. Few studies investigated these changes, namely the role of CO. Knee-chest (KC) positioning following induction reduces venous return and CO.²⁻⁴ The goal of this study was to quantify CO changes during induction and KC positioning and its relationship with propofol plasmatic concentrations in spine surgery patients.

Methods: Patients scheduled for lumbar spinal surgery in the KC position were studied. IRB approval and informed signed consent were obtained. Monitoring included BIS. A radial artery catheter was placed before induction and LIDCO rapid was connected to extract cardiac output (CO), stroke volume (SV), systemic vascular resistances (SVR), DeltaPP (PPV) and DeltaSV (SVV) every second. Induction was with Remifentanyl by TCI to achieve a target Ce of 2.5 ng/mL and propofol 1% at 200 mL/h until loss of consciousness (LOC), at which moment Propofol estimated Ce was noted (Schnider Pk model was used) and propofol was switched to TCI at a effect site target equal to the Ce at LOC. Three study moments were considered at which arterial blood samples were collected for propofol measurements: M0 (before induction), M1 (10' after induction) and M2 (10' after positioning). At each study moment hemodynamic variables were and Propofol plasmatic concentrations (Cp) were compared with Cp estimated. Rugloop software was used to collect data every 5" and to drive Remifentanyl and Propofol pumps. Statistics used Student *t* test for paired samples and $P < 0.05$ for significance. Results are mean $\pm$ SD.



Results: Nine consecutive ASA 1 to 2 patients, 5 females, age 33 to 63 years, were included. Propofol and remifentanyl Cp increased from M1:M0 and decreased from M2:M1 (Fig. 1A). CO, SV, Sys and BIS were significantly decreased from supine to KC position (Fig. 1B). SVR and HR increased from M1:M0 and increased from M2:M1. PPV and SVV increased from M2:M1. As illustrated in Figure 1C, Propofol Cp measured from blood samples and estimated did not exhibit significant differences ($P > 0.20$ for M1 and M2) and were found to be highly correlated ($r = 0.92$ for M1 and $r = 0.79$ for M2).

Discussion: CO decreased 25.6% after induction and 38.5% after knee chest positioning. KC positioning was associated not only to a greater decrease in HD parameters but also in propofol requirements (Figs. 1A, B). DeltaPP and DeltaSV increased, suggesting that patients in KC position behaved as if hypovolemic; this could justify the HD changes. The same could also explain the lower BIS values, without correlation with propofol Cp. Other studies determined significant differences between real and estimated Cp in the knee chest position,^{5,6} however contrast, we found good correlation between them and no differences (Fig. 1C). Our study suggests that KC position associated to anesthesia maintenance with propofol could potentiate HD changes without altering propofol pharmacokinetics.

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A50 Foot Temperature Changes May Reflect Nociception/Anti-nociception Balance During Anesthesia and Surgery: A Clinical Study Using Thermography

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Introduction: Recent research has attempted at developing objective parameters to measure nociception or surgical stress in patients under general anesthesia and surgery. EEG derived (like BIS variability) or autonomic nervous system derived parameters, have been proposed. A recently marketed new index (PSI from General Electric), uses pulse oximetry (size of the pulse wave of the plethysmography and rate variation). Skin conductance changes are the basis of another marketed device (Stress Detector). We explored the possibility that sympathetic activity changes related to nociception/anti-nociception balance could cause changes in skin temperature detectable at the extremities, and used a thermographic camera to assess such changes.

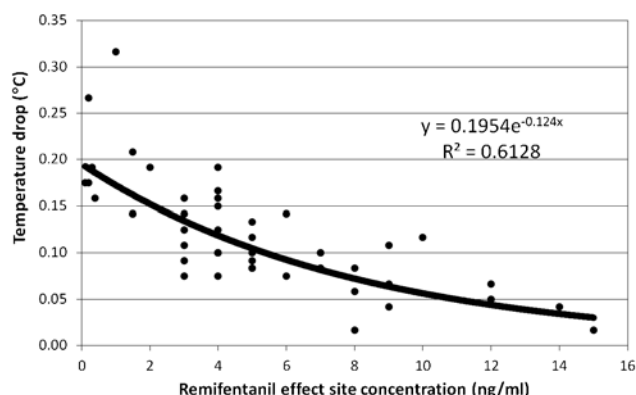
Methods: Neurosurgical patients subjected to brain surgery were studied. Anesthesia was with TCI of Remifentanyl (Pk Minto) and Propofol (Pk Schnider), using effect site target. Rocuronium was used for muscle paralysis. Monitoring included invasive arterial pressure, CVP, esophageal and finger-tip temperature and BIS. A thermographic camera was

used (FLIR A 325), with an accuracy of 0.068°C , a resolution of 320×240 pixels and a frame-rate of 30 Hz. The camera was used to film the patients right foot (plantar face) and continuously record skin temperature. For each patient temperature was measured at four 1-pixel points (first and second toe and their respective metatarsal) and two 20-pixel areas around the metatarsal points. The postprocessing of the recorded temperature values was a simple moving average (last out, first in) of 15 measurements. During the period between intubation and surgical incision, titanic stimuli were applied while remifentanyl effect site concentrations were increased by 1 or 2 ng/mL increments, or decreased, while keeping propofol constant. If HR or BP increased or dropped by more than 25%, Remifentanyl was set back to the initial concentration. IRB approval was obtained. Statistics used polynomial regression.

Results: Data was collected from 5 patients, age 44 to 66, weight 55 to 83, ASA II. A total of 45 assessments were made. Remifentanyl effect site concentrations varied between 0.2 and 15 ng/mL. In 1 patient it was possible to decrease Remifentanyl to very low concentrations; decrements were smaller than planned. Temperature drops were consistently recorded, normally occurring within 5 seconds after the tetanic stimulus. The duration of the temperature drop was not related to its magnitude. A highly significant negative correlation was found between the Remifentanyl effect site concentration and the temperature drop (Fig. 1).

Discussion: Nociceptive stimuli applied to the hand, consistently caused a brief (5s) drop in skin temperature measured on the foot. Vasoconstriction from sympathetic activity is the likely explanation. For the same stimulus, increasing Remifentanyl concentrations resulted in smaller drops in temperature. These results, preliminary data from a small number of patients, suggest that thermography can be a simple and noninvasive tool that can allow objective assessment of adequacy of analgesia during general anesthesia and surgery.

Acknowledgements: Sponsored by FCT-Fundação para a Ciência e a Tecnologia (project PTDC/EEA-ACR/75454/2006).



A51 Preoperative and Postoperative Characteristics of Grimacing and Nongrimacing Patients

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Introduction: We have previously described the incidence and associated anesthetic parameters of grimacing and nongrimacing patients during general anesthesia. Here we examine if there are preanesthesia or postanesthesia differences in patients who do and do not grimace.

Methods: Blinded evaluators selected grimacing from nongrimacing patients from a larger database. The 48 patient database was examined for preoperative and postoperative differences between these 2 groups. Data were analyzed using logistic regression after univariate exploration.

Results: Males grimaced more than females ($\chi^2 = 5.85$, $df = 1$, $P = 0.02$) and older patients grimaced more than younger patients ($r = 0.35$, $P = 0.02$). Length of surgery (LOS) also was positively associated with minutes of grimacing ($r = 0.36$, $P = 0.02$). Higher baseline heart rates predicted intra operative grimacing ($P = 0.009$). Minutes of grimacing (MinR2) in turn predicted higher PACU heart rates (Fig. 21) ($P = 0.01$). Grimacing patients had higher average PACU systolic blood pressures (131 mm/Hg) versus nongrimacing patients (120 mm/Hg) ($P = 0.006$). There were trends towards initially higher PACU pain scores among patients who grimaced during surgery (median NPS = 2.3) compared with nongrimacing patients (median NPS = 0). The results of this study indicate that higher baseline heart rates predict whether and how much grimacing will follow during anesthesia. In turn, minutes of grimacing during anesthesia predicted PACU heart rates. Sex, age and length of surgery were associated with grimacing.

Conclusions: Vital signs do not change during facial grimacing during intra-abdominal surgeries. Grimacing to incision is not related to preoperative clinical fentanyl loading or end tidal gas concentration. Male sex, older age and higher resting heart rates predicted minutes of grimacing. Grimacing during surgery predicts higher PACU heart rates. Grimacing during general anesthesia can be inferred from relative activations of facial muscle areas. The signal is easily obtained as these are surface recordings of an immediate source signal just beneath the skin. Technically correct recordings and an accurate baseline are required. Patterned facial expressions represent an area of investigation of brain function during anesthesia.

TABLE 1. Demographics of Grimacing and Nongrimacing Patients

	Gender		Age		LOS		Minutes R2		PACU HR		Median NPS
	% Male	% Female	Average	SD	Average	SD	Average	SD	Average	SD	
GRIM	90%	47%	46.7	14	131	60	21.8	20.7	82.8	14.4	2.3
NON-GRIM	10%	53%	42.8	13	117.6	40	0.5	0.8	73.5	10.2	0

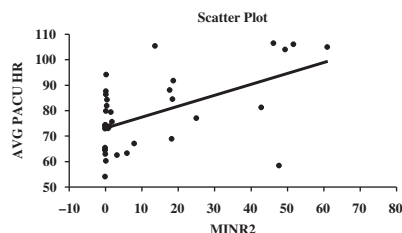


FIGURE 1. Plot of minutes of intra operative grimacing and PACU heart rates.

A52 Effects of Pregabalin on Postoperative Pain and Preoperative Anxiety in Patients Undergoing Spine Surgery

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Background: Pregabalin is a gabapentinoid which has been shown to possess analgesic and anxiolytic properties. We presumed that pregabalin premedication may reduce surgical anxiety and postoperative pain. Hence, the present study was aimed to assess the efficacy of a single dose of pregabalin on postoperative pain and preoperative anxiety in patients undergoing lumbar spine surgery.

Methods: Sixty patients posted for elective lumbar laminectomy and discectomy were randomly assigned to 1 of the 3 groups; Group C: received placebo capsules, Group P1: received pregabalin 150 mg and Group P2: received Pregabalin 300 mg capsules 2 hours before induction of anesthesia. The levels of anxiety and pain were assessed with respective 100 mm visual analog scale (VAS). Postoperative fentanyl consumption and the occurrence of side effects were recorded. All patients were observed for 8 hours, postoperatively.

Results: There was a significant reduction in preoperative anxiety in both pregabalin groups as compared with control group ($P = 0.001$). VAS score for pain (at rest and during movement) was significantly decreased at all time intervals in both P1 and P2 groups as compared with control ($P < 0.05$). Fentanyl consumption (μg) in the postoperative period was significantly reduced in P1 (213 ± 112.9) and P2 (190.8 ± 45.3) groups as compared with the control group (339.1 ± 88.0) ($P = 0.001$). The level of sedation was higher in P2 group in the first 4 hours ($P < 0.05$). The occurrence of postoperative nausea and vomiting was significantly more in control group as compared with the pregabalin groups ($P = 0.018$). The incidence of dizziness and blurring of vision was significantly more in P2 group in comparison to other 2 groups.

Conclusions: Single preoperative dose of pregabalin 150 mg or 300 mg was effective in reducing preoperative anxiety, postoperative pain, and total fentanyl consumption in patients undergoing lumbar laminectomy and discectomy. However, the incidence of postoperative side effects like sedation, dizziness and visual blurring were more in patients who received pregabalin 300 mg.

A53 Risk Factors for Delayed Extubation in Major Thoracolumbar Spine Surgery

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Introduction: Delayed extubation after major spine surgery is sometimes necessary because of the potential upper airway compromise secondary to edema and swelling. Risk factors for delayed extubation in cervical spinal surgery have been investigated.¹ However, risk factors for delayed extubation in major thoracolumbar spine surgeries have not been investigated yet. This study retrospectively reviewed 136 patients who underwent major thoracolumbar spine surgery to determine the risk factors responsible for delayed extubation.

Methods: Patients who underwent multi-level thoracolumbar spinal surgery including posterior spine decompression and fusion with instrumentation from 2006 to 2009 were included in this study. Surgeries were performed by 1 spine surgeon. Total of 220 charts were reviewed; patient's demographic data, surgical time, anesthesia technique, blood loss, blood transfusion and fluid infusion information were recorded. One hundred thirty-six patients were included for analysis. The patients were divided into 2 groups: patients who were extubated after surgery in operative room (nonintubated group, NIG, $n = 108$) and patients who remained intubated postoperatively (Intubated group, IG, $n = 28$). All patients received general anesthesia either total intravenous anesthesia (TIVA) comprised of propofol and fentanyl, or propofol and fentanyl infusion with low dose inhalation agent. There were no antifibrilics used during the surgery.

Results: There were no differences in age, weight and BMI between NIG and IG groups. The cited reason for delayed extubation was potential airway compromise. The risk factors for delayed extubation are related to lengthy surgical duration, increased fluids replacement and blood loss

Factors	NIG n=108	IG n=28
Surgical time (min)	312±7.4	399±25.5 *
Crystalloid replacement (ml/hr)	809.6±19.3	908.2±55.7*
Total Blood loss (ml)	836.2±46.7	2133.7± 100.2*
PRBC transfusion (Units)	0.65±0.1	3.6±0.2*
Time from Incision to transfusion (min)	199.1±9.6	110.9±12.7*

(* Indicates: p<0.01)

(see Table 1). The patients who received blood transfusion within 2 hours after incision ($< 110.9 \pm 12.7$ min) remained intubated while those patients who were extubated received blood transfusion much later ($> 199.1 \pm 9.6$ min after surgery started).

Conclusion: Our results suggest that longer surgical time, increased blood loss and crystalloid infusion, and earlier blood transfusions might contribute to delayed extubation in major thoracolumbar spine surgery.

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A54 Development and Validation of a Generalizable Model for Predicting Major Transfusion During Spine Fusion Surgery

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Introduction: Surgery for posterior spine fusion often requires major transfusion. A Predictive Model of Transfusion in Spine Surgery (PMTSS) based on age, preoperative hemoglobin, instrumentation of more than 2 levels, and the use of transpedicular osteotomies has been previously developed by others.¹ The aim of this study was to identify and validate independent predictors of the need for major transfusion in any spine fusion surgery of ≥ 3 levels to assist with preoperative risk stratification.

Methods: After IRB approval, data from 548 patients who underwent spine fusion of ≥ 3 levels between January 1, 2003, and May 30, 2009, were extracted. Major blood transfusion was defined as 4 or more total units of red cells, which included packed red cells and reinfusion from cell salvage. The surgical complexity was categorized as anterior surgery only, posterior cervical and/or thoracic fusion, lumbar posterior fusion, or any surgery including transpedicular osteotomies. A univariate analysis identified variables to be included in the initial multiple logistic regression analysis. Multiple logistic regression analysis with stepwise backward elimination of variables was used to develop the final predictive model. The model was validated with a separate dataset of patients who underwent surgery from June 1, 2009, through September 30, 2010. This patient dataset was also used to externally validate the ability of a PMTSS > 3 to predict major transfusion. McNemar test was used to compare the sensitivity and specificity of both models.

Results: Variables in the final multiple logistic regression model as predictors of major transfusion were operation duration; number of posterior levels instrumented; surgical complexity score; and pre-incision hemoglobin. Although there was no difference in the sensitivity of our model with a cut off probability of $> 80\%$ and the PMTSS > 3 in predicting major transfusion ($P = 0.359$), our model had an increased specificity in predicting major transfusion compared with the PMTSS > 3 ($P = 0.043$) (Table 1).

Discussion: Our model identified some of the same major predictors identified in PMTSS—pre-incision hemoglobin and number of posterior levels instrumented.¹ In contrast, we demonstrated that surgical duration and the number of surgical levels are each independent determinants of major transfusion. This extends the PMTSS by discretely examining the influence of the number of levels instrumented

MODEL	Sensitivity	Specificity	Positive Predictive Value	Negative Predictive Value
Current Model	70.5%	82.4%	87.8%	60.9%
PMTSS > 3	63.9%	58.8%	73.6%	47.6%

and the duration of surgery. We validated both models with a separate dataset. Our model had an increased specificity and similar sensitivity for predicting the probability of major transfusion compared with a PMTSS > 3 . In addition, our model is generalizable to all types of spine fusion surgery, and accounts for the complexity of surgical instrumentation, the number of levels instrumented, and the predicted duration of surgery as independent variables.

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A55 Postoperative Visual Loss After Spine Surgery—A Case Control Comparison From Single Institution

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Vision loss is a rare but a devastating complication that can be seen after both ocular and nonocular surgeries. The exact incidence is unknown. The incidence is generally thought to be in the range of 0% to 0.2% after spine surgery. We undertook a retrospective chart review and analysis of all patient undergoing complex spine surgery at the Cleveland Clinic between December 1995 and December 2010 to identify the incidence and quantify the risk factors for postoperative visual loss after surgery. Six patients with a confirmed loss of vision were identified by review of their postoperative records and form the basis of this study. After an IRB approval, charts of all patients were reviewed. Each case was matched to 5 controls based on age, sex, height, weight, body mass index, systolic, diastolic and mean pressure, and hematocrit. Potential controls were selected from the Cleveland Clinic Perioperative Health Database System (PHDS maintained by the Cleveland Clinic Division of Anesthesiology. Total, 2532 potential controls undergoing spinal fusion surgery between August 2005 and December 2009 were identified. However we had to exclude 784 patients due to missing data on 1 or more variables. Intraoperative measurements were compared between cases and matched controls using Wilcoxon rank-sum test: duration of anesthesia, estimated blood loss, total crystalloids, colloids, red blood cells, total fluids), minimum systolic, diastolic and minimum mean arterial pressure. For our tests we used a significance level of 0.01 to adjust for multiple comparisons. Cases and controls were matched for age, sex, height and weight, BMI, preoperative systolic, diastolic and mean blood pressures and preoperative hemoglobin. Patients who had lost their vision during surgery had significantly greater blood loss ($P = 0.002$) and they had been transfused with greater volume of packed RBCs. ($P = 0.006$). No other intraoperative measures were significantly different from each other between the 2 groups as shown in Table 1.

In conclusion, the amount of blood loss and transfusion continue to be important predictors of development of POVL. Neither anesthetic duration nor intraoperative hemoglobin levels was found to be significantly different between cases and controls. Patients and controls were not different in the degree and duration of hypotension between them. Anemia had been shown to be a risk factor for ION. However our review did not find any association with anemia.

Limitations: Our pool of potential controls excluded a relatively large proportion of spinal fusion cases (31%) due to missing data, leaving the possibility of selection bias. Furthermore, the retrospective nature of the study rendered us unable to control for unavailable confounders.

Summary: In summary, in our study, the amount of blood loss and transfusion continue to be important predictors of development of POVL. None of the other risk factors like anesthetic duration, intraoperative hemoglobin levels nor the degree and duration of hypotension had been shown to be a risk factor for ION.

TABLE 2. Comparison of Intraoperative Measurements between Six Vision Loss Cases and 30 Matched Controls. *P*-values from Wilcoxon's Rank Sum Test

	Case #1	Case #2	Case #3	Case #4	Case #5	Case #6	Matched Controls ^A	<i>P</i> -Value
Duration of Anesthesia (minutes)	445	362	539	83	735	529	123 / 253 / 320 / 372 / 841	0.10
Estimated Blood Loss (cc)	4285	850	3600	800	57370	4000	50 / 112 / 275 / 400 / 6923	0.002
Total Fluids ^B (cc)	10090	4000	9504	4450	57650	11816	1300 / 3825 / 5200 / 5950 / 14300	0.12
Crystalloids (cc)	5600	3000	6600	4100	32000	7500	1300 / 3000 / 3800 / 5025 / 12200	0.08
Colloids (cc)	2470	1000	1000	350	1500	1000	0 / 125 / 1000 / 2000 / 3000	0.68
Red Blood Cells (cc)	2020	0	1904	0	24150	3316	0 / 0 / 0 / 0 / 2709	0.006
Minimum Systolic BP (mmHg)	63	80	70	85	80	85	35 / 62 / 78 / 84 / 103	0.67
Minimum Diastolic BP (mmHg)	30	50	30	60	30	50	32 / 37 / 44 / 50 / 57	0.52
Minimum MAP (mmHg)	41	60	42	77	47	62	32 / 47 / 57 / 62 / 71	0.77

^A Summary statistics given as 'minimum / 1st quartile / median / 3rd quartile / maximum'^B Total fluids calculated as the sum of crystalloids, colloids, and red blood cells.**A56 A Prospective Study of a Modified WHO Checklist in the Neurosurgical Operating Room**

Shaheen F. Shaikh, MD, Andrew Cocchiarella, MD, Julie Pilitsis, MD, U Mass Memorial Medical School.

Object: Recent studies in cardiac and general surgery have demonstrated that individually both the implementation of a WHO preoperative checklist and the presence of dedicated OR teams reduces complications, patient morbidity, mortality and health care expenditures. We prospectively examined the effects of implementation of a modified WHO checklist and development of a designated neuroanesthesia team on defined operating room metrics in an academic neurosurgical practice.

Methods: Prior to any intervention, we collected prospective data on OR times, timely administration of antibiotics, attitudes about safety and 30 day outcomes. During implementation of the checklist and of the neuroanesthesia team, the same data was collected and compared with pre-implementation data. Post-hoc analysis examined results in relation to compliance with the changes.

Results: Timely administration of antibiotics improved from 70% to 91%. OR times and safety attitudes scores however remained unchanged. A post-hoc analysis of compliance revealed that the presence of a core neuroanesthesia provider in 76% of cases correlated with significantly shorter surgical times and total OR times ($P = 0.047$, $P = 0.05$ respectively). The greatest improvement in total minutes in the OR was seen in cases where both the checklist was used and a neuroanesthesia provider was present ($P = 0.02$).

Conclusions: We provide objective prospective evidence that the presence of a designated neuroanesthesia providers and the use of the checklist improves timely administration of antibiotics and correlates with shorter OR times. Enhancing compliance is essential to real world effectiveness.

A57 Development of a Scale to Assess Noxious Stimuli Intensity During Anesthesia and Surgery Using Rasch AnalysisAna Castro, BS, Pedro Amorim, MD, Catarina S. Nunes, PhD, Fernando Gomes de Almeida, PhD. *Investigacao Clinica do Servico de Anestesiologia, King's College London, Faculdade de Engenharia da Universidade do Porto.*

Background: During surgery and anesthesia patients are subjected to various sources of noxious stimulation. Due to the impossibility of

obtaining direct evaluations of patients' perceived pain, we decided to investigate anesthesiologists' perception of anesthetic and surgical stimulus intensity.

Methods: A survey was designed to evaluate anesthesiologists' perception of noxious stimulus intensity, in an ordinal scale from 0 (non-noxious stimulus) to 10 (severe noxious stimulus), considering stimulus derived from the anesthetic procedure, surgical procedure pre-incision and incision, and from the surgical procedure postincision (in total 35 stimuli, Figure 1). The perception on which is the most intense stimulus, laryngoscopy/intubation or incision for appendectomy was also evaluated. The designed survey was disseminated in the anesthesiologist's community using Google Docs. Rasch analysis was performed to allow the construction of a pain intensity measure in a linear scale,^{1,2} providing both stimulus and rater location, using Minifac (Facets student version 3.66.3, Linacre 2010).

Results: Fifty-seven anesthesiologists (19 male) answered the survey. Data from 5 respondents were excluded due to deviations or rating scale misunderstanding. Total 74% of the respondents considered laryngoscopy/intubation the most intense stimulus when compared with abdominal incision for appendectomy, later confirmed in the Rasch analysis, were laryngoscopy was the third most intense stimulus, only surpassed by abdominal incision 15cm supra-umbilical and nephrectomy incision. Figure 1 presents the results of the Rasch analysis considering stimulus intensity location. The most painful stimuli considered were: abdominal incision 15cm supra-umbilical, nephrectomy incision, laryngoscopy, prostatectomy incision, lumbar incision, intubation, incision for drain or small incision for laparoscopic surgery and incision for craniotomy. In the considered stimulus the lowest intensity perception reported was for the following stimuli: venous catheter, radial artery catheter, gastric drainage tube insertion or temperature probe, epidural puncture, Foley catheter and periods in surgery with no surgical stimulus.

Conclusions: A stimulus intensity measure under general anesthesia was obtained based on anesthesiologists' perception. The results obtained in this study may be used in the search for relations between different stimulus intensity, analgesic levels and physiological responses to stimulation, and allow clinicians an overview of stimuli intensity they encounter in their daily activity, and the inference of the relations between them.

Acknowledgements: FCT; UISPA-IDMEC, Portugal.

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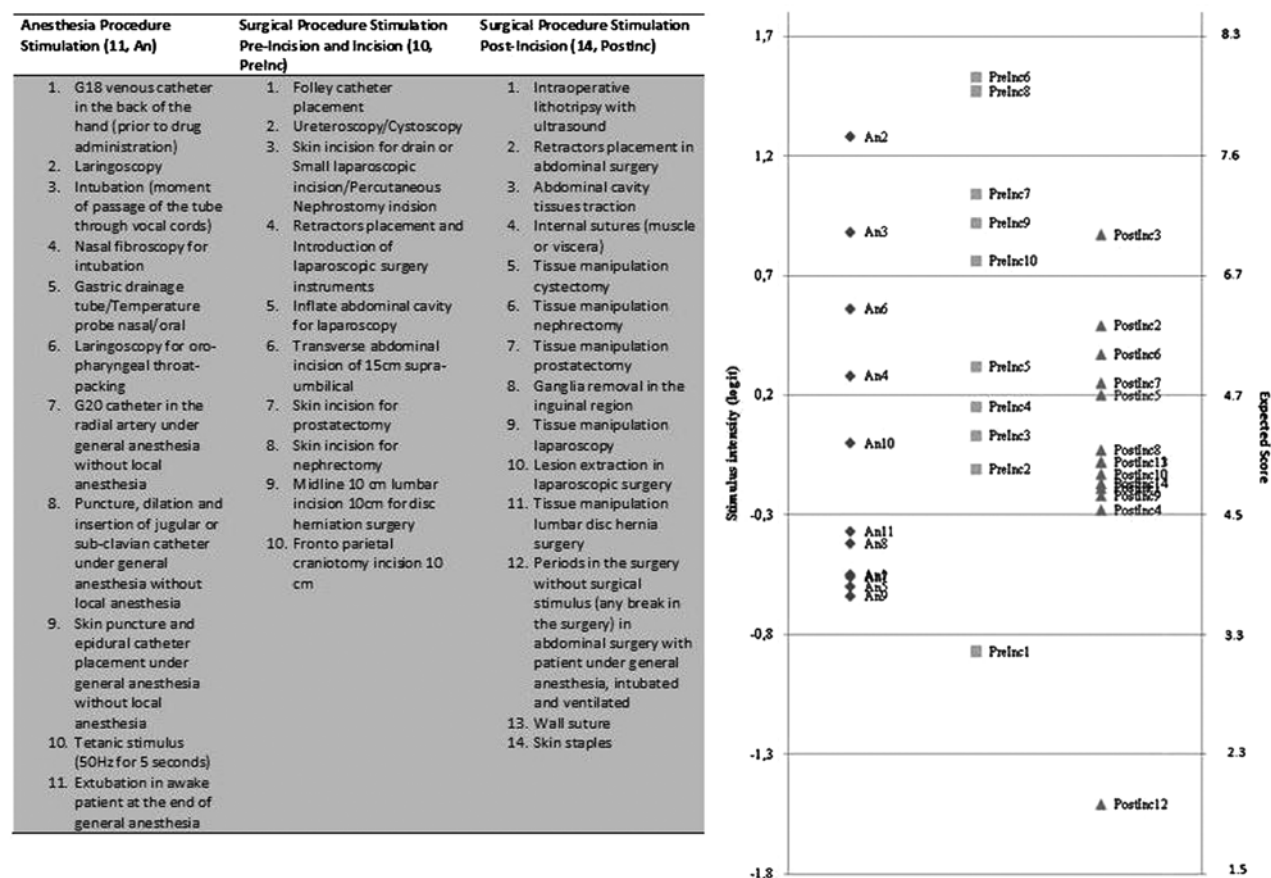


FIGURE 1. List of stimuli analysed in the study, divided according to three groups; stimuli derived from the anesthesia procedure, stimuli derived from the surgical procedure, pre-incision and incision; and stimuli derived from the surgical procedure post-incision (left). Stimulus intensity in logits, and corresponding expected score in the original rating scale (right).

A58 Relationship Between Peripheral Blood Glucose Concentration and Cerebrospinal Fluid Glucose Concentration

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Introduction: Under normal physiological condition, there is a dynamic equilibrium between cerebrospinal fluid glucose concentration (SFGC) and peripheral blood glucose concentration (PBGC). Changes in PBGC induce parallels changes in the SFGC with a normal SFGC/PBGC ratio between 0.6 and 0.8, in healthy subjects this ratio remains constant unless blood glucose increases above 270 to 300 mg/dL.^{1,2} Both PBGC high and low values (hyperglycemia and hypoglycemia) are frequent in critically ill neurological patients and possibly associated with increased morbidity and mortality and in this scenario the changes induced in SFGC can exert critical effects on brain metabolism.^{3,4} Aim of the study was to establish relationship between SFGC and PBGC in critically ill neurological patients with acute subarachnoid hemorrhage (SAH).

Materials and Methods: We simultaneously sampled SFGC and PBGC in patients with SAH admitted in neurocritical care, with external ventricular drains in-situ.

Results: In 10 patients we have recorded 50 paired measures, 50 PBGC and 50 SFGC, over the first 5 days of ICU stay. Mean PBGC on day 1st was 149.8 ± 12 mg/dL and a SFGC was 96.7 ± 6 mg/dL with mean SFGC/PBGC ratio of 0.64. Mean PBGC on day 5th was 167.4 ± 9 mg/dL and a SFGC was 86.2 ± 7.2 mg/dL with mean SFGC/PBGC ratio of 0.52 (Table 1, Figs. 1, 2).

Discussion: Our results demonstrated that in SAH patients over the first 5 ICU stay the SFGC/PBGC ratio progressively lowers up to a ratio 0.52 and reach values below the normal range of 0.6 and 0.8. This evidence can be possibly explained by the metabolic stress of an acutely ill brain and exposes patients to an increased risk of low glucose availability. If these data will be confirmed in larger case series the PBGC management in SAH patients should be designed according the specific need and the unique metabolic conditions.

TABLE 1. Mean Values of PBGC, SFGC and SFGC/PBGC Ratio

DAY	PBGC	SFGC	SFGC/PBGC ratio
DAY 1	149.8	96.7	0.64
DAY 2	163.9	94.6	0.591
DAY 3	150.5	79.7	0.56
DAY 4	136.8	77.8	0.57
DAY 5	167.4	86.2	0.52

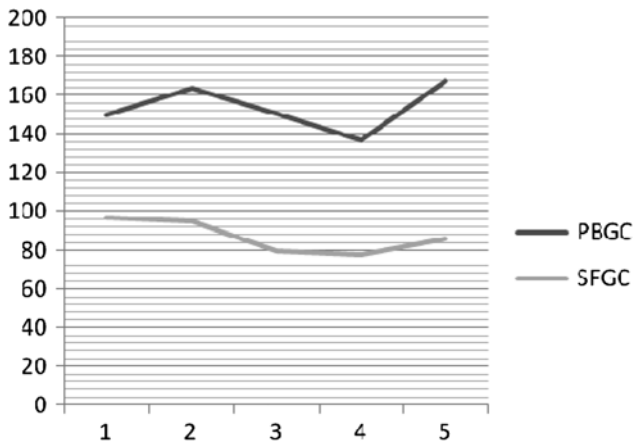


FIGURE 1. Mean PBGC.

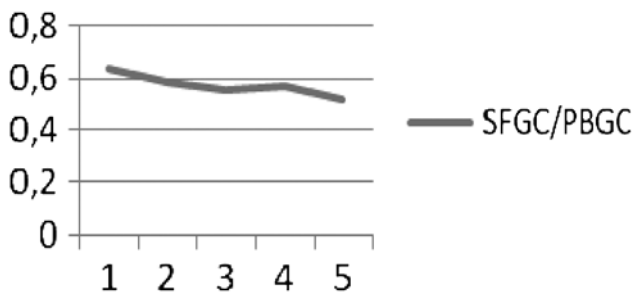


FIGURE 2. Mean SFGC/PBGC ratio.

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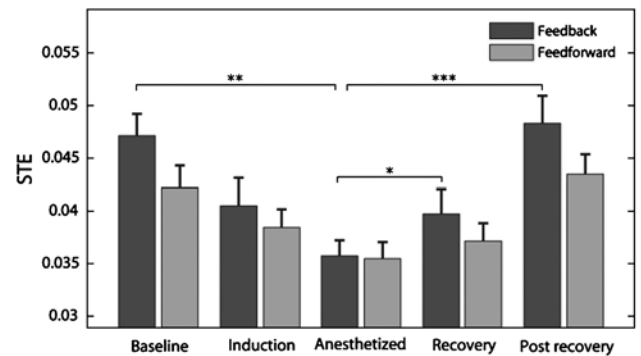
A59 Inhibition of Frontal-to-Parietal Feedback Connectivity is a Neural Correlate of General Anesthesia in Surgical Patients

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Background: The precise mechanism and optimal measure of anesthetic-induced unconsciousness has not been elucidated. Inhibition of feedback connectivity from frontal to parietal brain networks is 1 potential mechanism, but has only been demonstrated in animals or in limited conditions with healthy volunteers.

Method: We recruited 18 patients presenting for surgery under general anesthesia; electroencephalography of the frontal and parietal regions was acquired during (i) baseline consciousness, (ii) anesthetic induction with either propofol or sevoflurane, (iii) general anesthesia, (iv) recovery and (v) postrecovery states. We used a computational method, symbolic transfer entropy, to analyze frontoparietal connectivity.

Result: The dominant feedback connectivity of the baseline state was significantly inhibited after anesthetic induction and during general



anesthesia, resulting in reduced asymmetry of feedback and feed-forward connections in the frontoparietal network. Dominant feedback connectivity returned when patients regained consciousness (Significance was denoted with standard error in Figure 1, $*P < 0.05$, $**P < 0.01$, $***P < 0.001$: by repeated measure 1-way analysis of ANOVA and a post-hoc multi-comparison test). Both classes of anesthetics demonstrated similar results in this heterogeneous population of surgical patients; these results were further supported by an independent measure of effective connectivity.

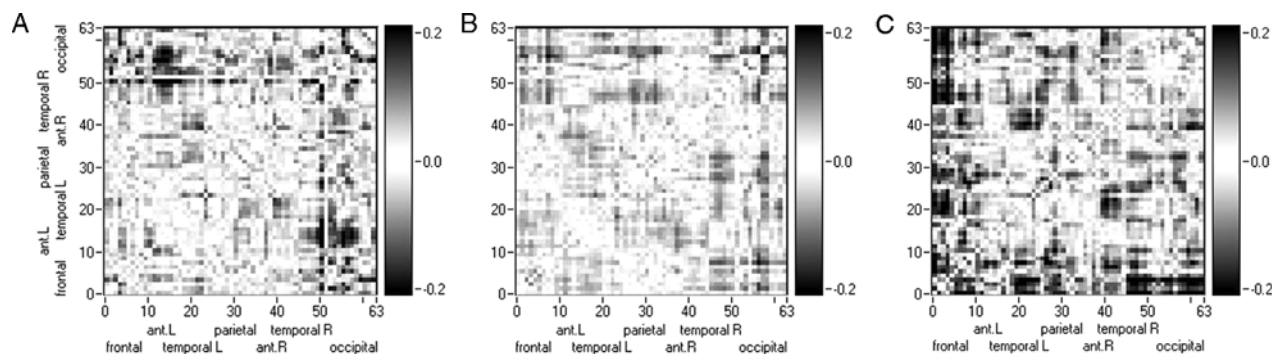
Conclusions: The disruption of dominant feedback connectivity in the frontoparietal network is a common neural correlate of general anesthesia. This study represents a key translational step from the underlying cognitive neuroscience of consciousness to more sophisticated monitoring of anesthetic effects in human surgical patients.

A60 Electroencephalographic Symbolic Transfer Entropy Indicates Changes in Cortical Information Flow During Propofol Sedation and Unconsciousness

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Introduction: Recently, a decreased connectivity of higher cortical networks under propofol induced loss of consciousness was reported using functional connectivity analysis of resting state networks (RSN).¹ The neurophysiologic correlates of observed changes in cerebral blood flow through imaging techniques on the system information processing level remain unclear. In the present study, effects of propofol induced sedation and unconsciousness in EEG information flow were investigated. Therefore, multichannel EEG was analyzed using symbolic transfer entropy (STEn) as a measure to quantify interdependences which underlay physiological processes.²

Methods: Approved by the ethics committee, 15 volunteers were enrolled into the study. After a resting period, volunteers were instructed to relax and close eyes while 64-channel EEG baseline (BL) recordings were performed for 10 minutes. Subsequently propofol was infused until loss of consciousness (LOC) using a target controlled infusion (TCI) pump. TCI concentration was maintained for 10 minutes and EEG measured. After that, a phase of sedation (0.5LOC) was maintained during 10 minutes at 50% of LOC concentration. Nonparametric STEn quantifies the mutual information flow between 2 signals and was computed over all channel pair combinations using EEG of 10 seconds length (0.5 to 30 Hz total bandwidth, 200 Hz sampling frequency) and 50 ms time delay reflecting mainly information processing within the beta-band (embedding dimension $m = 3$, transfer delay $d = 10$). Thereby information is coded by amplitude order sequences, where the degree of prediction of actual information from past signal content is reflected by a generalized Markov property.² Ninety-five percentile bootstrap confidence intervals (CI) based on the area under the curve (AUC) indicated effects of propofol ($P < 0.05$).³



Results: The figure shows density of STEN at BL (A), 0.5LOC (B) and LOC (C) for each electrode combination. Thereby, lower absolute values of STEN indicate balanced information exchange (BL and particularly 0.5LOC) and higher values unidirectional coupling (mainly LOC). Effects from BL to LOC can be observed in frontal-parietal (AUC CI 0.71 to 0.97), frontal-temporal (0.73 to 0.98) and frontal-occipital (0.72 to 1.00) combinations.

Conclusions: STEN reveals effects related to the information flow between cortical areas. A decrease of balanced flow in EEG could be caused by loss of feedback connection during propofol induced unconsciousness which was already reported in field potentials in rats during visual stimulation.^{4,5} Present results of spontaneous EEG show that long range communication pathways according to default and executive control networks seem to be particularly affected, whereas local interactions are mainly preserved during unconsciousness. In contrast, sedation seems even to intensify long range exchange of information. This may be explained by a reported extension of the default mode network during propofol sedation.⁶

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filtering routine that allows filtering with zero phase shift. Symbolic transfer entropy STEN (dimension $m = 3$, transfer delay $d = 6$) was calculated for all hipp vs. ncx channel combinations and for all combinations within these areas. STEN is a modification to the transfer entropy algorithm. In STEN the analyzed time is coded to a vector of different symbols as in the Permutation Entropy algorithm. Absolute STEN values of each animal were averaged and control versus propofol STEN was compared with a paired Wilcoxon test. Significance level was set to $P < 0.05$.

Results: Absolute STEN(alpha) significantly decreases with propofol, indicating decreased information transfer between hipp and ncx. Within hipp, STEN(alpha) shows a strong, decreasing trend, ie, less information is processed. There is no change in alpha-information processing in the prefrontal cortex. STEN(theta) between hipp and ncx nonsignificantly decreases but $P < 0.05$ in the performed signed test shows a strong trend, whereas there is no unitary change in STEN (theta) within the areas. The figure displays STEN for all channel combinations at control conditions (left column) and after delivery of propofol (right) in ncx (A), hippocampus (B) and between both areas (C). Darker shades of gray indicate less information flow.

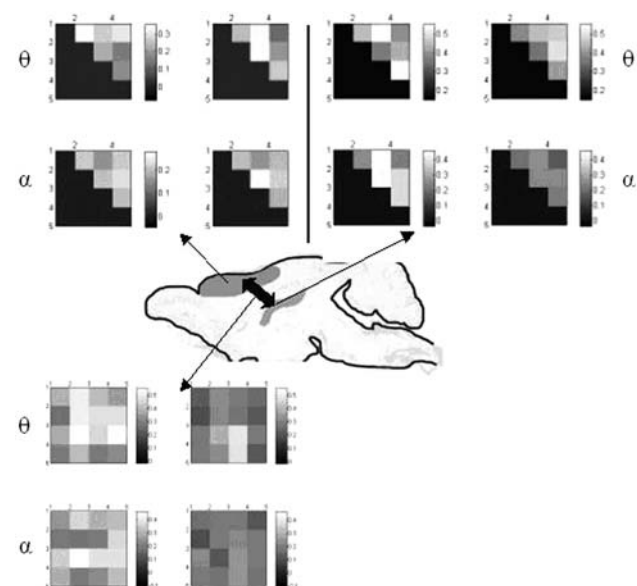
Discussion: With STEN, it is possible to reveal propofol induced reduction of information processing between hipp and ncx in the relevant alpha and theta frequency range, responsible for communication between both areas. Reduction of information transfer in the relevant pathway might be a main cause for anesthesia induced

A61 Effects of Propofol on Information Processing Between Neocortex and Hippocampus

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Introduction: Interaction between hippocampal (hipp) and prefrontal cortex (ncx) is performed by synchronic oscillations in the theta (4 to 8 Hz) and alpha band (8 to 12 Hz).^{1,2} Slowing of these oscillations may contribute to suppression of learning and memory, that is, a drug-induced modulation of these frequencies may contribute to anesthesia.³ We used the nonlinear, bivariate measure of symbolic transfer entropy STEN⁴ to quantify the propofol induced change in information processing within and between ncx and hipp areas. If local field potential LFP activity within the recording areas or between hippocampus and prefrontal cortex is affected by propofol, interaction between these areas may be perturbed and this effect should be detected by STEN.

Methods: LFP sequences of 9 mice were simultaneously recorded from 4-channel electrode arrays placed in ncx and hipp regions. An artifact free 5 seconds LFP sequence was extracted from each recording at control conditions and under hypnotic concentrations of propofol. All sequences were down sampled to 50 Hz and band pass filtered to 4 to 8 (theta) and 8 to 12 Hz (alpha) with a double reverse Butterworth



unconsciousness. With STEn an anesthetic affect on hippocampal-neocortical information transfer, by means of nonlinear signal analysis could be shown for the first time.

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3. *Anesthesiology*. 2010;113:1299–1309.
4. *Phys Rev Lett*. 2008;100:158101.

A62 Functional Connectivity Changes of Thalamus During Sufentanil Infusion

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Background: Thalamus has been indicated to play an important role in central analgesic function of opioids in previous studies. However, few

studies have investigated the effect of sufentanil on functional connectivity in thalamus. The purpose of this study was to investigate the changes of resting-state functional connectivity of thalamus during sufentanil target controlled infusion (TCI), thus to explore the analgesic mechanisms of sufentanil.

Methods: Seventeen (8 male, 9 female) healthy right-handed volunteers participated in this study. They were administered with crystalloid fluid, monitored with electrocardiography, pulse oximetry, and noninvasive blood pressure measurement, and relaxed for 30 minutes before the scan started. The subjects underwent functional magnetic resonance imaging (fMRI) scan twice with 3.0 T MRI. All the subjects received the fMRI scan before and after sufentanil administration respectively. The TCI concentration of sufentanil was set at $0.2 \text{ ng} \times \text{mL/L}$. Thalamus was selected as the region of interest (ROI). All the processing works were carried out using the statistical parametric mapping.

Results: Compared with the basic state, significantly greater functional connectivity was mainly found in the bilateral frontal cortex, insula, cingulate gyrus, claustrum, the left middle temporal gyrus, and the right middle occipital gyrus ($P < 0.05$).

Conclusions: The results show that the increased functional connectivity regions of thalamus include not only the cortical area but also the subcortical structure after sufentanil is administered. It indicates the effect of sufentanil may partially due to improvement the relation of thalamus with frontal cortex, insula, cingulate gyrus, claustrum, the left middle temporal gyrus and the right middle occipital gyrus. These regions may also play an important role in the analgesic effect of sufentanil, which should be explored in future.

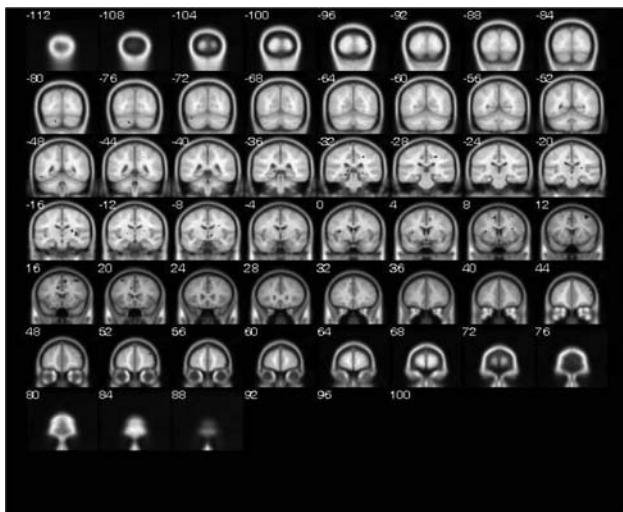


FIGURE 1. Coronal view of functional connectivity of thalamus.

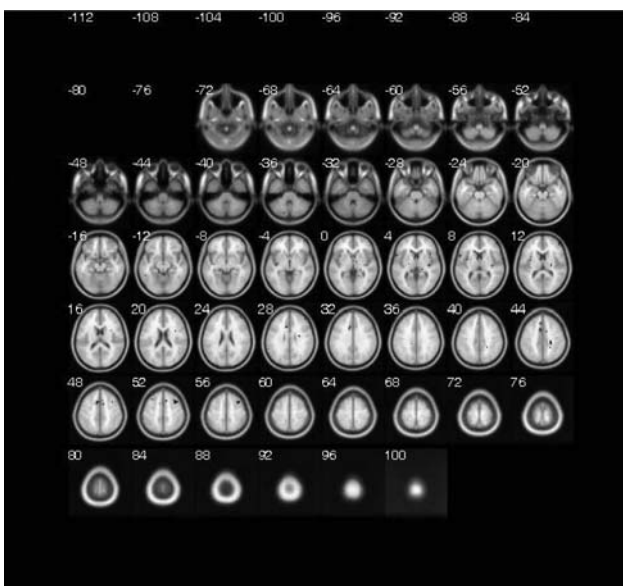


FIGURE 2. Transverse view of functional connectivity of thalamus.

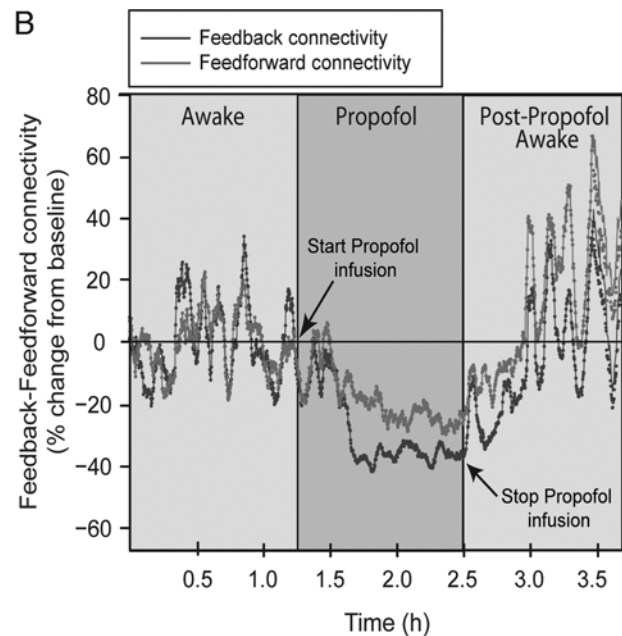
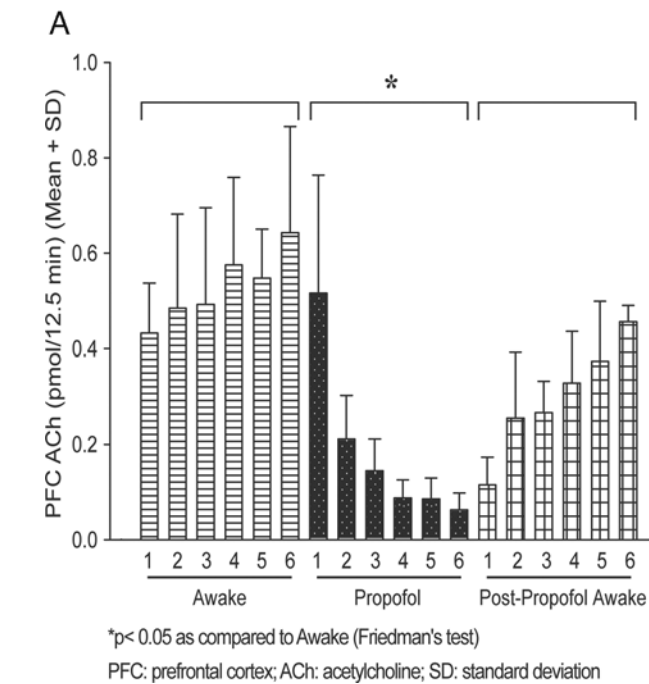
A63 Propofol Reduces Antero-Posterior Feedback Connectivity and Acetylcholine (ACh) Levels in Prefrontal Cortex (PFC) on a Similar Time Scale in Sprague Dawley Rat

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Background and Hypothesis: Anesthetic-induced unconsciousness is characterized by a selective inhibition of antero-posterior feedback connectivity while feed-forward connectivity is less suppressed.^{1,2} General anesthetics also suppress cortical ACh release.^{3,4} We hypothesize that the differential suppression of antero-posterior feedback connectivity and the decrease in PFC ACh are related events occurring on a similar time scale.

Methods: Male Sprague Dawley rats ($N = 3$, 300 to 350 g, Charles River) were instrumented with (i) electrodes to record monopolar electroencephalogram (EEG) from frontal, parietal and occipital cortices and, (ii) a guide tube aimed at PFC for microdialysis and electrochemical measurement of ACh. In addition, a chronic intravenous line for propofol infusion ($800 \mu\text{g/kg/min}$) was secured through a catheter positioned in the jugular vein. After 7 to 10 days of postsurgical recovery and habituation to the recording/microdialysis set-up, ACh levels along with EEG were measured during (i) awake, (ii) propofol infusion, and (iii) postpropofol awake conditions. Each condition lasted 75 minutes yielding 6 samples of $25 \mu\text{L}$ dialysate. To characterize the changes in feed-forward-feedback connectivity, EEG was analyzed using an evolutionary map approach based on phase dynamics. Mean ACh levels across 3 experimental conditions were statistically compared using Friedman test.

Results: The mean ACh levels during propofol infusion decreased by 65.09% ($P < 0.05$) as compared with the awake condition (Fig. 1A). As compared with propofol infusion, there was a nonsignificant (38.3%) increase in the mean ACh levels during postpropofol awake condition (Fig. 1A). The changes in antero-posterior feedback connectivity closely approximated changes in ACh levels (Fig. 1B). There was a sharp decline of approximately 35% in feedback connectivity at the onset of anesthesia. The feedback connectivity remained suppressed during the anesthetized state and started recovering to the basal levels at the return of consciousness, as measured by the righting reflex (Fig. 1B).



Conclusions: These preliminary data suggest an interrelationship between anesthesia-induced changes in brain functional connectivity and ACh levels. Confirmation of a mechanistic relationship between changes in ACh levels and functional connectivity during anesthesia would be a significant step toward linking molecular anesthetic mechanisms with network-level effects.

Acknowledgments: Funding: Supported by Departmental funds and NIH KL2 RR024987-01 (GAM), HL40881, HL65272 (RL).

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2. Lee U, et al. *Conscious Cogn.* 2009;18:1069–1078.
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A64 Propofol Alters Connectivity in Higher Order Sensory Regions—fMRI Study in Volunteers

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Introduction: Functional connectivity reflects intrinsic organization of the neuronal network in the brain. Synchronized neuronal activity in functionally related regions has been identified in the baseline resting state of the brain. This is identified by low frequency BOLD (blood oxygen level dependent contrast—qualitative measure of neuronal activity) signals (< 0.1 Hz) in related regions. Default mode connectivity (DMN) refers to networks that are active in the awake resting state. With task related activity DMN gets deactivated. DMN supports functions unique to humans like thinking, imagination, daydreaming, etc. There is evidence of alteration in connectivity in disease states like Alzheimer disease, schizophrenia etc. With 0.5 MAC sevoflurane we reported that while higher order sensory regions are affected DMN is not altered. Our aim in this protocol was to study the alteration in connectivity during 0.5 MAC equivalent propofol anesthesia.

Methods: Study subjects were 19 ASA¹ volunteers in the age group of 19–35 years. Basic monitoring as per ASA standard was carried out. Propofol was administered through a target controlled infusion (TCI) device set to a plasma level of 2 μ g/mL. Baseline BOLD signals were measured in a 3 Tesla Siemens TRIO. Two cycles of imaging were

carried out—awake and anesthesia. During the imaging cycle respiratory and plethysmographic trace was recorded continuously. These tracings aid in identifying variation in BOLD related to respiration and cardiac pulsation.

Regions with Decrease in Connectivity During 2 mcg/ml Propofol			
Region	Subregion	Talairach Coordinates	ICC decrease ml/100gm/min
Frontal Lobe	Sub Frontal G. R	(10 39 49)	-0.24
	Sub Frontal G. L	(-12 35 46)	-0.18
	Mid Frontal G. R (BA 9)	(33 40 36)	-0.2
	Mid Frontal G. L (BA 9)	(-40 38 31)	-0.26
	(BA 10)	(-45 47 14)	-0.25
	Mid Frontal G. R	(8 48 4)	-0.2
		(22 50 -7)	-0.37
	Mid Frontal G. L	(-14 50 4)	-0.22
		(-9 45 -11)	-0.43
		(-5 33 -13)	-0.33
Parietal Lobe	Precentral G. R (BA 4)	(61 -13 31)	-0.24
	Precentral G. L (BA 4)	(-62 -9 27)	-0.37
	Precentral G. L (BA 6)	(-57 4 27)	-0.35
Parietal Lobe	Postcentral G. R (BA 40)	(60 -23 17)	-0.29
Occipital Lobe	Mid. Occip. G. R	(49 -70 -8)	-0.29
	Inf. Occip. G. L	(-21 -89 -8)	-0.31
Temporal Lobe	Sup Temp. G. R	(59 -12 3)	-0.32
	Sup Temp. G. L (BA 22)	(-54 -12 9)	-0.27
	(BA 38)	(-49 7 -16)	-0.26
	Mid Temp. G. R (BA 21)	(52 4 -13)	-0.31
Putamen R		(16 13 -5)	-0.41
Putamen L		(-24 13 -5)	-0.25
Insula L		(-41 2 4)	-0.26
Brainstem L		(-4 -17 -9)	-0.31
Brainstem R		(11 -16 -9)	-0.26
Lentiform Nucleus R		(14 1 -2)	-0.28
Lentiform Nucleus L		(-12 0 -2)	-0.33
ParaHippoCampa G. L		(-25 0 -8)	-0.28
ParaHippoCampa G. R		(25 0 -8)	-0.26

Results: Connectivity was measured by the voxel based intrinsic connectivity contrast (ICC) technique. The change in connectivity with propofol was calculated. There was an increase in connectivity in the anterior cingulate cortex (ACC) while there was a decrease in connectivity in the insula and prefrontal cortex. Parahippocampal gyrus had regions with both increase and decrease in connectivity. In the thalamus and posterior cingulate cortex (PCC) there was no evidence of change in connectivity. Primary cortical areas had regions with an increase as well as decrease in connectivity.

Discussion: With propofol there was no connectivity alteration in PCC (primary region in DMN), but higher order association regions (insula, ACC, parahippocampal gyrus) were affected. While this observation is similar to our sevoflurane study, primary cortices were not affected by sevoflurane.

Reference:

1. Martuzzi R, Ramani R, Qiu M, et al. "Functional connectivity and network organization in baseline brain state in humans." *Neuroimage*. 2010;49(1):823–834.

A65 Dexmedetomidine Does not Increase Risk of Trigemino-Cardiac Reflex During Microvascular Trigeminal Decompression in Cases of Trigeminal Neuralgia

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Background: Trigemino-cardiac reflex (TCR), a well-recognized brainstem reflex that consists of bradycardia, arterial hypotension, apnea and gastric hypermotility, is a major concern for anesthesiologist during microvascular trigeminal decompression. Dexmedetomidine, an α_2 -adrenoceptor agonist, has been recommended as an adjunct to anesthesia and for the delivery of sedation and perioperative hemodynamic stability, if the anesthesiologist is alert for development of bradycardia. However, whether Dexmedetomidine is also beneficial to the hemodynamic stability, or just increases the incidence of TCR during microvascular trigeminal decompression, is not known and therefore constitutes the aim of this study.

Methods: Eighteen patients with trigeminal neuralgia and scheduled for microvascular decompression were randomized into 2 groups. Exclusion criteria included a history of neurologic or mental disease, serum active hepatic or nephric disease, cardiac or pulmonary dysfunction, endocrine disease, or metabolic disease. Each patient received either a saline placebo or a dexmedetomidine infusion (1 $\mu\text{g}/\text{kg}$) before the anesthesia induction. Target-controlled infusion with propofol and remifentanyl were used for anesthesia induction maintenance. TCR was defined as a drop in mean arterial blood pressure (MABP) and the heart rate (HR) of more than 20% to the baseline values before the stimulus and coinciding with the manipulation of the trigeminal nerve.

Results: Patients having dexmedetomidine achieved an excellent outcome that 12.5% (1 case) of them showed evidence of TCR during manipulation at the trigeminal radix by separation from microvascular structures, when compared with 37.5% (3 cases) of the saline controls. No significant difference in the HR rate or the MABP fell existed between 2 groups. All TCR reflex ceased automatically after cessation of manipulation of the trigeminal nerve. It should be noted there were 3 cases of bradycardia (37.5%) during the infusion of dexmedetomidine compared with 0 in the saline controls.

Conclusions: We showed for the first time that dexmedetomidine can ameliorate the incidence of TCR during manipulation of the central part of the trigeminal nerve during microvascular trigeminal decompression in cases of trigeminal neuralgia. It is likely that the properties of dexmedetomidine in hemodynamic stability and in anesthesia adjunct contribute to this beneficial effect in microvascular trigeminal decompression. A large randomized trial of α_2 -adrenergic agonists in microvascular trigeminal decompression is therefore warranted.

A66 Incidence of BIS Below 40 During General Anesthesia for Noncardiac Surgery

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Introduction: Adequacy of depth of anesthesia has deserved wide attention: too light anesthesia results in awareness and too deep anesthesia may be harmful. BIS, an EEG derived parameter, allows the assessment of adequacy of anesthesia. Recently, cumulative time of excessive anesthesia, defined as BIS below 45, has been associated with increased postoperative mortality. However, few studies have assessed the incidence of low BIS in anesthesia practice. The objective of the present study was to assess the incidence of low BIS in our clinical practice.

Methods: This was a prospective study with IRB approval. Twice a day, on weekdays, investigators (medical students) visited all the operating rooms at 1 pm and 6 pm, addressed the anesthesiologist in charge at each room and asked for participation in the study. After acceptance to participate, a printed record of the computerized anesthesia chart, containing BIS data in a graphical trend format, was obtained. The paper sheets were clipped with a scissors, so that information about the identity of the patient or the anesthesiologist was not kept. The anesthesia record was later examined to extract the total anesthesia time and the percentage of that time with BIS values below 40. Other information obtained was: type of anesthesia, age and sex of the patient, experience of the anesthesiologist and ASA grade. Statistics used Student *t* test and linear regression. Data are mean $\pm$ SD.

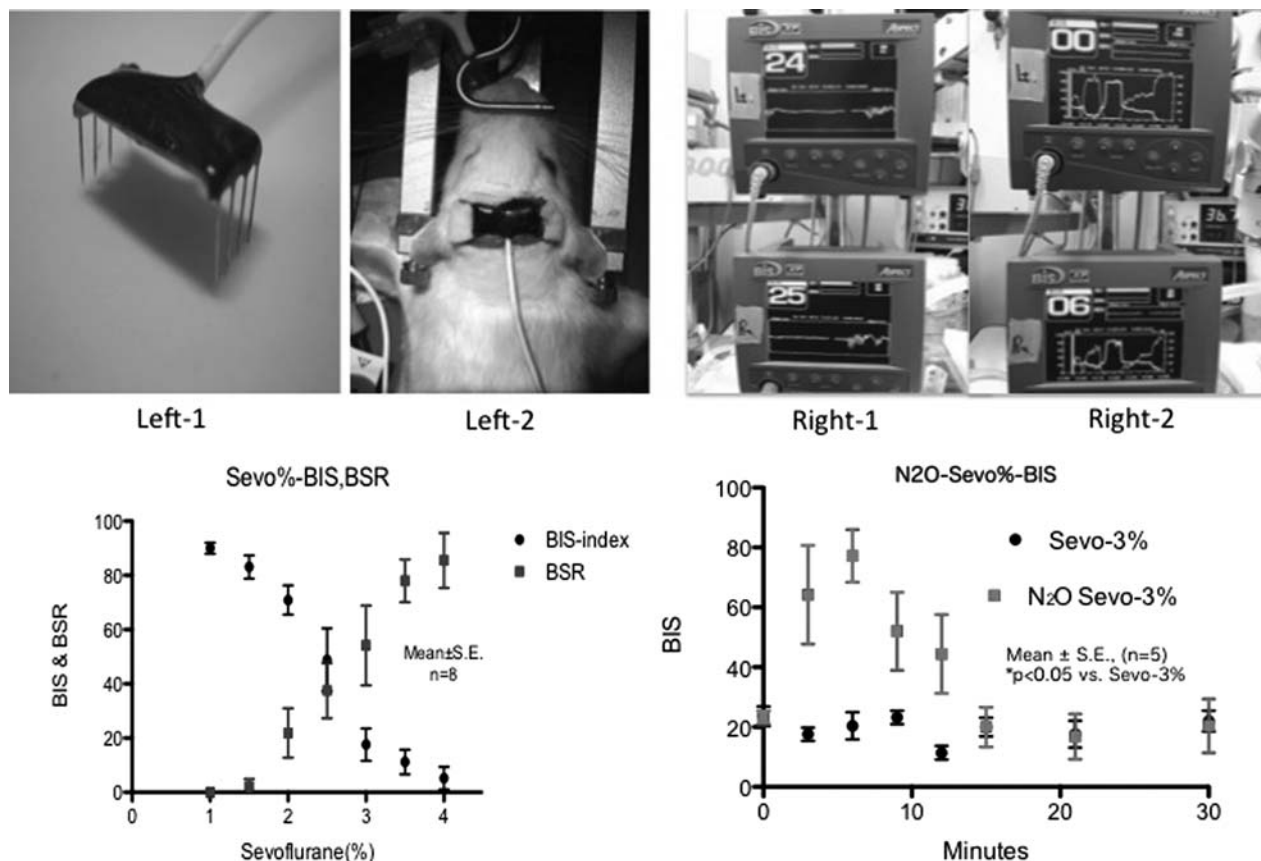
Results: Preliminary results are presented here for a total of 118 patients. Overall incidence of BIS below 40 was $41.5\% \pm 27.7\%$. BIS below 40 in at least 10% of the anesthesia time was observed in 21% of cases; BIS below 40 in 50% of anesthesia time was observed 42% of cases. Incidence of BIS below 40 did not differ between female (42.2%) and males (40.5%). The type of anesthesia did not have an influence: intravenous (36.4%) and balanced (44.1%). ASA grade and incidence of BIS below 40 were as follows: ASA I—37.2%, ASA II 41.9%, ASA III 38.9% and ASA IV 49.4%. There were no correlations between patient's age or anesthesia duration and incidence of BIS below 40.

Conclusions: Our results show an incidence of BIS below 40 as percentage of anesthesia time of 42%, a value that should be considered quite high. Anesthesiologists managing these cases were aware of the fact that a study aimed at evaluating the incidence of BIS below 40 was under way, therefore if anything the study has a bias towards underestimating the incidence of low BIS. It is remarkable that the current awareness about the need to adequately control depth of anesthesia (BIS between 45 and 60), still, even at an institution with several years of BIS guided anesthesia experience, such high incidence of low BIS is present. The fact that patient's characteristics and anesthesia type or duration are not related to incidence of low BIS suggests that maybe the individual anesthesiologist's attitude may be the decisive factor. Our results suggest that either the current evidence regarding the impact of excessive anesthesia on outcome is perceived as weak by anesthesiologists or a lack of active action towards keeping BIS within the recommended limits.

A67 Bispectral Index Monitoring in Rats can Mimic the Relative Changes With the Human Under the Volatile Anesthesia and can Detect the Cerebral Hypoxia

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Basic animal experiments are essential to explore the pharmacodynamics and interactions of anesthetic drugs and the pathology of cerebral ischemia. The analysis of the electroencephalogram (EEG) is one of the important tools for these purposes. The use of processed EEG has been simplified with the introduction of the bispectral index (BIS) monitoring in clinical cases. The BIS value has been statistically based on clinical cases, empirically derived complex parameter. However, if the BIS value



in rats could mimic the changes of the human, it might have absolute benefits where drugs and dose combinations beyond the clinical use could be studied. Several reports have been already published in various animals with commercially supplied BIS sensors. In this study, we have innovated a novel BIS electrode for a rat, consists of bipolar stainless needles settled in the device (Fig. upper left-1) fitting on the parietal bone (Fig. 1. upper left-2), with which enable to connect directory to the 2 sets of BIS monitors (Fig. 1. upper right-1, 2). The present study is the first report of its use whether the newly innovated bilateral BIS monitor could; 1. Detect the concentration-effects during the administration of volatile anesthesia in rats 2. Detect the loss of integrity by cerebral hypoxia in rats.

Experiment 1: The BIS monitoring during inhaled anesthesia with or without noxious stimulation. The adult Sprague-Dawley rats were anesthetized and maintained with control ventilation of inhaled anesthesia. After equilibration, the BIS, burst suppression ratio (BSR) and spectral edge frequency (SEF) were recorded. These values and mean blood pressure (mBP) were manually recorded with or without noxious stimulation at various concentrations of sevoflurane or isoflurane. These differences were tested using a 1-sample t test and the Bonferroni correction for multiple comparisons.

Results 1: The EEG signals and BIS values were symmetrically detected (Fig. 1 upper right). The BIS values at 0.5, 1.0, 1.5 and 2.0 MAC sevoflurane anesthesia were 86 ± 10 , 83 ± 9 , 36 ± 9 and 22 ± 5 , respectively (Fig. 1 lower left). Burst and suppression pattern were observed at 2.0 MAC (Fig. 1 upper Right-1). Additional nitrous oxide showed the biphasic pattern on the time course (Fig. 1 lower right). The noxious stimulation increased the BIS values at 1.5 MAC anesthesia.

Experiment 2: Bilateral BIS monitoring for cerebral hypoxia. The rats were deoxygenated or exposed to middle cerebral artery occlusion (MCAO). The BIS values, mBP and rCBF were recorded under the condition, of which 100% nitrogen was substituted for room air

transiently up to 240 seconds, or MCAO were established by filament method. The bilateral BIS values and unilateral rCBF were recorded continuously of MCAO.

Results 2: The BIS value decreased to 0 within 4 minutes after the onset of anoxic exposure. In MCAO model, the raw EEG wave showed low voltage patterns and the BIS values decreased within a few minutes at ischemic side after the steep drop of rCBF. However, the EEG signal was not changed at intact side.

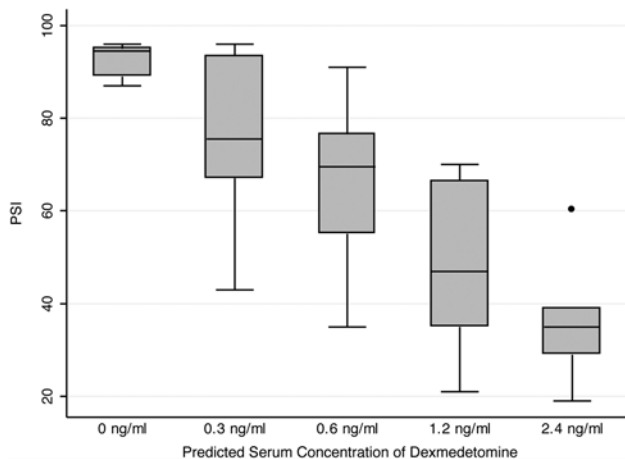
Conclusion: The bilateral BIS monitoring in rats could obtain the symmetrical EEG signal and BIS related values with and without noxious stimulation under volatile anesthesia and detect the cerebral dysfunction by the hypoxia.

A68 Patient State Index During Sedation With Dexmedetomidine in Healthy Volunteers

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Background: The Patient State Index (PSI) (SEDLine, Masimo Inc., Irvine, CA) estimates depth of hypnosis using processed frontal electroencephalogram (EEG) recordings. Little is known about the behavior of PSI during sedation with dexmedetomidine. Our objectives were to (1) evaluate the relationship between PSI and predicted serum concentration of dexmedetomidine, and (2) evaluate the relationship between PSI and Observer's Assessment of Alertness/Sedation (OAA/S), Ramsay Sedation Scale (RSS) and Visual Analog Scale (VAS) scores during a stepwise target-controlled infusion of dexmedetomidine.

Methods: After informed consent, dexmedetomidine was administered intravenously to healthy volunteers using a stepwise computer-controlled infusion (STANPUMP) to target plasma levels of 0.3, 0.6, 1.2



and 2.4 ng/mL. Each step was infused for 15 minutes. The PSI value (0 to 100) was recorded at baseline and the end of each infusion step. After the PSI value was recorded the OAA/S (1 to 5) and RSS (1 to 6) scores were noted and each subject was asked to provide a VAS score (0 to 10) describing their level of sedation. Spearman rank correlation coefficients (r) were used to measure the association between variables with a $P < 0.05$ considered significant.

Results: Eight healthy volunteers were recruited with a median age of 26.5 years (range, 19 to 40). The PSI correlated with the predicted serum concentration of dexmedetomidine ($r = -0.60$, $P = 0.0001$, Fig. 1), OAA/S ($r = 0.83$, $P < 0.0001$), RSS ($r = -0.52$, $P = 0.0008$) and VAS score ($r = -0.47$, $P = 0.003$). All subjects remained arousable for VAS testing with PSI values in the recommended range for general anesthesia.

Conclusion: The PSI correlates with the predicted serum concentration of dexmedetomidine during sedation with a stepwise target-controlled infusion and with objective (OAA/S and RSS) and subjective (VAS) sedation scores. Low PSI values should be interpreted with caution in patients sedated with dexmedetomidine, as all subjects remained arousable despite PSI values in the recommended range for general anesthesia. Further research is required to validate the use of PSI during administration of dexmedetomidine.

A69 Monitoring Bispectral Index in Small Animals: Some Interesting Observations

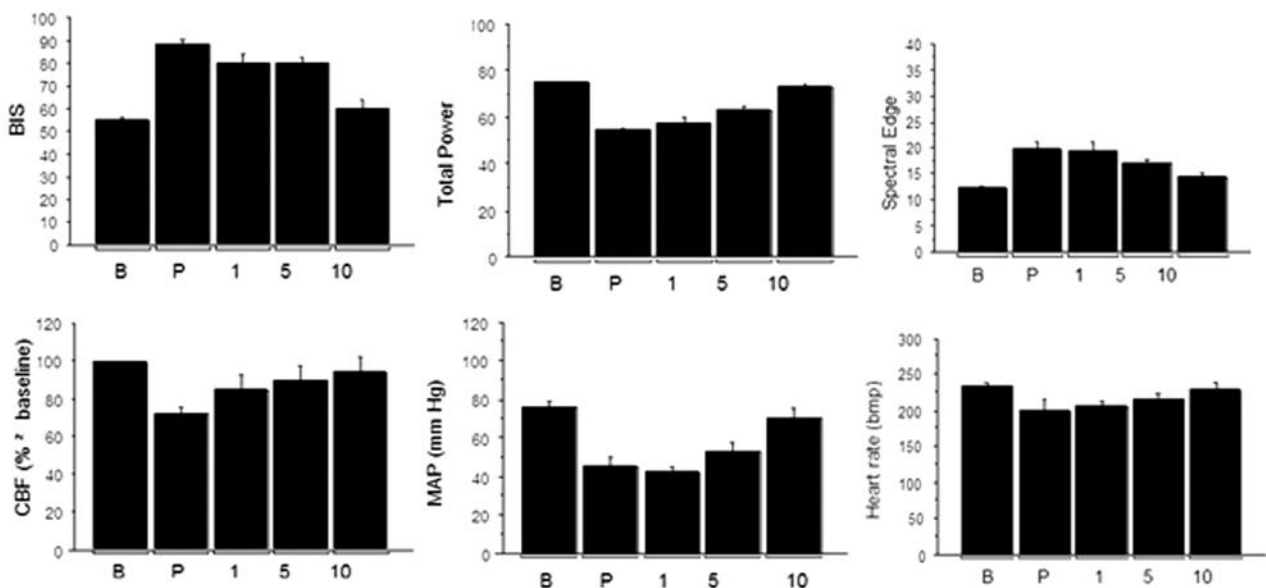
Shailendra Joshi, MD, James P. Lowe, MS, Mei Wang, MPH, Sachin Agarwal, MD, Rahul Rai, Columbia University, Emory University.

Introduction: Bispectral Index (BIS) monitoring is a well-accepted tool for monitoring the depth of anesthesia particularly during propofol based total intravenous anesthesia. Anecdotal reports support the use of BIS in a wide range of animals. However, the use of BIS as standard depth of anesthesia monitoring tool has not been widely investigated in laboratory animals. We regularly use propofol based anesthesia to sedate rabbits for drug delivery studies and we monitor anesthetic depth using standard bi-hemispheric electroencephalography. Our model therefore provides a platform to assess BIS monitoring and compare it with standard and independently recorded EEG activity.

Material and Methods: We adapted the sensor for rabbits by soldering the wires to a standard aspect XP BIS sensor. We used needle electrodes that could be screwed to the skull in locations that were analogous to the human BIS lead placements. Aspect XP BIS automatically checks of electrode impedance. Impedance was unaffected by such modification to the sensor. Under an IACUC approved protocol we monitored BIS with concurrent EEG, cerebral and systemic hemodynamic parameters. In rabbits under baseline ketamine (50 mg/kg) and propofol (30 to 50 mg/kg/h) anesthesia. A bolus propofol injection of 15 mg/kg produces transient EEG silence under these conditions. We compared changes in the following parameters: Bispectral index (BIS), Suppression ratio (SR), spectral edge frequency (SEF) and total power (TP). At baseline (B), onset EEG silence with propofol bolus (P) and at 1, 5, and 10 minutes after injection, Figure 1.

Results: Data from 7 rabbits (mean weight 1.9 ± 0.3 kg) is shown in figure. During propofol-induced EEG silence cerebral blood flow, mean arterial pressure and heart rate decreased significantly. The resting EEG activity in all anesthetized animals yielded a BIS value between 52 to 61. Propofol-induced EEG silence was associated with a paradoxical increase in BIS values despite independently recorded EEG silence. Total power of the EEG showed a significant decrease and SEF increased, while suppression ratio was unaltered. On recovery BIS and other parameters returned to baseline values.

Discussion and Conclusions: BIS has been used in veterinary medicine to monitor depth of anesthesia in dogs, cats, pigs, horses, chicken and dolphins. These reports describing the use of BIS across animal species are of interest to neuroscience research. Automatic quantification of EEG and precise control of depth of anesthesia would have a great impact on



translational neuroscience research. In our experience, we have found New Zealand white rabbits to be unusually tolerant to propofol, requiring as much as 50 mg/kg/h for produce stable anesthesia. Stable EEG pattern that we have long accepted to be the target of propofol infusion produced BIS values in 30 to 60. Yet when the depth of anesthesia was further deepened to EEG silence there was a paradoxical increase in BIS. It is unclear if the failure of BIS to monitor deeper levels of anesthesia induced by propofol in rabbits is specific to the animal species, instrument, signal processing or anesthetic drug.

A70 Effect of Different Ranges of End-tidal Sevoflurane on EEG-derived Indexes in Pigs

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Background: Entropy-based methods have been used as tools to address EEG based depth of anesthesia. This study analyses the performance of different EEG derived parameters applied to distinct intervals of end tidal sevoflurane in pigs.

Methods: Nine male pigs 30.2 ± 3.7 kg were anesthetized using sevoflurane anesthesia. Induction of anesthesia was performed by 8% sevoflurane using a facial mask, followed by endotracheal intubation and mechanical ventilation to keep EtCO₂ between 35 and 40 mm Hg. End-tidal sevoflurane (Etsevo - %) was then randomly adjusted to different levels. EEG was recorded at 256 Hz using BIS XP monitor; the Bispectral Index (BIS), electromyographic activity (EMG), spectral edge frequency 95% (SEF) suppression ratio (SR) and total power (TP) were also registered. Data recorded in the fully awake animals, and after end-tracheal intubation were organized in ranges of Etsevo. After raw EEG filtering (0.5 to 32 Hz), the signal was processed offline for derivation of approximate entropy (AE) and permutation entropy (PE) in 8 seconds epochs. Spearman correlation analysis was performed between Etsevo and each of the studied parameters: BIS, EMG, SEF, TP, AE and PE, using all data, and the data in each of the Etsevo range defined: 0% to 3% [(0 to 3)], 3% to 6% [(3 to 6)], 0% to 1% [(0 to 1)], 1% to 2% [(1 to 2)], 2% to 3% [(2 to 3)], 3% to 4% [(3 to 4)], 4% to 5% [(4 to 5)], and 5% to 6% [(5 to 6)].

Results: EEG frequencies decreased during sevoflurane administration, but burst suppression did not occur. The higher correlations were observed in the 0 to 3% Etsevo interval. Data correlations are shown in Table 1. Conclusions: In general, PE and SEF were the EEG derived parameters that showed a better correlation with Etsevo. The EEG changes reflected in PE and SEF are mainly observed during the 0 to 3%

Etsevo range. At higher sevoflurane concentrations (3 to 6%), the parameters studied do not reflect the changes in EEG. The study of smaller Etsevo ranges did not yield considerable correlations, besides that observed in the (0 to 1) interval.

Acknowledgments: Financial support by COMPETE: FCOMP-01-0124-FEDER-009525 and FCT (PTDC/CVT/101999/2008).

A71 SEPs May Objectively Translate Nociception/Anti-nociception Balance With Anesthetics

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Background and Goal of Study: SEPs wave amplitudes and latencies are related to stimulus intensity and analgesic drug attenuation.^{1,2} We decided to investigate the use of SEPs in volunteers and the impact of anesthetic drugs on evoked responses and pain ratings.

Material and Methods: After institutional approval and written informed consent, healthy adult volunteers were enrolled in the study. SEPs evaluated using an Endeavour Nicolet-Viasys monitor. Median nerve electrical stimulation (3.1 Hz) was adjusted according to volunteers' individual painful threshold (PT), and baseline SEPs obtained prior drug administration. EEG collected using scalp needles with a C3'-Fz montage (contralateral to site of stimulation). Remifentanyl and propofol were administered using TCI with effect-site concentration (Ce) steering with increasing remifentanyl steps (0, 1, 2, +0.5 ng/mL steps), followed by propofol increasing steps (0, 1.2, 2, +0.5 µg/mL), with remifentanyl constant Ce at 1.0 ng/mL, and finally remifentanyl increasing steps (1, 2, +0.5 ng/mL), with propofol Ce constant at 1.2 µg/mL; stop increasing drug's concentrations when BP or HR decrease below 20% baseline values, by clinical indication, or on OAAS/S = 2 (loss of response to verbal command). For each drug combination a $1.3 \times$ PT stimulation was administered, and volunteers asked to rate pain intensity in a numerical rating scale (NRS, 0-no pain, 10-worst possible pain). SEPs amplitude and latency were registered. Due to inter-individual variability, SEPs amplitudes were normalized by the amplitude registered in the PT stimulus prior drugs administration. Spearman correlation and Wilcoxon paired-sample test were used. Data presented as mean $\pm$ SD.

Results and Discussion: Six healthy volunteers were enrolled, 1 excluded due to technical issues: 3 women, 27.8 ± 4.0 years, 65.4 ± 11.0 kg and 172.2 ± 8.9 cm. Pain ratings in the NRS varied from 0 to 8, and $1.3 \times$ PT stimulation currents between 6 and 39 mA. A tendency towards decreasing SEPs amplitude with increasing drug doses was observed, both for propofol and remifentanyl. We also observed a decrease in SEPs amplitude with pain ratings (Fig. 1A). Increased latencies and decreased

EtSevo considered	AE	BIS	EMG (%)	PE	SEF (Hz)	TP
All data	-0,27*	-0,36*	-0,21*	-0,63*	-0,53*	0,52*
[0-3]	-0,59*	-0,65*	-0,39*	-0,77*	-0,77*	0,71*
[3-6]	0	0	-0,11*	0	0	-0,13*
[0-1]	-0,22*	-0,57*	-0,49*	-0,50*	-0,66*	0,56*
[1-2]	0,17*	0	0	0	-0,14*	0,10*
[2-3]	-0,17*	0	0	-0,24*	-0,26*	0,19*
[3-4]	0	-0,13*	-0,29*	-0,19*	0	-0,29*
[4-5]	0	0,24*	0,28*	0	0,1	0
[5-6]	0,38	-0,12	0	0,38	-0,36	-0,15

*Correlation significance at level $p < 0.05$.

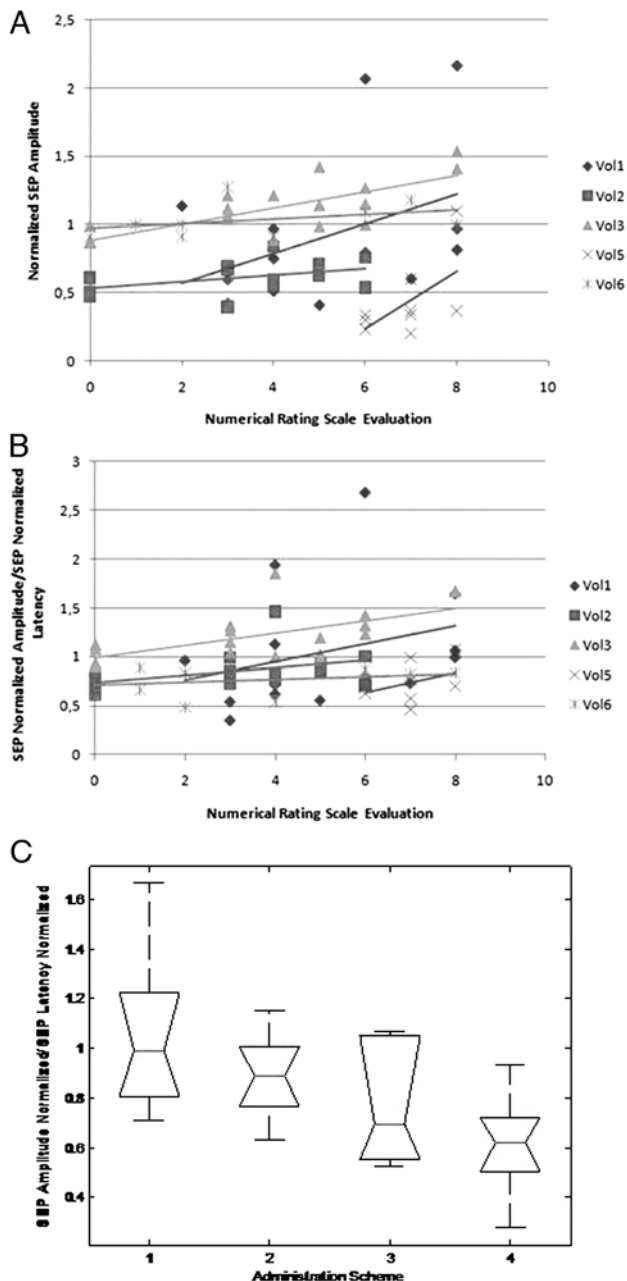


FIGURE 1. (A) Representation of the normalized SEPs amplitude versus pain ratings (NRS); (B) Representation of the ratio SEP normalized amplitude/SEP normalized latency versus pain ratings (NRS); (C) Boxplot of the SEP amplitude normalized/SEP latency normalized ratio for the maximum drug doses combination achieved in each volunteer, in each administration scheme: 1-No drugs; 2-Remifentanyl only; 3-Propofol increasing steps and remifentanyl; 4-Remifentanyl increasing steps and propofol.

amplitudes are related to drug SEPs attenuation^{1,2}; we decided to evaluate the relation between NRS and the ratio SEPs amplitude normalized/SEPs latency normalized, results presented in Figure 1B. We observed that volunteers' intervariability was reduced. Spearman correlations between SEPs ratio, drug doses and NRS were in the same direction (negative for

drug doses and positive for NRS), nevertheless only 2 volunteers had significant correlation coefficients, one with the NRS and other with remifentanyl Ce. The effect in the SEPs ratio of the maximum drug combination obtained in each administration scheme for all volunteers was compared (Wilcoxon test) to baseline prior drug administration (Fig. 1C): we observed a decrease in the ratio for every administration scheme comparing to baseline, however due to the reduced number of volunteers there was no significant difference (between baseline and propofol + remifentanyl scheme $P = 0.0625$).

Conclusions: SEPs amplitudes are depressed by the use of analgesic and hypnotic drugs.

A72 Perioperative Complications and its Effect on Patient Outcome in Brainstem Lesion Surgery

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Background: Brainstem tumors are lesions that occur in the region area between the aqueduct of Sylvius and the fourth ventricle that is midbrain, pons, and medulla. The important neurovascular structures and vital centre that regulate basic body functions of blood pressure, respiration, and swallowing as well as motor and sensory functions are affected by brainstem lesions. This study was planned to find out the preoperative risk factors, intraoperative adverse events, postoperative complication and there effects on outcome of patients operated for brainstem lesion and its association with each other.

Materials and Methods: Retrospective review of records of 63 patients who underwent surgery for brainstem lesions at AIIMS over a period of 8 years from 2002 through 2009. Preoperative data collected included age, sex, weight, height, neurological status, and history of cardiac or respiratory disease. Intraoperative data collected included anesthetic technique, monitoring, position, hemodynamic changes, occurrence of VAE, blood replacement, and tumor location and pathology. Post-operative data collected included neurologic status, cardiopulmonary complications, surgical complications, and status at discharge

Results: Factors associated with unfavorable outcome are:

- Reintubation due to any reason
- Chest infection in postoperative period
- GCS deterioration
- Development of postoperative hydrocephalus
- Seizure in postoperative period
- Hyponatremia as electrolyte imbalance
- Tracheostomy

Conclusions: Postoperative outcome in patients operated for brainstem lesions is solely dependent on intraoperative surgical insult to vital centers, cranial nerves and postoperative complication arising as a result of these. Anesthetic technique has no role in causation of complication and outcome. For better outcome, monitoring of vital centers and cranial nerve is advisable.

Demographic Characteristics

Characteristics	Total number	Percentage %
MEDIAN AGE (RANGE)	19 (5 - 65)	
SEX M/F	42 / 21	67% / 33%
ASA 1/2/3	59 / 2 / 2	94% / 3% / 3%
WEIGHT (mean $\pm$ SD)	40 $\pm$ 20	
RECURRENT TUMOR	17	27%
DURATION OF SURGERY (HR) (mean $\pm$ SD)	4.1 $\pm$ 1.2 (2.5, 6)	
BLOOD LOSS (ml) (mean $\pm$ SD)	428 $\pm$ 408 (50, 2500)	
ICU STAY (DAYS) (mean $\pm$ SD)	4 $\pm$ 2 (2, 16)	
HOSPITAL STAY (DAYS) (mean $\pm$ SD)	14 $\pm$ 13 (2, 60)	

A73 Characterization of Ischemia Thresholds as Evidenced by Optically Measured NADH Concentrations and Microdialysis Study During Transient Cerebral Hypoperfusion in Anesthetized Rabbits

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Introduction: Previous studies suggested that transient global cerebral hypoperfusion (CHP) results in anaerobic metabolism leading to an increase in optically measured brain tissue NADH levels and markers of cerebral energy metabolism (glucose, lactate, and pyruvate) and neurotransmitters (glutamate). The lower critical limit of cerebral blood flow for the maintenance of cerebral energy metabolism varies among different individuals and different species.

Methods: New Zealand White rabbits were anesthetized with ketamine and mechanically ventilated. The monitoring included a 5 mm diameter right temporal craniotomy for the NADH probe, bilateral EEG monitoring, laser Doppler probes to measure cerebral blood flow (CBF), and temperature probe in tympanic membrane. Microdialysis probes (CMA Microdialysis, Stockholm, Sweden), 4 mm membrane length, perfusion rate of 1 μ L/min in a 2.5 mL syringe were placed through a hole drilled 2 mm anterior and 2 mm lateral to the bregma on either side. CHP was achieved with bilateral internal carotid artery occlusion and pharmacologically induced systemic hypotension for 3 to 5 minutes. Samples collected every 10 minutes during the whole experiment were analyzed for Glucose, Lactate, Pyruvate and Glutamate. The differences between baseline and peak Hypoperfusion values were used to calculate area under the curve which was further used to evaluate the critical thresholds of cerebral blood flow causing rise in NADH and Microdialysis measures. A *P* value of ≤ 0.05 was considered to be significant.

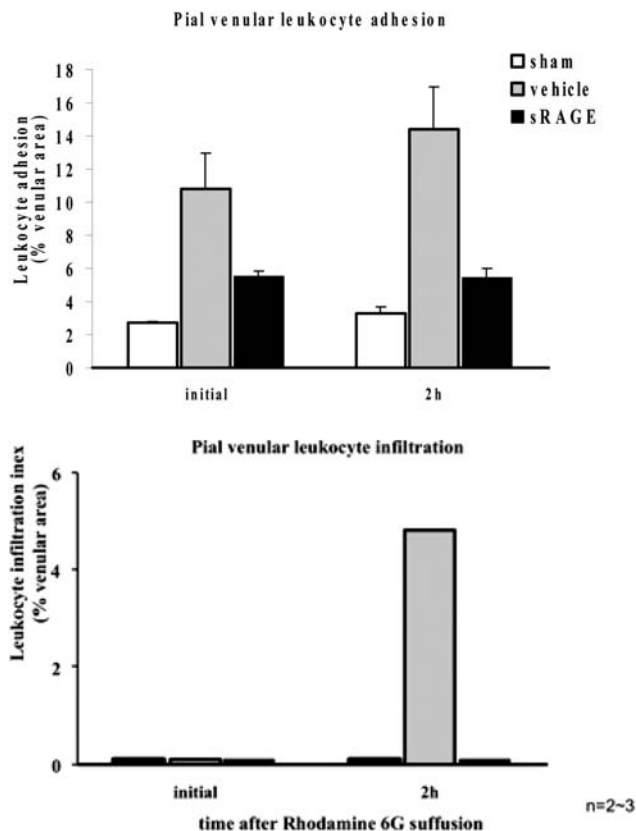
Results: Data on 20 animals and 50 unique runs were collected. Significant change in Hemodynamics parameters after CHP (Heart rate: 116 ± 51 ; mean arterial pressure (MAP): 64.5 ± 20 mm Hg; end-tidal CO₂ (EtCO₂): 12.5 ± 5.5) along with NADH (0.37 ± 0.13), average drop in CBF (122 ± 57 mL/100 g/min) were found with statistical significance (*P* < 0.001). Lowest average CBF was 31 ± 19 mL/100 g/min. The earliest event after CHP was a significant drop in glucose (0.13 ± 0.11 mmol/L) followed by elevation in Lactate (0.37 ± 0.25 mM/L) and Lactate-Pyruvate ratio (0.05 ± 0.1) and then a rise in glutamate (1.1 ± 1.3 μ M/L). Cerebral blood flow at 135 mL/100 g/min was considered to be the optimal value causing significant increase in Lactate (AUC = 0.80), glutamate (AUC = 0.76) and NADH (AUC = 0.82) adjusting for the relative drop in MAP during CHP. Correlation analysis showed a significant relationship between NADH rise and glucose drop (coeff. 0.35; *P* = 0.023), peak lactate (coeff. 0.20; *P* = 0.006), and change in CBF (coeff. 0.002; *P* < 0.001).

Conclusions: We observe a significant change in hemodynamic parameters including CBF leading to a significant rise in the levels of early ischemia markers after transient global cerebral Hypoperfusion. We found that a reduction of 135 mL/100 g/min of CBF is the critical value at which significant release of markers of disturbed energy metabolism and neurotransmitters occurred. There was a significant correlation between cerebral blood flow, microdialysis and NADH measurements using optical probe. The trend monitoring of NADH signals in the tissue could provide information about the metabolic state of the mitochondria.

A74 Inhibition of RAGE-related Pathway Prevents Subarachnoid Hemorrhage (SAH)-associated Neuroinflammation and Neurologic Dysfunction in Rats

Chanannait Paisansathan, MD, Dale Pelligrino, PhD, Haoliang Xu, MD, PhD, Francesco Vetri, MD, PhD. *University of Illinois at Chicago.* Recent study from our laboratory indicated that, in SAH, activation of the receptor for advanced glycation end-products (RAGE)-related pathway contributes to cerebral arteriolar dilating dysfunction. As RAGE activation is linked to the occurrence and exacerbation of

inflammatory reactions, and neuroinflammation is a contributor to cerebrovascular dysfunction, we tested the following hypotheses: (1) RAGE is involved in post-SAH neuroinflammation; and (2) preventing RAGE activation, using the RAGE decoy protein, soluble RAGE (sRAGE), attenuates SAH-associated neuroinflammation, and recovers neurological function. Rats were randomized into 3 groups: sham surgical controls, vehicle-treated SAH controls, and sRAGE-treated SAH rats. The SAH model involved suture perforation of the anterior cerebral artery. Neuroinflammation was assessed by pial venular leukocyte adhesion (PVLA) on day 2 post-SAH. PVLA, viewed through a closed cranial window, was expressed as the % pial venular area occupied by adherent rhodamine-6G-labeled leukocytes. Neurobehavioral function, which included spontaneous activity, muscle tone, and neurologic reflex was evaluated during the post-SAH recovery period (up to 3 d). Compared to the sham surgical group, a marked increased PVLA was observed in vehicle-treated animals ($10.78\% \pm 2.20\%$ vs. $2.72\% \pm 0.09\%$ in sham), with a profound leukocyte extravascular infiltration revealed during extended observation period (infiltration index was $4.83\% \pm 2.82\%$ of the venular area). No sign of infiltration was found in the sham surgical group. sRAGE treatment significantly decreased SAH-associated PVLA to $5.53\% \pm 0.29\%$, with complete suppression of leukocyte extravascular migration. Additionally, a significant improvement in SAH-related neurological deficits was found in the presence of sRAGE. In summary, RAGE plays an important role in mediating post-SAH neuroinflammation, which may contribute to SAH-associated neurological dysfunction.



A75 Perioperative Metoprolol Increases Risk of Stroke After Noncardiac Surgery

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Background: Patients undergoing noncardiac surgery are at risk for stroke and stroke-associated mortality. Numerous risk factors have been identified, but there are conflicting data regarding the role of beta blockade in general and metoprolol in particular.

Methods: We retrospectively screened 54,853 consecutive patients for radiologic evidence of stroke within 30 days after noncardiovascular, non-neurosurgical procedures using our perioperative electronic information system. Incidence of stroke and associated risk factors were assessed. Parsimonious logistic regression was used to generate a risk model.

Results: The incidence of perioperative stroke was 43/54,853 (0.08%). Preoperative metoprolol was associated with an approximately 4-fold increase in perioperative stroke ($P < 0.001$; 95% CI 2.0-9.3); no associations were found for preoperative atenolol, bisoprolol, carvedilol, nadolol, propranolol, or sotalol. Preoperative metoprolol, increased age, and history of stroke or transient ischemic attacks were independent predictors of perioperative stroke; preoperative atenolol was not an independent predictor. Furthermore, intraoperative metoprolol was associated with a 3.8-fold increase in perioperative stroke ($P = 0.02$; 95% CI, 1.5-9.8); no association was found for intraoperative esmolol or labetalol.

Conclusions: Routine use of preoperative metoprolol, but not atenolol, is an independent predictor of stroke after noncardiac surgery. Intraoperative metoprolol, but not esmolol or labetalol, is associated with increased risk for perioperative stroke. Based on these data and the results of the POISE trial, agents other than metoprolol should be used during the perioperative period if beta blockade is required.

A76 Open Versus Endovascular Abdominal Aortic Procedures is a Risk Factor for Perioperative Stroke

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Introduction: Perioperative stroke is a devastating complication of abdominal aortic procedures with an incidence of 0.5% to 1.9%.^{1,2} Few studies have examined stroke as a primary outcome in this population. Studies examining stroke in either open or endovascular procedures are limited by small numbers and a low frequency of stroke. Our objective was to compare the risk of perioperative stroke after open versus endovascular aortic procedures using a large prospectively gathered dataset derived from the American College of Surgeons National Surgical Quality Improvement Program (ACS-NSQIP).

Methods: Patients undergoing nonemergent open or endovascular abdominal aortic surgery were included in the analysis. Perioperative stroke is defined by the ACS-NSQIP as an embolic, thrombotic, or

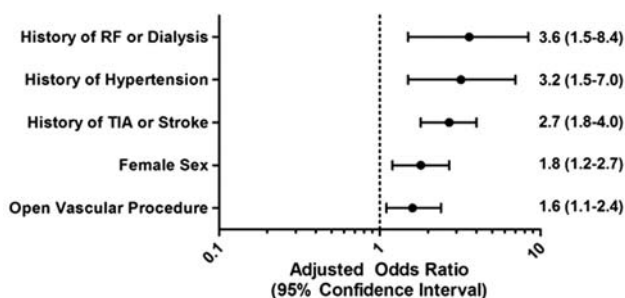
hemorrhagic cerebrovascular event with motor, sensory, or cognitive dysfunction that persists ≥ 24 hours within 30 days of the procedure. Descriptive statistics were performed on all continuous and dichotomous preoperative co-morbidities using a Pearson χ^2 , Fisher exact test or Student t test to determine association with perioperative stroke. Preoperative comorbidities and the type of surgical procedure (open vs. endovascular) were entered into a nonparsimonious logistic regression model to determine independent predictors of postoperative stroke.

Results: Total, 17,990 patients met our inclusion criteria and demonstrated an overall 0.6% incidence of perioperative stroke. Open procedures demonstrated a 0.8% incidence of perioperative stroke and endovascular procedures demonstrated a 0.5% incidence. The logistic regression demonstrated the following independent predictors for stroke: female sex, history of transient ischemic attack or stroke, history of renal failure or current dialysis, hypertension, and open vascular procedures. The ROC AUC was 0.70 ± 0.03 . Univariate analysis demonstrated patients with perioperative stroke had a 5-fold increase in 30-day mortality ($P < 0.001$). Patients with perioperative stroke were more likely to receive ≥ 4 units of PRBC ($P = 0.002$), had longer operative times ($P < 0.001$), and had a longer surgical stay ($P < 0.001$).

Conclusion: In patients undergoing nonemergent abdominal aortic surgery, open procedures are an independent predictor for perioperative stroke as compared with endovascular procedures. Endovascular procedures may be the preferred surgical approach for patients at particularly high risk of perioperative stroke.

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A77 Predictors of Thromboembolic Complications After Endovascular or Surgical Treatment of Posterior Communicating Artery Aneurysms

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Background/Objective: Posterior communicating aneurysms are one of the more common types of intracranial aneurysms. This study is a retrospective review to determine predictive factors of thromboembolic complications after procedure.

Methods: Data was collected using electronic medical records including demographics, procedural characteristics, and outcomes. The primary outcome was defined as cerebral infarction or subarachnoid hemorrhage by CT imaging or clinical signs and symptoms of stroke in the periprocedural period. Univariate analysis was performed to investigate patient factors and aneurysm characteristics that may be associated with complications.

Results: At Duke University Hospital between January 2000 and February 2011, 188 patients were treated for posterior communicating aneurysms. This cohort consisted of 91 ruptured and 97 nonruptured aneurysms undergoing either surgical clipping (98) or endovascular coiling (90). Fifteen of the 188 patients (8%) demonstrated imaging or clinical evidence of new thromboembolic event. Univariate analysis indicated 4 factors of significance (P value < 0.05) for predicting thromboembolic complication: wide neck ($P = 0.03$), abnormal shape ($P = 0.05$), hypertension ($P = 0.003$) and diabetes ($P = 0.01$).

Conclusion: The results suggest that aneurysm shape and neck width may be predictive for the risk of postoperative thromboembolic event. Patient comorbidities, including hypertension and diabetes, may be associated with complication. However, statistical modeling was limited by the low incidence of thromboembolic complication, the outcome of interest.

	Normal (PaCO ₂ > 38) N=32	Mild (38 ≤ PaCO ₂ < 33) N=38	Moderate (33 ≤ PaCO ₂ > 28) N=10	Severe (PaCO ₂ < 28) N=4
Gender				
Male	25	27	7	3
Female	6	10	3	1
Race				
Native American or Pacific Islander	7	2	0	0
Other	24	34	9	4
Unknown	1	2	1	0
Death				
Yes	7	12	7	2
No	25	26	3	2
GCS	9.3	8.8	7.4	9.2
Odds Ratio of Death	Reference	1.65 (0.56, 4.86)	8.33 (1.70, 40.91)	2.38 (0.33, 17.17)

A78 Short-term Outcomes in Patients With Severe TBI: An Analysis of Mild Hyperventilation

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Hyperventilation reduces ICP via a hypocarbia-induced vasoconstriction that results in decreased cerebral blood volume. When autoregulation is intact the effect is transient. Furthermore, when extended beyond 8 to 24 hours, hyperventilation offers no additional benefits and has been shown to worsen outcome, with systemic effect of hyperventilation likely contributing factors. Severe hyperventilation (PaCO₂ < 28 mm Hg) is typically reserved for a short period of time in patients who are pending herniation and when other modalities such as CSF drainage are ineffective or unavailable. While the use of prolonged severe hyperventilation has been curtailed, mild hyperventilation (PaCO₂ 33 to 38 mm Hg) remains commonly used in most patients with TBI. The underlying theory for the prophylactic use of mild hyperventilation assumes that mild hypocapnia provides the benefit of cerebral vasoconstriction and lowered ICP, but not at the expense of decreasing CPP as seen with long-term severe hyperventilation. However, the current literature on the survival impact of mild hyperventilation remains unclear.

Methods: A retrospective cohort study was performed by statistical analysis of a database comprised of 84 patients with severe TBI (GCS ≤ 8) at Shands Hospital at the University of Florida, Gainesville during the period 2006 to 2010. Patients were categorized into groups based on the mean arterial carbon dioxide level (PaCO₂) measured while sedated and mechanically ventilated without spontaneous ventilation. Data were analyzed for comparison and outcome analysis in 4 predetermined groups: normoventilation (PaCO₂ > 38 mm Hg), mild hyperventilation (38 ≤ PaCO₂ < 33 mm Hg), moderate hyperventilation (33 ≤ PaCO₂ > 28 mm Hg) and severe hyperventilation (PaCO₂ < 28 mm Hg). Logistic regression was performed with the primary outcome as death. Survival outcomes were calculated as odds ratio of death.

Results: Demographic information, outcome, and GCS score for the 84 patients are outlined in the Table 1. Odds ratios for death were calculated for severe, moderate and mild hyperventilation groups with the reference baseline being the event rate of death in the normoventilated group (Table 1). A statistically significant decrease in survival was seen in the moderate hyperventilation group [OR = 8.33 (1.70, 40.91)]. This decrease in mortality was not found in the mild hyperventilation group [OR = 1.65 (0.56, 4.86)] with a clear trend towards worse outcome. A significant decrease in survival was not seen in the severely hyperventilated group [OR = 2.38 (0.33, 17.17)].

Discussion: The results of this study suggest no correlation between induced mild chronic hyperventilation in severe TBI patients and survival. On the contrary, moderate hyperventilation significantly reduced survival. Our data is consistent with current literature that

indicates patients hyperventilated to a PaCO₂ less than 32 mm Hg have a worse outcome when compared with patients who are kept normocapnic.

Conclusions: The data from this 84 patient cohort with severe TBI show that maintaining patients in a state of mild hyperventilation (38 ≤ PaCO₂ < 33 mm Hg) does not independently impact the probability of overall survival. Further analysis of additional outcome data is planned.

A79 Neuron-targeted Caveolin-1 Enhances Pro-survival Signaling, Promotes Dendritic Growth, and Protects Hippocampal Neurons Against Traumatic Brain Injury

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Introduction: Within the injured brain, the development of a growth inhibition environment mediated by inflammatory cytokines and myelin-associated glycoproteins significantly reduces neuronal growth. Strategies aimed at reducing this inhibition to facilitate neuronal growth have, unfortunately, met with limited success. In addition to the creation of a favorable environment in which growth inhibition is removed, augmentation of the intrinsic pro-growth pathways may be necessary to enhance regeneration. In the injured brain, the expression of these pro-growth pathways is reduced substantially; it may be possible, however, to increase pro-survival and pro-growth signaling by increasing membrane/lipid rafts (MLR). MLR are plasmalemmal microdomains enriched in sphingolipids, cholesterol, and the protein caveolin (Cav) and are essential for neuronal signaling and synaptic development/stabilization. Cav-1 organizes glutamate and neurotrophin receptors, pro-survival kinases, and regulates cAMP formation. We hypothesized that neuron-targeted Cav-1 will initiate MLR formation and enhances pro-survival signaling pathways, thereby promoting neuronal survival, dendritic growth, and axonal sprouting.

Methods: Primary neurons in vitro (DIV4) were transfected with a viral vector encoding a synapsin promoter upstream of Cav-1 cDNA (SynCav1). Control neurons were transfected with a viral vector encoding a synapsin promoter upstream of red fluorescent protein cDNA (SynRFP). Expression of glutamate receptors, growth factor receptors and their downstream signaling molecules was evaluated by immunoblot. In separate neurons, the effect of inflammatory cytokines (IL-1β, TNFα) and myelin-associated glycoproteins (MAG, Nogo) on neuronal growth was evaluated. In vivo, SynCav1 (or SynRFP for control) was injected stereotactically into the hippocampus of mice 2 months before traumatic brain injury (TBI). TBI was induced by controlled cortical impact. Injury to the hippocampus was evaluated 7 and 21 days post-TBI.

Results: SynCav1 significantly enhanced MLR, PSD-95, NR2A, NR2B, AMPAR, and TrkB expression in comparison to SynRFP. BDNF-mediated activation of P-TrkB, P-Akt, and P-ERK1/2, NMDA-mediated activation of P-Src, P-CaMKII, and P-ERK1/2, and agonist stimulated cAMP formation was significantly increased by SynCav1 3 d posttransfection. SynCav1 led to a 4-fold increase in dendritic and axonal growth 21 d post-TBI. Even in the presence of inhibitory cytokines (TNFα, IL-1β) and myelin-associated glycoproteins (MAG, Nogo), SynCav1 promoted dendritic growth and arborization. In vivo, SynCav1 reduced the cortical lesion produced by controlled impact and protected hippocampal neurons 7 days and 21 days post-TBI.

Conclusions: Targeting of Cav-1 to neurons enhances pro-survival and pro-growth signaling and increases dendritic and axonal growth. This growth promoting effect can overcome growth inhibition mediated by cytokines and myelin-associated glycoproteins. In addition, SynCav1 protects neurons against TBI. As such, Cav1 over-expression may be a novel means to repair the injured brain.

A80 The Effect of Changes in Progesterone and Estrogen During Normal Pregnancy on Blood Glutamate Levels

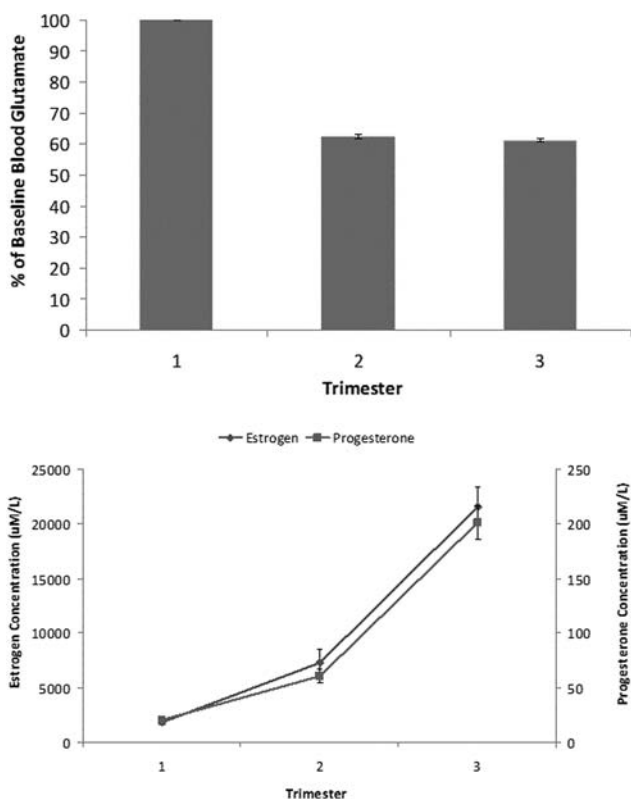
Shaun F. Gruenbaum, MD, Benjamin F. Gruenbaum, BA, Yoram Shapira, MD, PhD, Akiva Leibowitz, MD, Alexander Zlotnik, MD, PhD. *Yale University School of Medicine, Soroka University Medical Center.*

Introduction: It is well-established that acute neurological events are accompanied by abnormally elevated levels of glutamate (G) in the brain's fluids, which is neurotoxic and associated with a poor neurological outcome. Increased progesterone and estrogen levels have been shown to correlate with decreases in blood glutamate levels, thereby providing neuroprotection. In this study we examine whether the changes in progesterone and estrogen levels observed during normal pregnancy influence blood glutamate levels.

Materials and Methods: Total 150 pregnant women were divided into 3 groups based on gestational age: group 1 included women in their first trimester (until 14 wk), group 2 included women in their second trimester (14+1/7 wk until 28 wk), and group 3 included women in their third trimester (28+1/7 until 42 wk). A single venous blood sample was collected (10 mL) and analyzed for concentrations of estrogen, progesterone, glutamate-pyruvate transaminase (GPT), glutamate-oxaloacetate transaminase (GOT), and glutamate.

Results: Figure 1 displays the blood glutamate concentrations that were observed during each trimester. Compared with the first trimester, concentrations of blood glutamate was significantly lower during the second trimester ($P < 0.01$) and third trimester ($P < 0.001$). There were no significant differences in blood glutamate concentrations observed between the second and third trimester. The decreased concentrations of blood glutamate correlated with increases in plasma estrogen and progesterone during the second and third trimesters ($P < 0.001$, Fig. 2). There was a decrease in plasma GPT during the third trimester ($P < 0.001$).

Conclusions: Increases in progesterone and estrogen levels during advanced stages of pregnancy were associated with decreases in maternal blood glutamate concentrations. Despite higher levels of estrogen and progesterone during the 3rd trimester compared with the 2nd trimester, blood glutamate was not reduced additionally in 3rd trimester compared with the 2nd trimester. With the achievement of its maximal capacity, additional elevations in estrogen and progesterone levels do not further reduce glutamate. Progesterone was previously shown to decrease blood glutamate levels in a bell-shaped dose-response manner, with a disappearance of the effect observed at higher and lower concentrations.



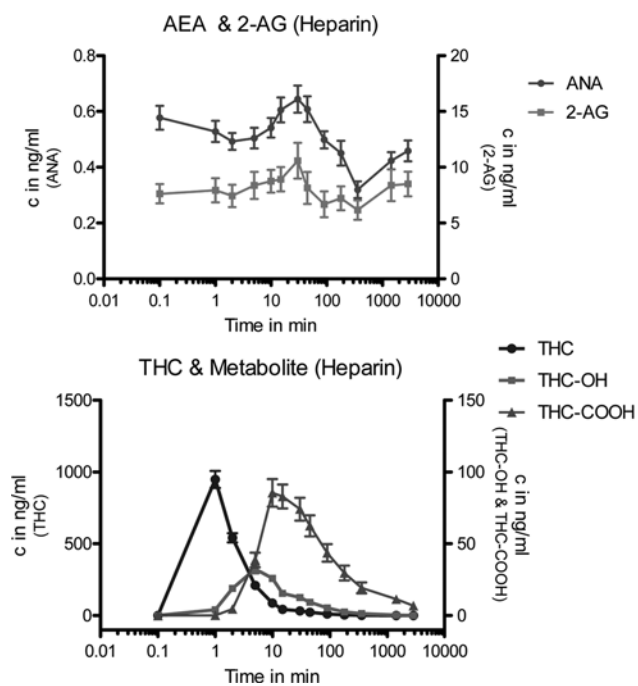
For this reason, with high concentrations of progesterone during the third trimester, a less pronounced effect on blood glutamate reduction was observed. An understanding of the physiological mechanisms by which blood glutamate is regulated may lead to the discovery of new therapeutic modalities to improve neurological outcome after an acute neurological event.

A81 Quantification of Anandamide (AEA) and 2-Arachidonoylglycerol (2-AG) Plasma Levels to Examine Potential Influences of Tetrahydrocannabinol (THC) Application on the Endocannabinoid Systems

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The aim of the presented sub-project consists in an examination of a potential tetrahydrocannabinol (THC) induced interference of the endocannabinoid (EC) system. Such a feedback to endogenous cannabinoids maybe assumed due to the well known fact that THC mediates its effects through competitive binding to cannabinoid receptors CB1 and CB2.

The global study was accomplished at the University of Bern and devoted to the examination of pharmacokinetic variations of the THC metabolism dependent on polymorphisms of CYP2C9 which is known to mediate the oxidative degradation of THC. Intravenous administration of single dosages of 0.1 mg/kg THC to volunteers and subsequent collection of plasma samples over 48 hours were approved by the ethical committee. Twenty-five volunteers were selected based on their genotypic profile for CYP2C9. Eleven blood samples were drawn during the first 6 hours and 2 more after 24 hours and 48 hours for analyzing the long-term effects. The arterial blood was simultaneously collected into heparin and EDTA tubes and centrifuged. Resulting plasma was separated and frozen at -20°C . THC and its metabolites THC-COOH and THC-OH and the EC's AEA and 2-AG were quantified by liquid chromatography-mass spectrometry (LC-MS/MS).



The highest mean THC plasma level of 949(SD = 299) ng/mL was observed right after application. THC-OH which is a strong psychoactive metabolite of THC reached its maximum of 32.5 (SD = 14.9) ng/mL after 5 minutes. The mean plasma level of the inactive metabolite THC-COOH reached its maximum of 85.6 (SD = 48.1) ng/mL after 10 minutes. THC (0.25 ng/mL), THC-OH (0.13 ng/mL) and THC-COOH (6.82 ng/mL) still exhibited significant plasma levels after 48 hours. Plasma levels of 2-AG and AEA showed a maximum at 30 minutes post-administration. There was a significant increase from mean 2-AG level 7.6 (SD = 4.3) ng/mL to 10.6 (SD = 8.1) ng/mL and an increase from mean plasma AEA level 0.58 (SD = 0.21) ng/mL to 0.64 (SD = 0.24) ng/mL. Six hours after THC injection plasma levels of both EC's decreased. 2-AG concentrations were reduced (compared with basal plasma levels) to 6.2 (SD = 4.4) ng/mL and AEA level to 0.32 (SD = 0.15) ng/mL and returned to normal after 24 hours. Comparison of plasma levels regarding the storage in either heparin or EDTA showed no significant difference in the AEA plasma levels. However plasma levels of 2-AG in heparin were 46% higher than the plasma levels in EDTA.

Administration of THC led to a slightly delayed reactive increase of EC plasma levels and later on to a suppression of its release. Changes in plasma levels occurred during the acute phase after THC injection while there was no effect detectable in the long-term analysis. The exact times of minimal and re-normalization of EC plasma levels remain uncertain as respective time spans were not accurately covered by the study design. Another aspect demonstrated by our study was the inconsistent depletion of 2-AG plasma levels bedded in EDTA. We have observed a reasonable short-term increase of the endocannabinoids AEA and 2-AG and a subsequent moderate suppression after intravenous administration of THC, which caused high plasma-levels and significant clinical impairments. This is not likely to suggest THC-induced long-term impairments due to a negative feedback or induction of the EC systems.

A82 The Effect of Hemodialysis on Blood Glutamate Levels in Patients With Chronic Renal Failure

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Introduction: Excess concentrations of glutamate (G) in the brain's fluids after traumatic brain injury (TBI) and stroke have been shown to be neurotoxic, and are associated with a poor neurological outcome. Reduced blood G levels were associated with an improved neurological outcome. The decrease in blood G levels would consequently increase the concentration gradient of G between the brain's fluids and the blood and accelerate the rate of the well-established brain-to-blood G efflux. This study examined whether blood G concentrations could be decreased in the peripheral blood via hemodialysis.

Materials and Methods: Blood samples from 47 patients with end stage renal disease on hemodialysis were collected immediately after beginning dialysis for baseline measurements, and at each hour thereafter (at $t = 0, 60, 120, 180$, and 240 min). Blood samples were also collected from 26 healthy patients of the same age, serving as a control group. The blood

was analyzed for concentrations of glutamate, glutamate oxaloacetate transaminase (GOT), and glutamate oxaloacetate transaminase (GPT).

Results: Baseline blood glutamate levels were 2 times higher compared with the healthy population in the same age (393 ± 115 vs. 178 ± 105 mM/l). Blood glutamate levels compared with baseline concentrations at each time point (Fig. 1). Concentrations of glutamate in the peripheral blood were significantly decreased at $t = 60$ minutes (86% of baseline concentration, $P < 0.001$), $t = 120$ minutes (80% of baseline concentration, $P < 0.001$), $t = 180$ minutes (80% of baseline concentration, $P < 0.001$), and $t = 240$ minutes (88% of baseline concentration, $P < 0.001$) compared with baseline levels ($t = 0$ min). The concentration of glutamate at $t = 240$ minutes was significantly higher than at $t = 180$ minutes (88% vs. 80%, $P < 0.05$). GOT levels measured 20 μ M/L and 25 μ M/L at $t = 0$ minutes and $t = 240$ minutes, respectively ($P < 0.001$). GPT levels measured 13 μ M/L and 15 μ M/L at $t = 0$ minutes and $t = 240$ minutes, respectively ($P < 0.01$).

Conclusions: This study demonstrated that glutamate can be cleared from the blood by hemodialysis. At 240 minutes however, concentrations of glutamate in the blood began to increase despite continued dialysis. This might be due to a compensatory release of glutamate from peripheral tissues as the concentrations in the blood drop. The higher blood glutamate levels in this population may contribute to the development of uremic encephalopathy in this population. This study suggests a potential therapeutic role for hemodialysis in reducing the excess glutamate levels in the brain's fluids that accompany TBI and stroke.

A83 Analysis of Sedation in the ICU by Drug and Time to Extubation. Post-Hoc Analysis of a Prospective Trial

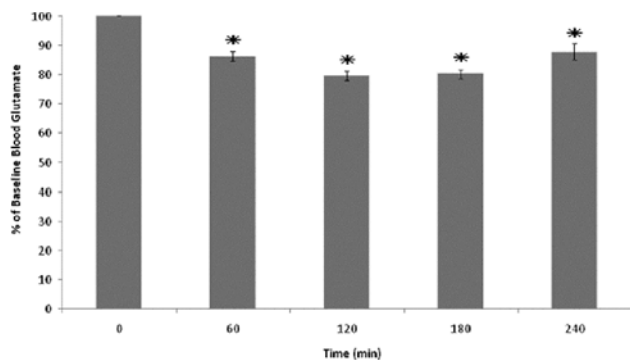
Quintin J. Quinones, MD, PhD, David L. McDonagh, MD, Daiwai M. Olson, PhD, RN. *Duke University Medical Center.*

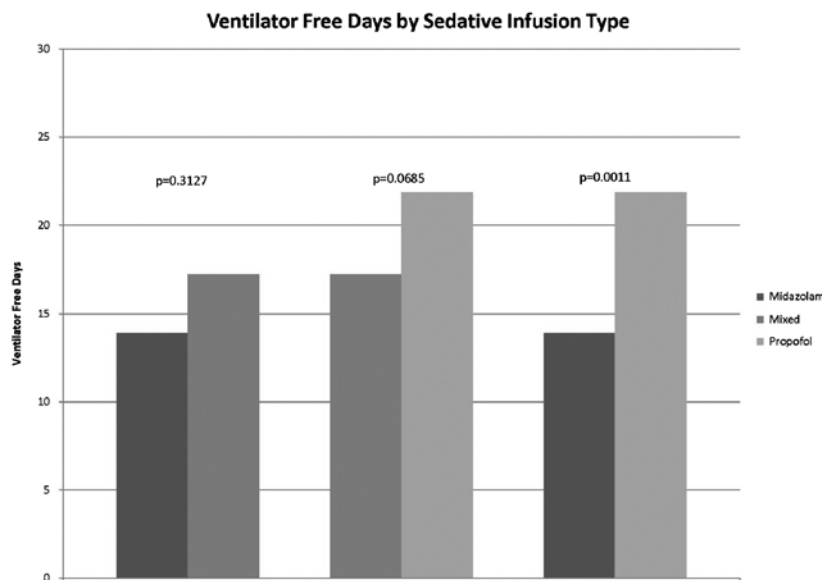
Introduction: Numerous choices exist for the sedation of mechanically ventilated critically ill patients. Although pharmacologic differences between drugs are known, the clinical impact of using 1 medication versus another remains unclear. To help address this question we examined the number of ventilator free days in patients sedated with midazolam infusion only, versus patients sedated with propofol only, versus a combination of sedative infusions including propofol, midazolam, and/or dexmedetomidine.

Methods: With IRB approval, patients were prospectively enrolled to examine monitoring in the ICU. Clinicians were free to choose method of sedation for mechanically ventilated patients. Data was gathered from 3 ICUs across 2 facilities within the Duke Health System. In a post-hoc analysis, patients were separated into 3 groups: those receiving midazolam only as a sedative hypnotic infusion, those receiving propofol only, and those receiving a combination of midazolam, propofol, and/or dexmedetomidine during mechanical ventilation. Ventilator free days (VFDs) were calculated as days the patient was extubated and spontaneously breathing subsequent to first intubation. VFDs were calculated up to a maximum of 30. Patients who died while being mechanically ventilated received zero VFDs. Statistical analysis was performed using single factor ANOVA and post-hoc Bonferroni corrected pair-wise comparison.

Results: The total number of patients studied was 194. The midazolam only group contained 68, the propofol only group contained 54, the mixed sedative group contained 72. Mortality in the midazolam only group was 39.7%, 18.5% in the propofol only group, and 20.8% in the mixed sedative group. The mean ventilator free days was 13.93 in the midazolam only group, 21.86 in the propofol only group, and 17.21 in the mixed sedative group. A 1-way between groups ANOVA was conducted to compare the effect of sedation type on VFDs in midazolam only, propofol only, and mixed sedative groups. There was significant effect of sedation type on VFDs at the $P = 0.0012$ level for the 3 groups [$F(2, 191) = 6.95$, $P = 0.0012$]. Post-hoc Bonferroni corrected pairwise comparison showed that the propofol infusion only group had more VFDs than the midazolam infusion only group $P = 0.0011$.

Conclusions: Our analysis of sedative infusion type in mechanically ventilated critically ill patients demonstrated a difference in VFDs in patient sedated with midazolam only, propofol only, or a combination





of midazolam, propofol, or dexmedetomidine infusions. Post-hoc Bonferroni corrected pairwise comparison demonstrated that a statistically significant difference exists between the propofol only infusion group and the midazolam only infusion group, although trends exist between all 3 groups. We believe that this correlation demonstrates the need for further study as numerous factors are at play. While the sedative infusion chosen likely plays a role in time to extubation it is also likely that the clinician's choice of sedative infusion demonstrates an expectation of time to extubation. Prospective randomization will be necessary to differentiate the impact of drug choice on outcome from that of other clinical factors influencing drug choice.

A84 Indirect Mirror Laryngoscopy to Predict the Difficult Airway

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Introduction: While cervical spine lesions can be safely managed with various techniques during intubation, concurrent management of the difficult airway clearly complicates this task. An unanticipated difficult airway in a patient with cervical insult poses inordinate risk of injury. The current state of the art for airway assessment includes a combination of examination techniques, creating a composite evaluation geared towards overcoming the limitations of each individual component. However, it remains clear that standard techniques of assessment, both singly and in composite, still cannot predict all difficult intubations. Indirect mirror laryngoscopy is a simple and well-tolerated physical exam technique which may improve the sensitivity of the composite airway examination.

Methods: Eighty-eight consecutive patients aged 18 years and older with planned general endotracheal anesthesia and lacking cervical instability or obvious airway deformity were prospectively enrolled for this study. Preoperative airway assessment included cervical range of motion, mouth opening, thyromental distance, Mallampati classification and upper-lip bite test. Indirect laryngoscopy was performed with the patient

protruding his or her tongue and a 14 mm dental mirror introduced into the pharynx to the level of the uvula. Visualization of any portion of the vocal cords was classified as a Grade I view, a view of any portion of the epiglottis classified as Grade II, and a view limited to the base of tongue as Grade III. Subsequent to induction of general anesthesia, direct laryngoscopy was performed and Cormack and Lehane grade determined, with Grade 3 and 4 views defined as a "difficult intubation."

Results: A preliminary analysis of the first 88 observations of this study was completed. Data were categorized as predictive of "easy" or "difficult" intubation. In this way Grade I and II views on indirect laryngoscopy were found to be predictive of easy intubation and Grade III of difficult intubation. A Fisher exact test was performed with a *P* value calculated as <0.0001 for indirect laryngoscopy, with the next significant value being that of the thyromental distance test with a *P* value of 0.0304. A kappa statistic was also calculated. Indirect laryngoscopy revealed a kappa statistic approaching 1.0, indicating near perfect agreement between poor indirect laryngoscopic view and difficult intubation as noted by Cormack and Lehane classification. Attached is a table summarizing calculated *P* values and kappa statistics.

Summary: Indirect mirror laryngoscopy is a simple and well-tolerated technique for examining the upper airway and aerodigestive tract. The initial results of this study have shown significant statistical promise. Further evaluation will show whether this technique can improve the predictive power of a multivariate airway exam for difficult intubation.

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A85 Laryngeal Morbidity Associated With Fiberoptic Intubation Under General Anesthesia With and Without Use of Muscle Relaxant

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Background: Awake fiberoptic technique is usually the best technique in difficult airway situations. Fiberoptic intubation under sedation is reported to have higher acute laryngeal injury score than conventional intubation by direct laryngoscopy under general anesthesia and muscle relaxation. Laryngeal morbidity is described to be less after using muscle relaxant but using muscle relaxant can be hazardous in patients with difficult airway. The aim of this study was to assess the effectiveness and rate of laryngeal morbidity during fiberoptic tracheal intubation

Screening Test	Fischer exact p-value	Kappa statistic
Mallampati Score	0.1406	0.1754
Thyromental Distance	0.0304	0.3378
Upper Lip Bite Test	0.1277	0.0000
Mouth Opening	0.5752	0.0256
Indirect Laryngoscopy	<0.0001	1.0000

	Group I (With MR) n=52	Group II (Without MR) n=48	p-value
Demographic and intraoperative data			
Age (yr)	37.6(12)	35.9(12)	0.5
Weight (kg)	62.3(11)	60(11)	0.5
Sex: Male	28(53.8)	30(62.5)	0.4
Female	24(46.2)	18(37.5)	
Duration of surgery (min)	287(73)	217(81)	0.5
Duration of attempts (sec)	58(19)	68.3(31)	0.0001
Number of attempts	1.1(0.3)	1.5(0.6)	0.0001
Total intubation time (sec)	60.1(37)	133(100)	0.0001
Airway pressure (cm H ₂ O)	16.4(2.5)	16.5(2.8)	0.82
End-tidal carbon dioxide	33.2(1.8)	33.7(2.1)	0.23
Surgical position			
Prone	47(90.4)	45(93.8)	0.60
Supine	4(7.7)	3(6.3)	
Others	1(1.9)	0(0)	
Details of laryngeal morbidity			
Grade of hoarseness			
0	25(48.5)	22(45.8)	0.46
1	21(40.4)	21(43.8)	
2	10(19.2)	5(10.4)	
3	1(1.9)	0(0)	
Location of VCIS			
Absent	38(73.1)	24(50)	0.05
Unilateral	1(1.9)	3(6.3)	
Bilateral	13(25)	21(43.8)	
Type of VCIS			
Rhynema	14(26.6)	20(41.7)	0.09
Edema	0(0)	3(6.3)	
Hematoma	0(0)	1(2.1)	

performed under general anesthesia with or without muscle relaxants in patients undergoing elective neurosurgical procedures.

Methods: A randomized, prospective study with 100 patients, scheduled for elective neurosurgical procedures, ASA I to II and age between 15 to 60 years was carried out. The patients with history of difficult intubation, patients with pathological laryngeal finding, ASA III to IV were excluded from the study. Patients were preoxygenated with 100% O₂ via face mask for 3 minutes. General anesthesia was induced with propofol 1.5 to 2 mg/kg and fentanyl 2 mcg/kg. Patients were assigned to 1 of the 2 groups (group I with muscle relaxant and group II without muscle relaxant) by a computer generated randomization chart. After confirmation of adequate mask ventilation, patients received either intravenous rocuronium 1 mg/kg or saline prepared in identical syringes and volume. Fiberoptic intubation was attempted 3 minutes later by an anesthesiologist with > 50 previous experiences of fiberoptic bronchoscopy. For male patients, we used 8 mm ID tracheal tubes, and for female patients 7.5 mm ID tracheal tubes. We stopped any attempt that lasts > 3 minutes. The number of failures, number of attempts, and their duration, total intubation time, and events during the whole procedure were noted. On the first postoperative day, an experienced ENT physician unaware of the patients' group assignment assessed hoarseness and vocal cords by indirect laryngoscopy. Data are presented as mean (SD) or number (%). The correlation between different variables was evaluated using Spearman rank test.

Results: Out of all 100 patients, 52 patients were in muscle relaxant group (group I) and 48 patients in without-muscle relaxant group (group II). Vocal cord sequelae was more in group II as compared with group I, that is using muscle relaxant. Hoarseness in the 2 groups was comparable. Study results are tabulated (Table 1).

Conclusions: Laryngeal morbidity associated with fiberoptic intubation under general anesthesia is more without using muscle relaxant as compared with using muscle relaxant.

A86 Failure of the Electronic Medical Record (EMR) to Record and Capture Secondary Brain Injury (SBI) Events

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Introduction: Intracranial pressure (ICP) data is required for clinical research in acute brain injury (ABI). In ICU's without continuous data acquisition (CDA), ICP must be obtained from the EMR. EMR data is typically recorded every 15 minutes or hourly by nursing. CDA appears to be the "gold standard" for experimental physiologic monitoring. However, EMR data is cleansed of erroneous values by nursing, while these are included in CDA. Prior studies have shown that mean ICP EMR values are valid, but they miss SBI events (ICP > 20 mm Hg).¹ EMR data may also be extrapolated to approximate CDA.² The goal of our

study is to examine the validity of EMR and CDA data as a research tool in ABI.

Methods: Five patients with ABI requiring ICP monitoring were randomly selected for CDA using ICU Pilot software sampling at 1 minute intervals for 24 hours. EMR ICP data, recorded by nursing who generally charted values every 15 minutes, was retrospectively collected for the same period after discharge. Mean ICP, ICP distribution, and SBI events (ICP > 20 mm Hg) were calculated and compared between CDA and EMR. Similar analysis was performed using SBP values with events recorded (> 2 min > goal BP).

Results: CDA and EMR data yielded a total of 7407 and 408 observations, respectively. The average number of observations per patient was 1481.4 and 81.6, respectively. The mean sampling rate of EMR was 1 observation per 20 minutes. The mean ICP values were 13.40 mm Hg and 13.38 mm Hg ($P = 0.993$). Analysis of ICP distribution was virtually identical. Importantly, the total number of SBI events recorded by CDA was 119 whereas EMR documented only 33. Similar results were found when examining systolic blood pressure goals.

Conclusions: The results of this pilot study indicate EMR data is sufficient to examine mean ICP and trends. However, EMR data significantly underreports the frequency of SBI's. Therefore, EMR data is insufficient for research data collection even at a sampling rate of every 20 minutes.

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A87 Jugular-to-Arterial Carbon Dioxide Difference and Post-operative Cognitive Dysfunction in Patients Undergoing Lung Surgery

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Background: During 1 lung ventilation (OLV) for lung surgery, systemic and cerebral oxygen and carbon dioxide balance can be impaired. Recent evidence indicated that venous-to-arterial carbon dioxide difference [P(cv-a)CO₂] is a useful tool in ICU-septic patients to identify persistent hypoperfusion and P(cv-a)CO₂ is larger in patients with complications than without complications in high-risk surgery.^{1,2} This study was conducted to investigate the association between jugular-to-arterial carbon dioxide difference (P(j-a)CO₂) during OLV and post-operative cognitive dysfunction (POCD) in patients undergoing lung surgery.

Methods: After the IRB approval and informed consent, 97 patients scheduled for elective lung surgery in the lateral position were enrolled. Anesthesia was maintained with propofol or sevoflurane combined with fentanyl and epidural anesthesia. The target blood concentration of propofol or expired concentration of sevoflurane was adjusted to keep a bispectral index (BIS) score of 45 to 60. Arterial and jugular bulb venous blood samples were obtained before OLV, 15 and 60 minutes after OLV, and 15 minutes after the termination of OLV. P(j-a)CO₂ and jugular bulb venous oxygen saturation (SjO₂) were measured at each time point. Cognitive function was assessed using the following neurocognitive tests 2 days before surgery, and 5 days and 3 months after the surgery: MMSE, Digit Span Forward, Digit Span Backward, Trail Making Test A, Trail Making Test B, and Grooved Pegboard Test (dominant and nondominant hands). Cognitive dysfunction was defined as a decline from preoperative testing of > 20% of test measures (at least 2 of 7 tests).

Results: POCD was noted in 25 of 94 (27%) patients at 5 days and in 14 of 80 (18%) at 3 months after the surgery. There were no significant differences in P(j-a)CO₂ values at each time point between the patients with and without POCD at 5 days and 3 months (Fig. 1). Between the patients with and without POCD, there were significant differences in age (72 ± 7 vs. 68 ± 10 ; $P < 0.05$) at 5 days, in duration of anesthesia

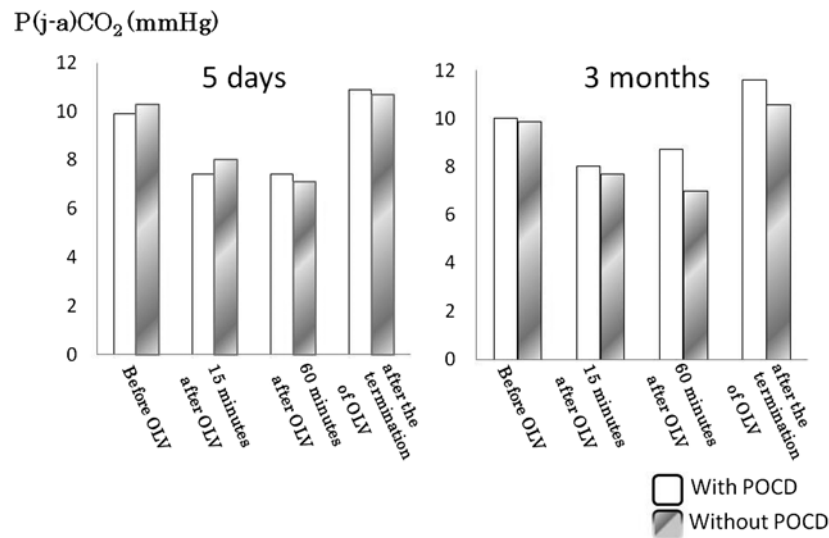


FIGURE 1. There were no significant differences in P(j-a)CO₂ values at each time point between the patients with and without POCD at 5 days and 3 months.

(331 ± 86 vs. 289 ± 68; $P < 0.05$) at 3 months, and in SjO₂ values at 60 minutes after OLV (71 ± 9 vs. 77 ± 11; $P < 0.05$) at 3 months.

Conclusions: The results in this study indicated that P(j-a)CO₂ values were not associated with the development of POCD after lung surgery with 1 lung ventilation.

Reference:

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A88 Cerebrovascular Autoregulation in Patients With Sepsis and Sepsis-associated Delir

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Background: Sepsis is often associated with neurologic deficits as part of the multi organ failure. This sepsis-associated delir (SAD) could be observed in up to 71% of the patients and increases mortality.¹ One of the potential factors leading to SAD might be an impaired cerebrovascular autoregulation (AR) in septic patients. Therefore, the present study assessed AR and markers of inflammation during the first 4 days in patients with sepsis, severe sepsis, and septic shock and then correlated these with the incidence of SAD.

Methods: After ethical care committee approval and informed consent, 30 adult patients with sepsis and treatment according to the international guidelines for management of severe sepsis and septic shock² at the ICU were included. The AR was measured using transcranial Doppler sonography daily during the first 4 days. The index of autoregulation Mx was calculated by correlating the arterial blood pressure and cerebral blood flow velocity over a period of 1 hour. An Mx > 0.3 indicates impaired AR.³ Delir was diagnosed using the confusion assessment method (CAM-ICU) and EEG. Furthermore, markers of inflammation (CRP, PCT) and neuronal damage (NSE, S100) were measured daily. The data were analyzed using mixed models, chi2-test and partial correlation (SPSS 18.0).

Results: Thirty patients (20 male) with sepsis and a mean APACHE II of 32 ± 6 were included. The AR index Mx was impaired in 83% of these patients with tendency to recover during the first 4 days. SAD was observed in 23 (77%) patients using the CAM-ICU and in 17 out of 18 patients (94%) using EEG. There was no statistically significant correlation between Mx and SAD (CAM-ICU: $P = 0.07$ $t = -1.88$; EEG: $P = 0.21$ $t = -1.30$). Mx, and thereby the status of AR, correlated

with markers of inflammation (CRP $P = 0.017$ and PCT $P \leq 0.001$) but not with the markers of neuronal damage (NSE and S100).

Conclusions: The Index of AR Mx was high during the first 2 days but decreases thereafter. Therefore, the AR was impaired in most of the septic patients at the onset of sepsis. The hypothesis, that the impaired AR alone causes SAD could not be proven. The origin of SAD is complex and may be associated with the potential neurotoxic effect of the increased cytokine-levels. This inflammation might then lead to impaired AR and cerebral hypoperfusion due to hypotension. This assumption is supported by the finding, that Mx correlates with the marker of inflammation.

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A89 The Effect of Inhalational Anesthetics Versus Propofol on Alzheimer Disease-related CSF Markers

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Multiple studies have linked anesthetic exposure to postoperative cognitive decline, particularly in geriatric patients. However, the underlying mechanisms of POCD remain unclear. One possibility is that anesthetic agents might accelerate the underlying neurobiological progression towards Alzheimer disease, the most common form of dementia.

The pathophysiology of Alzheimer disease involves the accumulation of extracellular amyloid beta peptide monomers into insoluble fibrils and plaques, a process thought to take years if not decades. Inhalational anesthetic agent exposure at clinically relevant concentrations increases whole brain amyloid beta content in vivo in animal models, while inhalational agents also polymerize A beta monomers into fibrils in vitro.

However, to our knowledge no study has measured amyloid beta levels in the human brain preanesthesia and postanesthesia. Here, we describe the measurement of amyloid beta levels in the CSF of patients undergoing aneurysm surgery with lumbar drain placement, and preanesthetic and postanesthetic exposure. To attempt to control for the possible direct effects of neurosurgery on CSF and brain amyloid beta, we are randomizing patients to receive either inhalational anesthesia with isoflurane or intravenous anesthesia with propofol. These studies should clarify whether inhalational anesthetic agents alter levels of Alzheimer disease related markers in human CSF.

A90 To Assess the Effects of Two Anesthetic Techniques (With and Without Nitrous Oxide) on Cognitive Function in Patients Undergoing Transsphenoidal Pituitary Surgery

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Introduction: Postoperative cognitive deficits (POCD) are common in adult patients of all ages at hospital discharge, which may increase morbidity and mortality. Patients with pituitary tumors form a special cohort where not only the location of pathology per se but also anesthetics may affect cognition. The addition of nitrous oxide to anesthetic regimen raises further concerns as effect of nitrous oxide on POCD has been conflicting. We studied the effect of nitrous oxide on cognitive functions during immediate postoperative period and at hospital discharge in patients undergoing transsphenoidal removal of pituitary tumors.

Methods: Ninety patients of ASA grade I to II, aged between 18 and 65 years of either sex, undergoing transsphenoidal surgery for pituitary tumor removal, were enrolled in the study. Preoperative cognitive functions were assessed by using Hindi Mini Mental State Examination (HMMSE) score 1 day before operation. General anesthesia was induced with fentanyl 2 mcg/kg and propofol 2 to 2.5 mg/kg, followed by rocuronium 1 mg/kg to facilitate tracheal intubation. Patients were randomly divided into 2 groups. Patient in Group A received oxygen, nitrous oxide (1:2) and patient in Group B received oxygen 40% in air. Propofol infusion was started and titrated according to BIS spectral index (BIS) value between 40 and 50. Heart rate, mean arterial pressure and BIS were recorded every 15 min interval starting from induction of anesthesia. Total amount of fentanyl and propofol consumed was noted and duration of surgery recorded. Postoperative cognitive functions were assessed using HMMSE score 1 hr after extubation, at 24 hours and at the time of discharge from hospital.

Results: Eighty-five patients completed the study. Five patients all in air group were excluded. HMMSE score in both the groups were found to be comparable at 1 hour, 24 hours and at hospital discharge (Table 1). A total of 25 patients in the nitrous oxide group (55.5%) and 21 patients in the air group (52.5%) experienced significant decline in cognitive function at 1 hour postoperatively. At 24 hours postoperatively, there were only 9 patients in the nitrous oxide group (20%) and 7 patients in the air group (17.5%) with persistent decline in their HMMSE scores. Importantly, 97.8% of patients in the nitrous oxide group and 100% in

the air group demonstrated completely normal cognitive function at hospital discharge (Table 1). Total requirement of propofol was similar in both the groups but consumption of fentanyl was more in air group (Table 1). Extubation time and duration of hospital stay in both groups were comparable.

Conclusion: Administration of nitrous oxide-propofol anesthesia does not increase the risk of postoperative cognitive dysfunction in early postoperative period and at hospital discharge in patients undergoing transsphenoidal removal of pituitary tumor. Maximum cognitive decline occurs in immediate postoperative period.

A91 Dexmedetomidine in Emergence Agitation in Children Undergoing General Anesthesia for Interventional Radiology

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Emergence delirium (ED) from general anesthesia is a common problem in the pediatric population with a reported incidence of up to 80%. ED is associated with accidental removal of surgical dressings and drains, intravenous and intra-arterial catheters, increased nursing care, extended recovery room stays, and delayed reunion with parents. Numerous pharmacologic agents, including benzodiazepines, opioids, and clonidine, have been studied as prophylactic agents for ED but have met with varying success. We studied the use of dexmedetomidine for prevention of emergence delirium in the pediatric population undergoing neurointerventional procedures.

This FDA and institutional IRB-approved double-blind randomized prospective study examined children between the ages of 6 months and 13 years old, ASA I to III. Monitoring included ECG, pulse oximetry, temperature, urine output and noninvasive blood pressure monitoring. Induction of anesthesia included mask induction with oxygen, sevoflurane and nitrous oxide. Maintenance of anesthesia included sevoflurane, oxygen and air as well as an intravenous infusion of remifentanyl. Prior to extubation, patients received intravenous hydromorphone for postoperative pain. Dexmedetomidine was used as an initial loading dose of 1.0 mg/kg given over 10 minutes followed by a continuous infusion at 0.4-0.7 mg/kg/h beginning approximately 1 hour before end of surgery and continuing for 1 hour of recovery in the PACU and the PICU.

Emergence delirium was assessed using objective and subjective measures. Cole Scale allows for a score of ED (1 = sleeping, 2 = awake, calm, 3 = irritable, crying, 4 = inconsolable, crying, 5 = severe restlessness, disorientation). The PAED scale consists of 5 variables including subjective assessment of eye contact with caregiver, purposeful movement, and awareness of surroundings, restlessness and inconsolability. In addition, ActiCal ActiWatch (Mini Mitter, Inc, Bend, OR), which measures and records the degrees a limb is moving using an accelerometer for 24 hours was used. Fourteen drug and 14 placebo study patients provided complete data. Strong evidence for efficacy in avoiding emergence agitation (EA) was seen in the drug group. The placebo group showed typical agitation by the OPS and Cole scales beginning 5 minutes after extubation. Statistical significance was seen at the $P < 0.05$ level for the Cole scale between 10 and 30 minutes, the period of greatest EA. Associated variables of intraoperative blood pressure and especially heart rate were similar. Two adverse events were noted: severe bradycardia in 1 patient required unblinding of drug condition; the patient had received placebo normal saline. The second adverse event was excessive sedation and the drug was discontinued in the PICU resulting in slow arousal and a normal recovery.

Dexmedetomidine is highly selective for the 2A subtype of the central presynaptic α_2 adrenergic receptor which is well tolerated without respiratory or cardiovascular depression. Treatment of ED in pediatric neurosurgical patients involves balancing the need for smooth emergence with the need for accurate neurological exams. In this FDA approved limited study, we have shown appreciable benefits in the pediatric population.

TABLE 1. Baseline Parameters, Intraoperative Variables, Anesthetic Requirements, Cognitive Decline and Postoperative Cognition Scores in Both Groups

	Group A (n=40) (Nitrous oxide)	Group B (n=40) (Air)	p-value
Baseline parameters			
Heart rate	81.22(19.80)	83.08(16.85)	0.06
Mean BP	98.8(14.3)	103.5(14.1)	0.13
BIS	97.2(1.9)	97.1(2.1)	0.78
Cognition 0	28.8(1.1)	28.8(1.1)	0.94
Intraoperative variables			
Propofol (mg)	1422.2(1364.3)	1550(480.3)	0.53
Fentanyl (mcg)	116.1(88.3)	184.4(93.6)	<0.001
Duration (min)	162.5(40.1)	172.3(47.7)	0.08
Extubation time (min)	5.1(3.1)	4.8(2.6)	0.68
Cognitive decline (2 or more decrease in HMMSE score)			
Cognition 1	25(55.6%)	21(52.5%)	
Cognition 2	9(20%)	7(17.5%)	
Cognition 3	1(2.2%)	0(0%)	
Postoperative cognition and hospital discharge			
Cognition 1	27.0(0.27)	27.8(0.23)	0.11
Cognition 2	28.0(0.21)	28.4(0.19)	0.21
Cognition 3	28.2(0.18)	28.5(0.19)	0.22
Discharge (days)	4.3(1.0)	4.4(1.6)	0.96

n = number of patients; BIS = Bispectral index; Cognition 0: baseline cognition score; Cognition 1: cognition at 1 hr; Cognition 2: cognition at 24 hr; Cognition 3: cognition at discharge; HMMSE = Hindi Mini Mental State Examination

A92 Toward a New Class of Amnestic Agents: Blocking de Novo Synthesis of PKM ζ Potently Prevents the Consolidation of Long-term Memory

Panayiotis Tsokas, PhD, Changchi Hsieh, MA, Ira S. Kass, PhD, Todd C. Sacktor, MD, James E. Cottrell, MD. *SUNY Downstate Medical Center.* Amnesia during the perioperative period is an important constituent of the anesthetic state. Understanding the mechanisms of memory formation and storage may lead to novel, highly specific and potent amnestic agents. We have therefore devised an antisense oligodeoxynucleotide (ODN) sequence targeting the synthesis of protein kinase M ζ (PKM ζ), a brain-specific, constitutively active PKC isoform that plays an essential role in the formation and maintenance of long-term memory (LTM). During spatial learning and the induction of protein synthesis-dependent LTP (late-LTP or L-LTP, a cellular model of memory), PKM ζ is synthesized in the hippocampus from a PKM ζ mRNA, which encodes the catalytic domain of PKC ζ without a regulatory domain. Using the selective ζ inhibitor ZIP or a dominant negative mutation of the PKM ζ gene, previous studies have shown that the persistent enzymatic activity of PKM ζ is required for the maintenance of LTM and L-LTP. These agents, however, disrupt previously established memories by interfering with PKM ζ activity, including that of both newly synthesized and preexisting pools of the kinase. To block new synthesis of PKM ζ without affecting its basal levels, we briefly administered an antisense ODN sequence targeting the translation start site of PKM ζ . This construct may prove to be a specific amnestic agent that blocks only new memory formation.

Rats received single hippocampal injections of either antisense (AS) ODN or scrambled (SCR) ODN via bilaterally implanted cannulae. Thirty minutes after ODN injection, rats received 8 trials of active place avoidance training and were sacrificed immediately after the LTM retention test, 24 hours later. The hippocampi were frozen and lysed for immunoblotting. Alternatively, acute hippocampal slices were prepared 1 hour after ODN injection. Two hours later, field EPSPs were recorded from CA3-CA1 stratum radiatum synapses. L-LTP was induced by tetanization of the Schaffer collaterals/commissural fibers. Slices were either recorded for 120 minutes, or frozen 30 minutes posttetanus. Frozen CA1 regions were excised, lysed, and probed for PKM ζ and actin, as a control.

While SCR-ODN treatment had no effect on the de novo, activity-dependent synthesis of PKM ζ during L-LTP induction, AS-ODN completely blocked new PKM ζ synthesis. Neither treatment affected the basal levels of PKM ζ , presumably because of the short duration of ODN treatment and long half-life of PKM ζ . AS-ODN blocked L-LTP: the tetanization induced only a decremental form of LTP (early-LTP), with potentiation returning to basal levels within 2 hours posttetanus. SCR-ODN had no effect on L-LTP. Similarly, AS-ODN completely blocked PKM ζ synthesis during active place avoidance and potently prevented LTM consolidation, as assayed both during the acquisition of LTM in the course of training, and in the retention testing 24 hours later. Short-term memory was unaffected. These results demonstrate that in response to spatial learning or L-LTP induction, PKM ζ is synthesized de novo, and this pool of PKM ζ is specifically required for consolidation of new memories and L-LTP. Optimization of this approach may lead to the development of a new class of highly specific amnestic agents, without the side effects of conventional anesthetics, such as hypotension.

A93 Physostigmine For Emergence Delirium During Awake Craniotomy: A Case Series

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Introduction: Awake intraoperative mapping is the most precise method to locate eloquent cortex and associated subcortical tissue during brain resection. We have used a number of strategies to attempt to improve conscious cooperation during the awake mapping portion of the procedure including: dexmedetomidine, clonidine, midazolam, propranolol, and low dose remifentanyl. Patients who become agitated and uncooperative on emergence from anesthesia present a risk to themselves both from physical injury and from incomplete tumor resection. We have noted that these patients have an agitated delirium, in similar presentation to the central anti-cholinergic syndrome. Physostigmine has long been used to treat central anticholinergic syndrome in patients emerging from anesthesia. It is hypothesized that physostigmine provides benefit in treating emergence delirium by raising central acetylcholine levels. We present a series of 4 patients who were treated with physostigmine in an attempt to provide adequate conscious cooperation for awake cortical mapping.

Results: Four patients received physostigmine for emergence delirium in doses ranging from 0.5 mg to 2.75 mg (Table 1). It is notable that 3 of the 4 patients had received the centrally acting anticholinergic agent oxybutynin preoperatively to reduce bladder spasm from the urinary catheter. In the first case, the patient required 3 re-inductions after initial emergence; only after physostigmine administration did delirium as well as hypocholinergic symptoms of mydriasis, confusion, and agitation abate. In the second case, the patient awakened on low dose remifentanyl delirious, agitated, and moving on the table. He was bolused with propofol and dexmedetomidine, remifentanyl infusion was restarted, and dexmedetomidine infusion was started. The patient again awakened delirious after downward titration of remifentanyl. After bolus dosing of propofol and dexmedetomidine, physostigmine was given with propofol, dexmedetomidine, and remifentanyl infusion. This time, the patient was more coherent and mapping could be attempted. In the third case, physostigmine was given with dexmedetomidine for initial emergence delirium, with modest but not dramatic improvement. In the fourth case, a patient with multiple medical comorbidities had a slow emergence complicated by a generalized tonic-clonic seizure that was treated with cold saline to the cortical surface and a subsequent propofol bolus. Agitation that followed failed to respond to treatment with propranolol, fentanyl, or physostigmine.

Discussion: Physostigmine may be useful in treating agitated delirium after emergence from propofol-remifentanyl general anesthesia in patients undergoing awake cortical mapping. On examination of 4 case records, we found that physostigmine successfully treated existing emergence delirium in 2 patients in whom other agents had failed, and in 1 case as initial treatment (albeit with dexmedetomidine). In 1 case physostigmine failed to abort delirium when given with other agents. Oxybutynin may contribute to creating an anticholinergic syndrome and should be avoided in awake craniotomy. Physostigmine should be studied prospectively in the treatment of delirium during awake craniotomy.

Table 1: Medication administration for awake craniotomy								
Physostigmine								
Patient No.	Age (y)	Weight (kg)	BMI (kg·m ⁻²)	Gender (M or F)	Total dose (mg)	Duration of craniotomy prior to emergence (min)	Dexmedetomidine total (mg)	Oxybutynin total (mg)
1	30	77	25.5	M	2.75	310	18	0
2	35	91	26	M	0.5	160	36.34	10
3	30	86	25.8	M	2	90	8	5
4	48	109	34.4	M	2	74	36	5

A94 Hyperglycemia During Craniotomy for Adult Traumatic Brain Injury

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Background: Hyperglycemia (HG) after traumatic brain injury (TBI) is associated with increased morbidity and mortality. Although a number of studies have investigated HG in adult TBI in different contexts, none has specifically addressed the intraoperative period. As HG is often attributed to a stress response from injury and blood glucose levels increase under anesthesia, it is possible that added stress during anesthesia and surgery may worsen HG. However, the association of intraoperative HG during craniotomy with mortality in adult TBI is unclear and the prevalence and risk factors of intraoperative HG are not known.

Primary Aim: To describe the prevalence and risk factors of intraoperative HG during craniotomy for adult TBI.

Secondary Aim: To describe relationship between intraoperative HG and outcomes [mortality and discharge Glasgow Coma Scale (GCS) score] and prevalence of hypoglycemia.

Design: Retrospective cohort study.

Setting: Harborview Medical Center (HMC)—Level-1 trauma center.

Subjects: Adults ≥ 18 years who underwent urgent/emergent craniotomy for TBI between Jan 2007 and May 2010.

Materials and Methods: After IRB approval, preoperative (within 24 h of anesthesia start) and intraoperative (during anesthesia) glucose data was abstracted. Medical and anesthetic records were used to abstract demographics, admission and discharge GCS, anesthetic details and in-hospital mortality.

Definitions: Hyperglycemia (HG): serum glucose ≥ 200 mg/dL Hypoglycemia: serum glucose < 60 mg/dL.

Statistical Analysis: Descriptive statistics, Spearman rank correlation, Student "t" test, Fisher exact and χ^2 , 1-way and 2-way ANOVA, multivariate regression analyses and Generalized Estimating Equations. Data as mean $\pm$ SD, AOR (95% CI), $P < 0.05$ significant.

Results:

- 185 patients (136 males), age 51 ± 21 years. Ninety-eight percent had isolated TBI. Most common lesions were subdural hematoma (64%), extradural hematoma (15%) and multiple lesions (12%).
- Forty-six percent had severe TBI (GCS ≤ 8), overall in-hospital mortality was 15%.
- Mean anesthetic duration was 201 (range 93 to 437) minutes.
- The average glucose sampling frequency was once every 66 minutes.
- Intraoperative HG occurred in 26 (15%) patients and hypoglycemia was not observed.
- There was significance between patient (79.5% variance; $P < 0.0001$) and within patient (20.5% variance; $P < 0.0001$) variability in intraoperative glucose values.

- Independent risk factors for intraoperative HG were age ≥ 65 years, admission GCS ≤ 8 , preoperative HG and isolated subdural hematoma (Table 1).
- Mean intraoperative glucose was highest in severe TBI patients ($P = 0.02$). Patients with intraoperative HG had higher in-hospital mortality than those with normoglycemia (31% vs. 13%, $P < 0.01$). The average maximum intraoperative glucose was higher in patients who died in the hospital than those discharged alive (175 mg/dL vs. 150 mg/dL, $P = 0.004$).

Discussion: (1) Intraoperative HG was common in adults undergoing urgent/emergent craniotomy for TBI and associated with in-hospital mortality. (2) Anesthesiologists can predict intraoperative HG during craniotomy based on TBI severity, the presence of SDH, preoperative HG, and age ≥ 65 years.

A95 Predictors of Ventriculitis After Extraventricular Drain Placement

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Introduction: Bacterial ventriculitis, inflammation of the brain ventricles from infection, is a devastating disease that may occur as a complication in patients requiring extraventricular drainage (EVD) of their cerebrospinal fluid (CSF). In this study, a retrospective review was performed to determine predictors of ventriculitis onset and outcomes.

Methods: The sample population included all DUHS Neuroscience ICU patients from January 2005 through May 2010 who had a CSF sample collected for microbiological analysis. Results from patients with EVDs were examined for positive blood, urine, sputum, or CSF culture; MRSA or VRE colonization; demographics; and duration of EVD placement. Outcomes include mortality, modified Rankin Score, Glasgow Outcome Score, and hospital and ICU length of stay.

Results: Univariate analysis indicated 4 conditions with possible significance for predicting ventriculitis onset (P value < 0.10); of these 4 (age, EVD days, sex, and positive blood culture), EVD days retained significance in a multivariable logistic regression model. In this model, each additional day of EVD placement expanded the likelihood of ventriculitis by 0.044. Secondary results associating ventriculitis with outcome was predictive of an increase in hospital length of stay and suggestive of an increase in ICU length of stay.

Conclusions: The length of EVD insertion was correlative with bacterial ventriculitis and the presence of ventriculitis increased the hospital length of stay. Notably, the presence of ventriculitis did not increase mortality in this patient population, though this may be attributed to the high mortality rate already established by patients with EVDs.

TABLE 1. Independent Predictors of Intraoperative Hyperglycemia. Data are adjusted for age and gender

Risk Factors	Adjusted odds ratio (95% CI)	P
Age ≥ 65 years	3.9 (1.4–10.3)	0.007
Admission Glasgow Coma Scale (GCS) Score ≤ 8	4.9 (1.6–15.1)	0.006
Preoperative Hyperglycemia	4.4 (1.7–11.6)	0.003
Isolated Subdural Hematoma	5.6 (1.4–22.2)	0.02

A96 The Role of Extraventricular Drainage Rates in Surgical Intervention for Intracranial Pathologies in an ICU Setting

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Purpose/Objectives: The standard of care for hydrocephalus includes the use of extra-ventricular drains (EVDs) to monitor and control increased intracranial pressure (ICP) for various neurosurgical pathologies, including obstructive brain tumors and traumatic subarachnoid hemorrhage. However, while parameters such as ventricular diameter, subarachnoid bleed grade, caudate diameter, and CSF protein levels have been implicated in predicting the need for surgical intervention after EVD placement, such as CSF shunt placement, to date little information has been published or discussed investigating the role EVD drainage rates have. The purpose of the study was to investigate the effects of extra-ventricular drainage rates on the need for surgical intervention in a range of intracranial pathologies.

Materials and Methods: A retrospective review of 70 patients with extra-ventricular drains over 5 years in an ICU setting of a major academic center was conducted. Patients were categorized into the low pressure group if the majority of drainage days were set at 10 mm Hg or less. The high pressure group included those with a majority of drainage days set

at greater than 10 mm Hg. Eventual need for shunt insertion was analyzed.

Results: Of the 70 patients included in the study, 38 (54%) patients were included in the low drainage group, and 32 (46%) patients were included in the high drainage group. Six total patients, or approximately 9%, eventually had a shunt inserted; of those 83% (5/6) were in the low pressure group ($P = 0.2088$), and 1 patient who eventually had a shunt inserted was in the high pressure group. All shunts inserted were ventriculoperitoneal shunts. Average length of EVD insertion was 10.2 days.

Conclusions: Our data shows a trend towards greater need for eventual shunt insertion for those who are drained at low pressures for acute hydrocephalus, as defined by our study. Draining at lower pressures, or higher drainage rates, may provide for greater flow through arachnoid granulations, preventing blockage. Future directions include comparing clinical outcomes of specific intracranial pathologies, such as traumatic subarachnoid hemorrhage versus obstructive tumor as well as solute trends in drawn CSF samples.

A97 Trigemino-Cardiac Reflex During Surgery for Trigeminal Schwannoma

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Background: The trigemino-cardiac reflex (TCR) is a brainstem reflex that is initiated by stimulation of any of the sensory branches of the trigeminal nerve during eye, craniofacial or skull base surgery. Clinically, TCR manifest with the sudden onset of dysrhythmia (bradycardia or even asystole) and arterial hypotension. Here, we prospectively investigate the occurrence of TCR during surgery for trigeminal schwannoma and discuss the relationship between the induction of TCR and manipulation site of the fifth cranial nerve.

Methods: A total of 146 patients with trigeminal schwannoma underwent surgical treatment during recent 2 years. The tumor was accessed and removed via middle or combined with posterior cranial fossa approach. TCR was recognized as the sudden onset of bradycardia (lower than 60 beats per minute) and hypotension (a drop of 20% or more in mean arterial pressure). After the hemodynamic parameters returned to normal levels, the maneuver and location of manipulation relative to the induction of TCR was recorded in detail and evaluated with neurosurgeon postoperatively.

Results: TCR was identified in 15 patients (10.3%, 15/146) intraoperatively. In 12 patients, TCR was elicited by noxious compression and retraction of the dura of middle cranial base, which covers the distribution of 3 divisions of the trigeminal nerve. In 3 patients, TCR was found when dissecting and retracting the lateral wall of the Meckel cave which contains the Gasserian ganglion. TCR was not observed in stimulation of intracranial segment of the trigeminal nerve regardless of retraction, compression and other manipulations. The hemodynamic parameters return spontaneously to normal levels with cessation of stimulation in 8 patients and administration of atropine was need in the remains. The postoperative course is uneventful in all 15 patients.

Conclusions: To our knowledge, this is the first report concerning TCR during the surgery of trigeminal schwannoma. Furthermore, we found that the occurrence of TCR was exclusively relative to manipulation of the extracranial course of the trigeminal nerve in middle cranial fossa. Our findings may be helpful to prevention of the TCR during surgery for trigeminal schwannoma.

Keywords: Trigemino-cardiac reflex; trigeminal schwannoma; trigeminal nerve; bradycardia; hypotension.

A98 A Comparison of Two Glucose Sampling Frequencies for an Intensive Insulin Protocol During Craniotomy in Nondiabetic Patients: How Efficiently and Safely Can We Maintain Target Glucose Levels?

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Background: Patients undergoing craniotomies have increased levels of serum glucose as a result of both surgical stress and dexamethasone administration. In the IHASt patients, a blood glucose concentrations > 129 mg/dL was associated with an 18% to 70% incidence of neuropsychological deficits and a blood glucose concentration > 152 mg/dL was associated with an 18-39% incidence of gross neurologic deficits.¹ Intensive insulin protocols based on 60-minute interventions are slow to achieve target glucose levels and produce a high incidence of hypoglycemia. We hypothesized that an intensive insulin protocol based on 30-minute interventions would more rapidly achieve and more efficiently maintain normoglycemia while avoiding hypoglycemia.

Methods: IRB approved informed consent was obtained in nondiabetic patients undergoing craniotomy for the surgical treatment of tumors or intracranial aneurysms. Serum glucose levels were measured at 15-minute intervals. Patients were block randomized to have the insulin dose adjusted every 30 or 60 minutes to maintain intraoperative normoglycemia (serum glucose between 90 and 110 mg/dL). The 2-stage median absolute performance error (MDAPE), a measure of efficiency of target-controlled infusions, was the primary outcome.² The 2-stage median performance error (MDPE) was also calculated as a measure of the bias of the insulin protocol. Data are presented as medians (range). The Mann-Whitney U Rank Signed Test was used to determine differences between groups with a $P < 0.0002$, $P < 0.012$, or $P < 0.046$ considered significant at the a priori planned interim analyses of $n = 60$, 120, or 200 total patients. The χ^2 test was used to determine if there was a difference in the incidence of hypoglycemia between groups with a $P < 0.05$ considered significant at each interim analysis.

Results: This interim analysis ($n = 120$ patients total), demonstrated no differences in the efficiency (MDAPE) for the insulin protocol for target glucose levels of 110 or 90 mg/dL for either group (Table). There was also no difference in the bias of the insulin protocol for either group (MDPE). The incidence of hypoglycemia was similar between groups (11.7% vs. 6.7%, $P = 0.527$, 30 vs. 60-minute interventions).

Discussion: At the second interim analysis, there were no differences in efficiency or bias of blood glucose control between the 30-minute and 60-minute intervention groups. Furthermore, there was no difference in the incidence of hypoglycemia between intervention groups. However, the majority of hypoglycemic events occurred within 45 minutes of initiating a change in the insulin infusion. Because the difference in the efficiency (MDAPE) of the 2 groups is less than 5%, there is no benefit in continuing this trial. These data support the use of intensive insulin protocols in future operating room trials of intensive insulin therapy, such as DeLiT, with 60-minute interventions, rather than 30-minute interventions, and 15-minute monitoring for hypoglycemia.³

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Performance Parameter	Glucose Target [mg/dL]	Q 30 MIN median (range)	Q 60 MIN median (range)	P
MDPE (bias)	110 mg/dL	-1.8% (-12.3%, 19.1%)	0.9% (-18.6%, 42.7%)	0.135
	90 mg/dL	20.0% (1.5%, 45.6%)	22.8% (-0.6%, 74.4%)	0.090
MDAPE (efficiency)	110 mg/dL	6.1% (0.9%-21.8%)	6.8% (1.8%, 42.7%)	0.069
	90 mg/dL	20.0% (1.5%, 45.6%)	22.8% (2.2%, 74.4%)	0.105

A99 Effectiveness of Focused Neuroanesthesia Education Using Dance Cards in Improving the Performance of Anesthesiology Residents on Standardized Test

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Introduction: Providing effective subspecialty education to anesthesiology residents in a time constrained framework is challenging. After a formal needs assessment, a new curriculum was introduced for Neuroanesthesiology rotation at our institution. The major component was the “dance cards” (cards listing essential topics to be discussed in one-on-one intraoperative teaching). The important implementation strategies included: providing detailed orientation to rotation with clear goals and objectives and information on testing, covering essential topics in one-to-one teaching, providing selected reading material, and administration of standardized web-based tests immediately before the start of rotation, immediately after and 3 months after end of rotation.

Primary Aim: To assess the effectiveness of focused Neuroanesthesia education using “dance cards” in improving the performance of anesthesiology residents on standardized tests administered immediately after and 3 months after completion of the rotation.

Secondary Aims: To analyze resident evaluation of rotation and effectiveness of “dance cards.”

Methods: After institutional IRB approval, this retrospective review (May 2010 to Jan 2011) involved abstraction from departmental educational records for resident scores on standardized test before, immediately after and 3 months after the rotation, number of topics signed off on each “dance card” and resident responses to survey on evaluation of rotation and effectiveness of “dance cards.” The *t* test was used to compare score of standardized tests with significance at $P < 0.05$.

Results:

- Twenty residents underwent Neuroanesthesiology rotation, all took tests and 26 completed evaluation of the rotation.
- Mean number of topics signed off on the “dance card” = 7 ± 1 (range 5 to 9)
- Mean score immediately after and 3 months after rotation was significantly higher than the score before rotation ($P = 0.0005$ and $P = 0.0004$ respectively) while the scores immediately after and 3 months after rotation were similar ($P = 0.06$) (Fig. 1).
- Scores of 15 (54%) residents immediately after rotation were higher than their scores before the rotation.
- 3 months after the rotation, the scores of 13 (46%) and 11 (39%) residents were same and lower than their score immediately after rotation, respectively.
- Median improvement in score was 20% immediately after rotation compared with before rotation, 10% 3 months after rotation

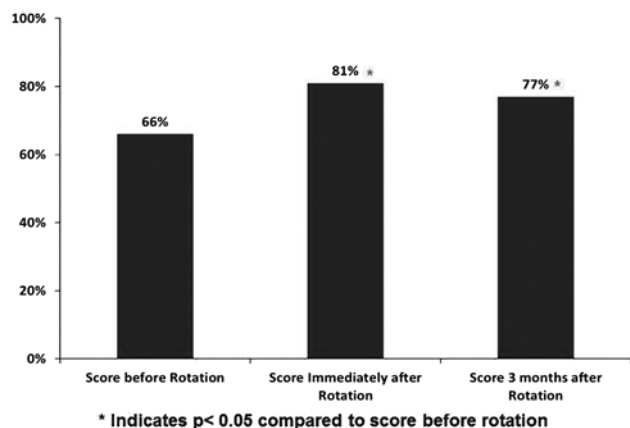


FIGURE 1. Scores on standardized tests before & after the neuroanesthesiology rotation.

compared with before rotation and 0% 3 months after rotation compared with immediately after rotation.

- Quality, quantity of intraoperative teaching, overall learning experience and overall effectiveness of rotation was rated as good/ outstanding by 100%, 77%, 92% and 92% residents, respectively. Total 70% residents found the tests useful and 77% found “dance cards” helpful in keeping focus during rotation and in making effective use of time.

Discussion: Despite its limitations, this study shows that focused neuroanesthesia education for anesthesiology residents using “dance cards” improves their scores on standardized tests with high degree of resident satisfaction of intraoperative teaching. This model may be useful for other subspecialty rotations to improve overall effectiveness of education.

A100 Dexmedetomidine for Awake Craniotomy-A Comparison With General Anesthesia for Supratentorial Brain Tumors

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Introduction: Awake craniotomy remains the gold standard for identifying functional areas during brain surgery. The rationale is to provide accurate resection of tumor maximizing surgical clearance of the cancer but minimizing functional impairment to motor and language function. In this study, we hypothesized that patients undergoing awake craniotomies have better immediate perioperative outcomes (pain, hypertension in the PACU) without suffering major respiratory and hemodynamic complications during surgery.

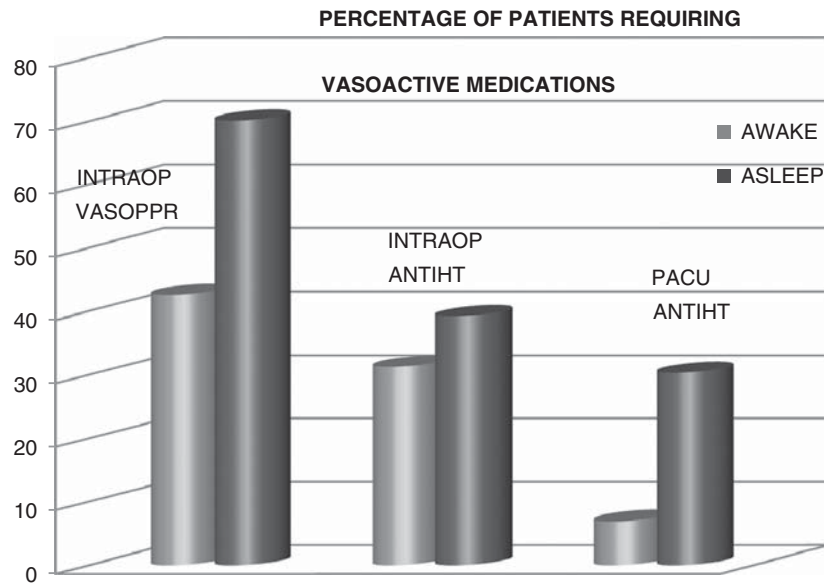
Materials and Methods: We retrospectively reviewed charts of patients who had awake (A) ($n = 115$) or asleep (S) ($n = 97$) craniotomies. Awake craniotomies had a continuous infusion of propofol (50 to 70 mcg/kg/min) and dexmedetomidine (0.4 to 0.7 mcg/kg/h). Oxygen was administered by a nasal cannula and a nasopharyngeal airway in most cases. The S group received general anesthesia with sevoflurane, remifentanyl (0.05 to 0.2 mcg/kg/min) infusion and muscle relaxation.

Results: Baseline characteristics were comparable in both groups. Respiratory and fluid data Episodes of desaturation ($A = 31$ vs. $S = 5$, $P < 0.0001$) although temporary and hypercarbia were more common in the awake group ($A = 43.75$ mm Hg vs. $S = 32.6$ mm Hg, $P < 0.001$). The administration of mannitol was comparable in both groups ($n = 41$ vs. $n = 35$, $P < 0.957$). Intraoperative fluids were also comparable between the 2 groups. Hemodynamic data: The mean arterial pressure during pinning and emergence were significantly lower in A compared with S patients (pinning 103.8 mm Hg vs. 92.6 mm Hg, $P < 0.005$) and (Emergence 97.47 mm Hg vs. 85.93, $P < 0.005$). The heart rate during these times were also lower in the A group. Patients requiring intraoperative vasopressors ($A = 42.6\%$ vs. $S = 70.10\%$, $P < 0.005$) and postoperative antihypertensives were much lower in A than S patients ($A = 12.9\%$ vs. $S = 30.3\%$, $P < 0.005$). Postoperative pain relief requirements of fentanyl were lower in A than S and visual analogue scores were significantly lower in A for first 15 minutes in PACU. Length of stay (LOS) in PACU (240 min each) and in the hospital (3.55 in A vs. 4.69 d in S) were comparable ($P < 0.097$).

Complications: In the A group, complications were delayed arousal ($n = 1$), seizures ($n = 2$), and airway obstruction requiring laryngeal mask airway insertion ($n = 1$). In the S group, complications included seizures ($n = 1$) and hypoxemia after extubation ($n = 2$).

Discussion: A higher incidence of respiratory complications in the awake group may be explained by a lack of controlled airway. A lower blood pressure and heart rate during pinning and emergence and possible better postoperative pain control in A could be explained by the hemodynamic and antinociceptive effect of dexmedetomidine combined with local anesthetic block performed by the surgeons before incision.

Conclusions: We conclude that anesthesia for awake craniotomy is safe with the judicious use of infusions of propofol and dexmedetomidine in the hands of a trained anesthesiologist and local anesthetic infiltration by the surgeon at the site of incision.



A101 Clopidogrel Resistance in Patients Requiring Endovascular Neurological Intervention

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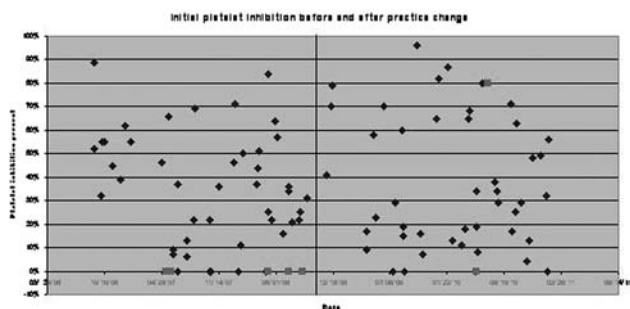
Introduction: Clopidogrel resistance, with a reported incidence of 52%, is increasingly important in vascular intervention because it may put patients at risk for perioperative thrombosis. We investigated the incidence of postoperative thrombosis in patients requiring endovascular carotid stent placement.

Methods: After IRB approval we retrospectively reviewed the charts of 96 patients requiring endovascular carotid stenting between 2006 and 2010. Each patient was treated with aspirin 325 milligrams (mg) daily and clopidogrel 75 mg daily for 5 days before intervention. On the day of the procedure, patients' blood was tested with a VerifyNow-P2Y12 clopidogrel assay (Accumetrics, San Diego, CA) to determine the effectiveness of clopidogrel. Patients were treated with an extra dose of clopidogrel 75 mg if they were found to be resistant (defined as <40% platelet inhibition) on the day of their surgery and the procedure commenced. After a number of thrombotic complications, practice changed in 2008 to try to increase the efficacy of clopidogrel in resistant patients. Therefore after August 2008, clopidogrel resistant patients were treated with increasing doses of clopidogrel daily until their assay showed >20% inhibition. In some patients, the assay never reached the 20% threshold, but only after the attempt to increase their percent platelet inhibition was their procedure performed. For purposes of data

analysis, clopidogrel resistance is defined as initial percent platelet inhibition of <20%.

Results: The main outcome variable was periprocedural thrombosis, which occurred in 7 patients. Statistically, it was insignificant whether the patient was treated before the practice change in 2008 or not. Prior to developing a logistic regression model, univariate analyses revealed that the following factors were associated with the outcome variable periprocedural thrombosis: use of statins ($P = 0.003$), clopidogrel resistance ($P = 0.004$), high diastolic blood pressure ($P = 0.04$), and history of stroke ($P = 0.09$). The logistic regression model ($P = 0.002$) identified the following variables as predictive of periprocedural thrombosis: clopidogrel resistance, lack of statin use, and higher diastolic blood pressure.

Conclusion: We show that clopidogrel resistance is associated with periprocedural thrombosis in patients after endovascular carotid intervention. Figure 1 graphically depicts all 96 patients as blue diamonds or pink squares, indicating their initial percent platelet inhibition by clopidogrel. The 7 pink squares depict those patients who experienced periprocedural thrombosis. The vertical line in the middle indicates the time of the practice change in 2008.



A102 The Application of Electro-Acupuncture Combined With Sevoflurane Anesthesia in Neurosurgery

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Object: To observe the effects of electro-acupuncture combined with sevoflurane anesthesia used in neurosurgery patients and the speed of recovery after surgery.

Methods: Eighty cases of supratentorial tumor resection were anesthetized with sevoflurane and randomly allocated into no treatment group (Group A), electro-acupuncture group (Group B). Han's acupoint nerve stimulator with 2/100 Hz frequency was used to stimulate the points. The patients in Group B received electro-acupuncture at FengChi (GB20) tou TianZhu (BL10) and CuanZhu (BL2) tou YuYao (EX-HN4) at the same side of the craniotomy before anesthesia induction. The stimulation is continued until the end of operation. All patients were induced with propofol 2 mg/kg, sufentanyl 0.3 µg/kg and vecuronium 0.1 mg/kg and maintained with sevoflurane. The end-tidal sevoflurane concentration, minimum alveolar concentration (MAC), bispectral index (BIS), changes of blood pressure and heart rate, and the speed of recovery were investigated in the 2 groups.

TABLE 2. End-tidal Sevoflurane Concentration Required to Maintain Stable Hemodynamics at Dierent Stages of Surgery

	Group A (n=40) (Acupuncture - Sevoflurane group)	Group B (n=40) (Sevoflurane group)	P	Reduction percentage (%)
Skin incision	1.60 ± 0.37	1.47 ± 0.33	0.214	8.12
Drilling skull	1.72 ± 0.43	1.56 ± 0.34	0.196	9.30
After skull drilling	1.74 ± 0.39	1.57 ± 0.36*	0.041	9.77
Opening the skull	1.75 ± 0.39	1.60 ± 0.40	0.056	8.57
Dura incision	1.79 ± 0.43	1.60 ± 0.36*	0.022	10.61
Intracranial period (10min)	1.78 ± 0.45	1.61 ± 0.32	0.057	9.55
Intracranial period (20min)	1.76 ± 0.44	1.61 ± 0.31*	0.046	8.52
Intracranial period (30min)	1.80 ± 0.44	1.67 ± 0.34*	0.037	7.22
Suturing dura	1.83 ± 0.39	1.69 ± 0.28	0.102	7.65
Sutuing skin	1.73 ± 0.37	1.66 ± 0.22	0.682	4.04

Data are mean ± SD or numbers. * $P < 0.05$ compared to the sevoflurane group.

Results: Hemodynamic variables were similar throughout surgery in both groups. The end-tidal concentration and MAC of sevoflurane were lower in Group B than in Group A, $P < 0.05$. The requirement of sevoflurane was reduced by an average of 8.34% in Group B. The BIS value was kept within 4050 in both groups, so as to insure a relatively anesthesia depth. The times to recovery of spontaneous respiration, extubation, opening eyes, voluntary movement and discharge from the recovery room were significantly shorter in Group B than in Group A, $P < 0.01$. Dysphoria, nausea and vomiting occurred in fewer patients in Group B.

Conclusion: Electro-acupuncture combined with sevoflurane used in neurosurgery can decrease the expired concentration and MAC of sevoflurane. It also speeds the recovery of anesthesia and improves the postoperative recovery of neurosurgery patients.

Keywords: Electro-acupuncture; Sevoflurane; Neurosurgery; Recovery.

A103 Combination of Standard Monitoring Parameters as a Reliable Measure of the Hypnotic Component of Anesthesia

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Introduction: Standard monitoring of anesthesia is mainly based on cardiovascular, respiration and to a certain extent motor response parameters, which provide information for assessing the hypnotic component of anesthesia. Specific parameter behavior and inter-patient variability requires a high level of the anesthesiologist experience for its evaluation. The present investigation evaluates ability and efficiency of the BIS and an indicator that combines different standard parameters using an adaptive fuzzy inference to indicate “depth of anesthesia.”

Methods: After ethics committee approval and written informed consent, 263 adult patients undergoing surgery under general anesthesia were included in a study conducted in 6 European centers.¹ Patients were randomly assigned to 1 of 11 anesthetic groups, consisting of different analgesic and hypnotic drugs for induction/maintenance phase. Standard parameters and EEG were continuously recorded and stored together with patient data (eg, age, weight, ASA, drug protocol). During anesthesia induction, Tunstall isolated forearm technique was performed to detect loss of consciousness (LOC). After skin incision, the anesthetic concentration was increased until EEG burst suppression (BS).

At the end of surgery, drugs were discontinued until return of consciousness (ROC). An adaptive fuzzy inference based on a Takagi Sugeno Kang model (TSK)² was used for data driven combination of

standard parameters (heart rate, blood pressure, inspiratory and expiratory gas concentrations, pulmonary peak pressure), patient data and drug protocol (threefold cross validation). The adaptive algorithm is able to provide indicator values even in case of partially available input data (eg, caused by measurement failures, equipment problems and human errors³). TSK and BIS (calculated offline⁴) were analyzed during consciousness (before LOC, after ROC), unconsciousness (after LOC, before ROC), general anesthesia and BS. Prediction probability (PK) including 95% bootstrap confidence intervals indicates the indicators ability to separate anesthetic levels.

Results: The TSK indicator leads to a PK of 0.90 (0.88 to 0.91) for separation of consciousness, unconsciousness, general anesthesia and BS. This is significantly higher than BIS with PK of 0.80 (0.78 to 0.81). Thereby, the TSK provides values even in case of false/missing standard parameter values (14.3% occurrence).

Conclusions: Single standard parameters provide partial information about the hypnotic component and have to be interrelated to assess “anesthetic depth.” The presented approach allows combining this individual information by a data driven adaptive fuzzy inference, so that the resulting indicator includes anesthesiologist’s expertise about parameter relevance. Results show that separation of consciousness, unconsciousness, general anesthesia and BS can be sufficiently revealed by the obtained indicator even though individual parameters are partially acquired. In particular, based on data of a multicenter study the indicator leads to a better separation than the electroencephalogram based BIS.

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A104 Intracerebral Hemorrhage—Does eGFR Affect Hospital Mortality?

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Background: Intracranial hemorrhage (ICH) is the second most common cause of stroke, with an incidence of 24.6 per 100,000 patient-years. The median case fatality is 40% at 1 month, with 12% to 39% of survivors independent of care after the stroke. Estimated glomerular filtration rate (eGFR) is a measure of renal function with an eGFR < 60 mL/min/1.73 m² diagnostic of chronic kidney disease. eGFR is associated with worse outcome after cardiac surgery and acute stroke. Our aim was to determine if an association between eGFR and outcome after ICH exists. **Methods:** This is a retrospective review of data for all patients admitted to our ITU with an ICH from 1.11.09 to 31.1.11. Data collected included: patient demographics, admission severity of illness score (APACHE II and SOFA), duration of mechanical ventilation and ITU stay, eGFR on admission. Survival (dead or alive) was determined at discharge from hospital. eGFR was calculated using the 4 variable Modification of Diet in Renal Disease formula. Sub-group analysis compared patients with: eGFR < 60 mL/min/1.73 m², 60 to 89 mL/min/1.73 m² and eGFR ≥ 90 mL/min/1.73 m².

Results: Sixty-nine patients were included. Mean age was 52.1 years (SD ± 14.6). There was no difference in demographics or admission APACHE II/SOFA score between the groups. Overall patient mortality was 31.9%. Total 35.3% of patients with eGFR < 90 mL/min/1.73 m² died compared with 28.5% with eGFR ≥ 90 mL/min/1.73 m² ($P = 0.3667$ for difference). Subgroup analysis also showed no difference in mortality (Table 1).

No statistically significant difference in median days ventilated [5 (0 to 29) vs. 3.5 (0 to 28) d, $P = 0.3584$], or median duration of ITU stay [5 (1 to 31) vs. 7 (1 to 29) d $P = 0.2111$] between the patients with eGFR ≥ 90 mL/min/1.73 m² and those with eGFR < 90 mL/min/1.73 m². Sub-group analysis showed no statistically significant difference between the 3 groups in duration of ITU stay or mechanical ventilation.

Conclusions: We found no significant difference in hospital mortality, duration of mechanical ventilation or ITU stay between ICH patients admitted to ITU with an eGFR ≥ 90 mL/min/1.73 m² and eGFR

TABLE 1. Differences among the groups of ITU Stay, Duration of Mechanical Ventilation and Survival of Discharge

	Estimated Glomerular Filtration Rate ml/min/1.73m ²			
	≥ 90	<90		
Percentage survival at hospital discharge	28.5%	35.3%		p=0.3667 Using Fisher's Exact test
Duration of mechanical ventilation median days (range)	5 (0-29)	3.5 (0-28)		p=0.3584 using Mann-Whitney U
Duration of ITU stay Median days (range)	5 (1-31)	7 (1-29)		p=0.2111 using Mann-Whitney U
Sub-group analysis	≥ 90	60-89	<60	
Median duration of ITU stay Days (range)	5 (1-31)	6.5 (1-28)	12 (1-29)	p=0.2559 using Kruskal-Wallis test
Median duration of mechanical ventilations Days (range)	5 (0-29)	2 (0-26)	5 (0-28)	p=0.6822 using Kruskal-Wallis test
Survival at hospital discharge Alive: Dead (number of patients)	25:10	13:11	9:1	

< 90 mL/min/1.73 m². The majority of patients who did not survive died early from severity of brain injury and were therefore less likely to be affected by co-morbidities, late complications of ICH and prolonged hospital stay possibly accounting for our results. As eGFR is a predictor of cardiovascular risk, it may therefore be more appropriate to compare eGFR in patients who survive an ICH and those who had a late death after an ICH. Although not statistically significant there was a trend towards a longer ITU stay in patients with an eGFR < 60 mL/min/1.73 m². A more adequately powered study is required to explore this further.

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A105 Infusion Dose Requirement of Rocuronium in Patients on Phenytoin Therapy in Comparison With Control Group

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Introduction: Rocuronium is a nondepolarizing neuromuscular blocking agent with intermediate duration of action and suitable for continuous infusion. Patients receiving chronic anticonvulsant therapy demonstrate

resistance to nondepolarizing muscle blockers.¹ Resistance to rocuronium and shorter recovery index in patients on chronic phenytoin therapy have been observed. There is also racial variation in the dose requirement of rocuronium.² The present study undertaken to determine the infusion dose requirements of Rocuronium in patients on phenytoin therapy in comparison with control group.

Methods: After obtaining institutional ethics committee approval and patient consent, 75 patients scheduled for neurosurgical procedure were included in the study. On the basis of preoperative serum phenytoin levels patients were divided into 2 groups: Gp I < 10 µg/mL (subtherapeutic levels (n = 27), Gp II > 10 µg/mL (therapeutic levels (n = 27)). Twenty-one patients not on phenytoin taken as control (Gp III). All patients received balanced general anesthesia. Train of four (TOF) mode of stimulation was used to assess the adequacy of neuromuscular block and recovery using Datex S5 NMT monitor. Tracheal intubation facilitated with rocuronium in the dose of 0.6 mg/kg. Infusion of rocuronium was started 30 minutes after the initial dose at the rate of 0.15 mg/kg/h and dose adjusted to maintain 90% block. Infusion was stopped 30 minutes before the end of the surgery. Time to recovery to 70% after stoppage of infusion was noted. All data reported as mean ± SD. Demographic data among the groups were compared using the paired Student *t* test. Comparison of neuromuscular response, recovery time and dose requirement were made between the groups by using analysis of variance with Post-hoc Tukey test. Values of *P* < 0.05 were considered as significant.

Results: Demographic data between the groups were similar. The plasma concentration of phenytoin in group I and group II was 4.32 ± 2.23 µg/mL and 18 ± 7.53 µg/mL. The total dose and infusion dose to maintain 90% to 95% blockade was significantly higher in phenytoin groups compared with control (*P* < 0.001). There was no difference in infusion dose between Gps I and II. There was no significant difference in recovery time between the 3 groups (Table 1). No statistical relationship could be established within the different levels of Phenytoin and infusion dose (*P* = 0.815).

Conclusions: Patients on phenytoin therapy require higher infusion dose of rocuronium bromide. Even subtherapeutic levels of phenytoin significantly increase the dose requirement compared with control. More patients to be included to determine the correlation of serum levels of phenytoin and infusion dose requirement.

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TABLE 1. Differences Among the Groups for Duration of ITU Stay, Duration of Mechanical Ventilation and Survival at Discharge

	Group I (phenytoin <10µg/ml)	Group II (phenytoin >10µg/ml)	Group III (Control)	P value
70% recovery (mins)	39.07±13.23	40.86±19.79	44.70±18.54	0.476
Total dose (mgs)	131.00±46.82	128.90±39.82	88.31±21.67	<0.001**
Infusion mg /kg/hr	0.429±0.205	0.429±0.176	0.265±0.147	0.001**

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A106 Perioperative Stress Response After Carotid Endarterectomy and Carotid Artery Stent

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Background: Arteriosclerotic change is frequently observed in both heart and brain simultaneously. As a result, postoperative cardiac event is a serious seizure after carotid endarterectomy (CEA) and carotid artery stenting (CAS). However, there was a discrepancy in reported incidence of postoperative cardiac event after CEA and CAS. After CEA more cardiac events were observed, but more cerebral event after CAS. Landesberg et al suggested that perioperative surgical stress may induce postoperative cardiac events.¹ Therefore, we reviewed patients who underwent CEA or CAS to investigate whether surgical stress led to the difference in the incidence of cardiac event after CEA and CAS.

Methods: One hundred and ninety-seven patients who underwent CAS or CEA between April 2007 and March 2010 were included in this study. Ethics Committee waived the need for patients' consent for this study and approval was provided. The incidence of postoperative non fetal cardiac event (myocardial infarction, percutaneous coronary intervention, and arrhythmia) was investigated. Candidate variables of perioperative stress and inflammatory response included postoperative peak value of blood sugar, C reactive protein (CRP), and creatine kinase (CK). Further, whether history of coronary artery intervention (coronary artery bypass surgery and percutaneous coronary intervention) had an effect on perioperative stress and inflammatory response was accessed.

Results: The incidence of postoperative cardiac events was slightly higher after CEA than after CAS (12 and 8%, respectively), but the difference did not reach to the significance. In regards to the perioperative stress response, there was no significant difference in blood sugar value between the 2 groups, but tended to be higher after CEA than after CAS.

TABLE 1. Post operative stress and inflammatory response

	CAS (n=48)	CEA (n=144)	p value
BS (mg/dl)	132±60	150±56	0.05
CRP (mg/dl)	2.1±4.0	6.4±6.7	<0.001
CK (IU/l)	100±80	213±344	0.02

BS: blood sugar; CRP: C reactive protein; CK: creatine kinase; CAS: carotid artery stent; CEA: carotid endarterectomy.

TABLE 2. Effect of history of ischemic heart disease on postoperative stress and inflammatory response

		IHD	non IHD	p values
BS (mg/dl)	CAS	126±36	133±65	0.54
	CEA	154±59	148±55	0.55
CRP (mg/dl)	CAS	1.2±0.9	2.4±4.6	0.13
	CEA	5.0±5.0	7.2±7.5	0.06
CK (IU/l)	CAS	90±47	110±91	0.50
	CEA	172±264	237±385	0.34

BS: blood sugar; CRP: C reactive protein; CK: creatine kinase; CAS: carotid artery stent; CEA: carotid endarterectomy; IHD: ischemic heart disease.

CRP and CK after CEA were significantly higher than those after CAS (Table 1). History of coronary artery intervention (old myocardial infarction, coronary artery bypass surgery, and percutaneous coronary intervention) was observed in 22% and 40% of patients after CAS and CEA, respectively. There was no significant difference in perioperative stress or inflammatory response with or without history of coronary events (Table 2).

Conclusions: After CEA inflammatory response was stronger than after CAS. History of coronary artery event did not have an impact on postoperative stress or inflammatory response after CEA or CAS.

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A107 An Uncommon Case of Rhabdomyolysis and Unexpected Lactic Acidosis in Neurosurgery, is the Prone Position the Most Secure Option?

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Prone position is frequently used for neurosurgical, orthopedic, and urological procedures. It provides excellent surgical access, the position itself, and because it reduces the extradural venous system and spinal fluid pressure, which is very advantageous in spinal and posterior cranial fossa surgery. Main complications associated, as described in literature, are: ocular and ear complications, musculoskeletal, venous air embolism, neurovascular injuries, skin ischemia in pressure points, changes in hemodynamics and ventilation, anxiety, and endotracheal tube or catheter loss. However, we do not find in the most recent literature, cases of visceral hypoperfusion and rhabdomyolysis associated in the same patient, as consequence of prolonged prone position.¹ We present a case of sudden intraoperative lactic acidosis and rhabdomyolysis during neurosurgical procedure after 7 hours in prone position.

Methods: Fifty-seven-year-old and 80 kg woman scheduled for lumbar discectomy and arthrodesis because of a herniated disc at this level. Medical background includes hypertension controlled with valsartan and pituitary Adenoma under replacement therapy with cabergoline, levothyroxine and hidroaltesona. Preoperative laboratory tests were normal. Under general anesthesia, using remifentanyl in continuous perfusion (0.05 to 0.2 Ug/Kg/L/min/L) and sevoflurane 1% for anesthetic maintenance, placement in prone position was careful and without incidents. After 6 hours of surgery, a blood gas analysis showed following values: lactate 9 mmol/L, Cl 102 mEq/L, pH 7.19 and HCO₃ 21, being the initial rigorously normal. Patient was hemodynamically stable with adequate urine output during all surgery, did not show hypoxemia episodes, always maintained body temperature above 36°C and didn't need blood transfusion. Transferred to recovery room under spontaneous ventilation, where hyperlactatemia of 9.3 mmol/L, a CPK of 4500 U/L and myoglobin in urine were detected. Two hours from arrival levels of lactate and acid-base balance were normalized, and 3 days later so did muscle enzymes (Table 1).

Discussion: As already described by McCormack et al, prone position, especially during prolonged general anesthesia, can be the cause of abdominal and pelvic vessels compression, resulting in visceral hypoperfusion and subsequent lactic acidosis, contributing to CPK increase.² Targa et al³ studied the appearance of rhabdomyolysis, in different positions during long surgeries. This is an unusual case where we find both events, hyperlactatemia and rhabdomyolysis. In absence of hypotension, hypoxemia, or severe anemia, we must think about hypoperfusion of intra-abdominal organs as a cause of hyperlactatemia, considering that compression of lower limbs arteries, may lead to rhabdomyolysis without other clinical signs; CPK elevation and myoglobinuria are usual, and usually the only clinical signs, but often not diagnosed.

	Arrival to PICU	Hours after admission to PICU		
		12	24	48
Blood pressure (mm Hg)	145 / 77	120 / 65	130 / 60	142 / 85
Hct (%)	32,7	33,4		
PH	7,17	7,29	7,37	7,38
pCO ₂ (mm Hg)	33	38	35	37
HCO ₃ (mmol/L)	15	23	27	28,4
Base excess (mmol/L)	-9,2	-5,9	-1,6	0,8
Lactate (mmol/L)	9,3	4,2	2,3	1,7
AST (U/L)	74	95	110	85
ALP (U/L)	56	76	90	90
CK (U/L)	3200	4500	2000	890
Creatinine (mg/dl)	0,9	1,5	1,3	1,0
Urine: myoglobin (mg/ml)	0,056	3,2	0,230	-

AST: aspartate transaminase, ALP: alkaline phosphatase, CK: creatine kinase.

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A108 Comparison of Cardiac Output, Stroke Volume, and Stroke Volume Variation Measurements Using Pulse Contour Versus Bioreactance-based Monitors in Neurosurgical Operations

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Background: Perioperative hemodynamic monitoring with stroke volume variation (SVV) analysis has emerged to be an important technique for fluid optimization and management. Its application in neurosurgical anesthesia has been reported. Conventional volume status monitoring using central venous pressure or pulse contour technique, for example, Vigileo, requires invasive venous or arterial line placement. Noninvasive

bioreactance technology, for example, NICOM, uses 75 kHz signal processing for its algorithm in calculation of hemodynamic data.

Objectives: We retrospectively reviewed a series of cases during which NICOM and Vigileo™ were simultaneously used.

Methods: Both Vigileo and NICOM monitors were applied in 3 skull base approach neurosurgical operations. Patients were monitored intraoperatively using somatosensory evoked potential, electroencephalograph and brainstem auditory evoked potential when applicable. Anesthesia management was balanced anesthesia of sevoflurane, propofol and remifentanyl infusion with transition to total intravenous anesthesia with targeted bispectral index below 60. Paired hemodynamic data including stroke volume (SV), cardiac output (CO) and SVV from both NICOM and Vigileo monitors throughout the cases were used for data analysis.

Results: Total 619 pairs of CO data, 209 pairs of SV data and 189 pairs of SVV data were analyzed with paired t test and correlation analysis. CO and SV values from Vigileo were significantly higher than those from NICOM ($P < 0.001$, $P < 0.001$). SVV values from NICOM were significantly higher than those acquired from Vigileo ($P < 0.001$). Correlation coefficients were 0.508 ($P < 0.001$) and -0.196 ($P = 0.005$), respectively, for CO and SV data from the 2 monitors. However, our data failed to demonstrate a linear correlation of SVV data between the 2 monitors (Fig. 1).

Conclusions: SVV acquired from NICOM had a tendency to underestimate the volume status compared with Vigileo.

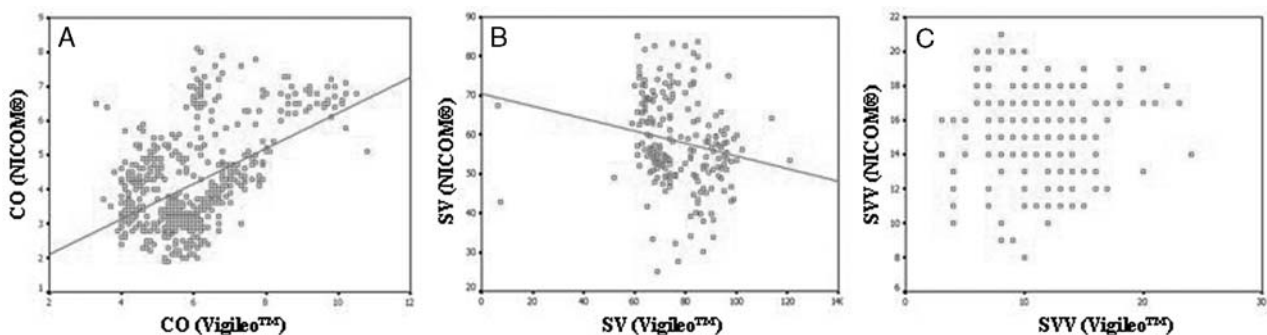


FIGURE 1. Comparison of cardiac output (CO), stroke volume (SV) and stroke volume variation (SVV) measured by NICOM^R and VigileoTM. (A and B) correlation coefficients were 0.508 and -0.196 , respectively, for CO and SV. (C) correlation coefficient for SVV data was 0.063 ($P=0.388$).

Discussion: SVV is a very useful clinical parameter for scalable fluid optimization and management in neurosurgical operations. NICOM monitor does not correlate with Vigileo monitor on SVV evaluation which indicates the need for further clinical validation study of its utility in neurosurgical anesthesia subspecialty.

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A109 Scalp Nerve Blockade Reduces Pain of Stereotactic Headframe Placement for Radiosurgery

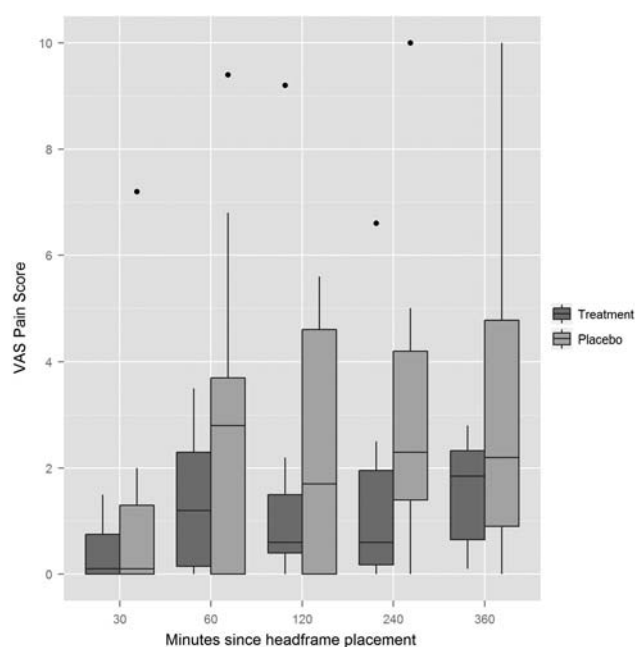
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Introduction: Radiosurgery for intracranial tumors and arteriovenous malformations uses tightly focused beams of radiation using a stereotactic head frame which is attached to the patient's skull with pins. Most commonly, before headframe placement, the anesthesiologist administers intravenous sedation and the surgeon infiltrates local anesthetic at the pin sites. After the headframe is placed and the patient awakens, generally the anesthesiologist is not in attendance for the duration of the treatment day, although the patient may have considerable discomfort associated with wearing the headframe. This randomized prospective study examined the effectiveness of the addition of scalp nerve blocks to relieve head pain in radiotherapy patients during the 4 to 12 hours they must spend conscious in the headframe. Prior clinical trials of craniotomy patients have found that a scalp block is superior to placebo and provides similar postoperative pain relief to intravenous morphine.

Methods: After institutional review board approval and written consent, adult patients presenting for elective radiosurgery of an intracerebral lesion requiring headframe placement were enrolled in this study. Patients were excluded if they were unable to give informed consent, understand the Visual Analogue Scale (VAS), had an allergy to local anesthetic, an incompletely healed craniotomy scar, or a known coagulopathy. We recruited 27 patients, 14 male and 13 female patients, aged 57 ± 14.8 years. After local anesthetic was infiltrated at pin sites and the headframe attached, the anesthesiologist performed bilateral supraorbital and greater occipital nerve blocks using a blinded syringe containing either saline or 0.5% bupivacaine with epinephrine. The patient recorded their head pain on a VAS at 30 minutes, 1 hour, 2, 4 hours, 6 hours, 8 hours, and 12 hours after headframe placement or until the headframe was removed. Patients were also able to receive lidocaine gel, lidocaine injections, and oral pain medication (oxycodone/acetaminophen or ibuprofen) as needed for pain during the treatment day.

Results: Of the 27 enrolled patients, 14 received placebo and 13 received local anesthetic for their nerve blocks. The placebo group had similar age, sex ratio, and ASA scores compared with the treatment group ($P > 0.05$ for each category.) Due to few procedures lasting more than 6 hours, pain scores after 6 hours were dropped from the analysis. The proportion of moderate-to-severe pain measurements ($VAS \geq 4$) was less in the nerve block group than the placebo group (3.13% vs. 27.4%; odds ratio, 0.091; 95% confidence interval 0.010–0.800; $P = 0.031$). Figure 1 displays the box plots of pain scores for each group up to the 6 hour time interval.

Conclusions: Cranial nerve blocks significantly decrease moderate to severe pain in radiosurgery patients. A 2009 study of craniotomy patients suggests that scalp blocks prevent neuropathic postcraniotomy pain, but whether this is a significant issue in radiosurgery patients has not been established. Future studies should examine whether neuropathic head pain is a significant cause of morbidity after radiosurgery and if so, whether scalp blocks are effective as a preventative measure.



A110 Intraoperative Blood Pressure Management During Microelectrode Recording in Deep Brain Stimulation Surgery for Parkinsonism

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Deep Brain Stimulator (DBS) placement is effective in improving the quality of life in Parkinsonian patients.¹

The procedure involves 2 stages: The insertion of the microelectrode(s) through burr hole in a target area found by intraoperative recording; and internalization of the lead(s) and implantation of the programmable pulse generator.² The testing requires an awake and co-operative patient to show clinical result as well as adverse effects of DBS placement. During this

TABLE 1. Demographic Data

No of patients	54
Age(mean±SD)	61.1±9.9
Wt (mean±SD)	87.055±16.9
Ht(mean±SD)	172.3±9.2
Sex(M:F)	35:19
ASA Grade	
ASA1 (n) %	(19) 35.1%
ASA2 (n)%	(35) 64.8
Comorbidity	
Hypertension	N = 28
Diabetes	51 %
	N = 11
	20.3 %

ASA=American Society of Anesthesiologists, F= Female, Ht= Height, M= Male, No.= number, SD= Standard deviation Wt=Weight.

period, a rise in blood pressure is common and thought to be hazardous in increasing the risk of intracerebral hemorrhage.³ The optimal level of blood pressure is not well defined; in our institution per surgeon request a systolic pressure of 130 mm Hg is used as a cutoff point. The fluctuating level of sedation necessary during the procedure can result in an unpredictable and changing blood pressure, necessitating the use of antihypertensives and vasopressors. Hypertension can be resistant and difficult to control requiring intravenous rapid acting antihypertensives. Fluctuation of the blood pressure can increase the risk of cerebral hematoma, cardiac ischemia and hypo and hyper perfusion of vital organs.⁴ In this retrospective analysis we reviewed the intraoperative pharmacological management of blood pressure in 54 Parkinsonian patients undergoing DBS placement.

TABLE 2. Baseline and Maximum Blood Pressure Measurements

	Baseline	Maximum
SBP mean $\pm$ SD(Range)	128.96 $\pm$ 16.2 (90-157)	146.12 $\pm$ 16.6 (110-190)
DBP mean $\pm$ SD(Range)	75.59 $\pm$ 9.8 (56-104)	80.25 $\pm$ 11.6 (60-110)
MAP mean $\pm$ SD(Range)	89.53 $\pm$ 10.2 (68-120)	96.35 $\pm$ 18 (85-130)

SBP= Systolic Blood pressure, DBP= Diastolic Blood Pressure, MAP= Mean Arterial Blood Pressure

TABLE 3. Number and Types of Antihypertensive Used Intraoperatively

	No of patients	Percentage of patients
Patients who needed antihypertensive medications	53/54	98.1%
Antihypertensive modalities used	0=1 1=13 2=24 3=11 4=5	1.8% 24% 44.5% 20.5% 9.2%
Type of antihypertensive medication:		
Labetalol		
Patients used	50	92.5%
Average dose mg $\pm$ SD, range	63.98 $\pm$ 42.37(0-175)	
Hydralazine		
Patients used	33	61.1%
Average dose mg $\pm$ SD(Range)	9.6 $\pm$ 9.4(0-6)	
Nitroglycerine		
Patients used	15	27.7%
Average dose mg, $\pm$ SD(Range)	434.6 $\pm$ 430.7 (80-1440)	
Esmolol		
Patients needed	10	18.5%
Average Dose (mg) $\pm$ SD (Range)	66 $\pm$ 43.5 (30-130)	

TABLE 4. Types and Doses of Vasopressors Used

Patients received (No), %	(28) 51.8%
Ephedrine:	
Patients received (No), %	(12), 22.2%
Average Dose (mg)	28.18 $\pm$ 15
Range	(5-50)
Phenylephrine	
Patients received	(16) 29.6%
Average	215.6 $\pm$ 185
Range	50-800

Methods: Retrospective chart review of 54 Parkinson patients during microelectrode recording (MER) of DBS placement. All patients had Monitored Anesthesia Care (MAC) along with injection of local anesthetics. The anesthetic regimen included Dexmedetomidine and/or Propofol infusion. Demographic data, number, types and doses of antihypertensive and vasopressor medications used to maintain an optimal blood pressure were recorded.

Results: Table 1 shows the demographic data of patients included in the chart review. Although only 28 (51%) were diagnosed with hypertension, 53 (98%) needed antihypertensive therapy. There was a wide range of maximum systolic as well as diastolic and mean blood pressure (Table 2). In the majority of cases 2 to 3 types of antihypertensives were used. Labetalol was the most common antihypertensive used followed by hydralazine (Table 3). In 28 patients (51.8%) at least 1 dose of vasopressor was used (Table 4).

Conclusions: There is a wide variation in blood pressure in patients undergoing DBS placement necessitating frequent use of antihypertensive as well as vasopressors at different stages of procedure. A well designed prospective study is needed to identify the risk factors causing blood pressure fluctuations.

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A111 Is Fast Track After Deep Brain Stimulation Surgery (DBS) a Safe Alternative to "Overnight Intensive Care"?

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Background and Goal of Study: A 24 hours-stay in the ICU is the common postoperative control for deep brain stimulation (DBS) procedures in many centers in our country. However, the low reported complications and limited postoperative care prompted us to analyze the impact of DBS fast-track postoperative care in our hospital.

Materials and Methods: We retrospectively analyzed patients who underwent DBS, divided in 2 periods: 2006 and 2007 to 2010 (when "fast track" started). Variables recorded: age, sex, comorbidity, diagnostic, procedure, anesthesia type, postoperative care (time, location), intra and postoperative complications and immediate postoperative brain imaging (BI).

Results and Discussion: Eighty-two patients (13 w, 69 m), age 56 $\pm$ 14 years underwent DBS, 19 during 2006 and 63 from 2007 to 2010. Diagnose: Parkinson disease (64), dystonia (10), essential tremor (7), corea (1). Bilateral DBS: 68 cases, unilateral: 14. Conscious sedation: 67 patients, general anesthesia: 6 and 9 both (for pacemaker implant), with remifentanyl and propofol TCI in 77 patients and remifentanyl in 5. Intraoperatively, 23 patients had high blood pressure values requiring treatment. One patient presented complete AV block requiring pacemaker, 2 patients developed nausea and another patient presented intraoperative hemiparesia. Patients from 2006 were admitted in an overnight recovery room for 18 $\pm$ 2 hours. One patient presented hemiparesia and 2 agitation. The patient with AV block remained in ICU for 3 days. Patients from 2007-2010 were admitted in the Postoperative Care Unit (PCU) for 8.8 $\pm$ 5 hours. The patient with intraoperative hemiparesia was admitted in the ICU. One patient developed hemiparesia during the first 6 hours and transferred to the ICU. Three patients remained in the PCU for more than 17 hours due to an elevated number of brain tracks or unavailability of immediate BI. One patient developed a hemiparesia 48 hours after the procedure and another developed deep venous thrombosis on the 3rd day.

A CT scan or a MR was performed in 45 and 37 patients respectively before they were transferred to the ward. Three patients showed a pallid (2) or subthalamic (1) hematoma and 26 a frontal pneumocephalus without clinical repercussion. The patient who developed a late hemiparesia had a normal postoperative CT-scan and showed a hematoma around the electrode on the CT performed when the neurological signs appeared 2 days later.

Conclusion(s): The small percentage of serious perioperative events occurred in the first 6 hours. In our experience a fast track protocol is a safe postoperative standard in DBS surgery.

A112 Factors Predicting Intraoperative Hypertension During Deep Brain Stimulator Insertion in Parkinson's Disease

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Background: Deep brain Stimulation (DBS) is a relatively new procedure for control of motor symptoms in Parkinson disease. The procedure is performed in a staged manner. Although many publications have certified the safety of micro electrode recording (MER) which is the first stage, it is possible that an increased risk of brain hemorrhage could occur secondary to acute hypertension intraoperatively.^{1,2} In this study we sought to evaluate preoperative factors predicting the requirements of antihypertensive medication intraoperatively. As withdrawal of anti-Parkinsonism medications has been thought to be associated with rebound hypertension, we sought to evaluate if there was a dose related effect.

Methods: We retrospectively reviewed the records of 54 MER placements for Parkinson disease. All patients were done under monitored anesthesia care and local anesthesia. We looked at presence of factors like essential hypertension based on Joint National Commission (JNC) 7 classification, Diabetes Mellitus, levodopa dose and the motor score for staging of the disease (mUDPRS).

Results: Fifty-three out of 54 patients needed antihypertensive medications intraoperatively. The commonest antihypertensives used were Labetalol, hydralazine, nitroglycerin and esmolol. The average number of boluses was 9. Presence of preoperative hypertension and Diabetes mellitus were independent factors which associated with the amount of antihypertensive medications used intraoperatively. Patients who had been on angiotension converting enzyme inhibitors and or ARBs had significantly ($P < 0.02$) more antihypertensive boluses (average of 16.5) intraoperatively when compared with those who were controlled with other medications. Patients in JNC 7 categories 3 and 4 received the highest number of boluses (average of 14.26). Patients who required more than 9 boluses (median) received an average of 870 mg of carbi-levodopa and those who received 8 or less boluses received an average of 707.285 mg which was significant. Patients older than 61 years (median age) needed significantly more boluses than those less than 61 years. There was no correlation between the mUDPRS or sex and the number of boluses of antihypertensive medications used. Standard t tests were used to compare the values between the groups.

Conclusions: Incidence of HTN in patients undergoing DBS for Parkinson disease is very high. Other factors associated with refractory intraoperative hypertension were diabetes and age above 61. Continuation

of ACE inhibitors in this subgroup of patients could be a useful recommendation. Intraoperative rebound hypertension is more likely associated with baseline high requirement of antihypertensive medications, and not necessarily to rebound from withdrawal from levodopa. Hence excellent control of blood pressure preoperatively would be very important in the management of these patients.

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A113 Prospective Reaudit of Postoperative Pain, Nausea, and Vomiting in Neurosurgical Patients

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Introduction: Effective pain management is a fundamental quality of care issue in surgical patients. Inadequate treatment of postoperative pain and postoperative nausea and vomiting (PONV) may result in physical and psychological outcomes¹ and decrease patient satisfaction with care.² Patient satisfaction is a key indicator of performance.³ To assess current practice and to compare with an audit done in September 2007 in our hospital we decided to conduct an audit on postoperative pain management in our hospital.

Methods: After registering with the audit department data was collected prospectively from elective neurosurgical patients in the first 24 hours over a period of 2 months from 13th of September until the 5th of November 2010. Data included demographic details, nature of operative procedure, analgesics and antiemetics administered, pain and postoperative nausea and vomiting (PONV) scores and patient satisfaction.

Results: We followed-up 206 patients in this period. Multimodal analgesia was used in all the patients. On arrival in recovery 34% of craniotomies, 30% of spines, 40% of endovascular patients were in moderate to severe pain as against 30%, 40%, and 20%, respectively in 2007. On discharge from recovery, 6% of craniotomies, 30% of endovascular, 1% of spines were in moderate pain as against 4.7%, 2%, 0%, 3 years ago. Patient satisfaction with analgesia in both audits averaged at 84.5%. Additional data exclusive to this audit include 26% of patients in the wards that were in moderate to severe pain in the first 24 hours postoperatively; PONV scores (scale 0 to 2, where 0 = no nausea and vomiting, 1 = nausea only, 2 = nausea and vomiting) of > 0 in 7% on arrival, 13% at 30 minutes and 5% on discharge from Recovery. The worst PONV scores on the ward were > 0 in 2% of patients. Total 72% of patients were satisfied with the management of PONV postoperatively.

Conclusions: The findings indicate that there has been little change in practice since 2007. It is clear that postoperative pain and PONV in neurosurgical patients are a significant entity in our center. Increased use of opioids, antiemetics, local infiltration blocks, possibly NSAIDs may have to be considered. We hope to reaudit in the future after implementing guidelines.

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A114 A Novel Method for IV Induction of General Anesthesia With Remifentanyl and Propofol in Neurosurgical Patients: Pharmacokinetic Models and Time Profile

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	HT YES NO	HT	DIABETES		ANXIETY	
NO OF PTS	28	25	11	43	17	38
AVG dose LABETALOL (mg)	67.86	59.8	96.82	55.58	100	64.2
AVG HYDRAL	12.46	6.63	15.23	8.29	20	8.92
AVG NITRO	144.28	95.38	234.54	91.62	1120	53.16
NO OF BOLUS	13.42	8.5	18.2	9.2	12.25	10.5
P VALUE	0.05		0.05		Not significant	
STD DEV	7.3	5.21	7.34	5.41	9.03	5.75

Pt = patients

	General results (n=50)	Brain Surgery (n= 21)	Spine Surgery (n=29)
Time for LOC (sec)	179 ± 48	139 ± 31	208 ± 35
Amount of Propofol (mg)	99 ± 27	77 ± 17	115 ± 19
Dose of propofol (mg/kg)	1.5 ± 0.4	1.2 ± 0.2	1.7 ± 0.3
Cp modified Marsh Pk Model (µg/mL)	4.4 ± 0.9	3.8 ± 0.9	4.8 ± 0.7
Effect-site Schnider Pk Model (µg/mL)	4.7 ± 1.2	3.6 ± 0.9	5.5 ± 0.8

Introduction: Classically, induction dose of propofol have been calculated based on the weight, sometimes adjusted to age. When using Target Controlled Infusion (TCI) of propofol, the target concentration that causes loss of consciousness depends of the pharmacokinetic model used and should be adapted to the patient, surgery and other covariates. In this study, the target concentrations of propofol calculated by 2 different PK models needed to induce LOC were determined in patients submitted to brain or spine surgery after a constant rate of propofol infusion.¹

Methods: Data from 50 patients (ASA 1-3, Glasgow 15) submitted to brain (B group) or spine (S group) surgery were prospectively recorded. Using Fresenius Primea pump, anesthesia induction was started with an effect-site (ES) TCI of remifentanyl of 2.5 ng/mL. One minute after the remi ES target was achieved, a 1% propofol infusion was started at a constant infusion of 200 mL/h until LOC. LOC was defined by lack of eye opening in response to name calling and to a tap on the forehead. Time since beginning of propofol infusion until LOC, the amount of propofol given and hemodynamic data were recorded. Tivatrainer 8.0 software was used to calculate the concentrations predicted by Schnider² and Marsh (with short thkeo)³ PK models.

Results: Surgeries were brain (21) and spinal (29). Both groups were similar regarding demographic covariates. Patients were 51 ± 17 years old, ASA 1-3, 24 female. Data are presented in Table 1. There weren't significant changes in the MAP and HR before induction and at LOC.

Conclusions: Although TCI anesthesia with propofol have been implemented many years ago, there are still some doubts about which target concentrations should be used to induce LOC and how much propofol is required to achieve it. Our study provides a guide for those who want to use TCI in brain and spinal surgery, using Schnider or modified Marsh PK models and also helps those who are using TIVA with manual-control infusion of propofol. These results also confirm previous evidence⁴ showing that patients submitted to brain surgery have less anesthetic requirements.

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A115 Magnesium Improves Neuronal Bioenergetics After Excitotoxic Glutamate Exposure

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Background: Extracellular glutamate is elevated in the brain during and following induced circulatory arrest and likely contributes to neurologic injury. Exposure of neurons in vitro to excitotoxic glutamate inhibits mitochondrial respiration which may contribute to neuronal death. This study tested the hypothesis that magnesium protects against excitotoxic neuronal metabolic impairment.

Methods: A Seahorse Bioscience Metabolic Flux Analyzer was used to measure respiration by primary cultures of rat cortical neurons in the absence or presence of 3mM magnesium sulfate. Cells were also pretreated with 3mM magnesium sulfate for 1 hour before measurements. After a baseline was obtained, cells were exposed to 100 µM glutamate for 30 minutes, followed by glutamate receptor antagonists.

Maximum respiration was then observed after the addition of a protonophore uncoupler and pyruvate.

Results: Exposure of neurons to glutamate for 30 minutes significantly reduced the maximal rate of O₂ consumption to 49% ± 5.6% of control ($P < 0.05$, $n = 7$). Magnesium sulfate pretreatment (3 mM) significantly protected against this glutamate-induced attenuation of maximal respiratory capacity (68% ± 8.3% of control in the presence of magnesium sulfate, $P < 0.05$, $n = 7$). In addition, exposure to glutamate caused an expected initial transient respiratory stimulation due to the increased ATP turnover required to restore ionic equilibrium following glutamatergic sodium and calcium influx. Pretreatment with magnesium sulfate enhanced the transient increase in O₂ consumption after glutamate exposure. Experiments performed in the absence or presence of the NMDA antagonist MK801 suggest that the stimulation of initial, glutamate-induced respiration by magnesium sulfate is not due to NMDA receptor antagonism.

Discussions: This study is the first to demonstrate that clinically relevant elevation of extracellular magnesium above normal levels protects cultured neurons against metabolic inhibition caused by excitotoxic levels of glutamate. In addition to protecting against loss of maximal respiratory capacity, magnesium also maximizes the immediate respiratory stimulation elicited by glutamate in a NMDA receptor independent manner. This work was supported by NIH R01NS064978 and NIH P01HD016596.

A116 Effect of Magnesium Sulfate on Subarachnoid Aneurysmal Hemorrhage Using Serum S100B Protein as a Prognostic Marker

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The beneficial mechanisms of magnesium on brain damage are multimodal. Increased S100B levels in cerebrospinal fluid (CSF) and serum are related to reactive gliosis and astrocyte death. Therefore, S100B has been used as a marker of glial damage in several neurological disorders.

Methods: Thirty patients treated by aneurysmal clipping. Patients were randomized within 4 days after subarachnoid hemorrhage (SAH). Divided into 2 groups of 15 patients each: group I received magnesium infusion and group II the control group. Magnesium sulfate therapy consisted of a bolus injection and continuous infusion over 24 hours. Started within 4 days after SAH and continued until 14 days after occlusion of the aneurysm. World Federation of Neurological Surgeons, Fisher, and Glasgow outcome scores in the intensive care unit, discharge and at 3 months were evaluated. Samplings of serum S100B were performed at admission, day 1, day 3 and day 7 postoperative.

Results: There is statistically significant difference between both groups as regard the second reading of the S100B (day 1 postoperative), ($P < 0.05$). There is no statistically significant difference between both groups as regards outcome at 3 months using clinical status and S100B values.

Conclusions: The use of magnesium sulfate in subarachnoid hemorrhage has a neural protective action but on the laboratory level. Further study is needed to check its effect on the outcome.

A117 Posttreatment With NH-Phenothiazine Does Not Improve Neuronal Outcome After Cerebral Ischemia in Rats

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Introduction: Pre-ischemic treatment with NH-phenothiazine has demonstrated protection against ischemia induced neurodegeneration in vivo.¹ The present study characterizes the effects of post-ischemic

treatment with NH-phenothiazine administered 30 minutes and 4 hours after cerebral ischemia in rats.

Methods: After institutional animal care committee approval, male Sprague-Dawley rats were randomly assigned to the following groups ($n = 13/\text{group}$): high-dose NH-phenothiazine (20 mg/kg), low-dose NH-phenothiazine (10 mg/kg), vehicle control, ischemia control, sham operation. Fasted rats were anesthetized with sevoflurane and cerebral ischemia was induced by bilateral common carotid artery occlusion and hemorrhagic hypotension (mean arterial pressure = 35 mm Hg for 14 min). NH-phenothiazine was administered intravenously 30 minutes and 4 hours after ischemia. On postoperative days 3, 5, 7, and 9 learning behavior was assessed with the Barnes maze test. On postoperative day 10 animals were euthanized in deep anesthesia and their brains were removed. Viable neurons in the hippocampus were counted in 7 μm coronal sections stained with cresyl violet. Statistics: Kruskal-Wallis Test, Mann-Whitney Test; $P < 0.05$.

Results: The number of viable neurons [Median (Q1/Q3)/ROI] was reduced after cerebral ischemia compared with sham-operated animals [112 (100/135) vs. 321 (267/357)]. NH-phenothiazine did not increase the number of viable neurons after ischemia compared with vehicle-control [high-dose 123 (119/179), low-dose 130 (111/159) vs. vehicle 117 (99/151)]. In line with this, NH-phenothiazine did not affect neurocognitive performance in the Barnes Maze test.

Discussion: NH-phenothiazine did not increase the post-ischemic number of viable neurons in the hippocampus after cerebral ischemia compared with vehicle treated controls. Likewise, post-ischemic learning behavior in the hippocampus-dependent Barnes maze test was not improved by NH-phenothiazine. In contrast, a previous study demonstrated neuronal protection by pre-ischemic treatment with 10 mg/kg NH-phenothiazine.¹ The discrepancy between the effects of pre-ischemic and post-ischemic treatment with NH-phenothiazine may be explained by in vitro-data that show an antioxidative mechanism of NH-phenothiazine.² Oxidative processes take part in the early ischemic signaling cascade. Therefore, the therapeutic benefit of NH-phenothiazine may be limited to pre-ischemic administration.

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A118 tPA Variant tPA-S481A Prevents Impairment of Cerebral Autoregulation During Hypotension After TBI By Protecting Cerebrovasodilation Mediated by NMDA Receptor Activation and Prostaglandins

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Introduction: Cerebrovasodilation induced by NMDA receptor activation is impaired after fluid percussion brain injury (FPI) due to up-regulation of endogenous tissue plasminogen activator (tPA). Impaired NMDA-induced dilation contributes to disturbed cerebral autoregulation during hypotension after FPI. Pial artery dilation during hypotension is also mediated by release of dilator prostaglandins (PG) such as PGE2 and PGI2; dilation to both PGs is blunted after FPI. We hypothesize that generation of a mutant tPA that competes with wild type tPA for binding to NMDA receptors and protects it from subsequent cleavage/activation will improve outcome after TBI. tPA-S481A is a variant with single mutation in the catalytic site that maintains its docking site and capacity to bind to the NMDA receptor, but cannot activate/cleave the receptor. This study investigated the ability of tPA-S481A to prevent vascular dysregulation after TBI.

Methods: Lateral FPI was produced in anesthetized piglets. Pial artery reactivity was measured via a closed cranial window. Data ($n = 5$) were analyzed by repeated measures ANOVA, with significance determined at P less than 0.05.

Results: NMDA induced pial artery dilation was reversed to vasoconstriction after FPI. tPA-S481A (1 mg/kg iv) administered 30 minutes after FPI blocked NMDA-mediated vasoconstriction and restored the response to vasodilation. Similarly, pial artery dilation during hypotension and in response to PGE2 and PGI2 was blunted after FPI but

protected by tPA-S481A. However, papaverine induced cerebrovasodilation was unchanged by FPI and tPA-S481A.

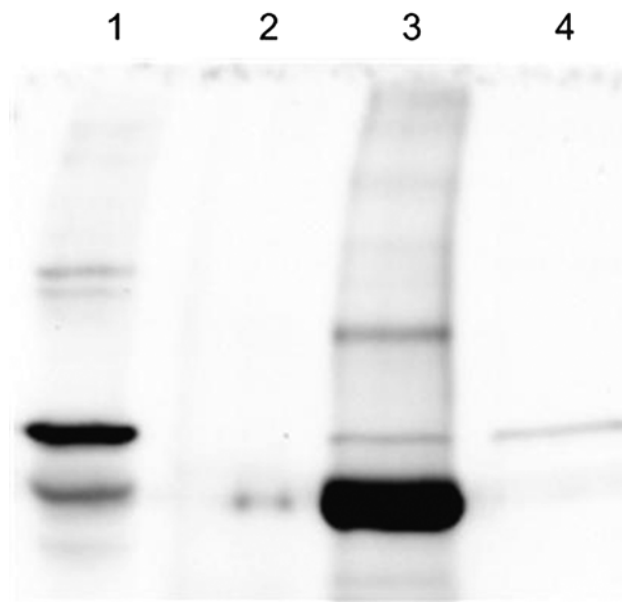
Discussion: These data indicate that tPA-S481A protects against impaired cerebrovasodilation induced by NMDA and PGs after FPI. As tPA-S481A also helped to maintain autoregulation during hypotension after FPI, these data provide insight into the underlying pathways involved in prevention of cerebrovascular dysregulation postinsult. In the neurovascular unit, normalization of cerebral hemodynamics after CNS injury is a key therapeutic target, as CBF contributes to outcome. These data suggest that administration of tPA-S481A may help provide neuroprotection when given after TBI.

A119 GCOM1-Interneixin- α Y2H Interaction Confirmed in Rat Brain Suggests Novel NMDA Receptor-Initiated Pathway

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Introduction: We are investigating the GCOM1/GRINL1A gene¹ as a likely new target for neuroprotection because anti-GCOM1 antibodies protected rat cortical neuronal cultures against NMDA toxicity.² Cloning and expression of a cDNA encoding a novel, large human Gcom15 protein (765aa) with NR1 NMDAR subunit cDNA in heterologous cells demonstrated reciprocal co-immunoprecipitation (co-IP) of NR1 protein by anti-GCOM1 Ab and vice versa.³ In parallel studies, at least 28 GCOM1-interacting genes (Gints) were discovered by yeast 2-hybrid (Y2H) screening with Gcom1 cDNA.⁴ We hypothesized that some GCOM1-Gint interactions would participate in novel CNS pathways, for example, NMDAR- > GCOM1- > Gints- > other mediators that might be relevant to neuroprotection and neurodegenerative diseases (NDDs). Therefore we performed IP and co-IP (in both directions) on rat brain cortical lysates with anti-GCOM1 and anti-interneixin- α (gene symbol INA) Ab.

Methods: Rabbit anti-Gcom1 (GenBank AY207007) polyclonal anti-peptide Ab were produced using newly synthesized T423-Q440



IP: GCOM1 (1585). Stain: INA

(GenScript, Piscataway, NJ). Rabbit anti-INA came from Millipore. Western blots with each Ab detected the expected bands in rat brain cortical lysates, 105 kDa for Gcom15 and 66 kDa for internexin- α . IP and co-IP of 100 μ g aliquots of rat brain cortical lysates were performed with G-Dynabeads. WBs were stained with appropriate Ab and detected with chemiluminescence. Y2H methods were described previously.⁴

Results: In initial Westerns (not shown), anti-Gcom1 T423-Q440 IgG from rabbit 1585 displayed strong IP of the 105 kDa Gcom15 bands from rat brain extracts and was used for subsequent co-IP studies. Unprecipitated brain lysate (50 μ g) revealed an intense 66 kDa INA band (Fig. 1, lane 1). Anti-Gcom1 pulled down substantial internexin- α immunoreactivity (lane 3, IP), although some INA appeared in the flow-through (lane 4). Incomplete co-IP may be due to suboptimal amounts of antibody in the IP step and/or INA variants that do not bind to Gcom15. G-Dynabeads alone (no Ab control) show no bands corresponding to INA (lane 2). Interestingly, anti-INA also co-IPs a larger, novel band (85 kDa) not present in the flow-through. The reverse co-IP (IP with anti-INA, stain with anti-GCOM1) revealed a weak Gcom15 band (not shown).

Discussion: The 105 kDa Gcom15 protein co-IPs with the NR1 NMDAR subunit, appears to enhance NMDAR functions (2,4) and has now been shown to interact with the Y2H Gint INA in rat brain, suggesting the pathway NMDAR- $\rightarrow$ GCOM1- $\rightarrow$ INA- $\rightarrow$ other mediators. It is critical to test for Y2H interactions in mammalian systems to confirm probable physiologic relevance and rule out artifact. INA occurred 9 times in Y2H screens (the most frequent Gint), forms a component of the axonal cytoskeleton, and plays roles in nervous system development and neuronal intermediate filament inclusion body disease,⁵ a NDD characterized by progressive early-onset dementia. Future studies will address whether and how this putative pathway mediates changes in gene expression seen after NMDAR activation.

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A120 The Effects of Insulin, Glucagon, Glutamate, and Glucose Infusion on Blood Glutamate and Plasma Glucose Levels in Naive Rats

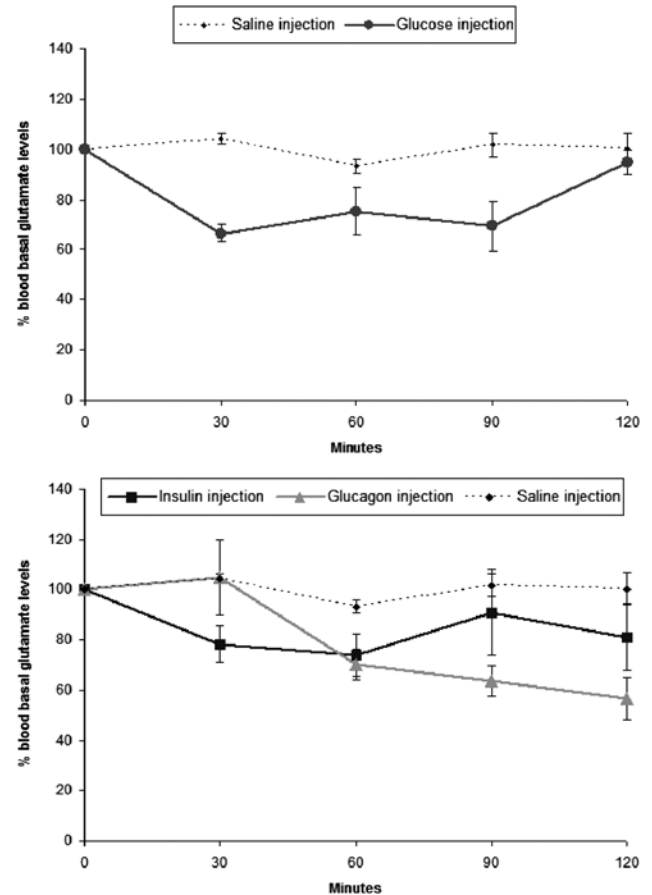
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Introduction: Elevated levels of glutamate in the brain's fluids, in the context of several neurodegenerative conditions, are associated with a worsened neurological outcome. A positive correlation between brain glutamate levels and blood glutamate levels was previously demonstrated, and stress was shown to influence blood glutamate levels. The purpose of this study was to investigate the effects of glucose, insulin, and glucagon on rat blood glutamate levels.

Materials and Methods: Rats were injected with either 1 ml/100 g intravenous isotonic saline (control), 150 mg/1 mL/100 g intravenous glucose, 75 mg/1 mL/100 g intravenous glutamate, 50 μ g/100 g glucagon intraperitoneally, or 0.2 UI/100 g insulin intraperitoneally. Blood samples were subsequently drawn at 0, 30, 60, 90, and 120 minutes for determination of blood glutamate and glucose levels.

Results: A significant decrease in blood glutamate levels was observed at 30 minutes after injection of glucose ($P < 0.05$, Fig. 1) and at 30 and 60 minutes after injection of insulin ($P < 0.05$, Fig. 2), and at 60 ($P < 0.01$), 90 and 120 minutes after injection of glucagon ($P < 0.05$; Fig. 2). As expected, plasma glucose levels were elevated after infusion of glutamate and glucose but were decreased after injection of insulin.

Conclusions: The results of this study demonstrate that glucose, insulin, and glucagon significantly reduce blood glutamate levels. The effect of insulin is immediate and transient while the effect of glucagon is delayed



but long-lasting, suggesting that the sensitivity of pancreatic glucagon and insulin-secreting cells to glutamate are dependent on glucose concentration. The results of this study provide insight into blood glutamate homeostasis, and may assist in the implementation of new therapies for brain neuroprotection from excess glutamate.

A121 Pharmacologic Perioperative Brain Neuroprotection: A Systematic Review and Meta-analysis of Randomized Clinical Trials

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Background: Perioperative cerebral damage may be associated with surgery and anesthesia. Several pharmacological and nonpharmacological therapies for perioperative brain neuroprotection have been tested in clinical trials with conflicting results.

Objective: To assess the efficacy of tested therapies in terms of reduction of new neurologic deficit and/or cognitive decline and mortality rate.

Methods: We searched the Cochrane Library; Ovid MEDLINE; Ovid EMBASE (1980 to December 2010) and reference lists of all retrieved articles. Prospective randomized controlled clinical trials using any pharmacologic therapies for perioperative brain neuroprotection were selected.

Results: Of 5904 identified studies, 25 randomized trials met our inclusion criteria. Tested therapies were: lidocaine, thiopental, S(+)-ketamine, propofol, nimodipine, GM1 ganglioside, lexicapant, glutamate/aspartate and xenon remacemide, atorvastatin, magnesium, erythropoietin, piracetam, rivastigmine, pegorgotein and 17 β -estradiol. Global RR analysis showed no differences in the rate of new neurologic

deficit/cognitive decline between treated and nontreated patients. Drug-based RR analysis showed that magnesium sulfate effectively reduces postoperative neurologic deficit/cognitive decline (RR: 0.48 95%CI 0.27-0.85, I² = 0%, P = 0.01).

According to the results reported in the original studies, evidences of benefits are reported for 5 out of the 17 drugs tested: lidocaine, remacemide, erythropoietin, piracetam and atorvastatin. Thiopental, ketamine, propofol, nimodipine, GM1Ganglioside, lexipafant, glutamate/aspartate enriched crystalloid cardioplegia, xenon-based anesthesia, rivastigmine, pegorgotein and 17 β -estradiol have not shown neuroprotectant effects.

Conclusions: Magnesium sulfate can effectively reduces perioperative cerebral damage and some studies support a neuroprotectant effect of lidocaine, remacemide, erythropoietin, piracetam and atorvastatin. More extensive research and wider consensus on the methodological approach for perioperative neurocognitive functional evaluation is needed.

A122 Anesthetic Neurotoxicity in Hippocampal Neurons is Caused by Unfolded Protein Stress

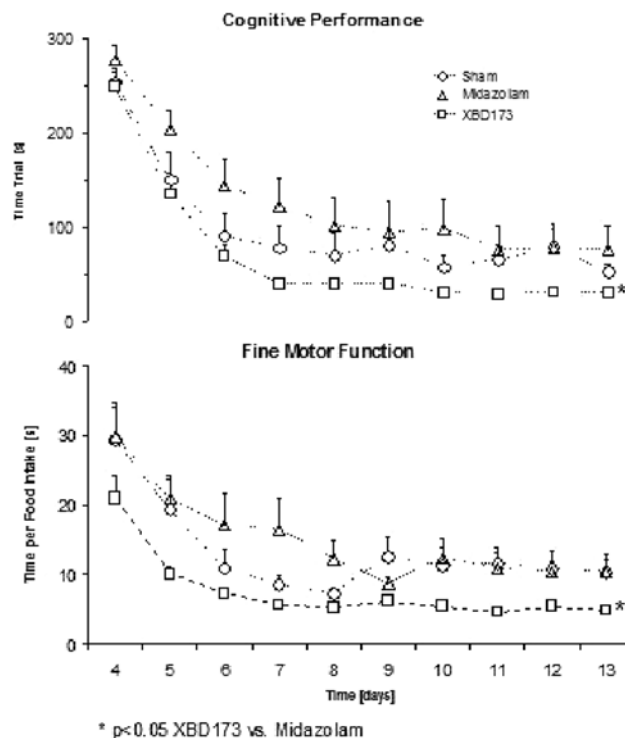
Alexandra M. Birch, BS, Pablo Gabatto, BS, David Polhemus, BS, Philip Bickler, MD, PhD. *University of California, San Francisco.*

The commonly used anesthetics isoflurane, sevoflurane and propofol have both neuroprotective and neurotoxic properties, with dose and durations of exposure determining pro-survival or pro-apoptotic outcome. We suggest that volatile anesthetic neurotoxicity is caused by endoplasmic reticulum stress, manifesting as accumulation of unfolded proteins, protein ubiquitinylation and neurodegeneration caused by a pro-apoptotic unfolded protein response (UPR). To test this hypothesis we exposed hippocampal slice cultures (HSCs) from postnatal day 7 rats to clinically relevant doses of anesthetics and measured region-specific neuron death with uptake of the viability dyes propidium iodide or Sytox. Isoflurane (1.3 MAC, 3%), Sevoflurane (1.3 MAC, 6%) and Propofol (3 μ M, approx 1 MAC equivalent) all increased neuron death in CA1, CA3 and dentate regions for exposure durations of 6 hours. The anesthetic-exposed cultures expressed significantly higher levels of activated caspase-12, which is a pro-apoptotic marker specific to the ER. Western blots on slice culture total protein lysates probed for both lysine 48-linked polyubiquitin proteins specifically marked for degradation and mono/poly ubiquitinated proteins showed increased levels of both. Tunicamycin (10 μ M), a toxic causing unfolded protein stress, caused the cleavage of X-box RNA (part of the IRE1 signaling involved in the UPR), and caused protein ubiquitinylation. Taurourodeoxycholic acid (175 μ M), a bile salt that reduces ER stress, eliminated the toxicity of isoflurane and sevoflurane and also reduced the accumulation of ubiquitinated proteins. In addition, isoflurane-exposed slice cultures had increased levels of amyloid-like protein aggregates, as detected with fluorescence labeling with the amyloid-specific dye thioflavin T (1 μ M X 20 min labeling of PFA fixed cultures). We suggest that the neurotoxic qualities of anesthetics involve a primary stress to the ER that causes protein misfolding that can explain anesthetic-induced neurodegeneration. These findings may have significant implications for anesthetic exposures to cells that already express abnormal protein regulation, such as in Alzheimer disease and Parkinson disease.

A123 Neurotoxicity in Infant Rats: the Effects on the Neurocognitive, Behavioral and Histologic Outcome of XBD173 and Midazolam

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Background: Benzodiazepines, commonly used for their anxiolytic and sedative effects, are discussed to exert neurotoxic effects particularly on the developing brain.¹ In a recent translational study, the selective and



high-affinity translocator protein (18 kD, TSPO) ligand XBD173 enhanced GABAergic neurotransmission in brain slices via the induction of neurosteroidogenesis and showed antipanic and anxiolytic efficacy in rodents and in humans in the absence of sedation.² To elucidate whether XBD173 might be an alternative drug to the use of benzodiazepines in infants without showing neurotoxic effects, we investigated the effects of XBD173 and midazolam on neurocognitive and fine motor outcome in infant rats.

Methods: With institutional review board approval, 30 newborn self-bred male Sprague-Dawley rats were randomized to receive either midazolam (0.9 mg/kg), XBD173 (10 mg/kg) or a placebo (n = 10 for each group). The respective substance was administered orally for 14 days, starting on the 7th postnatal day. Starting on the 4th day after the last drug administration, rats were subjected to the modified hole board test for 10 days to determine cognitive and fine motor outcome. Data were analyzed with ANOVA ($P < 0.05$).

Results: Cognitive performance, assessed by the time to perform the trial, and fine motor function, assessed by the time for food intake, were significantly better after XBD173 compared with midazolam (see Fig. 1).

Conclusions: In this study in infant rats XBD173 improved cognitive and fine motor function compared with midazolam, suggesting that XBD173 does not exert neurotoxic effects on the developing brain. However, further investigations are needed to study the effect of XBD173 on the developing brain, also when combined with other anesthetics.

Summary: In this study in infant rats cognitive and fine motor function were better after XBD173 compared with midazolam, suggesting that TSPO ligands are promising candidates as anxiolytic and sedative drugs with less severe side effects than benzodiazepines.

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A124 Differential Neurotoxic Effects of Isoflurane and Ketamine in the Fetal and Neonatal Macaque

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We have found that exposure for 5 hours to isoflurane or ketamine anesthesia triggers widespread neuroapoptosis in the neonatal (postnatal day 6; P6) rhesus macaque brain.^{1,2} Slikker and colleagues have reported that exposure of the fetal (gestation age 120 d; G120) rhesus macaque to ketamine for 24 hours causes neuroapoptosis confined to the frontal cortex. Neuroapoptosis in other regions of the fetal brain was not detected, and shorter durations of exposure were not studied.³ We exposed fetal G120 macaque brains to either isoflurane or ketamine for 5 hours and compared the results with anesthesia-related neurotoxicity observed in P6 macaques.

Methods: With institutional approval, neonatal (P6) and pregnant (G120) rhesus macaques ($n \geq 4$ per treatment condition) were exposed for 5 hours to either ketamine or isoflurane, or to no anesthesia (controls). The anesthetized animals received endotracheal intubation and were mechanically ventilated, and vital signs were monitored and controlled. General anesthesia was performed using the same methods, precautions and monitors that would be employed for administering anesthesia to humans. Ketamine was administered by intravenous infusion and isoflurane by inhalation at a dose/concentration sufficient to maintain a surgical plane of anesthesia for the entire 5 hour period (no movement to mosquito-clamp pinch to all extremities q 30 min). Three hours later, neonatal and fetal (after cesarean section) brains were perfused with fixative and studied immunohistochemically using markers of apoptosis (activated caspase 3) and cell death (silver stain). A comprehensive quantitative evaluation was achieved by counting all apoptosis-positive profiles in a series of sections selected at approximately 2 mm intervals across the entire brain.

Results: In the G120 fetal brain, the number of apoptotic neurons after ketamine-exposure and isoflurane-exposure was 4.95 fold and 4.17-fold greater, respectively, than in G120 controls. In the P6 neonatal brain we found that the number of apoptotic neurons in ketamine-exposed and isoflurane-exposed neonatal brains was 3.82-fold and 13.45-fold greater, respectively, than in P6 controls. This indicates that the toxic impact of ketamine declined to a moderate degree between the fetal and neonatal period, whereas the neurotoxic impact of isoflurane increased dramatically during that same period. The pattern of neuroapoptosis in the fetal brain after 5 hours' exposure to either ketamine or isoflurane was widespread, which is in contrast to previous findings after a 24-hour exposure of the fetal brain to Ketamine.³

Conclusions: Exposure of the fetal or neonatal rhesus macaque brain to either ketamine or isoflurane for a duration of 5 hrs is sufficient to induce a significant neuroapoptosis response at either age. The severity of the response to ketamine decreased moderately between the fetal and neonatal period, whereas the severity of the response to isoflurane increased dramatically during that same age interval. The pattern of neuroapoptosis was widespread, especially in the fetal brains.

A125 Morphine Exposure During the Brain Growth Spurt Does Not Impair Dendritic Differentiation and Synaptogenesis in the Rat Medial Prefrontal Cortex

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Background: Mu opioid receptors are expressed on excitatory synapses of the cerebral cortex, and in vitro evidence suggests that morphine signaling through these receptors modulates synapse stability. Little is known, however, about the impact of morphine exposure on neuronal circuitry development during the brain growth spurt in vivo. In this study, we investigated this issue by focusing on dendritic arbor and spine development of layer 5 pyramidal neurons in the rat medial prefrontal cortex (mPFC).

Methods: Wistar rats received either single bolus or chronic exposure to morphine (1 or 10 mg/kg intraperitoneally). In the single bolus group, morphine was administered at postnatal day (PND) 7 or 15, corresponding respectively to early and peak phases of cortical synaptogenesis. Chronic administration of morphine consisted in daily injections of this drug either from PND 7 to PND 15 or from PND 15 to PND 20. The effects of these experimental paradigms on neuronal apoptosis were evaluated using activated caspase-3 immunohistochemistry. The impact on neuronal cytoarchitecture was assessed using iontophoretic injections of Lucifer Yellow into layer 5 pyramidal neurons of the mPFC. Tracing of dendritic tree was carried out using the NeuroLucida station, whereas dendritic spines were analyzed by confocal microscopy.

Results: Intraperitoneal injections of 1 mg/kg morphine induced light sedation, while exposure to 10 mg/kg resulted in moderate sedation, opioid-like changes in respiratory pattern and discrete respiratory acidosis at any developmental stages applied. Caspase 3 immunohistochemistry revealed no changes in the number of apoptotic cells in the cerebral cortex between control and morphine-treated animals when tested 4 hours post-administration at any stages and dosing regimens. Neither acute nor chronic administration of morphine affected intense dendritic arbor development or spinogenesis.

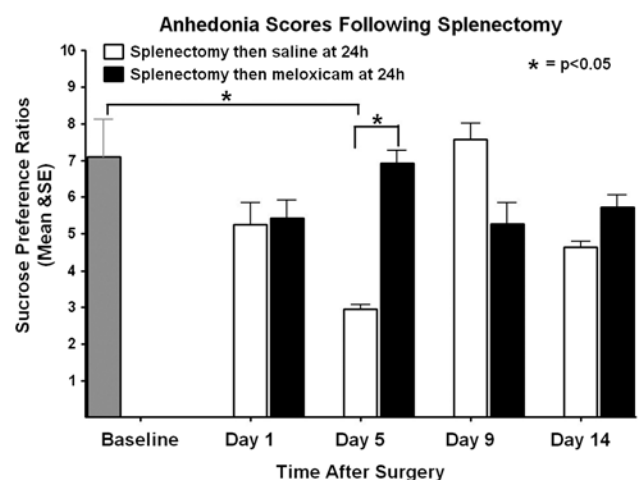
Conclusions: These new results suggest that, in contrast to the majority of general anesthetics, exposure to morphine at physiologically relevant concentrations during the brain growth spurt does not affect neuronal survival, differentiation and synaptogenesis.

A126 Depressive Behavior in Adult Mice After Splenectomy is Ameliorated by the COX-2 Inhibitor Meloxicam

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Introduction: Mood disorders are related to both cognitive deficits and inflammation. Previously we found that after splenectomy, adult mice developed neuro-inflammation and short-term memory dysfunction that were both relieved by the NSAID meloxicam (MEL) at 24 hours.¹ We tested the hypothesis that anhedonia, the inability to experience pleasure, would present in a similar manner after splenectomy. Anhedonia is a hallmark of depression and the sucrose preference test is a well standardized test for depressive behavior in rodents.

Methods: After IACUC approval 20 Swiss Webster male mice (35 to 45 g, 6 to 8 wk old) were randomized into 2 groups: (1) splenectomy then intraperitoneal saline (SAL) at 24 hours and (2) splenectomy then intraperitoneal MEL at 24 hours. All mice were trained before surgery according to our sucrose preference protocol. Mice were kept singly with



tap water freely available from 2 graduated cylinders. Every fourth night for 2 weeks animals were fluid restricted before being given a choice of either 2% sucrose or water from the cylinders for 1 hour in the morning. Baseline sucrose preference ratios were established as the ratio of sucrose to water consumption. On day 0 splenectomy was carried out under 1.5% isoflurane in 30% O₂/70% N₂ with a 1 cm incision, vessel ligation with 6-0 silk suture, and closure with 4-0 silk. Testing sucrose preference ratios were then established on days 1, 5, 9 and 14.

Results: We used the Kruskal Wallis test to reject the null hypothesis that the distributions for baseline, d5 MEL, d5 SAL were equal ($P = 0.02$). Then using pair wise tests (2-tailed t test), we rejected the null hypothesis that d5 MEL = d5 SAL ($P < 0.01$), and that baseline = d5 SAL ($P = 0.01$). Splenectomy caused anhedonia in a delayed transient manner. Sucrose preference was at baseline on day 1, was absent by d5 and restored to baseline by d9 and 14. MEL prevented anhedonia on day 5.

Conclusions: An inflammatory process along with its resolution and treatment may explain these findings. Splenectomized adult mice developed a time course of depressive behavior ameliorated by meloxicam that was similar to our previous findings for short-term memory dysfunction and neuro-inflammation.¹ Cytokines trigger and sensitize brain stress circuits and may impair mood related dopaminergic and serotonergic neurotransmission.² COX-2 therapy was effective in our study at 2 hours when cytokine release peaks after brain damage in rodents.³ These findings underscore the need for research into the association of mood disorders and delirium in the postoperative period.

References:

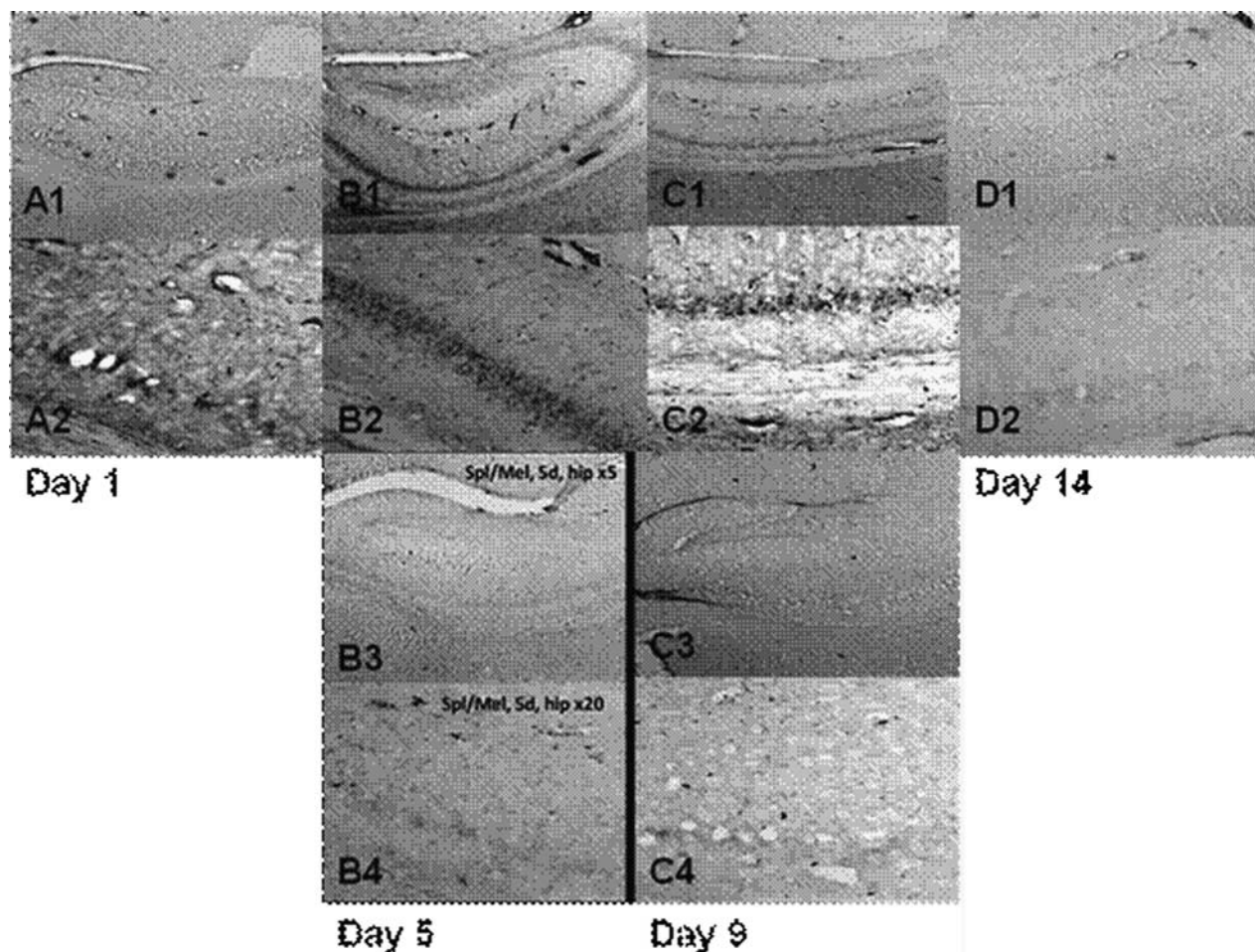
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A127 The COX-2 Inhibitor Meloxicam Ameliorates Short-term Memory Dysfunction and Modulates Microglia Activation After Splenectomy in Adult Mice

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Introduction: Surgery induced neuro-inflammation is implicated in cognitive disorders. We tested whether the NSAID meloxicam (MEL) 24 hours after splenectomy could suppress microglia activation and prevent short-term memory (STM) impairment in adult mice. Splenectomy causes neuro-inflammation and cognitive dysfunction.¹

Methods: After IACUC approval 63 Swiss-Webster mice (35 to 45 gm, 6 to 8 wk) randomized: (1) splenectomy, test on day 1; (2) splenectomy,



test d5; (3) splenectomy, test d9; (4) splenectomy, test d14; (5) splenectomy then intraperitoneal MEL at 24 hours, test d5; (6) splenectomy, MEL 24 hours, test d9. Splenectomy under isoflurane, 1 cm incision, ligation with 6-0 silk, closure with 4-0 silk. The object recognition test (ORT) was used to test STM. On d5 training, mouse placed in arena with identical objects for 15 minutes. For 3 minutes testing 1 hour later a novel object replaced familiar one. STM scores were defined as the ratio of time spent exploring the novel object to time spent exploring both objects. Statistics used Wilcoxon and Kruskal Wallis rank tests. Three additional mice per group had histology and histochemistry (Tomato lectin staining) to visualize activated microglia. **Results:** Hippocampal photomicrographs show microglia activation shown by staining and morphology changes: normal on d1 (A1, A2), inflammation on d5 (B1, B2) and d9 (C1, C2), resolved d14 (D1, D2). On d5 (B3, B4) and d9 (C3, C4), MEL significantly reduced activation (Fig. 1). STM was normal on d1, dysfunctional d5 through d9 and resolved by d14. MEL restored STM on d5 and d9.

Conclusions: Splenectomized mice developed neuro-inflammation, particularly in the hippocampus and concurrent short-term memory dysfunction. The NSAID meloxicam at 24 hours post-splenectomy resolved both. These effects may depend on the cytokine time course modulated by meloxicam via COX-2 inhibition. Cytokine release peaks at 24 hours in rodents after brain damage.² Cytokine activation in the periphery stimulates neuro-inflammation across the blood-brain barrier.³ Research into cytokine targeted therapies for postoperative delirium is warranted.

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